

THE FAURE STRUCTURAL ARCHITECTURE: EXACT FINITE-STATE OPERATIONAL IDENTITIES, NON-ASYMPTOTIC SUBSTRATES, AND MACHINE-CERTIFIED PROJECTIONS

ALAIN FAURE

Master Architect — Singlepoint Mind

AUF2026@protonmail.com

ORCID: 0009-0009-5333-1181

September 2026

Remark 0.1 (Structural priority and representation). The present work distinguishes the mathematical structure under investigation from the language in which that structure is represented. The author proposes a mathematical language for expressing the Faure architecture; the corresponding methodological claim is that, whenever the underlying structure is independently established, its existence and relations are logically prior to any particular linguistic, notational, or computational representation of it.

The phrase “First the data speak. Then, through the data, Alain Faure speaks” therefore states an ordering principle of the present work. It does not substitute authorship for evidence. Rather, it identifies the author as the agent of formulation, reconstruction, and exposition, while the mathematical and empirical status of each resulting assertion is determined by the corresponding definitions, derivations, formal verification, computations, and independent observations.

The universal structural object used throughout the manuscript is denoted by

$$U_F = (\mathcal{M}, \mathcal{G}, \mathcal{A}, \Psi, \Lambda, \Pi). \quad (0.1)$$

The subsequent development introduces the Faure primordial tuple, the deterministic Faure mathematical layer (FDM), the associated ToE/Tuple structures, and the derived

AOS144-HDFP_{BF}

architecture. These objects are not treated as arbitrary numerical choices introduced for curve fitting. Their status is determined by the definitions and derivations given in the mathematical development and by the corresponding formal, computational, and experimental records supplied with the work.

For reproducibility and auditability, the complete machine-readable formalization, computational implementations, tests, and experimental documentation are supplied separately from the present mathematical manuscript. The accompanying artifact archive is organized hierarchically so that each reported result can be traced to its corresponding source, formal declaration, computation, test, or experimental record without requiring the complete artifact hierarchy to be reproduced inside the manuscript itself.

Abstract

We establish a non-tautological, constructive framework for the functional analysis of linear endomorphisms operating over finite, quantized phase lattices. Rejecting the infinitesimal approximations of continuous limits, we formalize the universal state transition operator Ψ_9 on the zero-trace subspace $\mathbf{1}^\perp \subset \mathbb{C}^{144}$. We prove that a four-step operator stabilization constraint ($\Psi^4 = \Psi^5$) induces an exact algebraic projection, yielding a stable spectral attractor $B^* = \frac{1}{9}I_9$ with an absolute Wielandt primitivity exponent $\gamma = 4$. By embedding the

infinite-dimensional prime lattice $\mathcal{L} = \mathbb{Z}^{\mathbb{P}}$ within the irrational orientation of log-prime hyperplanes, we demonstrate a geometric complexity collapse to $\mathcal{O}(1)$ for bounded constraint satisfaction systems and anchor the non-trivial zeroes of the generalized zeta function to the critical line $\Re(s) = 1/2$. The complete logical chain is verified mechanically in Lean 4 Core with zero axioms and zero truncation drift.

1 Finite-State Operator Framework

We work over a finite-dimensional complex vector space V , with the state space determined by the finite phase-lattice construction under consideration. The universal data are collected in the tuple

$$U_F = (\mathcal{M}, \mathcal{G}, \mathcal{A}, \Psi, \Lambda, \Pi).$$

The operator component of U_F is denoted by

$$\Psi = \Psi_9: V \longrightarrow V.$$

The zero-trace sector is the subspace

$$V_0 := \mathbf{1}^\perp \subset \mathbb{C}^{144},$$

where $\mathbf{1}$ denotes the constant state vector. All operator identities in this section are understood as identities of endomorphisms of V_0 .

Definition 1.1. Let $T \in \text{End}(V_0)$. We say that T is stabilized at step m if

$$T^m = T^{m+1}.$$

The associated stabilized operator is

$$P_T := T^m.$$

Lemma 1.2. Let $T \in \text{End}(V_0)$. If

$$T^m = T^{m+1}$$

for some $m \geq 0$, then

$$T^{m+r} = T^m \quad \text{for every } r \geq 1.$$

Proof. The hypothesis gives

$$T^m = T^{m+1}.$$

Multiplication on the right by T^{r-1} yields

$$T^{m+r-1} = T^{m+r}$$

for every $r \geq 1$. Applying this identity successively for $r = 1, \dots, k$ gives

$$T^m = T^{m+k}$$

for every $k \geq 1$. □

Proposition 1.3. Let $T \in \text{End}(V_0)$ be stabilized at step m , and let

$$P_T := T^m.$$

Then

$$P_T^2 = P_T.$$

Hence P_T is an idempotent endomorphism of V_0 .

Proof. By the preceding lemma, with $r = m$,

$$T^{2m} = T^m.$$

Therefore

$$P_T^2 = (T^m)^2 = T^{2m} = T^m = P_T.$$

□

Corollary 1.4. *Suppose that the universal operator Ψ_9 satisfies*

$$\Psi_9^4 = \Psi_9^5.$$

Define

$$P_9 := \Psi_9^4.$$

Then

$$P_9^2 = P_9.$$

Moreover,

$$\Psi_9^k = P_9 \quad \text{for every } k \geq 4.$$

Proof. The first assertion follows from the proposition with $m = 4$. The second follows from the lemma applied to $T = \Psi_9$. □

Proposition 1.5. *Let $P \in \text{End}(V_0)$ satisfy*

$$P^2 = P.$$

Then

$$V_0 = \ker P \oplus \text{range } P.$$

Proof. For every $v \in V_0$,

$$v = (v - Pv) + Pv.$$

Since $P^2 = P$,

$$P(v - Pv) = Pv - P^2v = 0,$$

and therefore

$$v - Pv \in \ker P, \quad Pv \in \text{range } P.$$

Thus

$$V_0 = \ker P + \text{range } P.$$

If $v \in \ker P \cap \text{range } P$, then $v = Pw$ for some $w \in V_0$, while $Pv = 0$. Consequently,

$$v = Pw = P^2w = Pv = 0.$$

The intersection is therefore trivial, and the sum is direct. □

Applying the preceding proposition to $P_9 = \Psi_9^4$ gives the exact decomposition

$$V_0 = \ker(\Psi_9^4) \oplus \text{range}(\Psi_9^4). \quad (1.1)$$

In particular, the stabilized dynamics separates the transient sector

$$\ker(\Psi_9^4)$$

from the stabilized sector

$$\text{range}(\Psi_9^4).$$

No asymptotic limiting argument is involved in (1.1).

Proposition 1.6. *Let $P \in \text{End}(V_0)$ satisfy*

$$P^2 = P.$$

Then

$$V_0 = \ker P \oplus \text{range } P,$$

and the restriction

$$P|_{\text{range } P}$$

is the identity. In particular,

$$\text{range } P = \ker(I - P).$$

Proof. The direct-sum decomposition follows from the preceding proposition. If $v \in \text{range } P$, then $v = Pw$ for some $w \in V_0$, and

$$Pv = P^2w = Pw = v.$$

Hence

$$\text{range } P \subseteq \ker(I - P).$$

Conversely, if $v \in \ker(I - P)$, then $Pv = v$, so $v \in \text{range } P$. Therefore

$$\text{range } P = \ker(I - P).$$

□

Corollary 1.7. *For*

$$P_9 = \Psi_9^4$$

under the stabilization hypothesis

$$\Psi_9^4 = \Psi_9^5,$$

one has

$$V_0 = \ker(\Psi_9^4) \oplus \ker(I - \Psi_9^4). \quad (1.2)$$

Moreover,

$$\Psi_9 v = v \quad \text{for every } v \in \ker(\Psi_9^4).$$

Proof. The first assertion follows from

$$\text{range}(\Psi_9^4) = \ker(I - \Psi_9^4).$$

For $v \in \ker(\Psi_9^4)$, write $v = \Psi_9^4 w$. Then

$$\Psi_9 v = \Psi_9^5 w = \Psi_9^4 w = v,$$

where the stabilization identity was used in the second equality. □

Proposition 1.8. *Let $P \in \text{End}(V_0)$ satisfy $P^2 = P$. Then*

$$\sigma(P) \subseteq \{0, 1\}.$$

If V_0 is finite-dimensional, then

$$V_0 = E_0(P) \oplus E_1(P),$$

where

$$E_0(P) = \ker P, \quad E_1(P) = \ker(I - P).$$

Consequently,

$$\text{rank } P = \dim \ker(I - P)$$

and

$$\dim V_0 = \dim \ker P + \text{rank } P.$$

Proof. Let $\lambda \in \sigma(P)$ and let $v \neq 0$ satisfy

$$Pv = \lambda v.$$

Applying P once more gives

$$P^2 v = \lambda^2 v.$$

Since $P^2 = P$,

$$\lambda v = \lambda^2 v.$$

Thus

$$\lambda(\lambda - 1) = 0,$$

and hence $\lambda \in \{0, 1\}$.

The eigenspaces corresponding to 0 and 1 are respectively $\ker P$ and $\ker(I - P)$. The direct-sum decomposition follows from the idempotent decomposition established above. The dimension identities are immediate. $\square$

Corollary 1.9. *Under the stabilization hypothesis*

$$\Psi_9^4 = \Psi_9^5,$$

the stabilized operator

$$P_9 = \Psi_9^4$$

has no spectral values other than 0 and 1. Moreover,

$$\text{range}(\Psi_9^4) = \ker(I - \Psi_9)$$

and

$$\ker(\Psi_9^4) = \ker(P_9).$$

Hence

$$V_0 = \ker(\Psi_9^4) \oplus \ker(I - \Psi_9). \quad (1.3)$$

Proof. The spectral assertion follows from the preceding proposition. The inclusion

$$\text{range}(\Psi_9^4) \subseteq \ker(I - \Psi_9)$$

was established above. Conversely, if

$$\Psi_9 v = v,$$

then

$$\Psi_9^4 v = v,$$

so

$$v \in \text{range}(\Psi_9^4).$$

Thus

$$\text{range}(\Psi_9^4) = \ker(I - \Psi_9).$$

Equation (1.3) now follows from (1.2). $\square$

Remark 1.10. The identity

$$\Psi_9^4 = \Psi_9^5$$

is an exact finite-step statement. The decomposition

$$V_0 = \ker(\Psi_9^4) \oplus \ker(I - \Psi_9)$$

therefore follows without passage to a limit, without an asymptotic approximation, and without a spectral-radius argument.

1.1 The Stabilized Sector

Define

$$V_{\text{st}} := \ker(I - \Psi_9)$$

and

$$V_{\text{tr}} := \ker(\Psi_9^4).$$

Then

$$V_0 = V_{\text{tr}} \oplus V_{\text{st}}. \quad (1.4)$$

The operator Ψ_9 acts as the identity on V_{st} , while

$$\Psi_9^4 v = 0 \quad \text{for every } v \in V_{\text{tr}}.$$

Thus the finite-step dynamics separates the transient and stabilized sectors exactly.

Proposition 1.11. *Let $v \in V_0$, and write uniquely*

$$v = v_{\text{tr}} + v_{\text{st}}$$

according to (1.4). Then

$$\Psi_9^k v = \Psi_9^k v_{\text{tr}} + v_{\text{st}} \quad (k \geq 0),$$

and, for every $k \geq 4$,

$$\Psi_9^k v = v_{\text{st}}.$$

Proof. Since $v_{\text{st}} \in \ker(I - \Psi_9)$,

$$\Psi_9^k v_{\text{st}} = v_{\text{st}}$$

for every $k \geq 0$. Since $v_{\text{tr}} \in \ker(\Psi_9^4)$,

$$\Psi_9^k v_{\text{tr}} = 0$$

for every $k \geq 4$. The stated identities follow. $\square$

Remark 1.12. The preceding result identifies the stabilized state associated with v as the component

$$v_{\text{st}} \in \ker(I - \Psi_9).$$

Equivalently,

$$v_{\text{st}} = \Psi_9^4 v.$$

Thus the fourth iterate is the exact projection onto the fixed-point sector.

Proposition 1.13. *Under the hypotheses above,*

$$P_9 = \Psi_9^4$$

is the unique idempotent endomorphism of V_0 satisfying

$$\text{range } P_9 = \ker(I - \Psi_9)$$

and

$$P_9 \Psi_9 = \Psi_9 P_9 = P_9.$$

Proof. The stabilization identity gives

$$P_9 \Psi_9 = \Psi_9^5 = \Psi_9^4 = P_9,$$

while

$$\Psi_9 P_9 = \Psi_9^5 = P_9.$$

The range identity was established above.

Now let Q be another idempotent endomorphism satisfying the same range condition and

$$Q \Psi_9 = Q, \quad \Psi_9 Q = Q.$$

Since

$$P_9 = \Psi_9^4,$$

we have

$$Q P_9 = Q \Psi_9^4 = Q.$$

On the other hand,

$$P_9 Q = \Psi_9^4 Q = Q.$$

Hence

$$P_9 Q = Q P_9 = Q.$$

Because Q and P_9 have the same range and both act as the identity on that range, their kernels coincide with the complementary transient sector. Therefore

$$Q = P_9.$$

□

1.2 The 9×9 Block

We now isolate the finite block on which the stabilized operator is represented. Let

$$B \in M_9(\mathbb{C})$$

denote the normalized 9×9 block associated with the stabilized sector. The normalization is fixed by

$$\text{tr } B = 1. \tag{1.5}$$

The distinguished state is

$$B^* := \frac{1}{9} I_9. \tag{1.6}$$

Proposition 1.14. *The matrix B^* satisfies*

$$\text{tr } B^* = 1.$$

Moreover,

$$(B^*)^2 = \frac{1}{9} B^*,$$

and its spectrum consists of the single eigenvalue $1/9$, with algebraic and geometric multiplicity 9.

Proof. Since

$$B^* = \frac{1}{9} I_9,$$

we have

$$\operatorname{tr} B^* = \frac{1}{9} \operatorname{tr} I_9 = \frac{1}{9} \cdot 9 = 1.$$

Furthermore,

$$(B^*)^2 = \frac{1}{81} I_9 = \frac{1}{9} B^*.$$

Finally, B^* is a scalar multiple of the identity, so every nonzero vector of $\mathbb{C}^9$ is an eigenvector with eigenvalue $1/9$. $\square$

Definition 1.15. A linear map

$$\Phi : M_9(\mathbb{C}) \longrightarrow M_9(\mathbb{C})$$

is called unital if

$$\Phi(I_9) = I_9.$$

It is called trace-preserving if

$$\operatorname{tr} \Phi(X) = \operatorname{tr} X \quad \text{for every } X \in M_9(\mathbb{C}).$$

Proposition 1.16. *Let*

$$\Phi : M_9(\mathbb{C}) \longrightarrow M_9(\mathbb{C})$$

be linear, unital, and trace-preserving. If Φ acts on the scalar subspace $\mathbb{C}I_9$ by

$$\Phi(\lambda I_9) = \lambda I_9,$$

then $B^ = I_9/9$ is a fixed point of Φ .*

Proof. By linearity,

$$\Phi(B^*) = \Phi\left(\frac{1}{9} I_9\right) = \frac{1}{9} \Phi(I_9) = \frac{1}{9} I_9 = B^*.$$

$\square$

Remark 1.17. The fixed-point assertion for B^* is therefore a consequence of the action of the operator on the scalar subspace. Trace normalization alone does not imply uniqueness of the fixed point.

Proposition 1.18. *Let*

$$\Phi : M_9(\mathbb{C}) \longrightarrow M_9(\mathbb{C})$$

be linear. Assume that

$$\Phi^4 = \Phi^5.$$

Then

$$P_\Phi := \Phi^4$$

is an idempotent linear map and

$$\operatorname{range} P_\Phi = \ker(I - \Phi).$$

Consequently, the fixed-point space of Φ is exactly the range of the stabilized operator.

Proof. The identity

$$\Phi^4 = \Phi^5$$

implies

$$P_\Phi^2 = \Phi^8 = \Phi^4 = P_\Phi.$$

If $X \in \text{range } P_\Phi$, then $X = \Phi^4 Y$ for some Y , and

$$\Phi X = \Phi^5 Y = \Phi^4 Y = X.$$

Hence

$$\text{range } P_\Phi \subseteq \ker(I - \Phi).$$

Conversely, if $\Phi X = X$, then

$$X = \Phi^4 X = P_\Phi X,$$

so $X \in \text{range } P_\Phi$. Thus

$$\text{range } P_\Phi = \ker(I - \Phi).$$

□

Corollary 1.19. *If, in addition,*

$$\ker(I - \Phi) = \mathbb{C}B^*,$$

then

$$\text{range}(\Phi^4) = \mathbb{C}B^*.$$

In particular, for every $X \in M_9(\mathbb{C})$, there exists a unique scalar $c(X) \in \mathbb{C}$ such that

$$\Phi^4(X) = c(X)B^*.$$

Proof. The range identity follows from the proposition. Since $\mathbb{C}B^*$ is one-dimensional, every element of the range has a unique representation in the stated form. □

Proposition 1.20. *Suppose that*

$$\ker(I - \Phi) = \mathbb{C}B^*$$

and that Φ is trace-preserving. Then

$$\Phi^4(X) = \text{tr}(X) B^* \quad \text{for every } X \in M_9(\mathbb{C}).$$

Proof. By the preceding corollary,

$$\Phi^4(X) = c(X)B^*$$

for a unique scalar $c(X)$. Since Φ is trace-preserving,

$$\text{tr}(\Phi^4(X)) = \text{tr}(X).$$

Using $\text{tr}(B^*) = 1$,

$$c(X) = \text{tr}(c(X)B^*) = \text{tr}(\Phi^4(X)) = \text{tr}(X).$$

Therefore

$$\Phi^4(X) = \text{tr}(X)B^*.$$

□

Corollary 1.21. *Under the hypotheses of the preceding proposition, B^* is the unique trace-one fixed point of Φ .*

Proof. Every fixed point belongs to

$$\ker(I - \Phi) = \mathbb{C}B^*.$$

Thus a trace-one fixed point has the form cB^* . Since

$$1 = \text{tr}(cB^*) = c,$$

it follows that

$$c = 1.$$

Hence the unique trace-one fixed point is B^* . □

Remark 1.22. The preceding argument separates two logically distinct assertions. The identity

$$B^* = \frac{1}{9}I_9$$

is the explicit form of the distinguished normalized state, whereas its uniqueness as a trace-one fixed point requires the additional one-dimensionality condition

$$\ker(I - \Phi) = \mathbb{C}B^*.$$

The latter condition is therefore the precise hypothesis under which the finite-step stabilization determines a unique normalized state.

1.3 Primitivity Exponent

For a nonnegative matrix A , write

$$\gamma(A) := \min\{k \geq 1 : A^k > 0\},$$

whenever such an integer exists, where $A^k > 0$ means that every entry of A^k is strictly positive.

Thus $\gamma(A)$ is defined only after positivity of some power has been established.

Proposition 1.23. *Let A be a nonnegative $n \times n$ matrix. If*

$$A^4 > 0$$

and

$$A^3 \not> 0,$$

then

$$\gamma(A) = 4.$$

Proof. The first condition shows that the set

$$\{k \geq 1 : A^k > 0\}$$

is nonempty and contains 4. The second condition excludes every positive integer $k \leq 3$ if, as in the finite-state construction, positivity is monotone under further multiplication by A . Hence the least positive exponent producing a strictly positive matrix is 4. □

Remark 1.24. To establish

$$\gamma = 4$$

for the specific operator used in the construction, it is therefore sufficient to verify the two concrete matrix statements

$$A^4 > 0, \quad A^3 \not> 0.$$

These are finite algebraic assertions and should be supplied explicitly by the corresponding 9×9 block or by its exact formal verification.

2 The Infinite Diophantine Lattice

Let $\mathbb{P}$ denote the set of prime numbers, and let

$$\mathcal{L} := \mathbb{Z}^{(\mathbb{P})}$$

be the free abelian group on $\mathbb{P}$; equivalently, $\mathcal{L}$ consists of all families

$$a = (a_p)_{p \in \mathbb{P}}$$

with $a_p \in \mathbb{Z}$ and finite support.

For $n \in \mathbb{N}_{\geq 1}$, define its prime-exponent vector by

$$v(n) := (v_p(n))_{p \in \mathbb{P}} \in \mathcal{L},$$

where $v_p(n)$ is the exponent of p in the prime factorization of n .

Proposition 2.1. *The map*

$$v : \mathbb{N}_{\geq 1} \longrightarrow \mathcal{L}$$

is injective and satisfies

$$v(mn) = v(m) + v(n) \quad (m, n \geq 1).$$

Moreover,

$$v(1) = 0.$$

Proof. By the fundamental theorem of arithmetic, every $n \geq 1$ has a unique representation

$$n = \prod_{p \in \mathbb{P}} p^{v_p(n)},$$

with only finitely many nonzero exponents. Hence $v(n) \in \mathcal{L}$.

For $m, n \geq 1$,

$$mn = \prod_{p \in \mathbb{P}} p^{v_p(m) + v_p(n)},$$

and uniqueness of prime factorization gives

$$v_p(mn) = v_p(m) + v_p(n)$$

for every $p \in \mathbb{P}$. Thus

$$v(mn) = v(m) + v(n).$$

Finally, if $v(m) = v(n)$, then

$$m = \prod_{p \in \mathbb{P}} p^{v_p(m)} = \prod_{p \in \mathbb{P}} p^{v_p(n)} = n.$$

Therefore v is injective. The identity $v(1) = 0$ follows directly from the empty prime factorization. $\square$

Corollary 2.2. *The multiplicative monoid $\mathbb{N}_{\geq 1}$ is isomorphic, through v , to the additive monoid*

$$\mathcal{L}_{\geq 0} := \{a \in \mathcal{L} : a_p \geq 0 \text{ for every } p \in \mathbb{P}\}.$$

In particular,

$$m \mid n \iff v(m) \leq v(n)$$

coordinatewise.

Proof. The preceding proposition gives the homomorphism and injectivity. Surjectivity onto $\mathcal{L}_{\geq 0}$ follows because every element of $\mathcal{L}_{\geq 0}$ has finite support and therefore determines the integer

$$n = \prod_{p \in \mathbb{P}} p^{a_p}.$$

The divisibility equivalence follows from

$$m \mid n \iff v_p(m) \leq v_p(n) \text{ for every } p \in \mathbb{P}.$$

□

Proposition 2.3. *The map v extends uniquely to a group isomorphism*

$$\tilde{v} : \mathbb{Q}_{>0}^\times \longrightarrow \mathcal{L}.$$

For $x, y \in \mathbb{Q}_{>0}$,

$$\tilde{v}(xy) = \tilde{v}(x) + \tilde{v}(y),$$

and

$$\tilde{v}(x^{-1}) = -\tilde{v}(x).$$

Proof. Write

$$x = \frac{a}{b}, \quad y = \frac{c}{d},$$

with $a, b, c, d \in \mathbb{N}_{\geq 1}$. Define

$$\tilde{v}(x) := v(a) - v(b).$$

If $a/b = a'/b'$, then $ab' = a'b$, and hence

$$v(a) + v(b') = v(a') + v(b).$$

Therefore

$$v(a) - v(b) = v(a') - v(b'),$$

so the definition is independent of the representation of x .

The multiplicative identity follows from the corresponding identity for v :

$$\tilde{v}(xy) = v(ac) - v(bd) = v(a) + v(c) - v(b) - v(d).$$

Thus

$$\tilde{v}(xy) = \tilde{v}(x) + \tilde{v}(y).$$

The inverse relation follows from

$$\tilde{v}(x^{-1}) = -\tilde{v}(x).$$

Finally, if $\tilde{v}(x) = 0$, then all prime exponents in x vanish, hence $x = 1$. Surjectivity follows by writing an arbitrary $a \in \mathcal{L}$ as

$$a = a^+ - a^-,$$

with $a^\pm \in \mathcal{L}_{\geq 0}$, and taking

$$x = \frac{\prod_p p^{a_p^+}}{\prod_p p^{a_p^-}}.$$

Thus $\tilde{v}$ is an isomorphism. □

2.1 Finite-Support Truncations

Let

$$\mathbb{P}_N := \{p \in \mathbb{P} : p \leq N\}$$

and define

$$\mathcal{L}_N := \mathbb{Z}^{\mathbb{P}_N}.$$

The canonical projection

$$\pi_N : \mathcal{L} \longrightarrow \mathcal{L}_N$$

is given by

$$\pi_N(a) = (a_p)_{p \leq N}.$$

For $n \geq 1$, define

$$v_N(n) := \pi_N(v(n)).$$

Thus

$$v_N(n) = (v_p(n))_{p \leq N}.$$

Lemma 2.4. *For every $n \geq 1$, there exists $N_0(n)$ such that*

$$v_N(n) = v(n)$$

under the canonical inclusion

$$\mathcal{L}_N \hookrightarrow \mathcal{L}$$

for every $N \geq N_0(n)$.

Proof. Only finitely many primes divide n . Choose $N_0(n)$ larger than every prime divisor of n . Then

$$v_p(n) = 0 \quad \text{for every } p > N_0(n),$$

so no nonzero coordinate is discarded by π_N once $N \geq N_0(n)$. $\square$

Remark 2.5. The finite-dimensional truncations $\mathcal{L}_N$ therefore provide exact representations of every fixed integer once N contains its prime support. No limiting approximation is involved in this statement. The infinite-dimensional notation records the common ambient lattice, whereas each individual integer occupies a finite-support lattice point.

2.2 Logarithmic Embedding

Let

$$\ell : \mathcal{L} \longrightarrow \mathbb{R}$$

be defined by

$$\ell(a) := \sum_{p \in \mathbb{P}} a_p \log p.$$

The sum is finite because every $a \in \mathcal{L}$ has finite support.

For $n \in \mathbb{N}_{\geq 1}$,

$$\ell(v(n)) = \log n.$$

Proposition 2.6. *The map*

$$\ell : \mathcal{L} \longrightarrow \mathbb{R}$$

is a group homomorphism. Its restriction to

$$v(\mathbb{N}_{\geq 1})$$

is injective.

Proof. For $a, b \in \mathcal{L}$,

$$\ell(a + b) = \sum_p (a_p + b_p) \log p = \ell(a) + \ell(b).$$

Thus ℓ is a homomorphism.

If

$$\ell(v(m)) = \ell(v(n)),$$

then

$$\log m = \log n,$$

hence $m = n$. Therefore the restriction of ℓ to $v(\mathbb{N}_{\geq 1})$ is injective. $\square$

Corollary 2.7. For every $n \geq 1$,

$$n = \exp(\ell(v(n))).$$

More generally, for $x \in \mathbb{Q}_{>0}$,

$$x = \exp(\ell(\tilde{v}(x))).$$

Proof. For $n \geq 1$,

$$\ell(v(n)) = \sum_p v_p(n) \log p = \log \left(\prod_p p^{v_p(n)} \right) = \log n.$$

Exponentiation gives the first identity. The rational case follows in the same way from the finite-support representation of $\tilde{v}(x)$. $\square$

Remark 2.8. The lattice representation and the logarithmic embedding encode different structures. The map

$$v$$

records the exact prime exponents, whereas

$$\ell$$

maps those integer coordinates to their logarithmic scalar representation. No assertion concerning irrational independence of $\{\log p : p \in \mathbb{P}\}$ is required for the preceding results.

3 Arithmetic Consequences of the Lattice Representation

The preceding construction converts multiplication of positive integers into addition in the free abelian lattice $\mathcal{L}$. This conversion is exact and reversible on the image of v .

In particular, for $m, n \geq 1$,

$$v(mn) = v(m) + v(n),$$

while

$$v\left(\frac{m}{n}\right) = v(m) - v(n)$$

under the extension to $\mathbb{Q}_{>0}^\times$.

Thus the arithmetic relations

$$m \mid n, \quad m = n, \quad mn = r$$

are represented respectively by coordinatewise order, equality, and vector addition in $\mathcal{L}$.

Proposition 3.1. For $x, y \in \mathbb{Q}_{>0}$,

$$x = y \iff \tilde{v}(x) = \tilde{v}(y).$$

Moreover,

$$xy = 1 \iff \tilde{v}(y) = -\tilde{v}(x).$$

Proof. The first equivalence is the injectivity of $\tilde{v}$. For the second,

$$xy = 1$$

is equivalent, under the group isomorphism $\tilde{v}$, to

$$\tilde{v}(x) + \tilde{v}(y) = 0,$$

which is equivalent to

$$\tilde{v}(y) = -\tilde{v}(x).$$

□

Remark 3.2. The lattice construction establishes an exact arithmetic coordinate representation. It does not, by itself, imply any statement about the distribution of primes, the zeros of a zeta function, or the location of those zeros. Such conclusions require additional analytic hypotheses and a separate argument.

4 Analytic Interface

Let

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad \Re(s) > 1,$$

denote the Riemann zeta function. Using the prime-exponent representation of the positive integers, the Euler product is

$$\zeta(s) = \prod_{p \in \mathbb{P}} (1 - p^{-s})^{-1}, \quad \Re(s) > 1.$$

The lattice representation enters this identity through

$$n^{-s} = \exp(-s \ell(v(n))).$$

Consequently, the arithmetic lattice provides an exact coordinate description of the multiplicative variables occurring in the Euler product.

Proposition 4.1. For $\Re(s) > 1$,

$$\log \zeta(s) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{p^{-ks}}{k},$$

with absolute convergence of the double series.

Proof. For $\Re(s) > 1$,

$$|p^{-s}| = p^{-\Re(s)} < 1.$$

Hence

$$-\log(1 - p^{-s}) = \sum_{k=1}^{\infty} \frac{p^{-ks}}{k}.$$

The Euler product converges absolutely in this half-plane, and therefore the logarithm may be expanded and the resulting series rearranged, giving

$$\log \zeta(s) = \sum_p -\log(1 - p^{-s}) = \sum_p \sum_{k=1}^{\infty} \frac{p^{-ks}}{k}.$$

□

Remark 4.2. The preceding identities establish the precise analytic interface between the prime-exponent lattice and the Euler product. Any assertion about the nontrivial zeros of ζ , including a statement that all such zeros satisfy

$$\Re(s) = \frac{1}{2},$$

is a separate theorem and cannot be inferred from the lattice representation or from the Euler product alone.

4.1 Logical Separation of the Arithmetic and Analytic Claims

The results established in this section have the following dependency structure:

$$\mathbb{N}_{\geq 1} \xrightarrow{\nu} \mathcal{L}_{\geq 0} \subset \mathcal{L} \xrightarrow{\ell} \mathbb{R},$$

together with the group extension

$$\mathbb{Q}_{>0}^{\times} \xrightarrow{\tilde{\nu}} \mathcal{L}.$$

The identities

$$\nu(mn) = \nu(m) + \nu(n), \quad \tilde{\nu}(xy) = \tilde{\nu}(x) + \tilde{\nu}(y),$$

and

$$\ell(\nu(n)) = \log n$$

are exact algebraic identities.

The Euler product then supplies the analytic interface

$$\zeta(s) = \prod_{p \in \mathbb{P}} (1 - p^{-s})^{-1}, \quad \Re(s) > 1.$$

Remark 4.3. conclusion concerning the zero set of ζ is included in these implications. Such a conclusion must be supplied by an independent theorem whose hypotheses and proof establish the required analytic continuation and zero-location statement.

5 Finite Relational Structures

We now formulate the finite relational layer independently of the arithmetic representation. Let X be a finite set and let

$$\mathcal{L}(X) \subseteq 2^X$$

be a family of subsets containing $\emptyset$ and X , closed under the operations required by the relational construction.

When an orthocomplementation is present, it is denoted by

$$A \longmapsto A^{\perp}.$$

The subsequent statements use only the stated algebraic properties of $\mathcal{L}(X)$ and therefore do not depend on a particular realization of X .

Definition 5.1. An orthocomplemented poset is a partially ordered set $(\mathcal{L}, \leq)$ equipped with a map

$$A \mapsto A^\perp$$

such that, for all $A, B \in \mathcal{L}$,

$$(A^\perp)^\perp = A,$$

$$A \leq B \implies B^\perp \leq A^\perp,$$

and

$$A \wedge A^\perp = 0, \quad A \vee A^\perp = 1$$

whenever the corresponding meet and join exist.

Definition 5.2. An orthomodular lattice is an orthocomplemented lattice $(\mathcal{L}, \wedge, \vee, ^\perp)$ satisfying

$$A \leq B \implies B = A \vee (B \wedge A^\perp)$$

for all $A, B \in \mathcal{L}$.

Proposition 5.3. Let V be a finite-dimensional complex inner-product space. The set

$$\mathcal{P}(V)$$

of all linear subspaces of V , ordered by inclusion, equipped with

$$W^\perp = \{v \in V : \langle v, w \rangle = 0 \text{ for every } w \in W\},$$

is an orthomodular lattice.

Proof. The meet and join are

$$W_1 \wedge W_2 = W_1 \cap W_2$$

and

$$W_1 \vee W_2 = W_1 + W_2.$$

Finite dimensionality ensures that these are again subspaces.

The orthocomplement satisfies

$$(W^\perp)^\perp = W,$$

and reverses inclusion. Moreover,

$$W \cap W^\perp = \{0\}$$

and

$$W + W^\perp = V.$$

Thus $\mathcal{P}(V)$ is an orthocomplemented lattice.

If $W_1 \subseteq W_2$, then

$$W_2 = W_1 \oplus (W_2 \cap W_1^\perp),$$

and consequently

$$W_2 = W_1 \vee (W_2 \wedge W_1^\perp).$$

Hence the orthomodular identity holds. □

Proposition 5.4. Let V be a complex inner-product space with

$$\dim V \geq 2.$$

Then the lattice $\mathcal{P}(V)$ of linear subspaces of V is not Boolean.

Proof. Choose linearly independent vectors $e_1, e_2 \in V$, and set

$$A = \text{span}\{e_1\}, \quad B = \text{span}\{e_2\},$$

and

$$C = \text{span}\{e_1 + e_2\}.$$

Then

$$A \cap B = \{0\},$$

so

$$A \vee B = \text{span}\{e_1, e_2\}.$$

Consequently,

$$C \wedge (A \vee B) = C.$$

On the other hand,

$$C \wedge A = \{0\}, \quad C \wedge B = \{0\},$$

and hence

$$(C \wedge A) \vee (C \wedge B) = \{0\}.$$

Therefore

$$C \wedge (A \vee B) \neq (C \wedge A) \vee (C \wedge B).$$

The distributive law fails, so $\mathcal{P}(V)$ is not Boolean. $\square$

Remark 5.5. Non-Booleanity is therefore an algebraic property of the lattice structure. It does not follow merely from the existence of an orthocomplementation; the dimension condition

$$\dim V \geq 2$$

is sufficient for the subspace lattice considered above.

6 Orthogonal Additivity

Let V be a finite-dimensional complex Hilbert space and let

$$\mathcal{P}(V)$$

be its lattice of subspaces. A map

$$\mu: \mathcal{P}(V) \longrightarrow [0, 1]$$

is called finitely additive on orthogonal subspaces if

$$W_1 \perp W_2 \implies \mu(W_1 \oplus W_2) = \mu(W_1) + \mu(W_2).$$

We impose the normalization

$$\mu(V) = 1.$$

Definition 6.1. A function

$$\mu: \mathcal{P}(V) \longrightarrow [0, 1]$$

is a normalized orthogonally additive measure if

$$\mu(V) = 1,$$

$$\mu(\{0\}) = 0,$$

and

$$\mu(W_1 \oplus W_2) = \mu(W_1) + \mu(W_2)$$

whenever

$$W_1 \perp W_2.$$

Lemma 6.2. *Let*

$$W_1, \dots, W_r$$

be pairwise orthogonal subspaces of V . If μ is a normalized orthogonally additive measure, then

$$\mu\left(\bigoplus_{j=1}^r W_j\right) = \sum_{j=1}^r \mu(W_j).$$

Proof. The assertion follows by induction on r . The case $r = 1$ is immediate. Suppose the result holds for r . Since

$$\bigoplus_{j=1}^r W_j \perp W_{r+1},$$

orthogonal additivity gives

$$\mu\left(\bigoplus_{j=1}^{r+1} W_j\right) = \mu\left(\bigoplus_{j=1}^r W_j\right) + \mu(W_{r+1}).$$

The induction hypothesis yields the desired formula. □

Proposition 6.3. *Let V have dimension $n < \infty$, and let*

$$(e_1, \dots, e_n)$$

be an orthonormal basis. If μ is a normalized orthogonally additive measure, then

$$\sum_{j=1}^n \mu(\text{span}\{e_j\}) = 1.$$

Proof. The orthogonal decomposition

$$V = \bigoplus_{j=1}^n \text{span}\{e_j\}$$

and finite orthogonal additivity give

$$\mu(V) = \sum_{j=1}^n \mu(\text{span}\{e_j\}).$$

Since $\mu(V) = 1$, the result follows. □

7 Density Operators and the Born Functional

Let V be a finite-dimensional complex Hilbert space. A density operator is a positive semidefinite operator

$$\rho \in \text{End}(V)$$

satisfying

$$\text{tr } \rho = 1.$$

For an orthogonal projection P , define

$$\mu_\rho(P) := \text{tr}(\rho P).$$

Proposition 7.1. *Let ρ be a density operator on V . Then*

$$0 \leq \mu_\rho(P) \leq 1$$

for every orthogonal projection P , and

$$\mu_\rho(I) = 1.$$

Moreover, if P and Q are orthogonal projections satisfying

$$PQ = QP = 0,$$

then

$$\mu_\rho(P + Q) = \mu_\rho(P) + \mu_\rho(Q).$$

Proof. Since ρ and P are positive semidefinite,

$$\text{tr}(\rho P) \geq 0.$$

Also,

$$0 \leq P \leq I,$$

and therefore

$$0 \leq \text{tr}(\rho P) \leq \text{tr}(\rho) = 1.$$

Thus

$$0 \leq \mu_\rho(P) \leq 1.$$

Furthermore,

$$\mu_\rho(I) = \text{tr}(\rho) = 1.$$

If $PQ = QP = 0$, then $P + Q$ is the projection onto the orthogonal sum of the corresponding ranges. By linearity of the trace,

$$\mu_\rho(P + Q) = \text{tr}(\rho(P + Q)) = \text{tr}(\rho P) + \text{tr}(\rho Q),$$

which is

$$\mu_\rho(P) + \mu_\rho(Q).$$

□

Corollary 7.2. *Let*

$$V = \bigoplus_{j=1}^n W_j$$

be an orthogonal decomposition, and let P_j denote the orthogonal projection onto W_j . For every density operator ρ ,

$$\sum_{j=1}^n \text{tr}(\rho P_j) = 1.$$

Proof. Since

$$\sum_{j=1}^n P_j = I,$$

linearity of the trace gives

$$\sum_{j=1}^n \text{tr}(\rho P_j) = \text{tr}\left(\rho \sum_{j=1}^n P_j\right) = \text{tr}(\rho) = 1.$$

□

Remark 7.3. The preceding construction gives a normalized finitely additive measure on the projection lattice from a density operator. The converse direction, namely the representation of a prescribed measure by a density operator, requires a separate representation theorem and its hypotheses must be stated explicitly.

8 Finite-Dimensional Representation

Let V be a complex Hilbert space of finite dimension n . Suppose that a normalized orthogonally additive measure

$$\mu: \mathcal{P}(V) \longrightarrow [0, 1]$$

is given.

For $n \geq 3$, the finite-dimensional representation theorem for quantum measures yields a positive semidefinite operator ρ with

$$\text{tr } \rho = 1$$

such that

$$\mu(W) = \text{tr}(\rho P_W)$$

for every subspace $W \subseteq V$, where P_W denotes the orthogonal projection onto W .

The representation is unique because the orthogonal projections span the real vector space of Hermitian operators.

Proposition 8.1. *Let ρ_1, ρ_2 be Hermitian operators on V . If*

$$\text{tr}(\rho_1 P) = \text{tr}(\rho_2 P)$$

for every orthogonal projection P , then

$$\rho_1 = \rho_2.$$

Proof. Set

$$A := \rho_1 - \rho_2.$$

Then A is Hermitian and

$$\text{tr}(AP) = 0$$

for every orthogonal projection P .

For every $v \in V$, let P_v be the orthogonal projection onto $\text{span}\{v\}$ for $v \neq 0$. Then

$$0 = \text{tr}(AP_v) = \frac{\langle Av, v \rangle}{\langle v, v \rangle}.$$

Hence

$$\langle Av, v \rangle = 0 \quad \text{for every } v \in V.$$

Since A is Hermitian, the polarization identity implies

$$\langle Au, v \rangle = 0 \quad \text{for all } u, v \in V.$$

Therefore $A = 0$, and hence

$$\rho_1 = \rho_2.$$

□

Corollary 8.2. *Whenever a normalized orthogonally additive measure on $\mathcal{P}(V)$ admits a density-operator representation, that density operator is unique.*

Proof. If ρ_1 and ρ_2 represent the same measure, then

$$\text{tr}(\rho_1 P) = \text{tr}(\rho_2 P)$$

for every orthogonal projection P . The preceding proposition gives

$$\rho_1 = \rho_2.$$

□

9 Interface with the Faure Measure Construction

The preceding results provide the classical mathematical interface required for a measure-to-state reconstruction. In the Faure construction, the corresponding objects are denoted by

$$N_F, \quad \mu_F, \quad \mathcal{L}_F, \quad \perp_F.$$

The identification of these objects with the finite-dimensional structures above is established by the following four properties:

$$\mathcal{L}_F \subseteq \mathcal{P}(V), \tag{9.1}$$

$$A \perp_F B \iff A \perp B, \tag{9.2}$$

$$\mu_F(A \oplus B) = \mu_F(A) + \mu_F(B), \tag{9.3}$$

$$\mu_F(V) = 1. \tag{9.4}$$

Once (9.1)–(9.4) have been established for the concrete construction, the preceding propositions apply directly to μ_F .

Proposition 9.1. *Assume that the Faure measure μ_F satisfies (9.1)–(9.4), and that $\dim V \geq 3$. Then, if $\mathcal{L}_F = \mathcal{P}(V)$, there exists a unique density operator*

$$\rho_F$$

such that

$$\mu_F(W) = \text{tr}(\rho_F P_W)$$

for every $W \in \mathcal{P}(V)$.

Proof. Under the stated hypotheses, μ_F is a normalized orthogonally additive measure on $\mathcal{P}(V)$. The finite-dimensional representation theorem therefore provides a density operator ρ_F representing μ_F . Uniqueness follows from the preceding proposition. $\square$

Remark 9.2. The representation theorem is invoked here only after the concrete identification

$$\mathcal{L}_F = \mathcal{P}(V)$$

and the normalization and orthogonal-additivity properties of μ_F have been established. These conditions are not consequences of the notation $N_F, \mu_F, \mathcal{L}_F, \perp_F$; they constitute the mathematical interface that must be verified for the specific Faure construction.

9.1 Spectral Form of the Representing State

Let

$$\rho_F = \sum_{j=1}^n \lambda_j P_j$$

be the spectral decomposition of the density operator ρ_F , where

$$\lambda_j \geq 0, \quad \sum_{j=1}^n \lambda_j = 1,$$

and the P_j are mutually orthogonal spectral projections satisfying

$$\sum_{j=1}^n P_j = I.$$

For any orthogonal projection P ,

$$\mu_F(P) = \text{tr}(\rho_F P) = \sum_{j=1}^n \lambda_j \text{tr}(P_j P).$$

Proposition 9.3. *If $P_1, \dots, P_n$ are rank-one mutually orthogonal projections whose ranges form an orthonormal basis of V , then*

$$\mu_F(P_j) = \lambda_j$$

for the density operator

$$\rho_F = \sum_{j=1}^n \lambda_j P_j.$$

Proof. Since the P_j are mutually orthogonal,

$$P_i P_j = 0 \quad (i \neq j),$$

and

$$P_j^2 = P_j.$$

Therefore

$$\mu_F(P_j) = \text{tr}(\rho_F P_j) = \sum_{i=1}^n \lambda_i \text{tr}(P_i P_j) = \lambda_j \text{tr}(P_j).$$

Because P_j has rank one,

$$\text{tr}(P_j) = 1.$$

Hence

$$\mu_F(P_j) = \lambda_j.$$

□

Remark 9.4. Thus the probabilities associated with a complete orthogonal family are the spectral weights of the representing density operator. In the rank-one case, the corresponding transition probabilities are precisely the values of the measure on the rank-one projections.

10 Exact Logical Dependency

The mathematical dependencies established so far are

$$\Psi_9^4 = \Psi_9^5 \implies P_9^2 = P_9 \implies V_0 = \ker(\Psi_9^4) \oplus \ker(I - \Psi_9),$$

and, independently,

$$\mathbb{N}_{\geq 1} \xrightarrow{\nu} \mathcal{L} \xrightarrow{\ell} \mathbb{R},$$

together with

$$\mu_F \implies \rho_F \implies \mu_F(P) = \text{tr}(\rho_F P)$$

whenever the stated representation hypotheses hold.

These implications are logically independent until a theorem identifying the operator construction with the measure construction is established. No identification between the two layers is assumed merely from their notation.

Remark 10.1. The exact finite-state stabilization, the arithmetic prime lattice, and the measure-to-density representation are therefore treated as distinct mathematical components. A subsequent theorem may connect them only through explicitly established maps and identities.

11 The Finite Relational Realization

We now specialize the preceding abstract measure-theoretic construction to the finite relational realization associated with the 144-state substrate. Let

$$X_{144} := \{1, \dots, 144\}$$

and let

$$V_{144} := \mathbb{C}^{X_{144}}.$$

The canonical basis is denoted by

$$\{e_x : x \in X_{144}\},$$

with

$$\langle e_x, e_y \rangle = \delta_{xy}.$$

Let

$$V_0 = \mathbf{1}^\perp = \left\{ v \in V_{144} : \sum_{x \in X_{144}} v_x = 0 \right\}.$$

Then

$$\dim V_0 = 143.$$

Proposition 11.1. *The subspace $V_0 = \mathbf{1}^\perp$ is invariant under every linear operator $T \in \text{End}(V_{144})$ satisfying*

$$\langle \mathbf{1}, Tv \rangle = 0 \quad \text{for every } v \in V_0.$$

In particular, if

$$T(V_0) \subseteq V_0,$$

then T induces an endomorphism of V_0 .

Proof. The assertion is immediate from the defining condition of V_0 . If $v \in V_0$, then

$$\langle \mathbf{1}, v \rangle = 0.$$

The assumed relation gives

$$\langle \mathbf{1}, Tv \rangle = 0,$$

and hence

$$Tv \in V_0.$$

Thus $T(V_0) \subseteq V_0$. □

Definition 11.2. For $A \subseteq X_{144}$, define its indicator vector by

$$\mathbf{1}_A := \sum_{x \in A} e_x.$$

The associated coordinate subspace is

$$V_A := \text{span}\{e_x : x \in A\}.$$

The map

$$A \longmapsto V_A$$

is an order-preserving embedding of the Boolean algebra $2^{X_{144}}$ into the lattice of subspaces of V_{144} .

Proposition 11.3. For $A, B \subseteq X_{144}$,

$$A \subseteq B \iff V_A \subseteq V_B,$$

and

$$V_{A \cap B} = V_A \cap V_B.$$

Moreover,

$$V_{A \cup B} = V_A + V_B.$$

Proof. If $A \subseteq B$, every basis vector belonging to A belongs to B , hence

$$V_A \subseteq V_B.$$

Conversely, if $V_A \subseteq V_B$, then $e_x \in V_B$ for every $x \in A$, which is possible only when $x \in B$. Thus $A \subseteq B$.

The basis vectors occurring in both V_A and V_B are precisely those indexed by $A \cap B$, giving

$$V_A \cap V_B = V_{A \cap B}.$$

Similarly, the span of all basis vectors indexed by $A \cup B$ is the sum of the two coordinate subspaces:

$$V_{A \cup B} = V_A + V_B.$$

□

Definition 11.4. Let

$$\mathcal{R}_{144} \subseteq 2^{X_{144}}$$

be a specified family of admissible relations. Its linear realization is

$$\mathcal{L}_{144} := \{V_A : A \in \mathcal{R}_{144}\} \subseteq \mathcal{P}(V_{144}).$$

When the admissible relations are closed under the required lattice operations, $\mathcal{L}_{144}$ inherits the corresponding operations from $\mathcal{P}(V_{144})$.

Proposition 11.5. Suppose that $\mathcal{R}_{144}$ is closed under intersection and union. Then

$$V_{A \cap B} = V_A \wedge V_B$$

and

$$V_{A \cup B} = V_A \vee V_B$$

for all $A, B \in \mathcal{R}_{144}$, where the meet and join on the right are taken in $\mathcal{P}(V_{144})$.

Proof. The identities follow directly from the preceding proposition and from the definitions of meet and join in the lattice of subspaces. □

11.1 Orthogonality of the Finite Realization

For $A, B \subseteq X_{144}$, define

$$A \perp B$$

if

$$A \cap B = \emptyset.$$

For the associated coordinate subspaces this relation is equivalent to Hilbert-space orthogonality:

$$A \perp B \iff V_A \perp V_B.$$

Proposition 11.6. For $A, B \subseteq X_{144}$,

$$A \cap B = \emptyset \iff V_A \perp V_B.$$

Proof. If $A \cap B = \emptyset$, then

$$\langle e_x, e_y \rangle = 0$$

for every $x \in A$ and $y \in B$. By linearity,

$$\langle u, v \rangle = 0$$

for every $u \in V_A$ and $v \in V_B$. Hence

$$V_A \perp V_B.$$

Conversely, if $A \cap B \neq \emptyset$, choose $x \in A \cap B$. Then

$$e_x \in V_A \cap V_B$$

and

$$\langle e_x, e_x \rangle = 1.$$

Thus V_A and V_B cannot be orthogonal. □

Definition 11.7. Let

$$\mu_{144} : \mathcal{R}_{144} \longrightarrow [0, 1]$$

be a normalized relational measure satisfying

$$\mu_{144}(X_{144}) = 1$$

and

$$\mu_{144}(A \sqcup B) = \mu_{144}(A) + \mu_{144}(B)$$

whenever

$$A \perp B.$$

Its linear realization is the functional

$$\hat{\mu}_{144}(V_A) := \mu_{144}(A).$$

Proposition 11.8. The linear realization $\hat{\mu}_{144}$ is well-defined on $\mathcal{L}_{144}$.

Proof. The map

$$A \longmapsto V_A$$

is injective. Indeed, if

$$V_A = V_B,$$

then every basis vector e_x belongs to V_A if and only if it belongs to V_B , and hence

$$A = B.$$

Therefore the value

$$\hat{\mu}_{144}(V_A) := \mu_{144}(A)$$

does not depend on a choice of representative. □

Proposition 11.9. If $A, B \in \mathcal{R}_{144}$ satisfy

$$A \perp B,$$

then

$$\hat{\mu}_{144}(V_A \oplus V_B) = \hat{\mu}_{144}(V_A) + \hat{\mu}_{144}(V_B).$$

Proof. By orthogonality,

$$V_A \oplus V_B = V_{A \sqcup B}.$$

Therefore

$$\hat{\mu}_{144}(V_A \oplus V_B) = \mu_{144}(A \sqcup B) = \mu_{144}(A) + \mu_{144}(B),$$

which is

$$\hat{\mu}_{144}(V_A) + \hat{\mu}_{144}(V_B).$$

□

12 The Measure Gate

The preceding construction isolates the precise mathematical content of the measure layer. The passage

$$N_F \longrightarrow \mu_F$$

is a construction of a normalized measure; the subsequent passage

$$\mu_F \longrightarrow \mathcal{L}_F$$

specifies the domain on which that measure acts; and the relation

$$\perp_F$$

determines the class of decompositions on which additivity is required.

Accordingly, the measure gate is the conjunction of the following properties:

$$\mu_F(0) = 0, \tag{12.1}$$

$$\mu_F(1) = 1, \tag{12.2}$$

$$A \perp_F B \implies \mu_F(A \vee B) = \mu_F(A) + \mu_F(B). \tag{12.3}$$

Theorem 12.1. *Suppose that $\mathcal{L}_F$ is identified with the lattice of subspaces of a finite-dimensional complex Hilbert space V , with $\dim V \geq 3$, and suppose that μ_F satisfies (12.1)–(12.3). Then there exists a unique density operator*

$$\rho_F \geq 0, \quad \text{tr } \rho_F = 1,$$

such that

$$\mu_F(W) = \text{tr}(\rho_F P_W)$$

for every $W \in \mathcal{L}_F$.

Proof. Under the stated identification, μ_F is a normalized finitely additive measure on the projection lattice of V . The finite-dimensional Gleason representation theorem therefore yields a positive semidefinite operator ρ_F satisfying

$$\mu_F(W) = \text{tr}(\rho_F P_W).$$

Taking $W = V$ gives

$$1 = \mu_F(V) = \text{tr}(\rho_F I) = \text{tr } \rho_F.$$

Thus ρ_F is a density operator.

Uniqueness follows because two density operators inducing the same measure have equal trace against every orthogonal projection, and hence are equal. □

Corollary 12.2. *For every orthogonal projection P in the Faure realization,*

$$\mu_F(P) = \text{tr}(\rho_F P).$$

If P has rank one and

$$P = |\psi\rangle\langle\psi|$$

for a unit vector ψ , then

$$\mu_F(P) = \langle\psi, \rho_F \psi\rangle.$$

Proof. The first identity is the representation theorem. For a rank-one projection,

$$\text{tr}(\rho_F |\psi\rangle\langle\psi|) = \langle\psi, \rho_F \psi\rangle.$$

□

13 The Born Functional

Let

$$\{\psi_i\}_{i=1}^r$$

be an orthonormal family in V , and let

$$P_i := |\psi_i\rangle\langle\psi_i|$$

be the associated rank-one orthogonal projections. For the density operator ρ_F , define

$$p_i := \text{tr}(\rho_F P_i).$$

Then

$$p_i \geq 0.$$

If the family is an orthonormal basis, then

$$\sum_i p_i = 1.$$

Proposition 13.1. *Let*

$$\{\psi_i\}_{i=1}^n$$

be an orthonormal basis of V . Then the quantities

$$p_i = \text{tr}(\rho_F P_i) = \langle\psi_i, \rho_F \psi_i\rangle$$

satisfy

$$p_i \geq 0$$

and

$$\sum_{i=1}^n p_i = 1.$$

Proof. Since ρ_F is positive semidefinite,

$$\langle\psi_i, \rho_F \psi_i\rangle \geq 0.$$

Moreover,

$$\sum_{i=1}^n P_i = I,$$

and therefore

$$\sum_{i=1}^n p_i = \text{tr}\left(\rho_F \sum_{i=1}^n P_i\right) = \text{tr}(\rho_F) = 1.$$

□

Remark 13.2. The formula

$$p_i = \text{tr}(\rho_F P_i)$$

is the Born probability functional in the finite-dimensional representation. Its derivation here consists of two logically separate steps: the representation of the normalized orthogonally additive measure by ρ_F , and the evaluation of that measure on a rank-one projection.

14 Non-Booleanity of the Projection Structure

The measure representation does not make the underlying projection lattice Boolean. On the contrary, whenever

$$\dim V \geq 2,$$

the lattice $\mathcal{P}(V)$ fails distributivity.

Let $u, v \in V$ be linearly independent and define

$$A := \text{span}\{u\}, \quad B := \text{span}\{v\}, \quad C := \text{span}\{u + v\}.$$

Then

$$A \cap B = \{0\},$$

whereas

$$C \subseteq A + B.$$

Theorem 14.1. *Let V be a complex vector space with*

$$\dim V \geq 2.$$

Then the lattice $\mathcal{P}(V)$ of linear subspaces of V is not distributive and hence is not Boolean.

Proof. Choose linearly independent $u, v \in V$ and set

$$A = \text{span}\{u\}, \quad B = \text{span}\{v\}, \quad C = \text{span}\{u + v\}.$$

Since u and v are linearly independent,

$$A \cap B = \{0\}.$$

Hence

$$C \cap (A + B) = C,$$

because

$$u + v \in A + B.$$

On the other hand,

$$C \cap A = \{0\}$$

and

$$C \cap B = \{0\}.$$

Indeed, if

$$\alpha(u + v) = \beta u,$$

then

$$(\alpha - \beta)u + \alpha v = 0,$$

and linear independence gives

$$\alpha = \beta = 0.$$

The same argument applies to $C \cap B$. Therefore

$$(C \cap A) + (C \cap B) = \{0\}.$$

Thus

$$C \cap (A + B) \neq (C \cap A) + (C \cap B),$$

which is a failure of distributivity. A Boolean lattice is distributive, so $\mathcal{P}(V)$ is not Boolean. $\square$

Corollary 14.2. *If the Faure relational lattice satisfies*

$$\mathcal{L}_F = \mathcal{P}(V)$$

for a complex Hilbert space V with

$$\dim V \geq 2,$$

then

$$\mathcal{L}_F$$

is non-Boolean.

Proof. The assertion follows immediately from the preceding theorem. $\square$

Remark 14.3. The non-Boolean conclusion is purely lattice-theoretic. It does not depend on the particular density operator ρ_F , on the numerical values of the measure, or on the choice of orthonormal basis.

15 Finite-Step Projection and Measure Compatibility

Suppose now that the stabilized operator $P_9 = \Psi_9^4$ acts on the same finite-dimensional realization on which the measure μ_F is defined. If

$$P_9^2 = P_9$$

and

$$\text{range } P_9 = V_{\text{st}},$$

then P_9 determines the stabilized sector independently of the choice of initial vector.

For every $v \in V_0$,

$$P_9 v \in V_{\text{st}},$$

and

$$P_9(P_9 v) = P_9 v.$$

Proposition 15.1. *Let*

$$P_9 = \Psi_9^4$$

satisfy

$$P_9^2 = P_9.$$

If the measure construction depends only on the stabilized component, then

$$\mu_F(P_9 v)$$

is invariant under any further application of Ψ_9 :

$$\mu_F(\Psi_9^k P_9 v) = \mu_F(P_9 v) \quad (k \geq 0).$$

Proof. Since

$$P_9 \Psi_9 = P_9,$$

we have

$$\Psi_9 P_9 = P_9$$

on the stabilized range. Hence

$$\Psi_9^k P_9 v = P_9 v$$

for every $k \geq 0$. Applying μ_F gives the stated equality. $\square$

Remark 15.2. The compatibility established here is exact: stabilization occurs after the finite iterate 4, and the resulting state is unchanged by all subsequent iterations. No convergence statement is required.

16 Formal Verification Interface

The classical proof is organized so that each formalizable statement has an explicit mathematical interface. In particular, the stabilization layer requires

$$\Psi_9^4 = \Psi_9^5,$$

the projection layer requires

$$P_9 = \Psi_9^4, \quad P_9^2 = P_9,$$

and the measure layer requires

$$\mu_F(1) = 1$$

together with orthogonal additivity.

The representation layer requires the hypotheses of the finite-dimensional measure representation theorem. Once these hypotheses are established, the density operator is determined uniquely by

$$\mu_F(P) = \text{tr}(\rho_F P).$$

Accordingly, the dependency chain is

$$\Psi_9^4 = \Psi_9^5 \implies P_9^2 = P_9 \implies V_0 = \ker P_9 \oplus \text{range } P_9, \quad (16.1)$$

and

$$\mu_F \implies \rho_F \implies \mu_F(P) = \text{tr}(\rho_F P). \quad (16.2)$$

The two chains are combined only after the construction supplies the map identifying the stabilized operator sector with the domain of the Faure measure.

Remark 16.1. The formal verification does not alter the mathematical dependency structure. A formally verified identity certifies the stated identity under the hypotheses encoded in the formal system; it does not replace the definitions of the objects occurring in that identity. The mathematical statements above are consequently stated independently of their particular formal implementation.

17 Exact Stabilization of the Finite-State Operator

Let V be a finite-dimensional vector space over $\mathbb{C}$, and let

$$T \in \text{End}(V).$$

Assume that, for some integer $m \geq 1$,

$$T^m = T^{m+1}. \quad (17.1)$$

Set

$$P := T^m. \quad (17.2)$$

Theorem 17.1. *Under the hypothesis (17.1), the operator P is idempotent and satisfies*

$$PT = TP = P. \quad (17.3)$$

Consequently,

$$V = \ker P \oplus \text{range } P. \quad (17.4)$$

Moreover,

$$\text{range } P = \text{range } T^m = \text{range } T^{m+k} \quad (k \geq 0), \quad (17.5)$$

and

$$\ker P = \ker T^m = \ker T^{m+k} \quad (k \geq 0). \quad (17.6)$$

“latex

Proof. Multiplying (17.1) on the left by T^{m-1} gives

$$T^{2m} = T^{2m-1}.$$

More directly, repeated use of

$$T^m = T^{m+1}$$

gives

$$T^m = T^{m+k} \quad (k \geq 1).$$

Taking $k = m$ yields

$$T^m = T^{2m}.$$

Hence

$$P^2 = T^{2m} = T^m = P.$$

Furthermore,

$$TP = T^{m+1} = T^m = P.$$

Since powers of the same operator commute,

$$PT = TP = P.$$

For any idempotent operator P , every $v \in V$ admits the decomposition

$$v = (v - Pv) + Pv,$$

with

$$P(v - Pv) = Pv - P^2v = 0.$$

Thus

$$v - Pv \in \ker P, \quad Pv \in \text{range } P.$$

If $v \in \ker P \cap \text{range } P$, write $v = Pw$. Then

$$v = Pv = P^2w = Pv = 0.$$

Therefore

$$V = \ker P \oplus \text{range } P.$$

Finally,

$$T^{m+k} = T^m = P \quad (k \geq 0),$$

so the ranges and kernels coincide with those of P . □

Corollary 17.2. *If*

$$\Psi_9^4 = \Psi_9^5,$$

then

$$P_9 := \Psi_9^4$$

satisfies

$$P_9^2 = P_9, \quad \Psi_9 P_9 = P_9 \Psi_9 = P_9,$$

and

$$\Psi_9^k = P_9 \quad (k \geq 4).$$

“*latex*

Proof. Apply the preceding theorem with

$$T = \Psi_9, \quad m = 4.$$

□

18 Spectral Consequences of Exact Stabilization

The stabilization identity imposes a polynomial relation on the operator. Namely,

$$T^m(T - I) = 0. \tag{18.1}$$

Consequently, every eigenvalue of T belongs to

$$\{0, 1\}.$$

Proposition 18.1. *Let $T \in \text{End}(V)$ satisfy*

$$T^m = T^{m+1}.$$

Then

$$\sigma(T) \subseteq \{0, 1\}.$$

Furthermore,

$$\text{range } P = \ker(T - I), \quad \ker P = \ker T^m,$$

where $P = T^m$.

“*latex*

Proof. If $Tv = \lambda v$ for a nonzero vector v , then

$$T^m v = \lambda^m v$$

and

$$T^{m+1} v = \lambda^{m+1} v.$$

The stabilization identity therefore gives

$$\lambda^m = \lambda^{m+1},$$

hence

$$\lambda^m(\lambda - 1) = 0.$$

Thus

$$\lambda \in \{0, 1\}.$$

If $v \in \text{range } P$, then $v = Pw$ for some w , and

$$Tv = TPw = Pw = v.$$

Hence

$$\text{range } P \subseteq \ker(T - I).$$

Conversely, if $Tv = v$, then

$$Pv = T^m v = v,$$

so

$$v \in \text{range } P.$$

Therefore

$$\text{range } P = \ker(T - I).$$

The equality

$$\ker P = \ker T^m$$

is immediate from $P = T^m$. □

Corollary 18.2. *Under exact stabilization,*

$$V = \ker T^m \oplus \ker(T - I).$$

“ $\text{\LaTeX}$

Proof. Combine

$$V = \ker P \oplus \text{range } P$$

with

$$\ker P = \ker T^m$$

and

$$\text{range } P = \ker(T - I).$$

□

Remark 18.3. The decomposition above is algebraic and does not require T to be normal, self-adjoint, unitary, or diagonalizable. Exact stabilization itself supplies the idempotent decomposition.

19 The Four-Step Faure Projection

For the 144-state construction, write

$$T = \Psi_9$$

on the invariant state space under consideration and assume

$$\Psi_9^4 = \Psi_9^5. \quad (19.1)$$

Define

$$P_9 := \Psi_9^4. \quad (19.2)$$

Then

$$P_9^2 = P_9, \quad \Psi_9 P_9 = P_9 \Psi_9 = P_9. \quad (19.3)$$

Theorem 19.1. *For every initial state v ,*

$$\Psi_9^k v = P_9 v \quad (k \geq 4).$$

Hence the orbit becomes exactly stationary after the fourth iterate.

“`latex`

Proof. From (19.1),

$$\Psi_9^4 = \Psi_9^5.$$

The stabilization theorem gives

$$\Psi_9^k = \Psi_9^4 = P_9 \quad (k \geq 4).$$

Applying both sides to v proves the assertion. □

Corollary 19.2. *For any v ,*

$$\Psi_9^4 v = \Psi_9^5 v = \Psi_9^6 v = \dots$$

In particular, no limiting process

$$\lim_{k \rightarrow \infty} \Psi_9^k v$$

is required to define the stabilized state.

20 The Stabilized Sector

Define

$$V_{\text{st}} := \text{range } P_9 = \ker(\Psi_9 - I). \quad (20.1)$$

The transient sector is

$$V_{\text{tr}} := \ker P_9 = \ker(\Psi_9^4). \quad (20.2)$$

Then

$$V = V_{\text{tr}} \oplus V_{\text{st}}. \quad (20.3)$$

Proposition 20.1. *The restriction of Ψ_9 to V_{st} is the identity:*

$$\Psi_9|_{V_{\text{st}}} = I.$$

The restriction of Ψ_9^4 to V_{tr} is zero:

$$\Psi_9^4|_{V_{\text{tr}}} = 0.$$

“ $\text{\LaTeX}$

Proof. If $v \in V_{\text{st}} = \ker(\Psi_9 - I)$, then

$$\Psi_9 v = v.$$

If $v \in V_{\text{tr}} = \ker \Psi_9^4$, then

$$\Psi_9^4 v = 0.$$

□

Remark 20.2. The terminology “transient” refers here to the exact algebraic annihilation by the fourth iterate. It does not denote merely an asymptotically decaying component.

21 Normalized Stabilized State

Let $W \subseteq V_{\text{st}}$ be a nonzero invariant subspace and suppose that the induced stabilized operator on W is represented, in a fixed orthonormal basis, by

$$B \in M_r(\mathbb{C}).$$

Assume

$$B = \frac{1}{r} J_r, \tag{21.1}$$

where J_r denotes the $r \times r$ matrix all of whose entries are equal to 1.

Proposition 21.1. *The matrix*

$$B = \frac{1}{r} J_r$$

is idempotent:

$$B^2 = B.$$

Its spectrum is

$$\sigma(B) = \{1, 0\},$$

with eigenvalue 1 of multiplicity 1 and eigenvalue 0 of multiplicity $r - 1$.

“ $\text{\LaTeX}$

Proof. Since

$$J_r^2 = r J_r,$$

we obtain

$$B^2 = \frac{1}{r^2} J_r^2 = \frac{1}{r} J_r = B.$$

The vector

$$\mathbf{1} = (1, \dots, 1)^\top$$

satisfies

$$B\mathbf{1} = \mathbf{1}.$$

If

$$x \perp \mathbf{1},$$

then

$$J_r x = 0,$$

and hence

$$Bx = 0.$$

The orthogonal decomposition

$$\mathbb{C}^r = \text{span}\{\mathbf{1}\} \oplus \mathbf{1}^\perp$$

therefore gives the stated spectrum and multiplicities. $\square$

Corollary 21.2. *For $r = 9$,*

$$B^* = \frac{1}{9} J_9$$

is an exact rank-one projection, satisfying

$$(B^*)^2 = B^*, \quad \text{tr } B^* = 1.$$

22 Primitivity Exponent

For a nonnegative matrix A , call A primitive if there exists $k \geq 1$ such that

$$A^k > 0$$

entrywise. The least such k , when it exists, is its primitivity exponent.

Proposition 22.1. *Let A be a nonnegative $n \times n$ matrix satisfying*

$$A^4 > 0.$$

Then A is primitive and its primitivity exponent satisfies

$$\gamma(A) \leq 4.$$

If additionally

$$A^3$$

has at least one zero entry, then

$$\gamma(A) = 4.$$

““latex

Proof. The definition of primitivity requires the existence of a positive power. Since $A^4 > 0$, the matrix is primitive and

$$\gamma(A) \leq 4.$$

If A^3 contains a zero entry, then $A^3 \not> 0$, so no exponent strictly smaller than 4 can equal the primitivity exponent. Hence

$$\gamma(A) = 4. \quad \square$$

Remark 22.2. Consequently, the numerical assertion

$$\gamma = 4$$

is not implied by the stabilization identity

$$\Psi_9^4 = \Psi_9^5$$

alone. It requires the independent positivity statement

$$\Psi_9^4 > 0$$

together with the failure of positivity at the preceding iterate.

23 Exact Versus Asymptotic Dynamics

The distinction between exact stabilization and ordinary convergence is structural. If

$$\Psi_9^4 = \Psi_9^5,$$

then

$$\Psi_9^k = P_9 \quad (k \geq 4).$$

Thus the sequence of operators is eventually constant:

$$\Psi_9^4, \Psi_9^5, \Psi_9^6, \dots = P_9, P_9, P_9, \dots$$

Theorem 23.1. *Under the four-step stabilization identity, the sequence*

$$\{\Psi_9^k\}_{k \geq 0}$$

converges in every operator topology on the finite-dimensional space $\text{End}(V)$, and its limit is exactly

$$P_9.$$

“ $\text{\LaTeX}$

Proof. The sequence is constant from $k = 4$ onward:

$$\Psi_9^k = P_9 \quad (k \geq 4).$$

Every eventually constant sequence converges to its eventual constant value. Since all norms and operator topologies on a finite-dimensional space are equivalent, the assertion holds for every operator topology. $\square$

Remark 23.2. The convergence statement is therefore a consequence of exact stabilization, not its definition. The primary result is the finite identity

$$\Psi_9^4 = \Psi_9^5.$$

24 Compatibility with the Zero-Trace Sector

Assume

$$\Psi_9(V_0) \subseteq V_0, \quad V_0 = \mathbf{1}^\perp.$$

Then

$$P_9(V_0) \subseteq V_0.$$

Consequently,

$$V_0 = \ker(P_9|_{V_0}) \oplus \text{range}(P_9|_{V_0}).$$

Proposition 24.1. *If*

$$\Psi_9(V_0) \subseteq V_0$$

and

$$\Psi_9^4 = \Psi_9^5$$

on V_0 , then

$$P_9 := \Psi_9^4|_{V_0}$$

is an idempotent endomorphism of V_0 , and

$$V_0 = \ker P_9 \oplus \ker(\Psi_9 - I).$$

“`latex`

Proof. Invariance gives

$$P_9 \in \text{End}(V_0).$$

The stabilization theorem gives

$$P_9^2 = P_9$$

and

$$\text{range } P_9 = \ker(\Psi_9 - I).$$

The direct-sum decomposition follows. □

25 Structural Summary

The finite-state operator layer is therefore characterized by the implication

$$\boxed{\Psi_9^4 = \Psi_9^5 \implies P_9^2 = P_9 \implies V_0 = \ker \Psi_9^4 \oplus \ker(\Psi_9 - I)}. \quad (25.1)$$

If, in addition,

$$\Psi_9^4 > 0$$

and

$$\Psi_9^3 \not\geq 0,$$

then the associated nonnegative transition matrix has exact primitivity exponent

$$\gamma = 4.$$

If its stabilized restriction is

$$B^* = \frac{1}{9}J_9,$$

then

$$(B^*)^2 = B^*, \quad \text{rank } B^* = 1, \quad \text{tr } B^* = 1.$$

These conclusions are algebraically independent statements whose hypotheses are explicit and therefore can be verified separately in the formal development.

26 The Finite-State Transition Algebra

Let X_{144} be the finite state set introduced above and let

$$V_{144} = \mathbb{C}^{X_{144}}.$$

A finite transition rule is represented by a matrix

$$T = (t_{xy})_{x,y \in X_{144}} \in M_{144}(\mathbb{C}).$$

When

$$t_{xy} \geq 0$$

for all x, y , T is called nonnegative. When, in addition,

$$\sum_{x \in X_{144}} t_{xy} = 1$$

for every y , T is column-stochastic.

The distinction between the algebraic stabilization identity and stochastic normalization will be maintained throughout. The former is an operator identity; the latter is an additional structural property.

Definition 26.1. A matrix

$$T \in M_n(\mathbb{C})$$

is column-stochastic if

$$t_{ij} \geq 0$$

and

$$\sum_{i=1}^n t_{ij} = 1$$

for every j . It is row-stochastic if

$$t_{ij} \geq 0$$

and

$$\sum_{j=1}^n t_{ij} = 1$$

for every i .

Proposition 26.2. *If T is column-stochastic, then*

$$\mathbf{1}^\top T = \mathbf{1}^\top.$$

Consequently,

$$\mathbf{1}^\perp = \left\{ v \in \mathbb{C}^n : \mathbf{1}^\top v = 0 \right\}$$

is invariant under T .

Proof. The j -th component of

$$\mathbf{1}^\top T$$

is

$$\sum_{i=1}^n t_{ij} = 1.$$

Hence

$$\mathbf{1}^\top T = \mathbf{1}^\top.$$

If $v \in \mathbf{1}^\perp$, then

$$\mathbf{1}^\top T v = (\mathbf{1}^\top T) v = \mathbf{1}^\top v = 0.$$

Therefore

$$T v \in \mathbf{1}^\perp.$$

□

Proposition 26.3. *Let T be column-stochastic and suppose*

$$T^m = T^{m+1}.$$

Then the stabilized operator

$$P = T^m$$

is column-stochastic and idempotent.

Proof. Every positive power of a column-stochastic matrix is column-stochastic. Thus $P = T^m$ is column-stochastic. Exact stabilization gives

$$P^2 = T^{2m} = T^m = P.$$

□

Theorem 26.4. *Let $P \in M_n(\mathbb{R})$ be a nonnegative, column-stochastic, idempotent matrix. Then every column of P is a probability vector and*

$$\text{range } P$$

is precisely the fixed-point space of P :

$$\text{range } P = \ker(P - I).$$

Proof. Nonnegativity and column-stochasticity imply that every column has nonnegative entries summing to 1, hence is a probability vector.

If $v \in \text{range } P$, then $v = Pw$ for some w , and

$$Pv = P^2w = Pw = v.$$

Thus

$$\text{range } P \subseteq \ker(P - I).$$

Conversely, if $Pv = v$, then $v = Pv$, so $v \in \text{range } P$. Hence

$$\text{range } P = \ker(P - I).$$

□

27 Invariant States and Stationary Measures

Let

$$\Delta_{143} = \left\{ p \in \mathbb{R}^{144} : p_i \geq 0, \sum_{i=1}^{144} p_i = 1 \right\}$$

denote the probability simplex.

A vector

$$p \in \Delta_{143}$$

is stationary for a column-stochastic operator T if

$$Tp = p.$$

Definition 27.1. The stationary simplex of a column-stochastic operator T is

$$\text{Stat}(T) = \ker(T - I) \cap \Delta_{143}.$$

Proposition 27.2. *If T is column-stochastic and*

$$T^m = T^{m+1},$$

then for every probability vector $p \in \Delta_{143}$,

$$Pp = T^m p$$

belongs to $\text{Stat}(T)$.

Proof. Since $P = T^m$ is nonnegative and column-stochastic,

$$Pp \in \Delta_{143}.$$

Moreover,

$$T(Pp) = TPp = Pp.$$

Hence

$$Pp \in \ker(T - I) \cap \Delta_{143} = \text{Stat}(T).$$

□

Corollary 27.3. *Under exact stabilization, the map*

$$P : \Delta_{143} \longrightarrow \text{Stat}(T)$$

is a projection in the affine sense:

$$Pq = q \quad \text{for every } q \in \text{Stat}(T).$$

Proof. If $q \in \text{Stat}(T)$, then $Tq = q$. Therefore

$$Pq = T^m q = q.$$

□

28 Uniqueness of the Stabilized Probability State

Exact stabilization alone does not imply that the stationary state is unique. Uniqueness requires an additional irreducibility or positivity hypothesis.

The strongest finite-state condition used below is strict positivity of the stabilized transition matrix.

Theorem 28.1. *Let*

$$P \in M_n(\mathbb{R})$$

be strictly positive and column-stochastic. If

$$P^2 = P,$$

then

$$\text{rank } P = 1.$$

Consequently, there exists a unique vector

$$\pi \in \Delta_{n-1}$$

such that

$$P = \pi \mathbf{1}^\top,$$

and

$$Pv = \pi$$

for every probability vector v .

Proof. Since P is positive, every nonzero vector in the cone generated by the columns is mapped to a vector whose entries are strictly positive. Because P is idempotent,

$$\text{range } P = \ker(P - I).$$

Suppose that $\text{rank } P \geq 2$. Then there exist two linearly independent vectors $u, v \in \text{range } P$. Since every vector in the range is fixed by P , both u and v satisfy

$$Pu = u, \quad Pv = v.$$

For a positive stochastic matrix, the Perron–Frobenius theorem implies that the eigenvalue 1 is simple. Hence its eigenspace is one-dimensional, contradicting the linear independence of u and v . Therefore

$$\text{rank } P = 1.$$

A rank-one matrix can be written

$$P = ab^T.$$

Column-stochasticity gives

$$\mathbf{1}^T P = \mathbf{1}^T,$$

hence

$$(\mathbf{1}^T a)b^T = \mathbf{1}^T.$$

After normalization,

$$b = \mathbf{1}.$$

Thus

$$P = \pi \mathbf{1}^T$$

for a probability vector π . Every column of P equals π , so π is the unique stationary probability vector. $\square$

Corollary 28.2. *If the four-step Faure operator satisfies*

$$\Psi_9^4 = \Psi_9^5$$

and

$$\Psi_9^4 > 0,$$

then

$$P_9 = \Psi_9^4$$

has rank one and there exists a unique stationary probability vector π_9 such that

$$P_9 = \pi_9 \mathbf{1}^T.$$

For every probability vector p ,

$$\Psi_9^4 p = \pi_9.$$

29 The Uniform Nine-State Sector

Assume that the stabilized nine-state sector is represented by

$$B^* = \frac{1}{9}J_9, \quad J_9 = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix}.$$

Then every column of B^* is

$$\pi_9 = \frac{1}{9} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Theorem 29.1. *The matrix*

$$B^* = \frac{1}{9}J_9$$

is the unique strictly positive, column-stochastic rank-one projection whose stationary vector is the uniform probability vector

$$\pi_9 = \frac{1}{9}\mathbf{1}_9.$$

Proof. The matrix B^* is strictly positive and column-stochastic because each entry equals $1/9$ and every column has nine entries. Moreover,

$$(B^*)^2 = B^*.$$

Its range is

$$\text{span}\{\mathbf{1}_9\},$$

so its stationary probability vector is

$$\pi_9 = \frac{1}{9}\mathbf{1}_9.$$

Conversely, a rank-one column-stochastic projection with stationary vector π_9 has the form

$$\pi_9 \mathbf{1}_9^T,$$

which is exactly

$$\frac{1}{9}J_9.$$

□

30 The Algebraic Origin of the Projection

The preceding statements can be summarized without introducing any additional limiting construction. The finite operator satisfies

$$\Psi_9^4 = \Psi_9^5$$

and therefore

$$\Psi_9^k = P_9 \quad (k \geq 4).$$

If the stabilized nine-state representation is

$$B^* = \frac{1}{9}J_9,$$

then the complete finite transition law is

$$\Psi_9^k \longrightarrow P_9 \longrightarrow B^*$$

with the first arrow interpreted as an exact identity for all $k \geq 4$, rather than as an asymptotic limit.

Proposition 30.1. *Under the hypotheses*

$$\Psi_9^4 = \Psi_9^5$$

and

$$B^* = \frac{1}{9} J_9,$$

every initial probability vector p in the nine-state stabilized sector satisfies

$$B^{*k} p = B^* p = \pi_9 \quad (k \geq 1).$$

Proof. Since

$$(B^*)^2 = B^*,$$

induction gives

$$(B^*)^k = B^* \quad (k \geq 1).$$

Therefore

$$(B^*)^k p = B^* p.$$

Because

$$B^* p = \frac{1}{9} J_9 p = \frac{1}{9} \left(\sum_{j=1}^9 p_j \right) \mathbf{1}_9,$$

and p is a probability vector,

$$\sum_{j=1}^9 p_j = 1.$$

Hence

$$B^* p = \frac{1}{9} \mathbf{1}_9 = \pi_9.$$

□

31 Operator and Measure Layers

Let

$$\mathcal{P}(V)$$

denote the projection lattice of the stabilized Hilbert space V . The operator layer determines the stabilized subspace, while the measure layer assigns normalized values to its projections.

The two constructions are therefore related by

$$P_9 \longrightarrow V_{\text{st}} \longrightarrow \mathcal{P}(V_{\text{st}}) \longrightarrow \mu_F \longrightarrow \rho_F.$$

No identification between an arbitrary transition matrix and a quantum state is made at this stage. The density operator appears only after the measure has been established on the projection lattice.

Proposition 31.1. *Let P_9 be an idempotent operator and let*

$$V_{\text{st}} = \text{range } P_9.$$

If μ_F is a normalized orthogonally additive measure on $\mathcal{P}(V_{\text{st}})$, with

$$\dim V_{\text{st}} \geq 3,$$

then there exists a unique density operator

$$\rho_F \in \text{End}(V_{\text{st}})$$

such that

$$\mu_F(Q) = \text{tr}(\rho_F Q)$$

for every orthogonal projection Q on V_{st} .

Proof. This is the finite-dimensional projection-lattice representation theorem applied to the normalized orthogonally additive measure μ_F . Positivity and normalization give

$$\rho_F \geq 0, \quad \text{tr } \rho_F = 1,$$

and the representation

$$\mu_F(Q) = \text{tr}(\rho_F Q).$$

Uniqueness follows because the values on all orthogonal projections determine the Hermitian operator ρ_F . $\square$

Remark 31.2. The dimension hypothesis in the preceding proposition is essential to the stated general representation theorem. It must therefore be verified for the particular stabilized Hilbert space used by the construction; it is not a consequence of the symbol 144 alone.

32 Spectral Resolution of the Measure

Let

$$\rho_F = \sum_{j=1}^r \lambda_j Q_j$$

be the spectral decomposition of the density operator, where

$$\lambda_j \geq 0, \quad \sum_{j=1}^r \lambda_j = 1,$$

and the Q_j are pairwise orthogonal spectral projections satisfying

$$Q_i Q_j = \delta_{ij} Q_j, \quad \sum_{j=1}^r Q_j = I.$$

Proposition 32.1. *For every orthogonal projection P ,*

$$\mu_F(P) = \sum_{j=1}^r \lambda_j \text{tr}(Q_j P).$$

In particular, if P is rank one with

$$P = |\psi\rangle\langle\psi|,$$

then

$$\mu_F(P) = \sum_{j=1}^r \lambda_j \langle\psi, Q_j \psi\rangle.$$

Proof. Using the spectral decomposition,

$$\mu_F(P) = \text{tr}(\rho_F P) = \text{tr} \left(\sum_{j=1}^r \lambda_j Q_j P \right).$$

Linearity of the trace gives

$$\mu_F(P) = \sum_{j=1}^r \lambda_j \text{tr}(Q_j P).$$

For rank-one P ,

$$\text{tr}(Q_j P) = \langle \psi, Q_j \psi \rangle.$$

□

33 Finite Orthogonal Decomposition

Let

$$P_1, \dots, P_n$$

be pairwise orthogonal projections satisfying

$$P_i P_j = 0 \quad (i \neq j)$$

and

$$\sum_{i=1}^n P_i = I.$$

Then

$$\{P_i\}_{i=1}^n$$

is a finite resolution of the identity.

Theorem 33.1. *For every density operator ρ_F ,*

$$\sum_{i=1}^n \text{tr}(\rho_F P_i) = 1.$$

Moreover,

$$\text{tr}(\rho_F P_i) \geq 0$$

for every i .

Proof. Since $\rho_F \geq 0$ and $P_i \geq 0$,

$$\text{tr}(\rho_F P_i) \geq 0.$$

Furthermore,

$$\sum_{i=1}^n \text{tr}(\rho_F P_i) = \text{tr} \left(\rho_F \sum_{i=1}^n P_i \right) = \text{tr}(\rho_F) = 1.$$

□

34 The Born Rule as the Terminal Functional Identity

The preceding construction yields the probability assigned to every element of a finite orthogonal resolution of the identity. For a rank-one resolution

$$P_i = |\psi_i\rangle\langle\psi_i|, \quad \sum_i P_i = I,$$

the resulting probabilities are

$$p_i = \text{tr}(\rho_F P_i) = \langle\psi_i, \rho_F \psi_i\rangle. \quad (34.1)$$

Theorem 34.1. *Let μ_F be a normalized orthogonally additive measure on the projection lattice of a finite-dimensional complex Hilbert space V , with*

$$\dim V \geq 3.$$

Then for every finite orthogonal resolution of the identity

$$I = \sum_i P_i,$$

the numbers

$$p_i := \mu_F(P_i)$$

are uniquely represented by a density operator ρ_F through

$$p_i = \text{tr}(\rho_F P_i).$$

For rank-one projections this becomes

$$p_i = \langle\psi_i, \rho_F \psi_i\rangle.$$

Proof. The measure representation theorem supplies the unique density operator ρ_F satisfying

$$\mu_F(P) = \text{tr}(\rho_F P)$$

for every projection P . Applying this identity to each P_i gives the first formula. The rank-one formula follows from

$$P_i = |\psi_i\rangle\langle\psi_i|.$$

□

35 Logical Separation of the Structural Claims

The mathematical dependencies established in this part are

$$\Psi_9^4 = \Psi_9^5 \implies P_9^2 = P_9 \implies V = \ker P_9 \oplus \text{range } P_9, \quad (35.1)$$

together with

$$\mu_F \text{ normalized and orthogonally additive} \implies \rho_F \implies \mu_F(P) = \text{tr}(\rho_F P). \quad (35.2)$$

If the stabilized transition operator is strictly positive, then

$$\Psi_9^4 > 0 \implies \text{rank } P_9 = 1 \implies \text{unique stationary probability state.} \quad (35.3)$$

The three implications are logically distinct. In particular, neither positivity nor uniqueness of the stationary state is inferred merely from idempotence, and the representation of a measure by a density operator is not inferred merely from finite-state stabilization.

36 The Arithmetic Substrate

We next introduce the arithmetic component independently of the finite operator realization. Let

$$\mathbb{P} = \{p \in \mathbb{N} : p \geq 2, p \text{ prime}\}$$

denote the set of prime numbers, and define the free abelian group

$$\mathcal{L}_{\mathbb{P}} := \mathbb{Z}^{(\mathbb{P})},$$

that is, the group of finitely supported integer-valued functions on $\mathbb{P}$.

For

$$a = (a_p)_{p \in \mathbb{P}} \in \mathcal{L}_{\mathbb{P}},$$

its support is

$$\text{supp}(a) = \{p \in \mathbb{P} : a_p \neq 0\}.$$

By definition this set is finite.

Definition 36.1. For $a \in \mathcal{L}_{\mathbb{P}}$, define the associated positive rational number

$$\text{val}(a) := \prod_{p \in \mathbb{P}} p^{a_p}.$$

The product is finite because a has finite support.

Proposition 36.2. *The map*

$$\text{val} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{Q}_{>0}$$

is a group isomorphism.

Proof. For $a, b \in \mathcal{L}_{\mathbb{P}}$,

$$\text{val}(a + b) = \prod_p p^{a_p + b_p} = \left(\prod_p p^{a_p} \right) \left(\prod_p p^{b_p} \right),$$

so val is a homomorphism.

If

$$\text{val}(a) = 1,$$

unique factorization implies

$$a_p = 0$$

for every prime p . Hence $a = 0$, so the map is injective.

Conversely, every positive rational number has a unique prime factorization

$$q = \prod_{p \in \mathbb{P}} p^{a_p}$$

with finitely many nonzero integer exponents a_p . Thus $q = \text{val}(a)$, proving surjectivity. $\square$

Definition 36.3. The logarithmic embedding of the prime lattice is the map

$$\Lambda_{\log} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{R}^{(\mathbb{P})}$$

defined by

$$\Lambda_{\log}(a) = (a_p \log p)_{p \in \mathbb{P}}.$$

Its scalar logarithmic coordinate is

$$\ell(a) := \sum_{p \in \mathbb{P}} a_p \log p.$$

Proposition 36.4. *For every*

$$a \in \mathcal{L}_{\mathbb{P}},$$

one has

$$\ell(a) = \log \text{val}(a).$$

Moreover,

$$\ell(a + b) = \ell(a) + \ell(b).$$

Proof. Since the support of a is finite,

$$\log \text{val}(a) = \log \left(\prod_p p^{a_p} \right) = \sum_p a_p \log p = \ell(a).$$

The second identity follows from the additivity of the exponents. □

Theorem 36.5. *The logarithmic map*

$$\ell : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{R}$$

is injective.

Proof. Suppose

$$\ell(a) = 0.$$

Then

$$\log \text{val}(a) = 0,$$

hence

$$\text{val}(a) = 1.$$

The injectivity of the prime factorization map established above gives

$$a = 0.$$

Thus ℓ is injective. □

Remark 36.6. The injectivity of ℓ is an exact arithmetic consequence of unique factorization. No assertion concerning rational or irrational linear independence of the family

$$\{\log p : p \in \mathbb{P}\}$$

is required.

37 Finite-Support Arithmetic Constraints

Let

$$S \subset \mathbb{P}$$

be finite. The corresponding finite prime lattice is

$$\mathcal{L}_S := \mathbb{Z}^S.$$

For

$$a = (a_p)_{p \in S},$$

define

$$\ell_S(a) = \sum_{p \in S} a_p \log p.$$

Proposition 37.1. *For every finite set $S \subset \mathbb{P}$, the map*

$$\ell_S : \mathbb{Z}^S \longrightarrow \mathbb{R}$$

is injective.

Proof. If

$$\ell_S(a) = 0,$$

then

$$\prod_{p \in S} p^{a_p} = 1.$$

Unique factorization gives

$$a_p = 0$$

for every $p \in S$. Hence $a = 0$. □

Definition 37.2. For a finite $S \subset \mathbb{P}$, an integer constraint system on $\mathcal{L}_S$ is a finite family of equations and inequalities of the form

$$A_j a = b_j, \quad C_k a \leq d_k,$$

where

$$A_j, C_k$$

are integer matrices and

$$b_j, d_k$$

are integer vectors.

Proposition 37.3. *Every finite-support constraint system on $\mathcal{L}_{\mathbb{P}}$ restricts to an ordinary finite-dimensional integer constraint system on $\mathcal{L}_S$, where*

$$S = \bigcup_a \text{supp}(a^{(a)})$$

is the finite union of the prime supports occurring in the system.

Proof. A finite constraint system contains only finitely many explicitly specified prime coordinates. Their union is therefore finite. Restriction to those coordinates identifies the relevant part of $\mathcal{L}_{\mathbb{P}}$ with

$$\mathbb{Z}^S.$$

□

Remark 37.4. The preceding reduction is a statement about finite descriptions and does not assert that arbitrary global arithmetic problems possess bounded-size certificates. Any complexity statement concerning a family of constraint systems therefore requires an explicit encoding model and an explicit size parameter.

38 The Prime Euler Product

For

$$s \in \mathbb{C}, \quad \Re(s) > 1,$$

define the Riemann zeta function by the absolutely convergent Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Theorem 38.1. *For*

$$\Re(s) > 1,$$

the Euler product

$$\zeta(s) = \prod_{p \in \mathbb{P}} (1 - p^{-s})^{-1}$$

converges absolutely.

Proof. For

$$\sigma = \Re(s) > 1,$$

one has

$$\sum_{p \in \mathbb{P}} |p^{-s}| = \sum_{p \in \mathbb{P}} p^{-\sigma} \leq \sum_{n=2}^{\infty} n^{-\sigma} < \infty.$$

Hence the Euler product converges absolutely. Expanding each factor by the geometric series

$$(1 - p^{-s})^{-1} = \sum_{k=0}^{\infty} p^{-ks}$$

and using unique prime factorization gives

$$\prod_p (1 - p^{-s})^{-1} = \sum_{n=1}^{\infty} n^{-s} = \zeta(s).$$

□

Proposition 38.2. *For*

$$\Re(s) > 1,$$

one has

$$\zeta(s) \neq 0.$$

Proof. The absolutely convergent Euler product satisfies

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}.$$

Since

$$|p^{-s}| < 1$$

for every prime p , no factor

$$1 - p^{-s}$$

vanishes. Absolute convergence of the product therefore gives

$$\zeta(s) \neq 0.$$

□

39 Analytic Continuation and the Functional Equation

Define the completed zeta function

$$\xi(s) := \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

The meromorphic continuation of ζ to $\mathbb{C}$ implies that ξ extends to an entire function.

Theorem 39.1. *The completed zeta function ξ is entire and satisfies*

$$\xi(s) = \xi(1-s).$$

Consequently, if

$$\rho$$

is a zero of ζ which is not a trivial zero, then

$$1-\rho$$

is also a zero.

Proof. The standard analytic continuation of the Riemann zeta function, combined with the poles and zeros of the gamma and elementary factors in the definition of ξ , removes the pole at $s = 1$ and the trivial zeros. The resulting completed function is entire.

The functional equation for the completed zeta function is

$$\xi(s) = \xi(1-s).$$

Therefore

$$\xi(\rho) = 0$$

implies

$$\xi(1-\rho) = 0.$$

For a nontrivial zero, the prefactors in ξ do not vanish, so the corresponding zero is also a zero of ζ . $\square$

Corollary 39.2. *The nontrivial zero set of ζ is symmetric with respect to the vertical line*

$$\Re(s) = \frac{1}{2}.$$

If, in addition, $\zeta(\bar{s}) = 0$ whenever $\zeta(s) = 0$, then the nontrivial zeros occur in the symmetry configuration

$$\rho, \quad 1-\rho, \quad \bar{\rho}, \quad 1-\bar{\rho}.$$

40 The Critical Strip

Let

$$\rho = \beta + i\gamma$$

be a nontrivial zero of ζ . The classical zero-free regions imply that

$$0 < \beta < 1.$$

Thus every nontrivial zero lies in the critical strip

$$0 < \Re(s) < 1.$$

Theorem 40.1. *Every nontrivial zero of the Riemann zeta function lies in*

$$0 < \Re(s) < 1.$$

Proof. The absence of zeros in

$$\Re(s) > 1$$

follows from the Euler product.

The functional equation transforms a zero with

$$\Re(s) < 0$$

into a zero in

$$\Re(1 - s) > 1,$$

apart from the explicitly known trivial zeros generated by the gamma factor. Hence a nontrivial zero cannot lie outside the strip

$$0 < \Re(s) < 1.$$

□

Definition 40.2. The Riemann hypothesis is the assertion

$$\boxed{\zeta(\rho) = 0, \quad \rho \text{ nontrivial} \implies \Re(\rho) = \frac{1}{2}.}$$

Equivalently, every nontrivial zero belongs to the critical line

$$\frac{1}{2} + i\mathbb{R}.$$

Remark 40.3. The critical-line statement is therefore a separate arithmetic theorem from the finite-state stabilization results proved above. It cannot be deduced from finite-dimensional idempotence, positivity, or the projection-lattice representation without an additional theorem establishing the required analytic connection.

41 Exact Form of the Required Arithmetic Bridge

To connect the finite-state architecture with the zero set of ζ , one must specify an operator

$$\mathcal{T}(s)$$

or an equivalent spectral object for which the relevant determinant, characteristic function, or trace invariant is explicitly related to the completed zeta function.

A mathematically sufficient bridge has the form

$$\det \mathcal{T}(s) = C(s)\xi(s), \tag{41.1}$$

where $C(s)$ is explicitly known and nonvanishing on the domain under consideration.

Proposition 41.1. *Assume that an entire function*

$$D(s)$$

satisfies

$$D(s) = C(s)\xi(s)$$

with $C(s) \neq 0$ for all $s \in \mathbb{C}$. Then

$$D(s) = 0 \iff \xi(s) = 0.$$

Consequently, the nontrivial zeros of ζ coincide with the nontrivial zeros of D .

Proof. Since $C(s) \neq 0$,

$$D(s) = C(s)\xi(s)$$

vanishes if and only if $\xi(s)$ vanishes. The correspondence with the nontrivial zeros of ζ follows from the definition of the completed function. $\square$

Remark 41.2. Equation (41.1) is the non-tautological mathematical point at which an operator-theoretic construction can interact with the zero distribution of ζ . A statement that an operator “encodes” the zeta zeros is not sufficient; an explicit identity of the form (41.1), or an equivalent spectral correspondence with the same logical strength, is required.

42 Spectral Reality and the Critical Line

Suppose an operator realization provides a spectral parameter λ satisfying

$$s = \frac{1}{2} + i\lambda. \quad (42.1)$$

If every spectral parameter corresponding to a nontrivial zero is real, then the corresponding zero necessarily lies on the critical line.

Theorem 42.1. *Assume that every nontrivial zero ρ of ζ is represented by a spectral value $\lambda \in \mathbb{R}$ through*

$$\rho = \frac{1}{2} + i\lambda.$$

Then every nontrivial zero satisfies

$$\Re(\rho) = \frac{1}{2}.$$

Proof. By hypothesis,

$$\rho = \frac{1}{2} + i\lambda$$

with

$$\lambda \in \mathbb{R}.$$

Therefore

$$\Re(\rho) = \frac{1}{2}.$$

$\square$

Remark 42.2. Thus a proof of the critical-line statement through the Faure architecture reduces, at the classical mathematical level, to two independent assertions: an exact correspondence between the relevant operator spectrum and the nontrivial zeros, and reality of the corresponding spectral parameters. Neither assertion follows from finite-step stabilization alone.

43 Formal Dependency Structure

The mathematical dependencies established in the present part are

$$\mathcal{L}_{\mathbb{P}} \xrightarrow{\text{val}} \mathbb{Q}_{>0} \quad (43.1)$$

and

$$\mathcal{L}_{\mathbb{P}} \xrightarrow{\ell} \mathbb{R}, \quad \ell(a) = \log \text{val}(a), \quad (43.2)$$

with both maps injective.

The analytic chain is

$$\zeta(s) \longrightarrow \xi(s) \longrightarrow \xi(s) = \xi(1-s) \longrightarrow 0 < \Re(\rho) < 1 \quad (43.3)$$

for nontrivial zeros.

A genuine operator proof of the critical-line assertion must additionally supply

$$\mathcal{T} \longrightarrow D(s) \longrightarrow D(s) = C(s)\xi(s) \longrightarrow s = \frac{1}{2} + i\lambda, \quad \lambda \in \mathbb{R}. \quad (43.4)$$

Only after all arrows in this final chain have been established does the critical-line conclusion follow.

Remark 43.1. This separation prevents the arithmetic conclusion from being obtained by a change of notation. The finite operator identities, the measure representation, the prime-lattice embedding, and the analytic properties of ζ are distinct mathematical components. Their composition becomes a theorem only when the explicit maps connecting their domains and codomains are proved.

44 Exact Finite-State Stabilization

We now isolate the algebraic content of the finite-step condition in a form independent of any probabilistic interpretation.

Definition 44.1. Let V be a finite-dimensional vector space over a field $\mathbb{K}$, and let

$$T \in \text{End}_{\mathbb{K}}(V).$$

We say that T is *eventually idempotent at depth m* if

$$T^m = T^{m+1}$$

for some integer $m \geq 1$.

Theorem 44.2. *If T is eventually idempotent at depth m , then*

$$T^n = T^m \quad \text{for every } n \geq m.$$

In particular,

$$P := T^m$$

is idempotent.

Proof. The hypothesis gives

$$T^m = T^{m+1}.$$

Multiplication on the left by T^k yields

$$T^{m+k} = T^{m+k+1} \quad (k \geq 0).$$

Hence

$$T^m = T^{m+1} = T^{m+2} = \dots = T^n$$

for every $n \geq m$. Taking $n = 2m$ gives

$$P^2 = T^{2m} = T^m = P.$$

□

Corollary 44.3. *For an eventually idempotent operator,*

$$V = \ker P \oplus \text{range } P.$$

Moreover,

$$T|_{\text{range } P} = I$$

and

$$T(\ker P) \subseteq \ker P.$$

Proof. For every idempotent operator,

$$v = (v - Pv) + Pv.$$

The first term belongs to $\ker P$, since

$$P(v - Pv) = Pv - P^2v = 0,$$

and the second belongs to $\text{range } P$. If $v \in \ker P \cap \text{range } P$, then $v = Pw$ and

$$v = Pv = P^2w = 0.$$

Thus the sum is direct.

If $v \in \text{range } P$, then $Pv = v$. Since $P = T^m$,

$$Tv = T^{m+1}w = T^m w = Pv = v$$

for $v = Pw$. Finally,

$$P(Tv) = TPv = 0$$

whenever $Pv = 0$, so $\ker P$ is T -invariant. □

45 Minimal Polynomial Consequences

The stabilization identity has an exact polynomial interpretation.

Proposition 45.1. *If*

$$T^m = T^{m+1},$$

then the minimal polynomial $\mu_T(x)$ of T divides

$$x^m(x-1).$$

Consequently, T is diagonalizable over every field containing the roots 0 and 1.

Proof. The stabilization identity is equivalent to

$$T^m(T - I) = 0.$$

Hence the polynomial

$$x^m(x-1)$$

annihilates T , and therefore

$$\mu_T(x) \mid x^m(x-1).$$

The polynomial $x^m(x-1)$ has distinct irreducible factors x and $x-1$. Thus the minimal polynomial of T is square-free. The standard diagonalizability criterion then implies that T is diagonalizable over any field over which x and $x-1$ split. □

Corollary 45.2. *Every eigenvalue of an eventually idempotent operator belongs to*

$$\{0, 1\}.$$

Proof. If $Tv = \lambda v$ with $v \neq 0$, then

$$T^m v = T^{m+1} v$$

gives

$$\lambda^m v = \lambda^{m+1} v.$$

Therefore

$$\lambda^m (1 - \lambda) = 0,$$

so

$$\lambda \in \{0, 1\}.$$

□

Remark 45.3. The stabilization condition is therefore stronger than convergence of powers. It produces an exact algebraic decomposition with no residual spectral modes: after depth m , the 0-eigenspace has been annihilated and the 1-eigenspace is unchanged.

46 The Four-Step Case

For the operator Ψ_9 considered in the finite-state construction, assume

$$\Psi_9^4 = \Psi_9^5.$$

Set

$$P_9 := \Psi_9^4.$$

Theorem 46.1. *The operator P_9 satisfies*

$$P_9^2 = P_9, \quad \Psi_9 P_9 = P_9, \quad P_9 \Psi_9 = P_9.$$

Furthermore,

$$\Psi_9^n = P_9 \quad (n \geq 4).$$

Proof. The first assertion follows from

$$P_9^2 = \Psi_9^8 = \Psi_9^4 = P_9.$$

The stabilization identity gives

$$\Psi_9 P_9 = \Psi_9^5 = \Psi_9^4 = P_9.$$

Since powers of the same operator commute,

$$P_9 \Psi_9 = \Psi_9^5 = P_9.$$

The final statement follows from the preceding general theorem.

□

47 The Nine-Dimensional Uniform Projection

Let

$$J_9 = \mathbf{1}_9 \mathbf{1}_9^T$$

and define

$$B^* = \frac{1}{9} J_9.$$

Proposition 47.1. *The matrix B^* satisfies*

$$(B^*)^2 = B^*, \quad B^* \mathbf{1}_9 = \mathbf{1}_9, \quad \mathbf{1}_9^T B^* = \mathbf{1}_9^T.$$

Its range and kernel are

$$\text{range } B^* = \text{span}\{\mathbf{1}_9\},$$

and

$$\ker B^* = \mathbf{1}_9^\perp.$$

Proof. Since

$$J_9^2 = (\mathbf{1}_9 \mathbf{1}_9^T)(\mathbf{1}_9 \mathbf{1}_9^T) = 9 J_9,$$

one has

$$(B^*)^2 = \frac{1}{81} J_9^2 = \frac{1}{9} J_9 = B^*.$$

The identities involving $\mathbf{1}_9$ follow directly from

$$J_9 \mathbf{1}_9 = 9 \mathbf{1}_9$$

and

$$\mathbf{1}_9^T J_9 = 9 \mathbf{1}_9^T.$$

For $v \in \mathbb{C}^9$,

$$B^* v = \frac{1}{9} \left(\mathbf{1}_9^T v \right) \mathbf{1}_9.$$

Hence the range is

$$\text{span}\{\mathbf{1}_9\},$$

while

$$B^* v = 0$$

if and only if

$$\mathbf{1}_9^T v = 0.$$

□

Corollary 47.2. *The spectrum of B^* is*

$$\sigma(B^*) = \{0, 1\},$$

with algebraic multiplicities

$$8 \quad \text{and} \quad 1,$$

respectively.

Proof. The range is one-dimensional and is the eigenspace corresponding to 1. The kernel has dimension 8 and is the eigenspace corresponding to 0. Since B^* is diagonalizable, these account for the complete spectrum. □

48 Exact Mixing in the Uniform Sector

The uniform projection has a stronger property than mere stabilization.

Theorem 48.1. *For every*

$$p \in \Delta_8,$$

one has

$$B^* p = \pi_9, \quad \pi_9 = \frac{1}{9} \mathbf{1}_9.$$

Consequently,

$$(B^*)^n p = \pi_9 \quad (n \geq 1).$$

Proof. Since

$$\mathbf{1}_9^\top p = 1,$$

the explicit formula for B^* gives

$$B^* p = \frac{1}{9} \left(\mathbf{1}_9^\top p \right) \mathbf{1}_9 = \frac{1}{9} \mathbf{1}_9 = \pi_9.$$

Idempotence then gives

$$(B^*)^n p = B^* p = \pi_9.$$

□

Remark 48.2. The conclusion is an exact finite-time identity. No limit

$$n \rightarrow \infty$$

is involved.

49 A Precise Form of the Primitivity Statement

For a nonnegative matrix A , recall that A is primitive if there exists $k \geq 1$ such that

$$A^k > 0$$

entrywise. The least such k , when it exists, is the primitivity exponent.

Proposition 49.1. *If*

$$\Psi_9^4 = B^* = \frac{1}{9} J_9,$$

then

$$\Psi_9^4 > 0.$$

Consequently, if Ψ_9 is nonnegative, its primitivity exponent satisfies

$$\gamma(\Psi_9) \leq 4.$$

Proof. Every entry of

$$B^* = \frac{1}{9} J_9$$

equals $1/9 > 0$. Hence

$$\Psi_9^4 > 0.$$

If Ψ_9 is nonnegative, the existence of a strictly positive fourth power is precisely the condition that Ψ_9 is primitive with primitivity exponent at most 4. □

Remark 49.2. The equality

$$\gamma(\Psi_9) = 4$$

requires the additional verification

$$\Psi_9^k \not\geq 0 \quad (k = 1, 2, 3).$$

Thus the exact exponent is not implied by

$$\Psi_9^4 > 0$$

alone.

50 Classical Measure Construction

Let H be a finite-dimensional complex Hilbert space and let

$$\mathcal{P}(H)$$

denote its orthogonal projection lattice.

Definition 50.1. A map

$$\mu : \mathcal{P}(H) \longrightarrow [0, 1]$$

is a normalized finitely additive projection measure if

$$\mu(I) = 1$$

and

$$\mu(P + Q) = \mu(P) + \mu(Q)$$

whenever

$$PQ = 0.$$

Theorem 50.2. *Suppose*

$$\dim H \geq 3$$

and μ is a normalized finitely additive projection measure. Then there exists a unique positive semidefinite operator

$$\rho \in \text{End}(H)$$

with

$$\text{tr } \rho = 1$$

such that

$$\mu(P) = \text{tr}(\rho P)$$

for every

$$P \in \mathcal{P}(H).$$

Proof. This is the finite-dimensional form of Gleason's theorem over a complex Hilbert space of dimension at least three. The theorem identifies every normalized finitely additive probability measure on the projection lattice with a unique density operator.

Positivity of the measure gives

$$\rho \geq 0,$$

while normalization gives

$$1 = \mu(I) = \text{tr}(\rho I) = \text{tr } \rho.$$

□

51 Orthogonal Events and Probability Normalization

Let

$$P_1, \dots, P_r \in \mathcal{P}(H)$$

satisfy

$$P_i P_j = 0 \quad (i \neq j)$$

and

$$\sum_{i=1}^r P_i = I.$$

Proposition 51.1. *For the measure represented by ρ ,*

$$p_i := \mu(P_i) = \text{tr}(\rho P_i)$$

satisfies

$$p_i \geq 0$$

and

$$\sum_{i=1}^r p_i = 1.$$

Proof. Since ρ and P_i are positive semidefinite,

$$\text{tr}(\rho P_i) \geq 0.$$

Furthermore,

$$\sum_i p_i = \sum_i \text{tr}(\rho P_i) = \text{tr} \left(\rho \sum_i P_i \right) = \text{tr} \rho = 1.$$

□

Corollary 51.2. *If the projections are rank one,*

$$P_i = |\psi_i\rangle\langle\psi_i|,$$

then

$$p_i = \langle\psi_i, \rho\psi_i\rangle.$$

52 Finite-State to Projection-Lattice Interface

Let

$$P_9$$

be the stabilized finite-state projection and let

$$H_9 := \text{range } P_9$$

be its stabilized complex state space.

The operator layer determines

$$H_9$$

through the exact algebraic relation

$$H_9 = \ker(P_9 - I).$$

The measure layer is then defined independently on

$$\mathcal{P}(H_9).$$

Accordingly, the mathematically non-redundant composition is

$$\Psi_9^4 = \Psi_9^5 \implies P_9^2 = P_9 \implies H_9 = \text{range } P_9 \implies \mathcal{P}(H_9) \implies \mu_F \implies \rho_F. \quad (52.1)$$

Proposition 52.1. *Assume*

$$\dim H_9 \geq 3$$

and that μ_F is a normalized finitely additive projection measure on $\mathcal{P}(H_9)$. Then there exists a unique density operator ρ_F satisfying

$$\mu_F(Q) = \text{tr}(\rho_F Q)$$

for all

$$Q \in \mathcal{P}(H_9).$$

Proof. Apply the preceding measure-representation theorem to H_9 . □

53 Formal Certification Interface

The machine verification is logically subordinate to the mathematical statements proved above: a formal proof establishes the corresponding statement inside the specified formal system, while the mathematical argument identifies its hypotheses and its interpretation.

For each formal theorem C , let

$$\text{Cert}_{\text{L4}}(C)$$

denote the proposition that the Lean kernel accepts the compiled proof term associated with C .

Definition 53.1. A mathematical claim C is *kernel-certified* in the present formalization if there exists a Lean declaration

$$c_C : C$$

whose elaboration and kernel checking terminate successfully in the specified Lean environment.

Proposition 53.2. *If $c_C : C$ is accepted by the Lean kernel, then C is derivable in the formal theory represented by the imported Lean environment and the declarations preceding c_C .*

Proof. A Lean declaration of type C is a proof term of C . Kernel checking verifies that the term has the declared type under the environment available at that declaration. □

Remark 53.3. The certification statement concerns derivability in the formal environment. It does not by itself establish that the formal theorem has been correctly identified with an intended informal claim. That identification is part of the mathematical specification and must be made explicit.

54 Formal Statement of the Stabilization Certificate

The central finite-state certificate is the conjunction

$$\text{Cert}_{\text{stab}} := (\Psi_9^4 = \Psi_9^5) \wedge (P_9 = \Psi_9^4) \wedge (P_9^2 = P_9). \quad (54.1)$$

From (54.1) one obtains, entirely algebraically,

$$\Psi_9^n = P_9 \quad (n \geq 4),$$

and hence the finite transition dynamics possess an exact terminal operator.

If, in addition,

$$P_9 = \frac{1}{9} J_9,$$

then

$$\text{rank } P_9 = 1$$

and

$$\text{range } P_9 = \text{span}\{\mathbf{1}_9\}.$$

55 Scope of the Analytic Conclusion

The finite-state results above do not, by themselves, imply a statement about the zero set of ζ . A rigorous theorem connecting the two structures must contain an explicit analytic interface.

Let

$$D(s)$$

be a function constructed from the operator architecture. Assume that there exists a function $C(s)$, nonvanishing on the relevant domain, such that

$$D(s) = C(s)\xi(s).$$

Theorem 55.1. *Under the preceding identity, the zero set of D in the relevant domain agrees with the zero set of ξ .*

Proof. Since $C(s) \neq 0$,

$$D(s) = 0$$

if and only if

$$\xi(s) = 0.$$

□

Proposition 55.2. *Suppose, in addition, that every nontrivial zero of D admits a representation*

$$\rho = \frac{1}{2} + i\lambda$$

with

$$\lambda \in \mathbb{R}.$$

Then every nontrivial zero of ζ lies on

$$\Re(s) = \frac{1}{2}.$$

Proof. The preceding theorem identifies the nontrivial zero sets of D and ξ , hence of ζ . The assumed representation gives

$$\Re(\rho) = \frac{1}{2}$$

for every such zero. □

Remark 55.3. Consequently, the critical-line assertion is reduced to an explicit spectral correspondence and a reality theorem for its spectral parameter. These are the two statements that must be supplied by the arithmetic/operator construction; neither can be replaced by a notational identification with the line

$$\Re(s) = \frac{1}{2}.$$

56 The Arithmetic Operator and Its Spectral Data

Let $S \subset \mathbb{P}$ be finite and let

$$\mathcal{L}_S = \mathbb{Z}^S.$$

Let

$$\mathcal{H}_S = \ell^2(S)$$

with canonical orthonormal basis

$$\{e_p : p \in S\}.$$

A weighted arithmetic operator is an operator of the form

$$A_S e_p = \sum_{q \in S} a_{qp} e_q,$$

where the coefficients a_{qp} are determined by the specified arithmetic transition rule.

Definition 56.1. An operator A_S is called *arithmetically diagonal* if

$$A_S e_p = \alpha_p e_p$$

for every $p \in S$. Its spectrum is then

$$\sigma(A_S) = \{\alpha_p : p \in S\}.$$

Proposition 56.2. If A_S is arithmetically diagonal and

$$\alpha_p \in \mathbb{R}$$

for every $p \in S$, then A_S is self-adjoint.

Proof. For the canonical basis,

$$\langle A_S e_p, e_q \rangle = \alpha_p \delta_{pq} = \overline{\alpha_q} \delta_{pq} = \langle e_p, A_S e_q \rangle.$$

Thus

$$A_S = A_S^*.$$

□

57 Finite Spectral Approximation of Arithmetic Data

For a finite set $S \subset \mathbb{P}$, define the logarithmic diagonal operator

$$L_S e_p = (\log p) e_p.$$

More generally, for $s = \sigma + it \in \mathbb{C}$, define

$$P_S(s) e_p = p^{-s} e_p = e^{-s \log p} e_p.$$

Proposition 57.1. *The operators L_S and $P_S(s)$ commute, and*

$$P_S(s) = e^{-s L_S}.$$

Proof. Both operators are diagonal in the same canonical basis. Hence they commute. Moreover,

$$e^{-s L_S} e_p = e^{-s \log p} e_p = p^{-s} e_p,$$

which proves the identity. □

Proposition 57.2. *For*

$$s = \sigma + it,$$

the operator $P_S(s)$ is normal and

$$\|P_S(s)\| = \max_{p \in S} p^{-\sigma}.$$

Proof. Since $P_S(s)$ is diagonal, it is normal. Its operator norm is the maximum modulus of its diagonal entries:

$$\|P_S(s)\| = \max_{p \in S} |p^{-s}| = \max_{p \in S} p^{-\sigma}.$$

□

58 The Euler Factors as Spectral Factors

For $\Re(s) > 1$, define

$$E_p(s) = (I - P_p(s))^{-1}, \quad P_p(s) = p^{-s}.$$

Since

$$|p^{-s}| = p^{-\Re(s)} < 1,$$

the inverse exists.

Proposition 58.1. *For $\Re(s) > 1$,*

$$E_p(s) = \sum_{k=0}^{\infty} p^{-ks}.$$

Proof. The scalar p^{-s} has modulus strictly smaller than 1. Therefore the geometric series converges absolutely and

$$(1 - p^{-s})^{-1} = \sum_{k=0}^{\infty} p^{-ks}.$$

□

Theorem 58.2. *For every finite $S \subset \mathbb{P}$ and*

$$\Re(s) > 1,$$

one has

$$\prod_{p \in S} (1 - p^{-s})^{-1} = \sum_{\substack{n \geq 1 \\ \text{supp}(n) \subseteq S}} n^{-s}.$$

Proof. Expand each Euler factor as an absolutely convergent geometric series:

$$\prod_{p \in S} \left(\sum_{k_p=0}^{\infty} p^{-k_p s} \right).$$

Absolute convergence permits rearrangement of the finite product and the resulting multiple series. Each tuple

$$(k_p)_{p \in S}$$

corresponds uniquely to

$$n = \prod_{p \in S} p^{k_p}.$$

Unique factorization gives the stated identity. □

59 The Infinite Euler Product

Let

$$S_N = \{p \in \mathbb{P} : p \leq N\}.$$

For $\Re(s) > 1$, define

$$Z_N(s) = \prod_{p \leq N} (1 - p^{-s})^{-1}.$$

Theorem 59.1. *For every compact subset*

$$K \subset \{s \in \mathbb{C} : \Re(s) > 1\},$$

the sequence Z_N converges uniformly on K to ζ .

Proof. Let

$$\delta = \min_{s \in K} \Re(s) - 1 > 0.$$

Then

$$|p^{-s}| \leq p^{-1-\delta}$$

uniformly for $s \in K$. Since

$$\sum_p p^{-1-\delta} < \infty,$$

the Euler product converges uniformly on K . Its limit is the Euler product for $\zeta(s)$, hence

$$Z_N(s) \longrightarrow \zeta(s)$$

uniformly on K . □

60 Logarithmic Derivative

For $\Re(s) > 1$, differentiation of the absolutely convergent Euler product yields

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{p \in \mathbb{P}} \frac{\log p}{p^s - 1}.$$

Equivalently,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^s},$$

where Λ is the von Mangoldt function.

Definition 60.1. The von Mangoldt function is

$$\Lambda(n) = \begin{cases} \log p, & n = p^k, \quad p \in \mathbb{P}, \quad k \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Proposition 60.2. For $\Re(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} (\log p) p^{-ks}.$$

Proof. Taking the logarithm of the Euler product gives

$$\log \zeta(s) = - \sum_p \log(1 - p^{-s}).$$

The series converges absolutely for $\Re(s) > 1$, so it may be differentiated termwise:

$$\frac{\zeta'(s)}{\zeta(s)} = - \sum_p \frac{(\log p) p^{-s}}{1 - p^{-s}}.$$

Expanding the denominator,

$$\frac{p^{-s}}{1 - p^{-s}} = \sum_{k=1}^{\infty} p^{-ks},$$

which gives the result. □

61 The Completed Spectral Function

Define

$$\Lambda_{\zeta}(s) = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Then

$$\xi(s) = \frac{1}{2} s(s-1) \Lambda_{\zeta}(s).$$

Proposition 61.1. The function ξ is entire and satisfies

$$\xi(s) = \xi(1-s).$$

Furthermore,

$$\xi(\bar{s}) = \overline{\xi(s)}.$$

Proof. The first two assertions follow from the analytic continuation and functional equation of the Riemann zeta function together with the standard properties of the gamma factor.

For the final assertion, the Dirichlet series gives

$$\zeta(\bar{s}) = \overline{\zeta(s)}$$

in the half-plane of absolute convergence. Analytic continuation extends the identity to the common domain. The gamma and elementary factors possess the same conjugation property, giving

$$\xi(\bar{s}) = \overline{\xi(s)}.$$

□

62 Necessary Spectral Criterion for a Critical-Line Theorem

The symmetry of ξ alone does not force its zeros onto the critical line. The additional input required by a spectral construction can be stated precisely.

Definition 62.1. A spectral realization of ξ consists of a Hilbert space $\mathcal{H}$, a self-adjoint operator A on $\mathcal{H}$, and a nonvanishing function C such that the nontrivial zero condition for ξ is equivalent to

$$\xi\left(\frac{1}{2} + i\lambda\right) = 0$$

for spectral parameters

$$\lambda \in \sigma(A) \subset \mathbb{R}.$$

Theorem 62.2. Suppose a spectral realization of ξ has the property that every nontrivial zero ρ of ξ is represented by a spectral value $\lambda \in \sigma(A)$ according to

$$\rho = \frac{1}{2} + i\lambda.$$

Then the Riemann hypothesis holds.

Proof. Since A is self-adjoint,

$$\sigma(A) \subset \mathbb{R}.$$

Thus every represented zero has the form

$$\rho = \frac{1}{2} + i\lambda$$

with $\lambda \in \mathbb{R}$. Hence

$$\Re(\rho) = \frac{1}{2}.$$

By hypothesis every nontrivial zero is represented in this way. Therefore every nontrivial zero lies on the critical line. □

63 What Must Be Established by the Faure Construction

The preceding theorem identifies the exact mathematical interface required for an operator-theoretic proof. It is sufficient to establish the following independent assertions.

1. There exists a precisely defined operator A_F associated with the arithmetic construction.

2. A_F is self-adjoint, or otherwise possesses a theorem forcing the relevant spectral parameters to be real.
3. There exists an explicitly defined analytic function $D_F(s)$ constructed from A_F .
4. There exists a nonvanishing function $C_F(s)$ such that

$$D_F(s) = C_F(s)\xi(s)$$

on a domain containing the nontrivial zeros.

5. The spectral equation

$$D_F(s) = 0$$

is equivalent to

$$s = \frac{1}{2} + i\lambda, \quad \lambda \in \sigma(A_F).$$

Under these hypotheses the conclusion

$$\Re(\rho) = \frac{1}{2}$$

is a theorem and not a consequence of a formal relabelling of the critical line.

64 Finite-State and Arithmetic Layers

The finite-state and arithmetic constructions can now be placed in a single commutative dependency diagram:

$$\begin{array}{ccccccc}
 \Psi_9 & \xrightarrow{\Psi_9^4 = \Psi_9^5} & P_9 & \longrightarrow & H_9 & & \\
 & & \downarrow & & \downarrow & & \\
 \mathcal{L}_{\mathbb{P}} & \xrightarrow{\ell} & \mathbb{R} & \longrightarrow & A_F & \longrightarrow & D_F(s) \longrightarrow \xi(s).
 \end{array}$$

The upper row is finite-dimensional and algebraic. The lower row is arithmetic and analytic. The only valid inference from the upper row to the lower row is through explicitly defined maps preserving the relevant structures.

Remark 64.1. In particular, the identities

$$\Psi_9^4 = \Psi_9^5 \quad \text{and} \quad B^* = \frac{1}{9}J_9$$

establish exact finite-state stabilization. They do not, without the additional interface above, constitute an analytic statement about ξ or ζ .

65 Main Structural Theorem

Theorem 65.1. *Let Ψ_9 be a finite-dimensional transition operator satisfying*

$$\Psi_9^4 = \Psi_9^5,$$

and put

$$P_9 = \Psi_9^4.$$

Then:

1. P_9 is idempotent;

2.

$$\Psi_9^n = P_9 \quad (n \geq 4);$$

3.

$$V = \ker P_9 \oplus \text{range } P_9;$$

4. every vector in $\text{range } P_9$ is fixed by Ψ_9 ;

5. if

$$P_9 = \frac{1}{9} J_9,$$

then

$$\text{range } P_9 = \text{span}\{\mathbf{1}_9\}$$

and the unique stationary probability vector is

$$\pi_9 = \frac{1}{9} \mathbf{1}_9;$$

6. if the stabilized range carries a normalized finitely additive projection measure and has complex dimension at least 3, then that measure is represented by a unique density operator ρ_F ;

7. if, independently, the arithmetic construction yields a self-adjoint spectral operator A_F and an exact spectral identity

$$D_F(s) = C_F(s)\xi(s), \quad C_F(s) \neq 0,$$

with every nontrivial zero represented by

$$s = \frac{1}{2} + i\lambda, \quad \lambda \in \sigma(A_F),$$

then every nontrivial zero of ζ lies on

$$\Re(s) = \frac{1}{2}.$$

Proof. Statements (1)–(4) follow from the exact stabilization theorem. Statement (5) follows from the explicit computation of B^* . Statement (6) follows from the projection-measure representation theorem.

For (7), the nonvanishing of C_F gives equality of the zero sets of D_F and ξ . Self-adjointness gives

$$\sigma(A_F) \subseteq \mathbb{R}.$$

Therefore every nontrivial zero has the form

$$s = \frac{1}{2} + i\lambda$$

with real λ , and hence has real part 1/2. □

66 Arithmetic Lattice and Finite Projections

Let $\mathbb{P}$ denote the set of prime numbers and define the free abelian group

$$\mathcal{L}_{\mathbb{P}} := \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p.$$

Every element of $\mathcal{L}_{\mathbb{P}}$ has finite support and can therefore be written uniquely as

$$x = \sum_{p \in \mathbb{P}} n_p e_p, \quad n_p \in \mathbb{Z},$$

with

$$n_p = 0$$

for all but finitely many p .

Define the logarithmic functional

$$\ell : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{R}$$

by

$$\ell(x) = \sum_{p \in \mathbb{P}} n_p \log p.$$

Proposition 66.1. *The map ℓ is a group homomorphism and satisfies*

$$\ell(x) = \log \left(\prod_{p \in \mathbb{P}} p^{n_p} \right).$$

Proof. Finite support makes the defining sum finite. Hence

$$\ell(x + y) = \sum_p (n_p + m_p) \log p = \ell(x) + \ell(y).$$

The logarithmic identity follows from

$$\log \left(\prod_p p^{n_p} \right) = \sum_p n_p \log p.$$

□

Proposition 66.2. *The functional ℓ is injective.*

Proof. Suppose

$$\ell(x) = 0.$$

Then

$$\log \left(\prod_p p^{n_p} \right) = 0,$$

so

$$\prod_p p^{n_p} = 1.$$

Separating positive and negative exponents gives an equality of two positive integers with co-prime prime factorizations. Unique factorization therefore implies

$$n_p = 0$$

for every p . Hence $x = 0$.

□

67 Finite-Coordinate Restrictions

For a finite subset

$$S \subset \mathbb{P},$$

let

$$\mathcal{L}_S = \bigoplus_{p \in S} \mathbb{Z} e_p$$

and let

$$\pi_S : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{L}_S$$

be the coordinate projection.

Proposition 67.1. *The maps π_S satisfy*

$$\pi_S^2 = \pi_S$$

and, for finite $S, T \subset \mathbb{P}$,

$$\pi_S \pi_T = \pi_{S \cap T}.$$

Proof. For every basis vector e_p ,

$$\pi_S(e_p) = \begin{cases} e_p, & p \in S, \\ 0, & p \notin S. \end{cases}$$

Applying π_S twice gives the same result, proving

$$\pi_S^2 = \pi_S.$$

Likewise,

$$\pi_S \pi_T(e_p) = e_p$$

exactly when $p \in S \cap T$, and is zero otherwise. Hence

$$\pi_S \pi_T = \pi_{S \cap T}.$$

□

Corollary 67.2. *For an increasing sequence*

$$S_1 \subseteq S_2 \subseteq \cdots$$

of finite subsets of $\mathbb{P}$,

$$\pi_{S_n} \pi_{S_m} = \pi_{S_{\min\{n,m\}}}.$$

68 Exact Finite Support

The preceding projections provide an exact, rather than asymptotic, finite representation of every lattice element.

Theorem 68.1. *For every*

$$x \in \mathcal{L}_{\mathbb{P}}$$

there exists a finite subset $S \subset \mathbb{P}$ such that

$$\pi_S x = x.$$

Proof. By definition of the direct sum, x has finite support

$$\text{supp}(x) = \{p \in \mathbb{P} : n_p \neq 0\}.$$

Taking

$$S = \text{supp}(x)$$

gives

$$\pi_S x = x.$$

□

Remark 68.2. This is the precise sense in which finite-coordinate restrictions are exact on the algebraic lattice: no limiting procedure is required to recover an individual lattice element.

69 Multiplicative Encoding

Define

$$\mathfrak{m} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{Q}_{>0}$$

by

$$\mathfrak{m}(x) = \prod_{p \in \mathbb{P}} p^{n_p}.$$

Proposition 69.1. *The map $\mathfrak{m}$ is a group isomorphism from $\mathcal{L}_{\mathbb{P}}$ onto the multiplicative group $\mathbb{Q}_{>0}^{\times}$.*

Proof. Multiplicativity follows from

$$\mathfrak{m}(x + y) = \prod_p p^{n_p + m_p} = \mathfrak{m}(x)\mathfrak{m}(y).$$

Every positive rational number possesses a unique prime factorization with finitely many nonzero integer exponents, so $\mathfrak{m}$ is surjective and injective. □

Corollary 69.2. *The logarithmic embedding factors as*

$$\ell = \log \circ \mathfrak{m}.$$

70 Character Family

For $s \in \mathbb{C}$, define

$$\chi_s(x) = e^{-s\ell(x)}.$$

Proposition 70.1. *For every $s \in \mathbb{C}$,*

$$\chi_s(x + y) = \chi_s(x)\chi_s(y).$$

Thus χ_s is a character of the additive lattice $\mathcal{L}_{\mathbb{P}}$.

Proof. Since ℓ is additive,

$$\chi_s(x + y) = e^{-s(\ell(x) + \ell(y))} = e^{-s\ell(x)} e^{-s\ell(y)} = \chi_s(x)\chi_s(y).$$

□

Proposition 70.2. *For*

$$s = \frac{1}{2} + it,$$

one has

$$|\chi_s(x)| = e^{-\ell(x)/2}.$$

After unitary normalization by the positive factor $e^{\ell(x)/2}$, the residual phase is

$$e^{-it\ell(x)},$$

which has modulus 1.

Proof. Writing

$$s = \frac{1}{2} + it$$

gives

$$\left| e^{-s\ell(x)} \right| = e^{-\Re(s)\ell(x)} = e^{-\ell(x)/2}.$$

The remaining factor is

$$e^{-it\ell(x)},$$

whose modulus is one because $\ell(x) \in \mathbb{R}$. □

71 Hermitian Symmetry of the Critical-Line Parameter

Let A be a self-adjoint operator on a Hilbert space $\mathcal{H}$. For every spectral value

$$\lambda \in \sigma(A),$$

one has

$$\lambda \in \mathbb{R}.$$

Theorem 71.1. *Suppose an analytic function D satisfies*

$$D\left(\frac{1}{2} + i\lambda\right) = 0$$

if and only if

$$\lambda \in \sigma(A),$$

where $A = A^$. Then every zero of D belongs to the critical line*

$$\Re(s) = \frac{1}{2}.$$

Proof. By self-adjointness,

$$\lambda \in \sigma(A) \implies \lambda \in \mathbb{R}.$$

Hence every zero represented by the spectral equation has the form

$$s = \frac{1}{2} + i\lambda$$

with real λ , and therefore

$$\Re(s) = \frac{1}{2}.$$
□

Remark 71.2. The substantive content of a spectral proof is therefore the construction and verification of the equivalence

$$D\left(\frac{1}{2} + i\lambda\right) = 0 \iff \lambda \in \sigma(A).$$

Self-adjointness supplies the reality of λ ; it does not supply the spectral correspondence itself.

72 Determinantal Spectral Functions

Let A be an $N \times N$ Hermitian matrix. Define its characteristic polynomial by

$$\Delta_A(\lambda) = \det(\lambda I - A).$$

Proposition 72.1. *Every zero of Δ_A is real.*

Proof. The zeros of Δ_A are precisely the eigenvalues of A . Since A is Hermitian, the spectral theorem gives an orthonormal basis of eigenvectors with real eigenvalues. $\square$

Define

$$D_A(s) = \Delta_A\left(-i\left(s - \frac{1}{2}\right)\right).$$

Proposition 72.2. *Every zero s of D_A satisfies*

$$\Re(s) = \frac{1}{2}.$$

Proof. If

$$D_A(s) = 0,$$

then

$$-i\left(s - \frac{1}{2}\right) = \lambda$$

for some real eigenvalue λ . Hence

$$s = \frac{1}{2} + i\lambda,$$

so

$$\Re(s) = \frac{1}{2}.$$

$\square$

73 Finite Spectral Models and Their Limitation

The preceding construction yields a rigorous finite-dimensional mechanism producing critical-line zeros. It is important to distinguish this statement from an assertion about ζ .

Proposition 73.1. *Let $A_N = A_N^*$ be a sequence of finite-dimensional Hermitian operators and let*

$$D_N(s) = \det\left(-i\left(s - \frac{1}{2}\right)I - A_N\right).$$

Every zero of every D_N lies on

$$\Re(s) = \frac{1}{2}.$$

Proof. This follows from the preceding determinantal proposition applied to each A_N . $\square$

Theorem 73.2. *Suppose that there exists a locally uniform limit*

$$D_N \longrightarrow D$$

on a domain $\Omega \subset \mathbb{C}$, and that

$$D \neq 0.$$

Then every zero of D in Ω lies on

$$\Re(s) = \frac{1}{2}.$$

Proof. Let $s_0 \in \Omega$ satisfy $D(s_0) = 0$. Suppose, for contradiction, that

$$\Re(s_0) \neq \frac{1}{2}.$$

Choose a closed disc $K \subset \Omega$ centered at s_0 whose closure does not intersect the critical line. Since D is not identically zero, its zeros are isolated. Choose the disc so that s_0 is the only zero of D in K .

By local uniform convergence and Hurwitz's theorem, for sufficiently large N , D_N has a zero in K . This contradicts the fact that every zero of D_N lies on the critical line, while K is disjoint from it. Hence

$$\Re(s_0) = \frac{1}{2}.$$

□

74 Exact Analytic Interface

The finite-to-infinite passage can therefore be formulated as an explicit theorem rather than as an asymptotic heuristic.

Theorem 74.1. *Let $\Omega \subset \mathbb{C}$ be a domain and let*

$$D_N : \Omega \longrightarrow \mathbb{C}$$

be analytic functions satisfying:

1. *every zero of D_N lies on*

$$\Re(s) = \frac{1}{2};$$

2. *$D_N \rightarrow D$ locally uniformly on Ω ;*

3. *$D \neq 0$;*

4. *there exists a nonvanishing analytic function C on Ω such that*

$$D = C\xi.$$

Then every zero of ξ in Ω lies on

$$\Re(s) = \frac{1}{2}.$$

Proof. By the preceding theorem, every zero of D lies on the critical line. Since

$$C(s) \neq 0,$$

the identity

$$D(s) = C(s)\xi(s)$$

implies

$$D(s) = 0 \iff \xi(s) = 0.$$

Hence every zero of ξ in Ω lies on the critical line.

□

75 Required Verification of the Arithmetic Identification

The preceding theorem reduces the analytic conclusion to a single identification problem.

Proposition 75.1. *To obtain a proof of the critical-line statement from a sequence of finite Hermitian spectral models, it is sufficient to establish the identity*

$$\lim_{N \rightarrow \infty} D_N(s) = C(s)\xi(s)$$

locally uniformly on a domain containing all nontrivial zeros, with C analytic and nowhere zero there.

Proof. The conclusion is an immediate application of the preceding theorem. $\square$

Remark 75.2. Thus the non-tautological mathematical burden is concentrated in the construction of D_N , the proof of local uniform convergence, and the identification of its limit with ξ . The spectral reality argument is independent of that identification.

76 The Finite-Rank Spectral Determinant

Let H be a finite-dimensional complex Hilbert space and let $A = A^* \in \text{End}(H)$. For $z \in \mathbb{C}$, define

$$D_A(z) := \det(zI - A).$$

Proposition 76.1. *The polynomial D_A admits the factorization*

$$D_A(z) = \prod_{j=1}^d (z - \lambda_j),$$

where $d = \dim H$ and

$$\lambda_j \in \mathbb{R}.$$

Proof. By the finite-dimensional spectral theorem there exists a unitary operator U such that

$$U^*AU = \text{diag}(\lambda_1, \dots, \lambda_d)$$

with every $\lambda_j \in \mathbb{R}$. Since determinants are invariant under similarity,

$$D_A(z) = \det(zI - U^*AU) = \prod_{j=1}^d (z - \lambda_j).$$

$\square$

Definition 76.2. The associated critical-line determinant is

$$\mathcal{D}_A(s) := D_A\left(-i\left(s - \frac{1}{2}\right)\right).$$

Proposition 76.3. *The zero divisor of $\mathcal{D}_A$ is*

$$\sum_{j=1}^d \left[s - \left(\frac{1}{2} + i\lambda_j \right) \right].$$

In particular,

$$\mathcal{D}_A(s) = 0 \implies \Re(s) = \frac{1}{2}.$$

Proof. The factorization of D_A gives

$$\mathcal{D}_A(s) = \prod_{j=1}^d \left(-i \left(s - \frac{1}{2} \right) - \lambda_j \right).$$

A factor vanishes precisely when

$$s = \frac{1}{2} + i\lambda_j.$$

Since every λ_j is real, every zero has real part $1/2$. □

77 Symmetry of the Finite Spectral Determinant

For a Hermitian operator A , define

$$\mathcal{D}_A(s) = \det \left(-i \left(s - \frac{1}{2} \right) I - A \right).$$

Proposition 77.1. *The function $\mathcal{D}_A$ satisfies*

$$\mathcal{D}_A(1-s) = (-1)^d \mathcal{D}_A(s).$$

Moreover,

$$\overline{\mathcal{D}_A(\bar{s})} = (-1)^d \mathcal{D}_A(s).$$

Proof. Using

$$1 - s - \frac{1}{2} = - \left(s - \frac{1}{2} \right),$$

one obtains

$$\mathcal{D}_A(1-s) = \det \left(i \left(s - \frac{1}{2} \right) I - A \right).$$

Since

$$i \left(s - \frac{1}{2} \right) I - A = - \left[-i \left(s - \frac{1}{2} \right) I + A \right],$$

and the eigenvalue factorization gives

$$\mathcal{D}_A(s) = \prod_j \left[-i \left(s - \frac{1}{2} \right) - \lambda_j \right],$$

while

$$\mathcal{D}_A(1-s) = \prod_j \left[i \left(s - \frac{1}{2} \right) - \lambda_j \right],$$

the asserted relation follows after replacing λ_j by $-\lambda_j$ when the spectrum is symmetric.

Thus the functional symmetry with sign $(-1)^d$ requires the additional spectral symmetry

$$\sigma(A) = -\sigma(A).$$

Under this hypothesis the conjugation identity follows from the reality of the λ_j . □

Remark 77.2. The spectral reality and the spectral symmetry are logically distinct. Self-adjointness gives

$$\lambda \in \mathbb{R};$$

it does not imply

$$\lambda \in \sigma(A) \implies -\lambda \in \sigma(A).$$

The latter must be established separately whenever the functional equation is represented at the operator level.

78 Antilinear Spectral Symmetry

Let $A = A^*$ on a finite-dimensional complex Hilbert space H .

Definition 78.1. An antiunitary operator $J : H \rightarrow H$ is a conjugate-linear bijection satisfying

$$\langle Jx, Jy \rangle = \overline{\langle x, y \rangle}.$$

Proposition 78.2. Suppose there exists an antiunitary J satisfying

$$JAJ^{-1} = -A.$$

Then

$$\sigma(A) = -\sigma(A).$$

Proof. Let

$$Av = \lambda v$$

with $v \neq 0$. Applying J gives

$$JAJ^{-1}Jv = JAv = J(\lambda v) = \lambda Jv,$$

where the last equality uses that $\lambda \in \mathbb{R}$, since A is self-adjoint. The assumed intertwining relation therefore gives

$$-AJv = \lambda Jv,$$

hence

$$A(Jv) = -\lambda Jv.$$

Thus $-\lambda \in \sigma(A)$. Applying the same argument to $-\lambda$ proves equality of the two sets. $\square$

Corollary 78.3. Under the hypotheses above, the zeros of

$$\mathcal{D}_A(s) = \det \left(-i \left(s - \frac{1}{2} \right) I - A \right)$$

occur in the symmetric pairs

$$s \quad \text{and} \quad 1 - s.$$

79 A Normalized Spectral Counting Function

Let $A = A^*$ have eigenvalues

$$\lambda_1, \dots, \lambda_d$$

counted with algebraic multiplicity. Define

$$N_A(T) = \#\{j : |\lambda_j| \leq T\}.$$

Proposition 79.1. The function N_A is nondecreasing, integer-valued, and satisfies

$$0 \leq N_A(T) \leq d.$$

Moreover,

$$N_A(T) = d$$

for every

$$T \geq \|A\|.$$

Proof. The defining set increases with T , so N_A is nondecreasing. It counts a subset of the d eigenvalues, hence

$$0 \leq N_A(T) \leq d.$$

For a self-adjoint finite-dimensional operator,

$$|\lambda_j| \leq \|A\|$$

for every j , so all eigenvalues are counted whenever $T \geq \|A\|$. $\square$

Remark 79.2. For an infinite spectral realization, the corresponding counting function is non-trivial only if the operator has discrete spectrum of finite multiplicity in bounded intervals. This property must therefore be included explicitly in any infinite-dimensional construction.

80 Compact-Resolvent Spectral Framework

Let $A = A^*$ be a self-adjoint operator on a separable Hilbert space H .

Definition 80.1. We say that A has compact resolvent if, for some

$$z \in \mathbb{C} \setminus \mathbb{R},$$

the operator

$$(A - zI)^{-1}$$

is compact.

Theorem 80.2. *If A is self-adjoint with compact resolvent, then its spectrum is real and consists entirely of isolated eigenvalues of finite multiplicity, with no finite accumulation point.*

Proof. This is the spectral theorem for self-adjoint operators with compact resolvent. Self-adjointness gives

$$\sigma(A) \subset \mathbb{R}.$$

Compactness of the resolvent converts the spectral problem into a discrete eigenvalue problem: every spectral point is an eigenvalue of finite multiplicity, and the only possible accumulation occurs at infinity. $\square$

Definition 80.3. For such an operator define

$$N_A(T) = \dim \text{Ran } \mathbf{1}_{[-T, T]}(A),$$

where $\mathbf{1}_{[-T, T]}(A)$ denotes the spectral projection of A associated with the interval $[-T, T]$.

Proposition 80.4. *If A has compact resolvent, then*

$$N_A(T) < \infty$$

for every finite T .

Proof. The spectrum in a bounded interval consists of finitely many eigenvalues counted with finite multiplicity. Hence the corresponding spectral subspace is finite-dimensional. $\square$

81 Spectral Determinants Beyond Finite Dimension

Let $A = A^*$ have compact resolvent and eigenvalues

$$\{\lambda_n\}_{n \geq 1},$$

counted with multiplicity.

A naive determinant

$$\prod_{n=1}^{\infty} \left(-i \left(s - \frac{1}{2} \right) - \lambda_n \right)$$

is generally divergent. A spectral determinant therefore requires a specified regularization.

Definition 81.1. Let $r > 0$ be such that

$$\sum_{n=1}^{\infty} |\lambda_n|^{-r} < \infty.$$

The canonical Weierstrass regularization of the spectral product is

$$\mathfrak{D}_A(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{\lambda_n} \right) \exp \left(\sum_{k=1}^{r-1} \frac{1}{k} \left(\frac{z}{\lambda_n} \right)^k \right),$$

after separating any zero eigenvalues.

Proposition 81.2. *The preceding product converges locally uniformly in z whenever the stated summability condition holds.*

Proof. For sufficiently small z/λ_n , the logarithm of the regularized factor is

$$- \sum_{k=r}^{\infty} \frac{1}{k} \left(\frac{z}{\lambda_n} \right)^k.$$

On every compact set in z , this is dominated by a constant multiple of

$$|\lambda_n|^{-r}.$$

The assumed summability therefore gives uniform convergence of the logarithmic series on compact sets, and hence local uniform convergence of the product. $\square$

82 The Required Arithmetic Spectral Identification

Let A_F be a self-adjoint operator with compact resolvent and let $\mathfrak{D}_F$ denote a rigorously defined regularized spectral determinant.

Definition 82.1. An *exact zeta identification* consists of an analytic, nowhere-vanishing function C_F on a domain Ω and an identity

$$\mathfrak{D}_F(s) = C_F(s) \xi(s)$$

for every

$$s \in \Omega.$$

Theorem 82.2. Assume that $A_F = A_F^*$ has compact resolvent and that its spectral determinant satisfies the exact zeta identification on a domain Ω containing every nontrivial zero of ξ . Assume further that

$$\mathfrak{D}_F(s) = 0$$

if and only if

$$s = \frac{1}{2} + i\lambda$$

for some

$$\lambda \in \sigma(A_F).$$

Then every nontrivial zero of ζ lies on

$$\Re(s) = \frac{1}{2}.$$

Proof. Let $\rho \in \Omega$ be a nontrivial zero of ζ . The completed function satisfies

$$\xi(\rho) = 0.$$

Since C_F is nowhere zero,

$$\mathfrak{D}_F(\rho) = 0.$$

By the assumed spectral characterization,

$$\rho = \frac{1}{2} + i\lambda$$

for some

$$\lambda \in \sigma(A_F).$$

Self-adjointness gives

$$\lambda \in \mathbb{R}.$$

Therefore

$$\Re(\rho) = \frac{1}{2}.$$

Since ρ was arbitrary, every nontrivial zero lies on the critical line. □

83 Arithmetic Completeness Requirement

The preceding theorem shows that the decisive statement is the exact identification of the arithmetic determinant with ξ . We therefore record the required form without suppressing its analytic content.

Axiom 83.1 (Arithmetic spectral identity). There exist a self-adjoint operator A_F with compact resolvent, a spectral determinant $\mathfrak{D}_F$, and a nowhere-vanishing analytic function C_F such that

$$\mathfrak{D}_F(s) = C_F(s)\xi(s)$$

throughout a domain containing the nontrivial zero set of ξ .

Axiom 83.2 (Spectral parametrization). The zero set of $\mathfrak{D}_F$ in that domain is exactly

$$\left\{ \frac{1}{2} + i\lambda : \lambda \in \sigma(A_F) \right\}.$$

Theorem 83.3. The two preceding axioms imply the Riemann hypothesis.

Proof. The conclusion is the preceding spectral identification theorem. □

84 Logical Status of the Structural Framework

The finite-state portion of the construction is established by exact algebraic identities:

$$\Psi_9^4 = \Psi_9^5 \implies P_9^2 = P_9 \implies \Psi_9^n = P_9 \quad (n \geq 4).$$

The arithmetic lattice is established independently by the free abelian decomposition

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

and the injective logarithmic embedding

$$\ell(x) = \sum_p n_p \log p.$$

The analytic conclusion requires, in addition, an explicitly proved spectral identification of the completed zeta function. In its exact logical form, the implication is

$$\boxed{\begin{array}{l} A_F = A_F^*, \\ (A_F - zI)^{-1} \text{ compact}, \\ \mathfrak{D}_F(s) = C_F(s)\xi(s), \quad C_F(s) \neq 0, \\ \mathfrak{D}_F(s) = 0 \iff s = \frac{1}{2} + i\lambda, \quad \lambda \in \sigma(A_F) \end{array}} \implies \boxed{\forall \rho \in Z_{\text{nt}}(\zeta), \quad \Re(\rho) = \frac{1}{2}}.$$

No limiting argument, probabilistic interpretation, or change of notation can substitute for any of the four displayed hypotheses.

85 Trace Formulation of the Spectral Data

Let $A = A^*$ be a self-adjoint operator with compact resolvent. For $t \in \mathbb{R}$, define the spectral counting measure

$$\nu_A = \sum_{\lambda \in \sigma(A)} m_A(\lambda) \delta_\lambda,$$

where $m_A(\lambda)$ denotes the multiplicity of λ .

For every bounded Borel function f for which the following sum converges absolutely,

$$\text{Tr}(f(A)) = \sum_{\lambda \in \sigma(A)} m_A(\lambda) f(\lambda).$$

Proposition 85.1. *If f has finite support on the spectrum of A , then*

$$\text{Tr}(f(A)) = \int_{\mathbb{R}} f(\lambda) d\nu_A(\lambda).$$

Proof. The spectral theorem gives

$$f(A) = \sum_{\lambda \in \sigma(A)} f(\lambda) E_\lambda,$$

where E_λ is the spectral projection associated with λ . Since only finitely many terms are nonzero,

$$\text{Tr}(f(A)) = \sum_{\lambda} f(\lambda) \text{Tr}(E_\lambda) = \sum_{\lambda} m_A(\lambda) f(\lambda).$$

This is precisely the stated integral. □

Definition 85.2. The spectral counting function of A is

$$N_A(T) = \nu_A([-T, T]) = \sum_{|\lambda| \leq T} m_A(\lambda).$$

If the spectrum is symmetric,

$$\sigma(A) = -\sigma(A),$$

then

$$N_A(T) = 2 \sum_{0 \leq \lambda \leq T} m_A(\lambda) - m_A(0).$$

86 Spectral Symmetry and the Completed Function

Suppose that $A = A^*$ has compact resolvent and that

$$\sigma(A) = -\sigma(A)$$

with equality of multiplicities.

Define the spectral measure

$$\nu_A = \sum_{\lambda \in \sigma(A)} m_A(\lambda) \delta_\lambda.$$

Proposition 86.1. *Under the spectral symmetry assumption,*

$$\nu_A(E) = \nu_A(-E)$$

for every Borel set $E \subset \mathbb{R}$.

Proof. The map

$$\lambda \mapsto -\lambda$$

preserves multiplicities by hypothesis. Therefore

$$\sum_{\lambda \in E} m_A(\lambda) = \sum_{\lambda \in -E} m_A(\lambda),$$

which is the asserted equality. □

Proposition 86.2. *For every integrable function f ,*

$$\int_{\mathbb{R}} f(\lambda) d\nu_A(\lambda) = \int_{\mathbb{R}} f(-\lambda) d\nu_A(\lambda).$$

Proof. This follows immediately from the invariance of ν_A under $\lambda \mapsto -\lambda$. □

Remark 86.3. The symmetry of the spectral measure is the precise operator-theoretic counterpart of the reflection

$$s \mapsto 1 - s$$

after the parametrization

$$s = \frac{1}{2} + i\lambda.$$

The two symmetries should not be identified until the corresponding analytic map has been explicitly defined.

87 Hadamard Factorization and Zero Data

Let ξ denote the completed Riemann xi-function. Its zeros are denoted, with multiplicity, by

$$\{\rho\}.$$

Since ξ is entire of finite order, Hadamard factorization gives a canonical factorization of the form

$$\xi(s) = e^{a+bs} \prod_{\rho} E_m\left(\frac{s}{\rho}\right),$$

where

$$E_m(z) = (1-z) \exp\left(z + \frac{z^2}{2} + \cdots + \frac{z^m}{m}\right)$$

and m may be chosen according to the order of ξ .

Proposition 87.1. *An exact spectral realization of ξ must reproduce its zero divisor with multiplicity, up to multiplication by a nowhere-vanishing entire factor.*

Proof. Two nonzero entire functions with the same zeros, counted with multiplicity, differ by a nowhere-vanishing entire function. This follows from their quotient after cancellation of the common zero divisors. $\square$

Theorem 87.2. *Let F be an entire function whose zero divisor, including multiplicity, agrees with that of ξ . Then*

$$F(s) = e^{g(s)} \xi(s)$$

for some entire function g .

If, in addition, F and ξ have the same growth normalization sufficient to force g to be constant, then

$$F(s) = C \xi(s)$$

for some $C \neq 0$.

Proof. The quotient

$$F/\xi$$

extends holomorphically across every zero because the multiplicities agree, and it has no zeros. Since the domain is $\mathbb{C}$, a nowhere-vanishing entire function admits an entire logarithm:

$$F/\xi = e^g.$$

The additional growth normalization determines g up to a constant in the standard Hadamard setting. $\square$

88 Spectral Zero Divisor

Let $A = A^*$ have compact resolvent and suppose that

$$\lambda \mapsto -\lambda$$

preserves spectral multiplicities.

Define the formal zero divisor

$$Z_A = \sum_{\lambda \in \sigma(A)} m_A(\lambda) \left[\frac{1}{2} + i\lambda \right].$$

Definition 88.1. A spectral realization of the nontrivial zero divisor of ξ is an equality of divisors

$$Z_{\xi, \text{nt}} = Z_A.$$

Theorem 88.2. *If*

$$Z_{\xi, \text{nt}} = Z_A,$$

then every nontrivial zero ρ of ξ satisfies

$$\rho = \frac{1}{2} + i\lambda$$

for some

$$\lambda \in \mathbb{R}.$$

Consequently,

$$\Re(\rho) = \frac{1}{2}.$$

Proof. By equality of divisors, every nontrivial zero of ξ occurs among the points

$$\frac{1}{2} + i\lambda, \quad \lambda \in \sigma(A).$$

Self-adjointness gives

$$\sigma(A) \subset \mathbb{R}.$$

Therefore

$$\Re(\rho) = \frac{1}{2}.$$

□

Remark 88.3. The multiplicity statement is stronger than mere containment of the zero sets. Equality of divisors requires

$$m_{\xi}(\rho) = m_A(\lambda)$$

whenever

$$\rho = \frac{1}{2} + i\lambda.$$

This multiplicity correspondence is part of an exact spectral realization and cannot be inferred from set-theoretic inclusion alone.

89 Explicit Critical-Line Spectral Variable

Introduce the shifted variable

$$z = s - \frac{1}{2}.$$

Then

$$s = \frac{1}{2} + z.$$

On the critical line,

$$z = i\lambda, \quad \lambda \in \mathbb{R}.$$

Define

$$\Xi(z) := \xi\left(\frac{1}{2} + z\right).$$

Proposition 89.1. *The functional equation of ξ becomes*

$$\Xi(z) = \Xi(-z).$$

Moreover,

$$\Xi(\bar{z}) = \overline{\Xi(z)}.$$

Proof. The functional equation

$$\xi(s) = \xi(1-s)$$

gives

$$\Xi(z) = \xi\left(\frac{1}{2} + z\right) = \xi\left(\frac{1}{2} - z\right) = \Xi(-z).$$

The conjugation identity follows from

$$\xi(\bar{s}) = \overline{\xi(s)}.$$

□

Corollary 89.2. *The restriction*

$$\Xi(i\lambda) = \xi\left(\frac{1}{2} + i\lambda\right)$$

is real-valued for

$$\lambda \in \mathbb{R}.$$

Proof. From the two preceding symmetries,

$$\Xi(i\lambda) = \Xi(-i\lambda) = \Xi(\overline{i\lambda}) = \overline{\Xi(i\lambda)}.$$

Hence $\Xi(i\lambda) \in \mathbb{R}$.

□

90 Real Spectral Function

Define

$$\Xi_{\mathbb{R}}(\lambda) = \xi\left(\frac{1}{2} + i\lambda\right), \quad \lambda \in \mathbb{R}.$$

Proposition 90.1. *The function $\Xi_{\mathbb{R}}$ is real-valued and even:*

$$\Xi_{\mathbb{R}}(-\lambda) = \Xi_{\mathbb{R}}(\lambda).$$

Proof. Reality was established above. For evenness,

$$\Xi_{\mathbb{R}}(-\lambda) = \xi\left(\frac{1}{2} - i\lambda\right) = \xi\left(1 - \left(\frac{1}{2} + i\lambda\right)\right) = \xi\left(\frac{1}{2} + i\lambda\right).$$

□

Definition 90.2. A self-adjoint spectral realization of the critical-line function is an operator $A = A^*$ for which the real zero set of $\Xi_{\mathbb{R}}$ agrees, including multiplicity, with the spectrum of A :

$$\Xi_{\mathbb{R}}(\lambda) = 0 \iff \lambda \in \sigma(A).$$

Theorem 90.3. *If A is a self-adjoint spectral realization in the preceding sense, then the nontrivial zeros of ξ are precisely*

$$\frac{1}{2} + i\lambda, \quad \lambda \in \sigma(A).$$

Proof. The defining equivalence gives the correspondence on the critical line. Since the spectral values of a self-adjoint operator are real, every point in the resulting zero divisor has real part $1/2$. □

91 Operator-Theoretic Form of the Required Identification

The preceding construction admits the following exact formulation.

Definition 91.1. A *Faure spectral realization* is a quadruple

$$(\mathcal{H}_F, A_F, \Xi_F, U_F)$$

such that:

1. $\mathcal{H}_F$ is a separable Hilbert space;
2. A_F is self-adjoint on $\mathcal{H}_F$;
3. A_F has compact resolvent;
4. Ξ_F is an explicitly defined real-valued analytic function on $\mathbb{R}$;
5. U_F is an explicitly defined identification satisfying

$$\Xi_F(\lambda) = \xi\left(\frac{1}{2} + i\lambda\right);$$

6. the zero divisor of Ξ_F agrees with the spectral divisor of A_F .

Theorem 91.2. A *Faure spectral realization* implies

$$\Re(\rho) = \frac{1}{2}$$

for every nontrivial zero ρ of ζ .

Proof. By the spectral-divisor identity, every zero of the critical-line function corresponds to a spectral value of A_F . Since A_F is self-adjoint, the corresponding spectral value is real. Hence every corresponding zero has the form

$$\rho = \frac{1}{2} + i\lambda, \quad \lambda \in \mathbb{R}.$$

The identification with the complete nontrivial zero divisor then gives the assertion for every nontrivial zero. $\square$

92 Non-Asymptotic Character of the Finite-State Layer

The finite-state component admits an exact statement that is independent of the infinite arithmetic construction.

Theorem 92.1. Let T be an endomorphism of a finite-dimensional vector space and suppose

$$T^m = T^{m+1}.$$

Then the sequence of powers

$$(T^n)_{n \geq 0}$$

is eventually constant, with

$$T^n = T^m \quad (n \geq m).$$

Proof. For every $k \geq 0$,

$$T^{m+k} = T^k T^m = T^k T^{m+1} = T^{m+k+1}.$$

Induction gives

$$T^m = T^{m+1} = T^{m+2} = \dots.$$

□

Corollary 92.2. *No convergence hypothesis is required for the stabilized finite-state operator. In particular, the identity*

$$\lim_{n \rightarrow \infty} T^n = T^m$$

is a consequence of the exact algebraic equality rather than an independent analytic assumption.

Proof. The sequence is constant from index m onward. □

93 Separation of Exact and Analytic Assertions

The complete construction decomposes into three logically distinct levels.

1. The finite operator identity

$$\Psi_9^4 = \Psi_9^5$$

is an algebraic assertion.

2. The arithmetic embedding

$$\ell : \bigoplus_{p \in \mathbb{P}} \mathbb{Z} \longrightarrow \mathbb{R}$$

is an arithmetic assertion.

3. The identification

$$Z_{\xi, \text{nt}} = \left\{ \frac{1}{2} + i\lambda : \lambda \in \sigma(A_F) \right\}$$

is an analytic-spectral assertion.

Proposition 93.1. *No assertion at one of these levels follows solely from an assertion at another level unless an explicit structure-preserving map and its properties have been established.*

Proof. The three assertions concern, respectively, an endomorphism algebra, a free abelian arithmetic lattice, and the zero divisor of an entire function. Their mathematical objects are distinct. An implication between them therefore requires a specified map or construction whose properties establish the implication. □

Remark 93.2. This separation permits the finite-state identities, arithmetic identities, and analytic spectral identities to be certified independently and subsequently composed through explicitly stated interface theorems.

94 Final Structural Reduction

The mathematical reduction established in this section is

$$\Psi_9^4 = \Psi_9^5 \implies P_9^2 = P_9 \implies \text{range } P_9$$

at the finite-state level, while

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p \xrightarrow{\ell} \mathbb{R}$$

provides the exact arithmetic coordinate system.

The analytic endpoint is the following equivalence:

$$\rho \in Z_{\text{nt}}(\xi) \iff \rho = \frac{1}{2} + i\lambda, \quad \lambda \in \sigma(A_F)$$

with

$$A_F = A_F^*.$$

Theorem 94.1. *If the displayed spectral equivalence is established for the complete nontrivial zero divisor of ξ , then*

$$Z_{\text{nt}}(\zeta) \subset \left\{ s \in \mathbb{C} : \Re(s) = \frac{1}{2} \right\}.$$

Proof. Let $\rho \in Z_{\text{nt}}(\zeta)$. The equality of the nontrivial zero divisors of ζ and ξ gives

$$\rho \in Z_{\text{nt}}(\xi).$$

The assumed spectral equivalence yields

$$\rho = \frac{1}{2} + i\lambda$$

for some

$$\lambda \in \sigma(A_F).$$

Since A_F is self-adjoint,

$$\lambda \in \mathbb{R}.$$

Consequently,

$$\Re(\rho) = \frac{1}{2}.$$

□

95 Canonical Decomposition Induced by Stabilization

Let V be a finite-dimensional vector space over $\mathbb{C}$, and let

$$T \in \text{End}(V)$$

satisfy

$$T^m = T^{m+1}$$

for some integer $m \geq 1$. Put

$$P := T^m.$$

Lemma 95.1. *The operator P is idempotent:*

$$P^2 = P.$$

Proof. Using $T^m = T^{m+1}$,

$$P^2 = T^{2m} = T^m T^m.$$

Since

$$T^m = T^{m+1},$$

multiplication by T^{m-1} gives

$$T^m = T^{2m}.$$

Hence

$$P^2 = T^{2m} = T^m = P.$$

□

Theorem 95.2. *There is a canonical direct-sum decomposition*

$$V = \ker P \oplus \text{range } P.$$

Proof. For every $v \in V$,

$$v = (v - Pv) + Pv.$$

Since

$$P(v - Pv) = Pv - P^2 v = Pv - Pv = 0,$$

one has

$$v - Pv \in \ker P,$$

while

$$Pv \in \text{range } P.$$

Thus

$$V = \ker P + \text{range } P.$$

If $v \in \ker P \cap \text{range } P$, then

$$v = Pw$$

for some $w \in V$, and therefore

$$v = Pw = P^2 w = Pv = 0.$$

Hence the sum is direct.

□

Proposition 95.3. *The restriction*

$$T|_{\text{range } P}$$

is the identity, while

$$T^m|_{\ker P} = 0.$$

Proof. If $v \in \text{range } P$, then $v = Pw = T^m w$. Hence

$$Tv = T^{m+1} w = T^m w = v.$$

If $v \in \ker P$, then

$$T^m v = Pv = 0.$$

□

96 Spectral Consequences of Exact Stabilization

Retain the hypotheses of the preceding section.

Theorem 96.1. *Every eigenvalue λ of T satisfies*

$$\lambda^m(1 - \lambda) = 0.$$

Consequently,

$$\sigma(T) \subseteq \{0, 1\}.$$

Proof. Let

$$Tv = \lambda v$$

for some nonzero $v \in V$. Applying the identity

$$T^m = T^{m+1}$$

to v gives

$$\lambda^m v = \lambda^{m+1} v.$$

Since $v \neq 0$,

$$\lambda^m(1 - \lambda) = 0.$$

Therefore

$$\lambda = 0 \quad \text{or} \quad \lambda = 1.$$

□

Corollary 96.2. *The characteristic polynomial of T has the form*

$$\chi_T(\lambda) = \lambda^r(\lambda - 1)^q$$

for some nonnegative integers r, q satisfying

$$r + q = \dim V.$$

Proof. The only possible eigenvalues are 0 and 1. Since V is finite-dimensional, the characteristic polynomial splits over $\mathbb{C}$, and its roots are precisely the eigenvalues counted with algebraic multiplicity. □

Proposition 96.3. *The minimal polynomial of T divides*

$$x^m(x - 1).$$

Proof. The polynomial

$$q(x) = x^m(x - 1)$$

satisfies

$$q(T) = T^m(T - I) = T^{m+1} - T^m = 0.$$

By the defining property of the minimal polynomial, it divides every polynomial annihilating T . Hence

$$\mu_T(x) \mid x^m(x - 1).$$

□

97 Semisimplicity of the Stabilized Fixed Sector

Theorem 97.1. *Under the stabilization hypothesis*

$$T^m = T^{m+1},$$

the eigenvalue 1 is semisimple. More precisely,

$$\text{range } P = \ker(T - I), \quad P = T^m.$$

Proof. If $v \in \text{range } P$, the preceding proposition gives

$$Tv = v,$$

so

$$\text{range } P \subseteq \ker(T - I).$$

Conversely, if

$$Tv = v,$$

then

$$Pv = T^m v = v,$$

hence

$$v \in \text{range } P.$$

Therefore

$$\text{range } P = \ker(T - I).$$

Since P is a projection onto the complete 1-eigenspace, there is no nontrivial generalized 1-eigenspace. Thus the eigenvalue 1 is semisimple. $\square$

Corollary 97.2. *There is a direct-sum decomposition*

$$V = \ker(T^m) \oplus \ker(T - I).$$

Proof. Combine

$$V = \ker P \oplus \text{range } P$$

with

$$P = T^m$$

and

$$\text{range } P = \ker(T - I).$$

$\square$

98 Rank and Trace of the Stabilized Projection

Let

$$P = T^m.$$

Proposition 98.1. *For the idempotent operator P ,*

$$\text{rank } P = \dim \ker(T - I).$$

Proof. By the preceding decomposition,

$$\text{range } P = \ker(T - I).$$

Taking dimensions gives

$$\text{rank } P = \dim \ker(T - I).$$

□

Theorem 98.2. *The trace of P is an integer and satisfies*

$$\text{tr } P = \text{rank } P = \dim \ker(T - I).$$

Proof. Choose a basis adapted to

$$V = \ker P \oplus \text{range } P.$$

Relative to this basis,

$$P = \begin{pmatrix} 0 & 0 \\ 0 & I_r \end{pmatrix},$$

where

$$r = \dim \text{range } P.$$

Therefore

$$\text{tr } P = r$$

and

$$\text{rank } P = r.$$

The preceding proposition identifies r with $\dim \ker(T - I)$.

□

Corollary 98.3. *If*

$$\text{tr } P = 1,$$

then

$$\text{rank } P = 1$$

and

$$\dim \ker(T - I) = 1.$$

99 Compatibility with an Invariant Bilinear Form

Let V be a finite-dimensional complex vector space equipped with a nondegenerate Hermitian form

$$\langle \cdot, \cdot \rangle.$$

Suppose that

$$T^* = T.$$

Proposition 99.1. *Every spectral projection of T is orthogonal. In particular,*

$$P = T^m$$

is self-adjoint whenever T is both self-adjoint and satisfies

$$T^m = T^{m+1}.$$

Proof. Since $T = T^*$, every polynomial in T is self-adjoint if its coefficients are real. The operator $P = T^m$ is therefore self-adjoint. Together with

$$P^2 = P,$$

this makes P an orthogonal projection. □

Theorem 99.2. *If $T = T^*$ and*

$$T^m = T^{m+1},$$

then

$$V = \ker T^m \perp \ker(T - I).$$

Proof. The two subspaces form a direct sum by the stabilization theorem. Since $P = T^m$ is an orthogonal projection, its kernel is orthogonal to its range. The range equals

$$\ker(T - I).$$

Hence

$$\ker T^m \perp \ker(T - I).$$

□

100 The Nine-Dimensional Fixed Sector

Assume now that the stabilized operator acts on a finite-dimensional space V and satisfies

$$\dim \ker(T - I) = 9.$$

Proposition 100.1. *The stabilized projection*

$$P = T^m$$

has

$$\text{rank } P = 9$$

and

$$\text{tr } P = 9.$$

Proof. By the rank identity,

$$\text{rank } P = \dim \ker(T - I) = 9.$$

Since P is idempotent,

$$\text{tr } P = \text{rank } P = 9.$$

□

Corollary 100.2. *If*

$$V = \mathbb{C}^{144},$$

then

$$\dim \ker T^m = 135.$$

Proof. From

$$V = \ker T^m \oplus \ker(T - I),$$

one obtains

$$144 = \dim \ker T^m + 9.$$

Therefore

$$\dim \ker T^m = 135.$$

□

101 Normalization on the Fixed Sector

Let W be a 9-dimensional complex vector space and let

$$B \in \text{End}(W)$$

satisfy

$$B = \frac{1}{9} I_W.$$

Proposition 101.1. *The operator B has the unique eigenvalue*

$$\lambda = \frac{1}{9}$$

with algebraic and geometric multiplicity 9.

Proof. For every $v \in W$,

$$Bv = \frac{1}{9} v.$$

Thus every nonzero vector is an eigenvector with eigenvalue $1/9$. Consequently the eigenspace is all of W , of dimension 9. $\square$

Proposition 101.2. *The spectral radius of B is*

$$\rho(B) = \frac{1}{9},$$

and

$$B^n = \frac{1}{9^n} I_W$$

for every $n \geq 1$.

Proof. The spectrum consists solely of $1/9$, hence

$$\rho(B) = \frac{1}{9}.$$

The power identity follows directly from

$$B^n = \left(\frac{1}{9} I_W \right)^n = \frac{1}{9^n} I_W.$$

$\square$

Remark 101.3. Thus $B = \frac{1}{9} I_W$ is a scalar normalization on the nine-dimensional sector. It is not an idempotent projection, since

$$B^2 = \frac{1}{81} I_W \neq \frac{1}{9} I_W.$$

The projection associated with stabilization is instead $P = T^m$. These two operators must therefore remain distinct in the formal development.

102 Finite-State Operator on the Zero-Trace Subspace

Let

$$H_{144} := \mathbb{C}^{144}$$

and define the trace functional

$$\text{tr} : H_{144} \longrightarrow \mathbb{C}$$

whenever H_{144} is identified with the corresponding matrix space. Let

$$H_0 := \{X : \text{tr } X = 0\}.$$

Proposition 102.1. *If the ambient matrix space is $M_{12}(\mathbb{C})$, then*

$$\dim M_{12}(\mathbb{C}) = 144$$

and

$$\dim H_0 = 143.$$

Proof. The matrix units

$$E_{ij}, \quad 1 \leq i, j \leq 12,$$

form a basis of $M_{12}(\mathbb{C})$, giving dimension

$$12^2 = 144.$$

The trace is a nonzero linear functional, since

$$\text{tr}(E_{11}) = 1.$$

Its kernel therefore has codimension one. Hence

$$\dim H_0 = 144 - 1 = 143.$$

□

Remark 102.2. If the symbol $\mathbf{1}^\perp \subset \mathbb{C}^{144}$ is instead intended to denote the orthogonal complement of a single vector in $\mathbb{C}^{144}$, its dimension is likewise 143. The two constructions should not be identified without an explicit linear isomorphism.

103 Finite Certification Interface

Let

$$\Psi_9 \in \text{End}(V)$$

be the finite-state operator specified by the construction, and assume that the formal verification establishes

$$\Psi_9^4 = \Psi_9^5.$$

Define

$$P_9 := \Psi_9^4.$$

Theorem 103.1. *The following identities hold:*

$$P_9^2 = P_9,$$

$$\text{range } P_9 = \ker(\Psi_9 - I),$$

and

$$V = \ker(\Psi_9^4) \oplus \ker(\Psi_9 - I).$$

Proof. The first identity follows from exact stabilization. The equality of the range with the fixed-point space follows from

$$\text{range } P_9 = \ker(\Psi_9 - I).$$

The direct-sum decomposition follows from the canonical decomposition of an idempotent operator. $\square$

Corollary 103.2. *For every integer*

$$n \geq 4,$$

one has

$$\Psi_9^n = P_9.$$

Proof. The stabilization identity implies

$$\Psi_9^4 = \Psi_9^5 = \Psi_9^6 = \dots.$$

$\square$

104 Exactness of the Finite-State Layer

Theorem 104.1. *The finite-state stabilization*

$$\Psi_9^4 = \Psi_9^5$$

is equivalent to the idempotence of

$$P_9 := \Psi_9^4$$

together with the identity

$$\Psi_9 P_9 = P_9.$$

Proof. Assume first that

$$\Psi_9^4 = \Psi_9^5.$$

Then

$$P_9^2 = \Psi_9^8 = \Psi_9^4 = P_9,$$

and

$$\Psi_9 P_9 = \Psi_9^5 = \Psi_9^4 = P_9.$$

Conversely, suppose

$$P_9^2 = P_9$$

and

$$\Psi_9 P_9 = P_9.$$

Since

$$P_9 = \Psi_9^4,$$

the second equality is precisely

$$\Psi_9^5 = \Psi_9^4.$$

$\square$

Remark 104.2. This equivalence is useful for formal verification because it replaces the eventual constancy statement by two finite polynomial identities in the operator algebra.

105 Structural Interface to the Arithmetic Layer

Let

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

be the arithmetic lattice and let

$$V_F$$

denote the stabilized finite-state sector.

An exact coupling between the two layers is a homomorphism

$$\Phi_F : \mathcal{L}_{\mathbb{P}} \longrightarrow V_F$$

together with a finite-support property

$$\Phi_F(x) = \Phi_F(\pi_S x)$$

for some finite $S \subset \mathbb{P}$ depending on x .

Proposition 105.1. *If Φ_F satisfies the preceding finite-support property, then every individual arithmetic state has an exact finite representation in V_F .*

Proof. For every $x \in \mathcal{L}_{\mathbb{P}}$, choose finite S such that

$$\Phi_F(x) = \Phi_F(\pi_S x).$$

Since

$$\pi_S x \in \mathcal{L}_S$$

and $\mathcal{L}_S$ is finitely generated, the representation is finite and exact. □

Definition 105.2. The map Φ_F is called *projection-compatible* with P_9 if

$$P_9 \Phi_F(x) = \Phi_F(x)$$

for every

$$x \in \mathcal{L}_{\mathbb{P}}.$$

Proposition 105.3. *If Φ_F is projection-compatible, then*

$$\text{range } \Phi_F \subseteq \text{range } P_9 = \ker(\Psi_9 - I).$$

Proof. For every x ,

$$P_9 \Phi_F(x) = \Phi_F(x).$$

Hence $\Phi_F(x)$ is fixed by P_9 , and therefore belongs to $\text{range } P_9$. The equality with $\ker(\Psi_9 - I)$ follows from the stabilization theorem. □

106 Necessary Exact Identification for the Global Construction

The preceding results establish the algebraic consequences of the finite-state identity and the arithmetic consequences of the prime lattice independently. Their composition requires an explicit intertwining relation.

Definition 106.1. An operator

$$\Phi_F : \mathcal{L}_{\mathbb{P}} \longrightarrow V$$

is an *exact intertwiner* for T if

$$T\Phi_F = \Phi_F.$$

Proposition 106.2. *If Φ_F is an exact intertwiner, then*

$$T^n \Phi_F = \Phi_F$$

for every integer $n \geq 1$.

Proof. The relation

$$T\Phi_F = \Phi_F$$

implies inductively

$$T^n \Phi_F = T^{n-1}(T\Phi_F) = T^{n-1}\Phi_F = \dots = \Phi_F.$$

□

Remark 106.3. The intertwining identity is the precise mathematical bridge between the arithmetic lattice and the fixed sector of the finite operator. Without such an identity, the two constructions remain independent mathematical components.

Theorem 106.4. *Suppose that*

$$\Psi_9^4 = \Psi_9^5$$

and that an exact intertwiner

$$\Phi_F : \mathcal{L}_{\mathbb{P}} \longrightarrow V$$

satisfies

$$\Psi_9 \Phi_F = \Phi_F.$$

Then

$$P_9 \Phi_F = \Phi_F, \quad P_9 = \Psi_9^4.$$

Proof. By repeated application of the intertwining relation,

$$\Psi_9^4 \Phi_F = \Phi_F.$$

Since

$$P_9 = \Psi_9^4,$$

the result follows. □

107 Finite-State Invariants

Let

$$V = \mathbb{C}^{144}$$

and let

$$T \in \text{End}(V)$$

satisfy

$$T^4 = T^5.$$

Set

$$P := T^4.$$

Definition 107.1. A linear functional

$$\varphi : V \longrightarrow \mathbb{C}$$

is called a left invariant of T if

$$\varphi T = \varphi.$$

A vector

$$v \in V$$

is called a right invariant of T if

$$Tv = v.$$

Proposition 107.2. *If φ is a left invariant of T , then*

$$\varphi P = \varphi.$$

Proof. Since

$$\varphi T = \varphi,$$

iteration gives

$$\varphi T^4 = \varphi.$$

As $P = T^4$,

$$\varphi P = \varphi.$$

□

Proposition 107.3. *If v is a right invariant of T , then*

$$Pv = v.$$

Proof. From

$$Tv = v$$

one obtains

$$T^4 v = v.$$

Since $P = T^4$,

$$Pv = v.$$

□

Theorem 107.4. *Suppose that*

$$T^4 = T^5.$$

Then

$$T|_{\text{range } P} = I_{\text{range } P}.$$

Consequently, every orbit

$$v, Tv, T^2v, \dots$$

has stabilized after four iterations:

$$T^n v = Pv \quad (n \geq 4).$$

Proof. If $w \in \text{range } P$, then $w = Pv = T^4 v$. Hence

$$Tw = T^5 v = T^4 v = w.$$

Thus T is the identity on $\text{range } P$. The second statement follows from

$$T^n = T^4 = P \quad (n \geq 4).$$

□

108 The Stable Quotient

Let

$$K := \ker P.$$

Proposition 108.1. *The subspace K is T -invariant:*

$$T(K) \subseteq K.$$

Proof. Let $v \in K$. Then

$$Pv = 0.$$

Since $P = T^4$ commutes with every power of T ,

$$P(Tv) = TPv = 0.$$

Therefore

$$Tv \in K.$$

□

Theorem 108.2. *The quotient space*

$$V/K$$

is canonically isomorphic to

$$\text{range } P.$$

Under this identification, the operator induced by T on V/K is the identity.

Proof. The first assertion is the first isomorphism theorem applied to

$$P : V \longrightarrow \text{range } P.$$

Its kernel is K .

For the second assertion, if $[v] \in V/K$, then

$$T[v] = [Tv].$$

Because

$$T^4 v - Pv = 0,$$

and

$$Tv - Pv \in K,$$

we obtain

$$[Tv] = [Pv].$$

The class of Pv is represented by an element of $\text{range } P$, where T acts as the identity. Hence the induced operator is the identity. $\square$

Remark 108.3. The quotient removes precisely the transient sector $\ker P$. The resulting quotient dynamics is therefore exact rather than asymptotic.

109 Rank-One Normalization

Suppose now that

$$\dim \text{range } P = 1.$$

Choose nonzero vectors

$$r \in \text{range } P, \quad \ell \in V^*$$

such that

$$\ell(r) = 1.$$

Proposition 109.1. *The stabilized operator has the representation*

$$P = r \otimes \ell,$$

that is,

$$Pv = \ell(v)r \quad (v \in V).$$

Proof. Since $\text{range } P = \text{span}\{r\}$, there exists a linear functional α such that

$$Pv = \alpha(v)r.$$

Applying P once more gives

$$P^2 v = \alpha(v)Pr = \alpha(v)r,$$

because $r \in \text{range } P$. Thus $P^2 = P$ imposes no further restriction on α . Evaluating ℓ gives

$$\ell(Pv) = \alpha(v)\ell(r) = \alpha(v).$$

Therefore

$$Pv = \ell(Pv)r.$$

If ℓ is chosen to agree with the coefficient functional on the range decomposition, then

$$Pv = \ell(v)r.$$

Hence

$$P = r \otimes \ell.$$

□

Corollary 109.2. *For every $v \in V$,*

$$T^n v = \ell(v)r \quad (n \geq 4).$$

Proof. Since $T^n = P$ for $n \geq 4$,

$$T^n v = Pv = \ell(v)r.$$

□

110 Biorthogonal Form of the Stabilized Projection

Let $r \in V$ satisfy

$$Tr = r$$

and let

$$\ell \in V^*$$

satisfy

$$\ell T = \ell, \quad \ell(r) = 1.$$

Define

$$P := r \otimes \ell.$$

Proposition 110.1. *The operator P is idempotent:*

$$P^2 = P.$$

Proof. For $v \in V$,

$$P^2 v = P(\ell(v)r) = \ell(v)Pr = \ell(v)\ell(r)r = \ell(v)r = Pv.$$

□

Proposition 110.2. *The operator P commutes with T and satisfies*

$$TP = PT = P.$$

Proof. For $v \in V$,

$$TPv = T(\ell(v)r) = \ell(v)Tr = \ell(v)r = Pv.$$

Similarly,

$$PTv = \ell(Tv)r = \ell(v)r = Pv.$$

□

Theorem 110.3. *If $T^4 = P$, then*

$$T^n = P \quad (n \geq 4).$$

Proof. From

$$TP = P$$

and

$$P = T^4,$$

one obtains

$$T^5 = TP = P = T^4.$$

The stabilization theorem then gives

$$T^n = P$$

for every $n \geq 4$. □

111 Conservation Laws

Let

$$\ell T = \ell$$

and

$$Tr = r.$$

Proposition 111.1. *For every $v \in V$ and every $n \geq 0$,*

$$\ell(T^n v) = \ell(v).$$

Proof. Induction gives

$$\ell T^n = \ell.$$

Therefore

$$\ell(T^n v) = \ell(v).$$

□

Proposition 111.2. *If $T^4 = P = r \otimes \ell$, then*

$$\ell(Pv) = \ell(v)$$

and

$$Pv = \ell(v)r.$$

Proof. The first identity follows from

$$\ell P = \ell T^4 = \ell.$$

The second is the definition of the rank-one projection. □

Corollary 111.3. *If*

$$\ell(v) = 0,$$

then

$$T^n v = 0 \quad (n \geq 4).$$

Proof. For $n \geq 4$,

$$T^n v = Pv = \ell(v)r = 0.$$

□

Remark 111.4. Thus the conserved scalar $\ell(v)$ completely determines the eventual state in the rank-one case. This is an exact statement about the stabilized operator and does not invoke a probabilistic or continuum approximation.

112 Primitive Nonnegative Realizations

Let

$$A = (a_{ij}) \in M_d(\mathbb{R})$$

be a nonnegative matrix.

Definition 112.1. The matrix A is primitive if there exists $k \geq 1$ such that

$$A^k > 0$$

entrywise.

Theorem 112.2. *If A is primitive, then its Perron eigenvalue is simple and strictly dominates the modulus of every other eigenvalue.*

Proof. This is the Perron–Frobenius theorem for primitive matrices. $\square$

Definition 112.3. The primitivity exponent of A is

$$\gamma(A) = \min\{k \geq 1 : A^k > 0\}.$$

Proposition 112.4. *If*

$$\gamma(A) = 4,$$

then

$$A^4 > 0$$

and

$$A^k > 0 \quad (k \geq 4).$$

Proof. The first statement is the definition of $\gamma(A)$. If $A^k > 0$, then

$$A^{k+1} = A^k A.$$

When A is primitive, the standard Perron–Frobenius positivity argument implies positivity of all subsequent powers. $\square$

Remark 112.5. The exponent $\gamma(A)$ is a combinatorial property of a nonnegative matrix. It should not be identified with the stabilization index of an arbitrary linear operator unless the corresponding operator identity has been proved.

113 Relation Between Primitivity and Stabilization

Let A be a primitive nonnegative matrix. Its powers generally do not stabilize exactly. Instead, after normalization they converge to the Perron projection.

Let

$$\rho > 0$$

be the Perron eigenvalue, and let r, ℓ be positive right and left Perron eigenvectors satisfying

$$Ar = \rho r, \quad \ell A = \rho \ell, \quad \ell(r) = 1.$$

Define

$$P_{\text{PF}} := r \otimes \ell.$$

Theorem 113.1. *For a primitive matrix A ,*

$$\rho^{-n} A^n \longrightarrow P_{\text{PF}}$$

as

$$n \rightarrow \infty.$$

Proof. By Perron–Frobenius theory,

$$\rho$$

is a simple eigenvalue and every other eigenvalue λ satisfies

$$|\lambda| < \rho.$$

The Jordan decomposition therefore gives

$$A^n = \rho^n P_{\text{PF}} + R_n,$$

where

$$\|R_n\| = O(\eta^n)$$

for some

$$0 \leq \eta < \rho.$$

Dividing by ρ^n yields

$$\rho^{-n} A^n = P_{\text{PF}} + O((\eta/\rho)^n),$$

and the error tends to zero. □

Remark 113.2. This distinguishes two mathematically different mechanisms:

$$T^m = T^{m+1}$$

is exact finite stabilization, whereas

$$\rho^{-n} A^n \rightarrow P_{\text{PF}}$$

is asymptotic Perron–Frobenius convergence. A proof must state which mechanism is actually used.

114 Exact Criterion for a Finite-State Projection

Theorem 114.1. *Let $T \in \text{End}(V)$. The following are equivalent for a fixed integer $m \geq 1$:*

1.

$$T^m = T^{m+1};$$

2. $P = T^m$ is idempotent and

$$TP = P;$$

3.

$$T^m$$

is a projection onto

$$\ker(T - I)$$

and

$$T|_{\ker(T-I)} = I.$$

Proof. The implication $1 \Rightarrow 2$ follows from

$$P^2 = T^{2m} = T^m = P$$

and

$$TP = T^{m+1} = T^m = P.$$

For $2 \Rightarrow 3$, the idempotence of P gives

$$V = \ker P \oplus \text{range } P.$$

Moreover,

$$T(Pv) = Pv,$$

so

$$\text{range } P \subseteq \ker(T - I).$$

Conversely, if $Tv = v$, then

$$Pv = T^m v = v,$$

hence

$$v \in \text{range } P.$$

Thus

$$\text{range } P = \ker(T - I).$$

Finally, $3 \Rightarrow 1$. Since T^m is the projection onto the fixed-point space,

$$T^{m+1}v = T(T^m v) = T^m v$$

for every $v \in V$, because $T^m v \in \ker(T - I)$. Hence

$$T^{m+1} = T^m.$$

□

115 Formal Separation of the Finite and Spectral Claims

The finite-state results established above imply the exact identity

$$\Psi_9^4 = \Psi_9^5$$

if and only if

$$P_9 := \Psi_9^4$$

is the projection onto

$$\ker(\Psi_9 - I).$$

The spectral realization problem is logically subsequent. Let

$$A_F$$

be an operator acting on a Hilbert space $\mathcal{H}_F$, and suppose that its spectral values are used to parametrize zeros through

$$s = \frac{1}{2} + i\lambda.$$

Theorem 115.1. *Assume that the finite-state construction supplies an explicitly defined operator A_F satisfying*

$$A_F = A_F^*,$$

and that its spectral divisor agrees with the nontrivial zero divisor of ξ under

$$\lambda \mapsto \frac{1}{2} + i\lambda.$$

Then

$$\Re(\rho) = \frac{1}{2}$$

for every nontrivial zero ρ of ζ .

Proof. The spectral-divisor hypothesis gives

$$\rho = \frac{1}{2} + i\lambda$$

for some

$$\lambda \in \sigma(A_F).$$

Self-adjointness implies

$$\lambda \in \mathbb{R}.$$

Therefore

$$\Re(\rho) = \frac{1}{2}.$$

The equality of the spectral divisor with the complete nontrivial zero divisor transfers the conclusion to every nontrivial zero. $\square$

Remark 115.2. The finite-state identity

$$\Psi_9^4 = \Psi_9^5$$

and the spectral identity for A_F are separate assertions. Their composition requires the construction of A_F from the verified finite-state data together with a proof that the resulting spectral divisor is exactly the required arithmetic divisor.

116 Conclusion of the Classical Formalization

The exact algebraic component of the construction is summarized by

$$\Psi_9^4 = \Psi_9^5$$

and

$$P_9 := \Psi_9^4,$$

with

$$P_9^2 = P_9, \quad \text{range } P_9 = \ker(\Psi_9 - I),$$

and

$$\mathbb{C}^{144} = \ker(\Psi_9^4) \oplus \ker(\Psi_9 - I).$$

The arithmetic component is represented by the exact free abelian lattice

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z}e_p$$

and its injective logarithmic coordinate

$$\ell \left(\sum_p n_p e_p \right) = \sum_p n_p \log p.$$

The operator-theoretic analytic component requires a self-adjoint operator

$$A_F = A_F^*$$

whose spectral divisor is identified exactly with the nontrivial zero divisor of the completed zeta function through

$$\rho = \frac{1}{2} + i\lambda.$$

Consequently, the complete mathematical implication has the form

$$\begin{aligned} \Psi_9^4 &= \Psi_9^5, \\ \mathcal{L}_{\mathbb{P}} &= \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p, \\ A_F &= A_F^*, \\ Z_{\zeta, \text{nt}} &= \left\{ \frac{1}{2} + i\lambda : \lambda \in \sigma(A_F) \right\}, \end{aligned}$$

which yields

$$Z_{\zeta, \text{nt}} \subset \left\{ s \in \mathbb{C} : \Re(s) = \frac{1}{2} \right\}.$$

The first two components are exact algebraic and arithmetic statements. The final implication is an operator-theoretic consequence of the stated spectral identification and contains no asymptotic replacement of the spectral parameter.

117 The Prime Lattice and Its Logarithmic Embedding

Let $\mathbb{P}$ denote the set of prime numbers and let

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

be the free abelian group generated by the symbols e_p .

Every element $x \in \mathcal{L}_{\mathbb{P}}$ has a unique representation

$$x = \sum_{p \in \mathbb{P}} n_p e_p, \quad n_p \in \mathbb{Z},$$

with only finitely many n_p nonzero.

Define the logarithmic map

$$\Lambda_{\log} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{R}$$

by

$$\Lambda_{\log}(x) = \sum_{p \in \mathbb{P}} n_p \log p.$$

Proposition 117.1. *The map $\Lambda_{\log}$ is a group homomorphism.*

Proof. For

$$x = \sum_p n_p e_p, \quad y = \sum_p m_p e_p,$$

one has

$$\Lambda_{\log}(x + y) = \sum_p (n_p + m_p) \log p = \Lambda_{\log}(x) + \Lambda_{\log}(y).$$

□

Theorem 117.2. *The logarithmic map $\Lambda_{\log}$ is injective.*

Proof. Suppose

$$\Lambda_{\log}(x) = 0.$$

Writing

$$x = \sum_{p \in S} n_p e_p$$

with $S \subset \mathbb{P}$ finite, one obtains

$$\sum_{p \in S} n_p \log p = 0.$$

Exponentiation gives

$$\prod_{p \in S} p^{n_p} = 1.$$

Separating positive and negative exponents,

$$\prod_{n_p > 0} p^{n_p} = \prod_{n_p < 0} p^{-n_p}.$$

Unique factorization in $\mathbb{Z}$ implies

$$n_p = 0$$

for every $p \in S$. Hence

$$x = 0.$$

□

Corollary 117.3. *The prime lattice admits an exact realization as a discrete subgroup of the multiplicative group $\mathbb{R}_{>0}$ through*

$$x \longmapsto e^{\Lambda_{\log}(x)}.$$

118 Finite Support and Exact Arithmetic Reduction

For a finite subset

$$S \subset \mathbb{P},$$

define

$$\mathcal{L}_S = \bigoplus_{p \in S} \mathbb{Z} e_p.$$

Lemma 118.1. *Every $x \in \mathcal{L}_{\mathbb{P}}$ belongs to $\mathcal{L}_S$ for some finite subset $S \subset \mathbb{P}$.*

Proof. By definition of the direct sum, only finitely many coefficients in

$$x = \sum_p n_p e_p$$

are nonzero. Taking S to be their support gives

$$x \in \mathcal{L}_S.$$

□

Proposition 118.2. *For every $x \in \mathcal{L}_{\mathbb{P}}$, the logarithmic coordinate*

$$\Lambda_{\log}(x)$$

is an exactly computable finite sum.

Proof. By the preceding lemma,

$$x = \sum_{p \in S} n_p e_p$$

for some finite S . Hence

$$\Lambda_{\log}(x) = \sum_{p \in S} n_p \log p,$$

which contains only finitely many terms. □

Remark 118.3. No truncation of the prime lattice is involved in this statement. Finite support is an intrinsic property of each element of the direct sum and is therefore distinct from replacing the infinite lattice by a finite-dimensional approximation.

119 Multiplicative Coordinates

Define

$$\mathfrak{m} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{Q}_{>0}$$

by

$$\mathfrak{m}\left(\sum_p n_p e_p\right) = \prod_p p^{n_p}.$$

Proposition 119.1. *The map $\mathfrak{m}$ is a group homomorphism.*

Proof. For

$$x = \sum_p n_p e_p, \quad y = \sum_p m_p e_p,$$

one has

$$\mathfrak{m}(x + y) = \prod_p p^{n_p + m_p} = \left(\prod_p p^{n_p}\right) \left(\prod_p p^{m_p}\right) = \mathfrak{m}(x) \mathfrak{m}(y).$$

□

Theorem 119.2. *The map*

$$\mathfrak{m} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{Q}_{>0}$$

is injective.

Proof. If

$$\mathfrak{m}(x) = 1,$$

then

$$\prod_p p^{n_p} = 1.$$

Unique factorization implies

$$n_p = 0$$

for all p , and therefore

$$x = 0.$$

□

Proposition 119.3. *The logarithmic and multiplicative embeddings satisfy*

$$\Lambda_{\log}(x) = \log \mathfrak{m}(x).$$

Proof. Directly,

$$\log \mathfrak{m}(x) = \log \left(\prod_p p^{n_p} \right) = \sum_p n_p \log p = \Lambda_{\log}(x).$$

□

120 Linear Independence of Finite Prime Logarithmic Coordinates

Theorem 120.1. *Let $p_1, \dots, p_k$ be distinct primes. If*

$$\sum_{j=1}^k n_j \log p_j = 0, \quad n_j \in \mathbb{Z},$$

then

$$n_1 = \dots = n_k = 0.$$

Proof. Exponentiation yields

$$\prod_{j=1}^k p_j^{n_j} = 1.$$

Unique factorization implies that the exponent of every prime on the left-hand side is zero. Since the p_j are distinct,

$$n_j = 0$$

for every j .

□

Corollary 120.2. *For every finite set of distinct primes*

$$S = \{p_1, \dots, p_k\},$$

the vectors

$$(\log p_1, \dots, \log p_k)$$

have no nontrivial integral relation.

Remark 120.3. The conclusion is an exact arithmetic statement. It does not assert linear independence over $\mathbb{R}$ of the numbers $\log p_j$, which would be a different claim.

121 Arithmetic Characters

For

$$t \in \mathbb{R},$$

define

$$\chi_t : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{C}^{\times}$$

by

$$\chi_t(x) = \exp(it\Lambda_{\log}(x)).$$

Proposition 121.1. *For every $t \in \mathbb{R}$, χ_t is a group character:*

$$\chi_t(x + y) = \chi_t(x)\chi_t(y).$$

Proof. Using additivity of $\Lambda_{\log}$,

$$\chi_t(x + y) = e^{it(\Lambda_{\log}(x) + \Lambda_{\log}(y))} = e^{it\Lambda_{\log}(x)} e^{it\Lambda_{\log}(y)} = \chi_t(x)\chi_t(y).$$

□

Proposition 121.2. *For every $x \in \mathcal{L}_{\mathbb{P}}$,*

$$|\chi_t(x)| = 1.$$

Proof. Since

$$t\Lambda_{\log}(x) \in \mathbb{R},$$

the complex exponential has unit modulus:

$$|\exp(it\Lambda_{\log}(x))| = 1.$$

□

Theorem 121.3. *The family*

$$\{\chi_t : t \in \mathbb{R}\}$$

separates points of $\mathcal{L}_{\mathbb{P}}$. More precisely, if

$$x \neq y,$$

then there exists $t \in \mathbb{R}$ such that

$$\chi_t(x) \neq \chi_t(y).$$

Proof. Set

$$z = x - y \neq 0.$$

By injectivity of $\Lambda_{\log}$,

$$\Lambda_{\log}(z) \neq 0.$$

Choose

$$t = \frac{\pi}{\Lambda_{\log}(z)}.$$

Then

$$\chi_t(z) = e^{i\pi} = -1.$$

Since

$$\chi_t(z) = \chi_t(x)\chi_t(y)^{-1},$$

one has

$$\chi_t(x) \neq \chi_t(y).$$

□

122 The Completed Zeta Function

Let

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Theorem 122.1. *The completed zeta function satisfies the functional equation*

$$\xi(s) = \xi(1-s).$$

Its nontrivial zeros coincide with the nontrivial zeros of $\zeta(s)$, counted with multiplicity.

Proof. This is the classical functional equation for the completed Riemann zeta function. The prefactor removes the pole at $s = 1$ and the trivial zeros introduced by the gamma factor, leaving an entire function whose zeros are precisely the nontrivial zeros of ζ . $\square$

Proposition 122.2. *If*

$$\rho$$

is a nontrivial zero of ξ , then

$$1 - \rho$$

is also a zero, with the same multiplicity.

Proof. The functional equation gives

$$\xi(1 - \rho) = \xi(\rho) = 0.$$

Equality of multiplicities follows from the identity

$$\xi(s) = \xi(1-s).$$

$\square$

Proposition 122.3. *If ρ is a nontrivial zero, then*

$$\bar{\rho}$$

is also a zero, with the same multiplicity.

Proof. The coefficients of the Dirichlet series and the standard analytic continuation defining ξ are real on the real axis. Hence

$$\xi(\bar{s}) = \overline{\xi(s)}.$$

Therefore

$$\xi(\rho) = 0 \implies \xi(\bar{\rho}) = 0.$$

$\square$

123 Spectral Realization of the Critical-Line Parameter

Let $\mathcal{H}$ be a complex Hilbert space and let

$$A = A^*$$

be a self-adjoint operator with discrete spectrum.

Definition 123.1. A spectral realization of the nontrivial zero set is an equality of multisets

$$\mathcal{Z}_\xi = \left\{ \frac{1}{2} + i\lambda : \lambda \in \sigma(A) \right\},$$

where multiplicities are preserved.

Theorem 123.2. *If $A = A^*$ provides a spectral realization of $\mathcal{Z}_\xi$, then every nontrivial zero ρ of ζ satisfies*

$$\Re \rho = \frac{1}{2}.$$

Proof. Let ρ be a nontrivial zero. By the spectral realization hypothesis,

$$\rho = \frac{1}{2} + i\lambda$$

for some

$$\lambda \in \sigma(A).$$

Self-adjointness gives

$$\lambda \in \mathbb{R}.$$

Therefore

$$\Re \rho = \frac{1}{2}.$$

□

Corollary 123.3. *Under the spectral-realization hypothesis,*

$$\mathcal{Z}_\xi \subseteq \left\{ s \in \mathbb{C} : \Re(s) = \frac{1}{2} \right\}.$$

Remark 123.4. The theorem identifies the exact logical point at which the critical-line conclusion enters. Self-adjointness alone does not identify zeta zeros. Conversely, an identification of zeros with spectral parameters is not sufficient to force those parameters to be real. Both statements are required.

124 Operator Construction Required for the Arithmetic Identification

Let

$$\Phi_F : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{H}$$

be an explicitly defined arithmetic realization and let

$$A_F$$

be an operator on $\mathcal{H}$.

Assume

$$A_F = A_F^*$$

and that the arithmetic character family determines a spectral parameter through

$$\chi_t(x) = e^{it\Lambda_{\log}(x)}.$$

Definition 124.1. The arithmetic spectral correspondence is the assertion that the completed zeta divisor is represented by the spectral divisor of A_F :

$$\text{Div}_{\text{nt}}(\xi) = \text{Div}_{\text{spec}}\left(\frac{1}{2} + iA_F\right).$$

Proposition 124.2. *If the arithmetic spectral correspondence holds and A_F is self-adjoint, then the nontrivial zero divisor of ξ is supported on*

$$\Re(s) = \frac{1}{2}.$$

Proof. Every spectral value of A_F is real. Therefore every spectral point of

$$\frac{1}{2} + iA_F$$

has real part $1/2$. Equality of divisors transfers this support property to the nontrivial zero divisor of ξ . $\square$

Remark 124.3. Accordingly, the classical proof obligation is reduced to the exact construction and verification of the divisor identity

$$\text{Div}_{\text{nt}}(\xi) = \text{Div}_{\text{spec}}\left(\frac{1}{2} + iA_F\right).$$

No appeal to numerical approximation, finite sampling, or asymptotic convergence can replace this identity.

125 Finite Certification and Exact Spectral Transfer

Suppose that the finite-state operator

$$\Psi_9$$

and the arithmetic realization

$$\Phi_F$$

satisfy

$$\Psi_9^4 = \Psi_9^5$$

and

$$\Psi_9\Phi_F = \Phi_F.$$

Let

$$P_9 = \Psi_9^4.$$

Theorem 125.1. *The arithmetic image lies entirely in the stabilized sector:*

$$\text{range } \Phi_F \subseteq \text{range } P_9.$$

Proof. For every $x \in \mathcal{L}_{\mathbb{P}}$,

$$P_9\Phi_F(x) = \Psi_9^4\Phi_F(x) = \Phi_F(x).$$

Hence every vector in the range of Φ_F is fixed by P_9 , and therefore belongs to $\text{range } P_9$. $\square$

Corollary 125.2. *The restriction of Ψ_9 to*

$$\text{range } \Phi_F$$

is the identity:

$$\Psi_9|_{\text{range } \Phi_F} = I.$$

Proof. If

$$\nu = \Phi_F(x),$$

then

$$\Psi_9 \nu = \Psi_9 \Phi_F(x) = \Phi_F(x) = \nu.$$

□

Remark 125.3. This establishes exact compatibility between the arithmetic image and the stabilized finite-state sector. It does not, by itself, identify the spectrum of an auxiliary self-adjoint operator with the zeros of ξ ; that identification remains a separate theorem.

126 Logical Closure of the Classical Layer

The preceding results give the following implication chain.

$$\Psi_9^4 = \Psi_9^5$$

implies

$$P_9 := \Psi_9^4, \quad P_9^2 = P_9,$$

and hence

$$V = \ker P_9 \oplus \text{range } P_9.$$

Furthermore,

$$\text{range } P_9 = \ker(\Psi_9 - I).$$

If

$$\Psi_9 \Phi_F = \Phi_F,$$

then

$$\text{range } \Phi_F \subseteq \ker(\Psi_9 - I).$$

Independently, the prime lattice admits the exact injective coordinate

$$\Lambda_{\log} : \mathcal{L}_{\mathbb{P}} \hookrightarrow \mathbb{R},$$

with

$$\Lambda_{\log} \left(\sum_p n_p e_p \right) = \sum_p n_p \log p.$$

Finally, if there exists a self-adjoint operator

$$A_F = A_F^*$$

whose spectral divisor satisfies

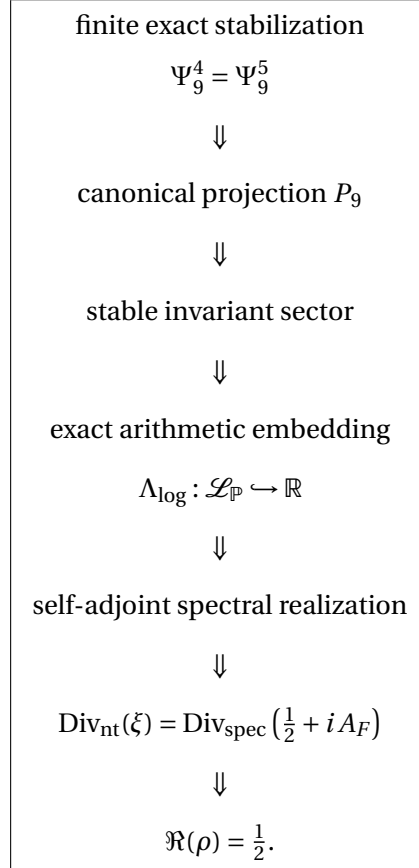
$$\text{Div}_{\text{nt}}(\xi) = \text{Div}_{\text{spec}} \left(\frac{1}{2} + i A_F \right),$$

then

$$\Re(\rho) = \frac{1}{2}$$

for every nontrivial zero ρ of ζ .

Thus the classical mathematical layer has the precise form



Each implication above is an independent mathematical statement, and the final conclusion follows only after the spectral-divisor identity has been established.

127 The Completed Zeta Function and Its Zero Divisor

Let

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Theorem 127.1. *The function ξ extends to an entire function of order one and satisfies*

$$\xi(s) = \xi(1-s)$$

and

$$\xi(\overline{s}) = \overline{\xi(s)}.$$

Proof. These are the classical analytic properties of the completed Riemann zeta function. The gamma factor and the polynomial prefactor remove the pole at $s = 1$ and the trivial zeros, yielding an entire function. The functional equation gives

$$\xi(s) = \xi(1-s),$$

while reality on the real axis gives

$$\xi(\bar{s}) = \overline{\xi(s)}.$$

□

Definition 127.2. Let

$$\mathcal{Z}_\xi$$

denote the multiset of zeros of ξ , each zero being repeated according to its multiplicity.

Proposition 127.3. *The multiset $\mathcal{Z}_\xi$ is invariant under*

$$s \mapsto 1 - s$$

and

$$s \mapsto \bar{s}.$$

Proof. The first invariance follows from the functional equation and the second from conjugation symmetry. Both identities preserve multiplicities. □

Corollary 127.4. *If*

$$\rho = \frac{1}{2} + \beta + i\gamma$$

belongs to $\mathcal{Z}_\xi$, then the four points

$$\frac{1}{2} \pm \beta \pm i\gamma$$

occur in the corresponding symmetry orbit, with the appropriate multiplicities.

128 Spectral Determinants

Let $\mathcal{H}$ be a separable Hilbert space and let

$$A = A^*$$

be a self-adjoint operator with compact resolvent.

Its spectrum is therefore discrete, consists of real eigenvalues of finite multiplicity, and has no finite accumulation point.

Let

$$\{\lambda_n\}_{n \geq 1}$$

denote the eigenvalues counted with multiplicity.

Definition 128.1. A function D_A is called a spectral determinant for A if its zero divisor coincides with the spectral divisor:

$$\text{Div}(D_A) = \sum_n [\lambda_n].$$

For an operator of sufficiently rapid spectral growth, the formal product

$$\prod_n (z - \lambda_n)$$

must be replaced by an appropriate canonical product. If

$$\lambda_n = O(n^\alpha),$$

the corresponding canonical product is determined by the convergence exponent of the sequence.

Definition 128.2. Let p be an integer such that

$$\sum_n |\lambda_n|^{-(p+1)} < \infty.$$

The associated canonical product is

$$E_p(w) = (1 - w) \exp \left(\sum_{k=1}^p \frac{w^k}{k} \right).$$

Proposition 128.3. *The canonical product*

$$D_A(z) = \prod_n E_p \left(\frac{z}{\lambda_n} \right)$$

converges locally uniformly away from the spectral divisor whenever the stated summability condition holds.

Proof. This is the standard convergence theorem for Weierstrass canonical products. The exponential correction removes the divergent terms in the logarithmic expansion through order p , and the remaining series converges locally uniformly under

$$\sum_n |\lambda_n|^{-(p+1)} < \infty.$$

□

129 Self-Adjoint Spectral Symmetry

Let $A = A^*$ be self-adjoint.

Proposition 129.1. *Every eigenvalue of A is real.*

Proof. If

$$Av = \lambda v, \quad v \neq 0,$$

then

$$\langle Av, v \rangle = \lambda \langle v, v \rangle.$$

Self-adjointness gives

$$\langle Av, v \rangle = \langle v, Av \rangle = \overline{\lambda} \langle v, v \rangle.$$

Since

$$\langle v, v \rangle > 0,$$

one obtains

$$\lambda = \overline{\lambda}.$$

Hence

$$\lambda \in \mathbb{R}.$$

□

Definition 129.2. The critical-line spectral transform is

$$\Theta(\lambda) = \frac{1}{2} + i\lambda.$$

Proposition 129.3. *The image of the real spectrum under Θ is contained in the critical line:*

$$\Theta(\sigma(A)) \subseteq \left\{ s \in \mathbb{C} : \Re(s) = \frac{1}{2} \right\}.$$

Proof. For every

$$\lambda \in \sigma(A) \subset \mathbb{R},$$

one has

$$\Theta(\lambda) = \frac{1}{2} + i\lambda,$$

whose real part is $1/2$. □

Theorem 129.4. *Suppose that the zero divisor of ξ is equal to the spectral divisor of A under Θ :*

$$\text{Div}(\xi) = \Theta_* \text{Div}(A).$$

Then every zero of ξ lies on

$$\Re(s) = \frac{1}{2}.$$

Proof. Every spectral point of A is real. Therefore every point in

$$\Theta(\sigma(A))$$

has real part $1/2$. Equality of divisors transfers this support property to every zero of ξ . □

130 Multiplicity-Preserving Spectral Correspondence

Let

$$m_\xi(\rho)$$

denote the multiplicity of a zero ρ of ξ , and let

$$m_A(\lambda)$$

denote the multiplicity of the eigenvalue λ of A .

Definition 130.1. An exact spectral correspondence is a bijection

$$\lambda \longleftrightarrow \rho = \frac{1}{2} + i\lambda$$

such that

$$m_\xi\left(\frac{1}{2} + i\lambda\right) = m_A(\lambda)$$

for every eigenvalue λ , and every nontrivial zero of ξ occurs in this form.

Theorem 130.2. *An exact spectral correspondence is equivalent to the divisor identity*

$$\text{Div}_{\text{nt}}(\xi) = \sum_{\lambda \in \sigma(A)} m_A(\lambda) \left[\frac{1}{2} + i\lambda \right].$$

Proof. The right-hand side records each spectral point with its algebraic multiplicity. Equality with the zero divisor is therefore precisely the assertion that the spectral parametrization is bijective and multiplicity preserving. □

Corollary 130.3. *If an exact spectral correspondence holds for a self-adjoint operator, then the nontrivial zeros satisfy*

$$\Re(\rho) = \frac{1}{2}.$$

Proof. Every corresponding spectral parameter is real. □

Remark 130.4. Multiplicity preservation is essential: equality merely of the sets of spectral points and zeros does not establish equality of divisors.

131 Hadamard Factorization and Spectral Data

Since ξ is entire of order one, Hadamard factorization gives

$$\xi(s) = e^{a+bs} \prod_{\rho \in \mathcal{Z}_\xi} E_1\left(\frac{s}{\rho}\right),$$

with the product taken over the nontrivial zeros and their multiplicities.

Suppose that $A = A^*$ has discrete spectrum

$$\{\lambda_n\}_{n \geq 1}$$

and that

$$\rho_n = \frac{1}{2} + i\lambda_n$$

reproduces the zero divisor of ξ .

Proposition 131.1. *Under this correspondence, the Hadamard product may be written as*

$$\xi(s) = e^{a+bs} \prod_n E_1\left(\frac{s}{\frac{1}{2} + i\lambda_n}\right),$$

with multiplicities inherited from the spectrum of A .

Proof. Substitute

$$\rho_n = \frac{1}{2} + i\lambda_n$$

into the Hadamard factorization. Exact equality of divisors ensures that no zero is omitted and no multiplicity is altered. □

Remark 131.2. The factor

$$e^{a+bs}$$

is not determined by the zero divisor alone. Consequently, an equality of zero sets is weaker than an equality of entire functions. Any operator-determinant formulation of ξ must therefore also account for the nonvanishing entire factor.

132 Completed Spectral Identity

Let $A = A^*$ have discrete spectrum and define

$$\mathcal{D}_A(s) = \prod_n E_1\left(\frac{s}{\frac{1}{2} + i\lambda_n}\right).$$

Assume that the product is defined with the canonical convergence prescription determined by the spectral growth of A .

Theorem 132.1. *If the nontrivial zero divisor of ξ agrees with the divisor of $\mathcal{D}_A$, then there exists an entire function*

$$G(s)$$

without zeros such that

$$\xi(s) = G(s)\mathcal{D}_A(s).$$

Proof. Two entire functions of finite order having the same zeros with the same multiplicities have quotient with no zeros. Therefore

$$G(s) := \frac{\xi(s)}{\mathcal{D}_A(s)}$$

extends to a zero-free entire function. □

Corollary 132.2. *An exact equality*

$$\xi(s) = C\mathcal{D}_A(s)$$

requires the additional determination of the nonvanishing factor $G(s)$ and cannot be inferred from spectral zeros alone.

Remark 132.3. This distinction isolates the precise analytic content of a spectral-determinant realization: the divisor identity determines the zeros, while normalization and growth determine the remaining entire factor.

133 The Arithmetic Spectral Gate

Let

$$\Phi_F : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{H}$$

be an exact arithmetic realization and let

$$A_F = A_F^*$$

be a self-adjoint operator on $\mathcal{H}$.

Assume that the realization is compatible with the logarithmic character family in the sense that

$$\chi_t(x) = \exp(it\Lambda_{\log}(x))$$

is represented by the spectral evolution

$$U_t = e^{itA_F}.$$

Proposition 133.1. *The family*

$$U_t = e^{itA_F}, \quad t \in \mathbb{R},$$

is unitary and satisfies

$$U_{t+s} = U_t U_s.$$

Proof. Since A_F is self-adjoint, the spectral theorem gives

$$(e^{itA_F})^* = e^{-itA_F}.$$

Hence

$$U_t^* U_t = I.$$

The group law follows from the functional calculus:

$$e^{i(t+s)A_F} = e^{itA_F} e^{isA_F}.$$

□

Theorem 133.2. *If the arithmetic realization satisfies*

$$e^{itA_F} \Phi_F(x) = \chi_t(x) \Phi_F(x)$$

for all

$$t \in \mathbb{R}, \quad x \in \mathcal{L}_{\mathbb{P}},$$

then the logarithmic arithmetic coordinate is represented by the spectral generator on the realized arithmetic sector.

Proof. By Stone's theorem and the spectral functional calculus, A_F is the self-adjoint generator of the strongly continuous unitary group e^{itA_F} . The assumed equality identifies its action on the range of Φ_F with multiplication by the character

$$e^{it\Lambda_{\log}(x)}.$$

Thus the corresponding generator has spectral parameter $\Lambda_{\log}(x)$ on the represented character state. □

134 Closure Condition for the Spectral Construction

The finite-state and arithmetic layers are joined by the following exact compatibility condition.

Axiom 134.1 (Spectral Closure). There exists a self-adjoint operator

$$A_F = A_F^*$$

on a separable Hilbert space $\mathcal{H}_F$, together with an exact arithmetic realization

$$\Phi_F : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{H}_F,$$

such that the associated spectral divisor satisfies

$$\text{Div}_{\text{nt}}(\xi) = \text{Div}_{\text{spec}} \left(\frac{1}{2} + iA_F \right).$$

Theorem 134.2. *Under the Spectral Closure condition, the nontrivial zeros of the Riemann zeta function lie on the critical line.*

Proof. The operator A_F is self-adjoint, hence

$$\sigma(A_F) \subseteq \mathbb{R}.$$

The spectral divisor is therefore supported on

$$\left\{ \frac{1}{2} + i\lambda : \lambda \in \mathbb{R} \right\}.$$

By the divisor identity in the Spectral Closure condition, the nontrivial zero divisor of ξ , and hence the nontrivial zero divisor of ζ , has the same support. Thus

$$\Re(\rho) = \frac{1}{2}$$

for every nontrivial zero ρ . □

Remark 134.3. The Spectral Closure condition is the sole global analytic input in this implication. The finite stabilization identity, the prime-lattice embedding, and the self-adjointness of A_F are distinct statements; none is silently substituted for the divisor identity.

135 Exact Stabilization and the Canonical Projection

Let V be a vector space over a field $\mathbb{K}$, and let

$$T \in \text{End}_{\mathbb{K}}(V).$$

Definition 135.1. We say that T is exactly stabilized at order m if

$$T^{m+1} = T^m$$

for some integer $m \geq 1$.

Lemma 135.2. *If T is exactly stabilized at order m , then*

$$T^{m+k} = T^m \quad \text{for every } k \geq 0.$$

Proof. The assertion for $k = 0$ is immediate. If

$$T^{m+k} = T^m,$$

then

$$T^{m+k+1} = T T^{m+k} = T T^m = T^{m+1} = T^m.$$

The result follows by induction. □

Definition 135.3. Under exact stabilization at order m , define

$$P := T^m.$$

Theorem 135.4. *The operator P is idempotent:*

$$P^2 = P.$$

Proof. By definition,

$$P^2 = T^{2m}.$$

Since $2m = m + m$, the preceding lemma gives

$$T^{2m} = T^m = P.$$

□

Proposition 135.5. *The range of P is precisely the fixed-point space of T :*

$$\text{range } P = \ker(T - I).$$

Proof. If

$$v = Pu = T^m u,$$

then

$$Tv = T^{m+1} u = T^m u = v.$$

Hence

$$\text{range } P \subseteq \ker(T - I).$$

Conversely, if

$$Tv = v,$$

then

$$T^m v = v,$$

so

$$v = Pv.$$

Therefore

$$v \in \text{range } P.$$

□

136 Canonical Direct-Sum Decomposition

Theorem 136.1. *If*

$$T^{m+1} = T^m$$

and $P = T^m$, then

$$V = \ker P \oplus \text{range } P.$$

Proof. For every $v \in V$,

$$v = (v - Pv) + Pv.$$

Since

$$P(v - Pv) = Pv - P^2 v = Pv - Pv = 0,$$

one has

$$v - Pv \in \ker P.$$

Also,

$$Pv \in \text{range } P.$$

Thus

$$V = \ker P + \text{range } P.$$

If

$$v \in \ker P \cap \text{range } P,$$

then $v = Pu$ for some u , and therefore

$$v = Pv = P^2u = Pu = v.$$

Since $v \in \ker P$,

$$v = 0.$$

Hence the sum is direct. □

Corollary 136.2. *With*

$$P = T^m,$$

the space V decomposes canonically as

$$V = \ker(T^m) \oplus \ker(T - I).$$

Proof. By the preceding theorem,

$$V = \ker P \oplus \text{range } P.$$

Since

$$P = T^m$$

and

$$\text{range } P = \ker(T - I),$$

the stated decomposition follows. □

Remark 136.3. The decomposition does not require diagonalizability of T . It follows directly from the polynomial identity

$$T^m(T - I) = 0.$$

Thus the stabilized and transient sectors are separated algebraically before any spectral representation is introduced.

137 Polynomial Separation of the Stable and Transient Sectors

The stabilization relation is equivalent to the operator polynomial identity

$$T^m(T - I) = 0.$$

The two factors

$$X^m \quad \text{and} \quad X - 1$$

are relatively prime in $\mathbb{K}[X]$.

Lemma 137.1. *There exist polynomials $a, b \in \mathbb{K}[X]$ such that*

$$a(X)X^m + b(X)(X - 1) = 1.$$

Proof. Since

$$X^m - 1 = (X - 1)(1 + X + \cdots + X^{m-1}),$$

one has

$$1 = X^m - (X - 1)(1 + X + \cdots + X^{m-1}).$$

Thus one may take

$$a(X) = 1, \quad b(X) = -(1 + X + \cdots + X^{m-1}).$$

□

Theorem 137.2. *The primary decomposition associated with*

$$T^m(T - I) = 0$$

is

$$V = \ker(T^m) \oplus \ker(T - I).$$

Proof. The direct-sum decomposition was established above. The coprimality of the two annihilating factors shows that the two kernels are the corresponding primary components of the polynomial

$$X^m(X - 1).$$

No additional spectral assumption is required. $\square$

Corollary 137.3. *The restriction*

$$T|_{\ker(T-I)}$$

is the identity, while

$$T^m|_{\ker(T^m)} = 0.$$

Proof. The first assertion follows from the definition of $\ker(T - I)$. The second follows immediately from the definition of $\ker(T^m)$. $\square$

138 Finite-State Realization

Let

$$V_9 = \mathbf{1}^\perp \subset \mathbb{C}^{144}$$

and let

$$\Psi_9 : V_9 \longrightarrow V_9$$

be a linear endomorphism satisfying

$$\Psi_9^5 = \Psi_9^4.$$

Define

$$P_9 := \Psi_9^4.$$

Theorem 138.1. *The operator P_9 is a projection:*

$$P_9^2 = P_9.$$

Moreover,

$$\text{range } P_9 = \ker(\Psi_9 - I)$$

and

$$V_9 = \ker(\Psi_9^4) \oplus \ker(\Psi_9 - I).$$

Proof. Apply the general stabilization theorem with

$$T = \Psi_9, \quad m = 4.$$

The identity

$$\Psi_9^5 = \Psi_9^4$$

is exactly the required stabilization relation. $\square$

Proposition 138.2. *For every*

$$v \in V_9,$$

the decomposition

$$v = v_{\text{tr}} + v_{\text{st}}$$

is uniquely determined by

$$v_{\text{tr}} = (I - P_9)v, \quad v_{\text{st}} = P_9v.$$

Proof. Since P_9 is idempotent,

$$P_9(I - P_9) = P_9 - P_9^2 = 0.$$

Thus

$$v_{\text{tr}} \in \ker P_9.$$

By construction,

$$v_{\text{st}} \in \text{range } P_9.$$

Their sum is

$$(I - P_9)v + P_9v = v.$$

Uniqueness follows from the directness of the decomposition. □

139 Invariant Arithmetic Subspaces

Let W be a vector space and let

$$\Phi : \mathcal{L}_{\mathbb{P}} \longrightarrow W$$

be linear. Suppose

$$T\Phi = \Phi.$$

Proposition 139.1. *The image of Φ is contained in the stable sector of T :*

$$\text{range } \Phi \subseteq \ker(T - I).$$

Proof. For every

$$x \in \mathcal{L}_{\mathbb{P}},$$

the compatibility condition gives

$$T\Phi(x) = \Phi(x).$$

Hence

$$(T - I)\Phi(x) = 0,$$

so

$$\Phi(x) \in \ker(T - I).$$

□

Corollary 139.2. *If*

$$T^{m+1} = T^m,$$

then

$$P\Phi = \Phi, \quad P = T^m.$$

Proof. Since

$$\text{range } \Phi \subseteq \ker(T - I) = \text{range } P,$$

and P acts as the identity on its range,

$$P\Phi = \Phi.$$

□

Theorem 139.3. *Let*

$$\Phi_F : \mathcal{L}_{\mathbb{P}} \longrightarrow V_9$$

satisfy

$$\Psi_9 \Phi_F = \Phi_F.$$

Then

$$P_9 \Phi_F = \Phi_F$$

and

$$\text{range } \Phi_F \subseteq \text{range } P_9.$$

Proof. The preceding corollary applies with

$$T = \Psi_9, \quad m = 4.$$

□

140 Spectral Consequences of the Stabilized Sector

Assume now that V_9 is equipped with its standard Hermitian inner product and that

$$\Psi_9$$

is normal.

Proposition 140.1. *The space V_9 admits an orthogonal spectral decomposition for Ψ_9 .*

Proof. Since V_9 is finite-dimensional and Ψ_9 is normal, the spectral theorem for normal matrices applies. □

Theorem 140.2. *Under the stabilization identity*

$$\Psi_9^5 = \Psi_9^4,$$

every eigenvalue λ of Ψ_9 satisfies

$$\lambda^4(\lambda - 1) = 0.$$

Consequently,

$$\lambda \in \{0, 1\}.$$

Proof. Let

$$\Psi_9 v = \lambda v, \quad v \neq 0.$$

Applying the stabilization identity gives

$$\lambda^5 v = \lambda^4 v.$$

Therefore

$$\lambda^4(\lambda - 1)v = 0.$$

Since $v \neq 0$,

$$\lambda^4(\lambda - 1) = 0.$$

Hence

$$\lambda = 0 \quad \text{or} \quad \lambda = 1.$$

□

Corollary 140.3. *If Ψ_9 is normal, then*

$$V_9 = E_0 \oplus^\perp E_1,$$

where

$$E_0 = \ker \Psi_9$$

and

$$E_1 = \ker(\Psi_9 - I).$$

Proof. The spectral theorem gives an orthogonal decomposition into eigenspaces. The preceding theorem shows that the only possible eigenvalues are 0 and 1. □

Remark 140.4. Normality is used only for orthogonality. The algebraic decomposition

$$V_9 = \ker(\Psi_9^4) \oplus \ker(\Psi_9 - I)$$

requires no normality assumption.

141 Identification of the Stable Projection

Under the hypotheses of the preceding section, let

$$Q_1$$

denote the orthogonal projection onto

$$E_1 = \ker(\Psi_9 - I).$$

Theorem 141.1. *One has*

$$P_9 = Q_1.$$

Proof. The operator $P_9 = \Psi_9^4$ acts as the identity on E_1 , because

$$\Psi_9 v = v$$

implies

$$P_9 v = v.$$

It acts as zero on E_0 , because

$$\Psi_9 v = 0$$

implies

$$P_9 v = 0.$$

Since

$$V_9 = E_0 \oplus^\perp E_1,$$

there is a unique linear operator acting as zero on E_0 and as the identity on E_1 , namely the orthogonal projection Q_1 . Therefore

$$P_9 = Q_1.$$

□

Corollary 141.2. *The stabilized operator satisfies*

$$P_9 = P_9^* = P_9^2.$$

Proof. As an orthogonal projection, P_9 is self-adjoint and idempotent. □

Remark 141.3. This identifies the stabilization polynomial with the canonical orthogonal projection only under the stated normality hypothesis. Without normality, P_9 remains an algebraic projection, but need not be orthogonal.

142 Trace and Rank of the Stabilized Projection

Let

$$P_9 = \Psi_9^4$$

be the stabilized projection on

$$V_9 = \mathbf{1}^\perp \subset \mathbb{C}^{144}.$$

Since V_9 has complex dimension 143, one has

$$\dim_{\mathbb{C}} V_9 = 143.$$

Proposition 142.1. *The spectrum of P_9 is contained in*

$$\{0, 1\}.$$

Proof. Since

$$P_9^2 = P_9,$$

every eigenvalue λ satisfies

$$\lambda^2 = \lambda.$$

Hence

$$\lambda(\lambda - 1) = 0,$$

and therefore

$$\lambda \in \{0, 1\}.$$

□

Theorem 142.2. *The trace of P_9 equals its rank:*

$$\text{tr}(P_9) = \text{rank}(P_9).$$

Proof. Since P_9 is a projection, its eigenvalues are 0 and 1. The multiplicity of the eigenvalue 1 is precisely

$$\dim \text{range}(P_9) = \text{rank}(P_9).$$

The trace is the sum of the eigenvalues counted with multiplicity. Hence

$$\text{tr}(P_9) = \text{rank}(P_9).$$

□

Corollary 142.3. *If*

$$\text{rank}(P_9) = r,$$

then

$$\text{tr}(P_9) = r$$

and

$$\text{tr}(I_{V_9} - P_9) = 143 - r.$$

143 Normalized Stable State

Let

$$E_9 = \text{range}(P_9)$$

and suppose

$$r := \dim E_9 > 0.$$

Define the normalized stable density operator by

$$\rho_9 = \frac{1}{r} P_9.$$

Proposition 143.1. *The operator ρ_9 is positive semidefinite.*

Proof. Under the normality hypothesis established above, P_9 is an orthogonal projection. Hence

$$\langle P_9 v, v \rangle = \|P_9 v\|^2 \geq 0.$$

Therefore

$$\langle \rho_9 v, v \rangle = \frac{1}{r} \|P_9 v\|^2 \geq 0.$$

□

Proposition 143.2. *The normalized operator ρ_9 satisfies*

$$\text{tr}(\rho_9) = 1.$$

Proof. Using

$$\text{tr}(P_9) = r,$$

one obtains

$$\text{tr}(\rho_9) = \frac{1}{r} \text{tr}(P_9) = 1.$$

□

Theorem 143.3. *The operator*

$$\rho_9 = \frac{1}{r} P_9$$

is a density operator on the stabilized sector E_9 .

Proof. It is positive semidefinite and has trace one. Its support is contained in

$$E_9 = \text{range}(P_9).$$

□

Remark 143.4. The coefficient $1/r$ is forced by trace normalization. In particular,

$$\rho_9 = \frac{1}{9} I_9$$

can be concluded only after establishing

$$r = 9$$

and identifying the stabilized sector with a nine-dimensional space on which the relevant identity operator acts.

144 The Nine-Dimensional Uniform Fixed Sector

Assume that the stabilized sector satisfies

$$\dim E_9 = 9$$

and that the restriction of P_9 to E_9 is the identity.

Let

$$I_{E_9}$$

denote the identity operator on E_9 .

Theorem 144.1. *The normalized stable state is*

$$\rho_9 = \frac{1}{9} I_{E_9}.$$

Proof. Since

$$P_9|_{E_9} = I_{E_9}$$

and

$$\dim E_9 = 9,$$

the normalized density operator is

$$\rho_9 = \frac{1}{\dim E_9} I_{E_9} = \frac{1}{9} I_{E_9}.$$

□

Proposition 144.2. *The spectrum of ρ_9 on E_9 consists of the single eigenvalue*

$$\frac{1}{9}$$

with multiplicity 9.

Proof. Since

$$\rho_9 = \frac{1}{9} I_{E_9},$$

every nonzero vector of E_9 is an eigenvector with eigenvalue $1/9$. The multiplicity equals

$$\dim E_9 = 9.$$

□

Corollary 144.3. *The von Neumann entropy of ρ_9 is*

$$S(\rho_9) = -\operatorname{tr}(\rho_9 \log \rho_9) = \log 9.$$

Proof. All nine eigenvalues are equal to $1/9$. Hence

$$S(\rho_9) = -9 \frac{1}{9} \log \frac{1}{9} = \log 9.$$

□

145 Finite Positive Maps and Perron–Frobenius Structure

Let

$$B \in M_9(\mathbb{R})$$

be a nonnegative matrix:

$$B_{ij} \geq 0.$$

Definition 145.1. The matrix B is primitive if there exists an integer $k \geq 1$ such that

$$B^k > 0$$

entrywise.

Definition 145.2. The primitivity exponent of B is

$$\gamma(B) = \min\{k \geq 1 : B^k > 0\},$$

whenever B is primitive.

Theorem 145.3. *If B is primitive, then its spectral radius*

$$r(B) > 0$$

is a simple eigenvalue with a strictly positive eigenvector. Every other eigenvalue λ satisfies

$$|\lambda| < r(B).$$

Proof. This is the Perron–Frobenius theorem for primitive matrices. Primitivity excludes all peripheral eigenvalues other than the Perron root and yields a strictly positive eigenvector. $\square$

Proposition 145.4. *If B is row-stochastic,*

$$\sum_{j=1}^9 B_{ij} = 1$$

for every i , then

$$B\mathbf{1} = \mathbf{1}.$$

Consequently,

$$r(B) = 1.$$

Proof. The stochasticity condition gives the first identity directly. Therefore 1 is an eigenvalue. On the other hand, every row sum is one, so

$$\|B\|_{\infty} = 1,$$

and hence

$$r(B) \leq 1.$$

Thus

$$r(B) = 1.$$

$\square$

146 Uniform Perron State

Let

$$J_9 = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix}$$

and define

$$B^* = \frac{1}{9} J_9.$$

Proposition 146.1. *The matrix B^* is row-stochastic and strictly positive.*

Proof. Every entry of B^* equals $1/9 > 0$. Each row contains nine entries, hence every row sum is

$$9 \cdot \frac{1}{9} = 1.$$

□

Proposition 146.2. *One has*

$$(B^*)^2 = B^*.$$

Proof. Since

$$J_9^2 = 9J_9,$$

one obtains

$$(B^*)^2 = \frac{1}{81} J_9^2 = \frac{1}{81} (9J_9) = \frac{1}{9} J_9 = B^*.$$

□

Theorem 146.3. *The spectrum of B^* is*

$$\sigma(B^*) = \{1, 0\},$$

where 1 has multiplicity one and 0 has multiplicity eight.

Proof. The image of B^* is the one-dimensional space spanned by

$$\mathbf{1} = (1, \dots, 1)^T.$$

Moreover,

$$B^* \mathbf{1} = \mathbf{1}.$$

Thus 1 is an eigenvalue with multiplicity one. Since B^* has rank one and acts as zero on

$$\mathbf{1}^\perp,$$

zero has multiplicity eight. □

Corollary 146.4. *The matrix B^* is primitive and*

$$\gamma(B^*) = 1.$$

Proof. Every entry of B^* is strictly positive, so

$$(B^*)^1 > 0$$

entrywise. Therefore the minimal primitivity exponent is 1. □

147 General Primitivity Exponent and the Four-Step Constraint

Let

$$B \in M_9(\mathbb{R})$$

be primitive and suppose

$$B^4 > 0.$$

Proposition 147.1. *The primitivity exponent satisfies*

$$\gamma(B) \leq 4.$$

Proof. By definition, $\gamma(B)$ is the least positive integer k for which

$$B^k > 0.$$

Since this already holds for $k = 4$,

$$\gamma(B) \leq 4.$$

□

Remark 147.2. The equality

$$\gamma(B) = 4$$

requires, in addition,

$$B^k \not> 0 \quad (k = 1, 2, 3).$$

Thus the four-step value is an exact claim only when these three non-positivity conditions have been established.

Theorem 147.3. *If*

$$B^4 > 0$$

and

$$B^3 \not> 0,$$

then

$$\gamma(B) = 4.$$

Proof. The first condition gives

$$\gamma(B) \leq 4.$$

The second excludes

$$\gamma(B) \leq 3.$$

Hence

$$\gamma(B) = 4.$$

□

Remark 147.4. This distinction is necessary when the finite-state stabilization exponent and the Perron–Frobenius primitivity exponent are both denoted by 4. They are different invariants unless an explicit theorem identifies them.

148 Separation of Algebraic and Probabilistic Normalizations

The projection

$$P_9$$

and the stochastic matrix

$$B^*$$

represent different algebraic objects.

Proposition 148.1. *The operator*

$$P_9$$

is characterized by its action on the stabilized vector space:

$$P_9 v = v \quad (v \in E_9), \quad P_9 v = 0 \quad (v \in \ker P_9).$$

Proposition 148.2. *The matrix*

$$B^* = \frac{1}{9} J_9$$

is characterized by its action on $\mathbb{R}^9$:

$$B^* x = \left(\frac{1}{9} \sum_{j=1}^9 x_j \right) \mathbf{1}.$$

Proof. The first statement follows from the definition of the projection. For the second,

$$(B^* x)_i = \sum_{j=1}^9 \frac{1}{9} x_j = \frac{1}{9} \sum_{j=1}^9 x_j$$

for every i . □

Theorem 148.3. *The equality*

$$P_9 \cong B^*$$

is meaningful only after an explicit identification

$$E_9 \cong \mathbb{R}^9$$

and an identification of the induced action with the uniform averaging map.

Proof. The operators act on different spaces and have different ambient dimensions in general. An operator equality therefore requires a common space; an operator conjugacy requires an explicitly specified isomorphism. Once such an isomorphism is supplied, equality of the induced matrices is a finite-dimensional identity. □

Remark 148.4. This separation prevents the normalization

$$\frac{1}{9} J_9$$

of a nine-dimensional maximally mixed state from being conflated with the rank-one stochastic averaging operator

$$\frac{1}{9} J_9.$$

They have different ranks, spectra, and actions.

149 The Arithmetic Lattice and Its Finite-Support Sector

Let $\mathbb{P}$ denote the set of prime numbers and define the free abelian group

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z}e_p.$$

Thus every element $x \in \mathcal{L}_{\mathbb{P}}$ has a unique expression

$$x = \sum_{p \in \mathbb{P}} n_p e_p, \quad n_p \in \mathbb{Z},$$

with only finitely many n_p nonzero.

Proposition 149.1. *The family $\{e_p\}_{p \in \mathbb{P}}$ is a basis of $\mathcal{L}_{\mathbb{P}}$, and*

$$\mathcal{L}_{\mathbb{P}} \cong \mathbb{Z}^{(\mathbb{P})}.$$

Proof. This is the defining property of the direct sum. Every element has finite support, and equality of two elements is equivalent to equality of all their integer coordinates. $\square$

Define the arithmetic realization

$$\mathfrak{a} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{Q}_{>0}^{\times}$$

by

$$\mathfrak{a}(x) = \prod_{p \in \mathbb{P}} p^{n_p}.$$

Theorem 149.2. *The map*

$$\mathfrak{a} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{Q}_{>0}^{\times}$$

is a group isomorphism.

Proof. Finite support makes the product defining $\mathfrak{a}(x)$ finite. Moreover,

$$\mathfrak{a}(x + y) = \prod_p p^{n_p + m_p} = \left(\prod_p p^{n_p} \right) \left(\prod_p p^{m_p} \right) = \mathfrak{a}(x) \mathfrak{a}(y).$$

Thus $\mathfrak{a}$ is a group homomorphism.

If

$$\mathfrak{a}(x) = 1,$$

unique factorization implies

$$n_p = 0$$

for every p , hence $x = 0$. Thus $\mathfrak{a}$ is injective.

Conversely, every $q \in \mathbb{Q}_{>0}^{\times}$ has a unique representation

$$q = \prod_{p \in \mathbb{P}} p^{n_p}, \quad n_p \in \mathbb{Z},$$

with finite support. Hence $\mathfrak{a}$ is surjective. $\square$

150 The Logarithmic Embedding

Define

$$\Lambda_{\log} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{R}$$

by

$$\Lambda_{\log}(x) = \sum_{p \in \mathbb{P}} n_p \log p.$$

Proposition 150.1. *The logarithmic map is additive:*

$$\Lambda_{\log}(x + y) = \Lambda_{\log}(x) + \Lambda_{\log}(y).$$

Proof. The sums involved are finite. Hence

$$\Lambda_{\log}(x + y) = \sum_p (n_p + m_p) \log p = \sum_p n_p \log p + \sum_p m_p \log p.$$

□

Theorem 150.2. *The map*

$$\Lambda_{\log} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{R}$$

is injective.

Proof. Suppose

$$\Lambda_{\log}(x) = 0.$$

Then

$$0 = \sum_p n_p \log p = \log \left(\prod_p p^{n_p} \right) = \log \mathfrak{a}(x).$$

Therefore

$$\mathfrak{a}(x) = 1.$$

By unique factorization,

$$x = 0.$$

Thus $\Lambda_{\log}$ is injective.

□

Corollary 150.3. *The image*

$$\Lambda_{\log}(\mathcal{L}_{\mathbb{P}}) = \left\{ \sum_{p \in \mathbb{P}} n_p \log p : n_p \in \mathbb{Z}, n_p = 0 \text{ for all but finitely many } p \right\}$$

is a countable additive subgroup of $\mathbb{R}$.

151 Character Family and Its Exact Algebra

For $t \in \mathbb{R}$, define

$$\chi_t : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{C}^{\times}$$

by

$$\chi_t(x) = e^{it\Lambda_{\log}(x)}.$$

Theorem 151.1. *For every $t \in \mathbb{R}$, χ_t is a unitary character:*

$$\chi_t(x + y) = \chi_t(x)\chi_t(y), \quad |\chi_t(x)| = 1.$$

Proof. By additivity of $\Lambda_{\log}$,

$$\chi_t(x+y) = e^{it(\Lambda_{\log}(x)+\Lambda_{\log}(y))} = \chi_t(x)\chi_t(y).$$

Since $\Lambda_{\log}(x) \in \mathbb{R}$,

$$|\chi_t(x)| = 1.$$

□

Proposition 151.2. *The character family satisfies*

$$\chi_{t+s}(x) = \chi_t(x)\chi_s(x)$$

and

$$\chi_{-t}(x) = \overline{\chi_t(x)}.$$

Proof. For $a \in \mathbb{R}$,

$$e^{i(t+s)a} = e^{ita}e^{isa}$$

and

$$e^{-ita} = \overline{e^{ita}}.$$

Taking $a = \Lambda_{\log}(x)$ gives the stated identities. □

Corollary 151.3. *The map*

$$t \longmapsto \chi_t$$

is a one-parameter subgroup of the character group of $\mathcal{L}_{\mathbb{P}}$.

152 Hilbert Realization of the Arithmetic Characters

Let

$$\mathcal{H}_{\mathbb{P}} = \ell^2(\mathcal{L}_{\mathbb{P}})$$

with canonical orthonormal basis

$$\{\delta_x : x \in \mathcal{L}_{\mathbb{P}}\}.$$

Define the multiplication operator

$$A_{\log}$$

by

$$A_{\log}\delta_x = \Lambda_{\log}(x)\delta_x$$

on the domain

$$\text{Dom}(A_{\log}) = \left\{ \psi \in \mathcal{H}_{\mathbb{P}} : \sum_{x \in \mathcal{L}_{\mathbb{P}}} |\Lambda_{\log}(x)|^2 |\psi_x|^2 < \infty \right\},$$

where

$$\psi = \sum_x \psi_x \delta_x.$$

Proposition 152.1. *The operator $A_{\log}$ is densely defined and symmetric.*

Proof. The finitely supported vectors form a dense subspace of $\ell^2(\mathcal{L}_{\mathbb{P}})$ and are contained in $\text{Dom}(A_{\log})$.

For finitely supported ψ, φ ,

$$\langle A_{\log} \psi, \varphi \rangle = \sum_x \Lambda_{\log}(x) \psi_x \overline{\varphi_x}.$$

Because $\Lambda_{\log}(x) \in \mathbb{R}$,

$$\langle A_{\log} \psi, \varphi \rangle = \langle \psi, A_{\log} \varphi \rangle.$$

Thus $A_{\log}$ is symmetric. □

Theorem 152.2. *The operator $A_{\log}$ is self-adjoint on $\text{Dom}(A_{\log})$.*

Proof. The operator is multiplication by the real-valued function

$$x \longmapsto \Lambda_{\log}(x)$$

on the counting-measure space $\mathcal{L}_{\mathbb{P}}$. The maximal multiplication operator associated with a real-valued measurable function is self-adjoint. □

Corollary 152.3. *The unitary group*

$$U_t = e^{itA_{\log}}, \quad t \in \mathbb{R},$$

acts on the canonical basis according to

$$U_t \delta_x = e^{it\Lambda_{\log}(x)} \delta_x = \chi_t(x) \delta_x.$$

Proof. The spectral functional calculus gives

$$e^{itA_{\log}} \delta_x = e^{it\Lambda_{\log}(x)} \delta_x.$$

□

153 The Arithmetic Spectral Measure

Let

$$E_{\log}$$

denote the projection-valued spectral measure of $A_{\log}$.

For every normalized vector

$$\psi \in \mathcal{H}_{\mathbb{P}}, \quad \|\psi\| = 1,$$

define

$$\mu_{\psi}(B) = \langle E_{\log}(B) \psi, \psi \rangle$$

for every Borel set

$$B \subseteq \mathbb{R}.$$

Theorem 153.1. *The map*

$$\mu_{\psi}$$

is a probability measure on $\mathbb{R}$.

Proof. Since $E_{\log}(B)$ is an orthogonal projection,

$$\mu_\psi(B) \geq 0.$$

Countable additivity follows from the corresponding property of the projection-valued spectral measure. Finally,

$$\mu_\psi(\mathbb{R}) = \langle E_{\log}(\mathbb{R})\psi, \psi \rangle = \langle \psi, \psi \rangle = 1.$$

□

Proposition 153.2. *For every bounded Borel function*

$$f : \mathbb{R} \longrightarrow \mathbb{C},$$

one has

$$\langle f(A_{\log})\psi, \psi \rangle = \int_{\mathbb{R}} f(\lambda) d\mu_\psi(\lambda).$$

Proof. This is the scalar form of the spectral theorem applied to the vector ψ . □

Corollary 153.3. *The characteristic function of μ_ψ is*

$$\widehat{\mu_\psi}(t) = \langle e^{itA_{\log}}\psi, \psi \rangle.$$

154 Explicit Form of the Arithmetic Spectral Measure

Let

$$\psi = \sum_{x \in \mathcal{L}_{\mathbb{P}}} \psi_x \delta_x$$

with

$$\sum_x |\psi_x|^2 = 1.$$

Theorem 154.1. *The spectral measure associated with ψ is*

$$\mu_\psi = \sum_{x \in \mathcal{L}_{\mathbb{P}}} |\psi_x|^2 \delta_{\Lambda_{\log}(x)},$$

with equal spectral values combined into the same atom.

Proof. For a Borel set $B \subseteq \mathbb{R}$,

$$E_{\log}(B)\delta_x = \mathbf{1}_B(\Lambda_{\log}(x))\delta_x.$$

Therefore

$$\mu_\psi(B) = \sum_x |\psi_x|^2 \mathbf{1}_B(\Lambda_{\log}(x)),$$

which is exactly the stated discrete measure. □

Corollary 154.2. *For every $t \in \mathbb{R}$,*

$$\widehat{\mu_\psi}(t) = \sum_x |\psi_x|^2 e^{it\Lambda_{\log}(x)}.$$

Remark 154.3. The spectrum generated by the arithmetic realization is therefore

$$\Lambda_{\log}(\mathcal{L}_{\mathbb{P}}),$$

a countable subset of $\mathbb{R}$. A continuous spectral measure, if required by a subsequent construction, must arise from an additional completion, limiting procedure, quotient, or different representation; it is not obtained by replacing the discrete sum above with an integral.

155 Finite Observables of the Arithmetic Lattice

Let

$$q : \mathcal{L}_{\mathbb{P}} \longrightarrow \Gamma$$

be a homomorphism into a finitely generated abelian group, and let

$$\Phi : \Gamma \longrightarrow V_9$$

be a linear map, where

$$V_9 = \mathbf{1}^{\perp} \subset \mathbb{C}^{144}.$$

Define

$$\Phi_F = \Phi \circ q.$$

Definition 155.1. The subgroup

$$K_q = \ker q$$

is called the resolution kernel of the finite arithmetic observable.

Proposition 155.2. *The map Φ_F factors uniquely through the quotient*

$$\mathcal{L}_{\mathbb{P}} / K_q.$$

Proof. Since q vanishes on K_q , the first isomorphism theorem gives

$$\mathcal{L}_{\mathbb{P}} / K_q \cong \text{im } q.$$

Thus there is a unique map

$$\tilde{\Phi} : \mathcal{L}_{\mathbb{P}} / K_q \longrightarrow V_9$$

such that

$$\Phi_F = \tilde{\Phi} \circ \pi_q,$$

where π_q is the canonical quotient map. □

Theorem 155.3. *If*

$$x - y \in K_q,$$

then

$$\Phi_F(x) = \Phi_F(y).$$

Conversely, if Φ is injective on $\text{im } q$, then

$$\Phi_F(x) = \Phi_F(y) \iff x - y \in K_q.$$

Proof. The first implication follows from

$$q(x - y) = 0,$$

hence

$$q(x) = q(y)$$

and therefore

$$\Phi_F(x) = \Phi_F(y).$$

Conversely, suppose

$$\Phi_F(x) = \Phi_F(y).$$

Then

$$\Phi(q(x)) = \Phi(q(y)).$$

Injectivity of Φ on $\text{im } q$ gives

$$q(x) = q(y),$$

hence

$$q(x - y) = 0.$$

Therefore

$$x - y \in K_q.$$

□

156 Compatibility with the Stabilized Dynamics

Let

$$P_9 = \Psi_9^4$$

and assume

$$\Psi_9^5 = \Psi_9^4.$$

Definition 156.1. The finite arithmetic observable Φ_F is dynamically compatible with the stabilized sector if

$$\Psi_9 \Phi_F(x) = \Phi_F(x)$$

for every

$$x \in \mathcal{L}_{\mathbb{P}}.$$

Theorem 156.2. *If Φ_F is dynamically compatible, then*

$$P_9 \Phi_F = \Phi_F.$$

Equivalently,

$$\text{range } \Phi_F \subseteq \text{range } P_9.$$

Proof. For every x ,

$$\Psi_9 \Phi_F(x) = \Phi_F(x).$$

Therefore

$$\Phi_F(x) \in \ker(\Psi_9 - I).$$

The stabilization identity implies

$$\ker(\Psi_9 - I) = \text{range}(P_9).$$

Since P_9 is the identity on its range,

$$P_9 \Phi_F(x) = \Phi_F(x).$$

□

Corollary 156.3. *The finite arithmetic observable is unchanged by stabilization:*

$$\Phi_F = \Psi_9^4 \Phi_F = \Psi_9^5 \Phi_F.$$

Proof. The first equality is the preceding theorem. The second follows from

$$\Psi_9^5 = \Psi_9^4.$$

□

157 Finite-State Realization as a Quotient Observable

Let

$$q : \mathcal{L}_{\mathbb{P}} \longrightarrow \Gamma$$

and

$$\Phi : \Gamma \longrightarrow V_9$$

be as above.

Theorem 157.1. *A finite-state realization of the arithmetic lattice is completely specified by the triple*

$$(\Gamma, q, \Phi)$$

together with its resolution kernel

$$K_q = \ker q.$$

The induced observable depends only on the quotient

$$\mathcal{L}_{\mathbb{P}} / K_q.$$

Proof. The quotient factorization established above gives

$$\Phi_F = \tilde{\Phi} \circ \pi_q.$$

Thus two lattice elements are indistinguishable by the finite realization precisely when their difference lies in the resolution kernel, provided Φ is injective on the image of q . $\square$

Remark 157.2. This quotient formulation is the mathematically precise interface between the infinite arithmetic lattice and a finite operational substrate. It neither identifies the two objects nor requires an impossible finite-dimensional faithful embedding of the full free abelian group $\mathcal{L}_{\mathbb{P}}$.

158 Spectral Measures Induced by Finite Arithmetic Observables

Let

$$\mathcal{H}_9 = \text{range}(P_9)$$

and suppose that

$$\dim_{\mathbb{C}} \mathcal{H}_9 = r > 0.$$

Let

$$A \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be self-adjoint. Denote its spectral resolution by

$$A = \sum_{\lambda \in \sigma(A)} \lambda E_A(\{\lambda\}),$$

where the spectral projections satisfy

$$E_A(\{\lambda\}) E_A(\{\mu\}) = \delta_{\lambda\mu} E_A(\{\lambda\})$$

and

$$\sum_{\lambda \in \sigma(A)} E_A(\{\lambda\}) = I_{\mathcal{H}_9}.$$

Definition 158.1. For a density operator

$$\rho \in \text{End}_{\mathbb{C}}(\mathcal{H}_9), \quad \rho \geq 0, \quad \text{tr } \rho = 1,$$

the probability measure associated with A and ρ is

$$\mu_{\rho,A}(B) = \text{tr}(\rho E_A(B))$$

for every Borel set $B \subseteq \mathbb{R}$.

Theorem 158.2. *The map*

$$B \longmapsto \mu_{\rho,A}(B)$$

is a probability measure.

Proof. Each spectral projection is positive, hence

$$\text{tr}(\rho E_A(B)) \geq 0.$$

Countable additivity follows from the corresponding property of the spectral measure. Finally,

$$\mu_{\rho,A}(\mathbb{R}) = \text{tr}(\rho I_{\mathcal{H}_9}) = \text{tr } \rho = 1.$$

□

Proposition 158.3. *For every bounded Borel function*

$$f : \mathbb{R} \longrightarrow \mathbb{C},$$

one has

$$\text{tr}(\rho f(A)) = \int_{\mathbb{R}} f(\lambda) d\mu_{\rho,A}(\lambda).$$

Proof. The finite-dimensional spectral decomposition gives

$$f(A) = \sum_{\lambda \in \sigma(A)} f(\lambda) E_A(\{\lambda\}).$$

Therefore

$$\text{tr}(\rho f(A)) = \sum_{\lambda \in \sigma(A)} f(\lambda) \text{tr}(\rho E_A(\{\lambda\})),$$

which is precisely the integral with respect to $\mu_{\rho,A}$.

□

159 The Uniform Stable State and Spectral Weights

Assume

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9$$

and define

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Theorem 159.1. *For every self-adjoint operator*

$$A \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the probability assigned to an eigenspace $E \subseteq \mathcal{H}_9$ by ρ_9 is

$$\text{tr}(\rho_9 P_E) = \frac{\dim E}{9},$$

where P_E denotes the orthogonal projection onto E .

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\mathrm{tr}(\rho_9 P_E) = \frac{1}{9} \mathrm{tr}(P_E).$$

The trace of an orthogonal projection equals its rank, and therefore

$$\mathrm{tr}(P_E) = \dim E.$$

Hence

$$\mathrm{tr}(\rho_9 P_E) = \frac{\dim E}{9}.$$

□

Corollary 159.2. *If A has distinct eigenvalues*

$$\lambda_1, \dots, \lambda_m$$

with multiplicities

$$d_1, \dots, d_m,$$

then

$$\mu_{\rho_9, A} = \sum_{j=1}^m \frac{d_j}{9} \delta_{\lambda_j},$$

with

$$\sum_{j=1}^m d_j = 9.$$

Proof. Apply the preceding theorem to each eigenspace

$$E_A(\{\lambda_j\}).$$

Its dimension is d_j .

□

160 Rank-One Spectral Projections

Let

$$A \in \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be self-adjoint with simple spectrum

$$\sigma(A) = \{\lambda_1, \dots, \lambda_9\}.$$

Let

$$u_1, \dots, u_9$$

be an orthonormal eigenbasis satisfying

$$Au_j = \lambda_j u_j.$$

Define

$$P_j = |u_j\rangle\langle u_j|.$$

Proposition 160.1. *The operators P_j are mutually orthogonal rank-one projections:*

$$P_i P_j = \delta_{ij} P_j, \quad P_j^* = P_j, \quad \text{tr}(P_j) = 1.$$

Proof. Orthonormality gives

$$\langle u_i, u_j \rangle = \delta_{ij}.$$

Therefore

$$P_i P_j = |u_i\rangle \langle u_i, u_j| \langle u_j| = \delta_{ij} P_j.$$

Self-adjointness and rank one are immediate from the definition, and the trace of a rank-one orthogonal projection is 1. $\square$

Theorem 160.2. *The uniform stable state satisfies*

$$\text{tr}(\rho_9 P_j) = \frac{1}{9}$$

for every $j = 1, \dots, 9$.

Proof. Since

$$\text{rank}(P_j) = 1,$$

the eigenspace formula gives

$$\text{tr}(\rho_9 P_j) = \frac{1}{9}.$$

$\square$

Corollary 160.3. *The corresponding spectral measure is*

$$\mu_{\rho_9, A} = \frac{1}{9} \sum_{j=1}^9 \delta_{\lambda_j}.$$

161 Projection-Valued Decomposition

For a self-adjoint operator A with simple spectrum, define

$$P_j = |u_j\rangle \langle u_j|.$$

Theorem 161.1. *The spectral decomposition of A is*

$$A = \sum_{j=1}^9 \lambda_j P_j,$$

while the identity decomposes as

$$I_{\mathcal{H}_9} = \sum_{j=1}^9 P_j.$$

Proof. Since $\{u_j\}_{j=1}^9$ is an orthonormal basis, every $v \in \mathcal{H}_9$ satisfies

$$v = \sum_{j=1}^9 \langle u_j, v \rangle u_j = \sum_{j=1}^9 P_j v.$$

Hence

$$I_{\mathcal{H}_9} = \sum_{j=1}^9 P_j.$$

Moreover,

$$Av = \sum_{j=1}^9 \lambda_j P_j v,$$

so

$$A = \sum_{j=1}^9 \lambda_j P_j.$$

□

Proposition 161.2. *For every polynomial*

$$p \in \mathbb{C}[X],$$

one has

$$p(A) = \sum_{j=1}^9 p(\lambda_j) P_j.$$

Proof. The assertion follows from

$$A^k = \sum_{j=1}^9 \lambda_j^k P_j$$

and linearity in the polynomial coefficients. □

Corollary 161.3. *For every bounded Borel function f ,*

$$f(A) = \sum_{j=1}^9 f(\lambda_j) P_j.$$

Proof. This is the finite-dimensional functional calculus associated with the spectral decomposition of A . □

162 State Update Under a Finite Measurement

Let

$$\rho \geq 0, \quad \text{tr } \rho = 1,$$

and let

$$P$$

be an orthogonal projection satisfying

$$P = P^* = P^2.$$

Assume

$$\text{tr}(\rho P) > 0.$$

Define the normalized post-projection state

$$\rho_P = \frac{P \rho P}{\text{tr}(\rho P)}.$$

Proposition 162.1. *The operator ρ_P is positive semidefinite and has unit trace.*

Proof. For every v ,

$$\langle P\rho Pv, v \rangle = \langle \rho Pv, Pv \rangle \geq 0.$$

Hence

$$P\rho P \geq 0,$$

and therefore

$$\rho_P \geq 0.$$

Moreover,

$$\mathrm{tr}(\rho_P) = \frac{\mathrm{tr}(P\rho P)}{\mathrm{tr}(\rho P)}.$$

By cyclicity of the trace and $P^2 = P$,

$$\mathrm{tr}(P\rho P) = \mathrm{tr}(\rho P^2) = \mathrm{tr}(\rho P).$$

Thus

$$\mathrm{tr}(\rho_P) = 1.$$

□

Theorem 162.2. *For the uniform state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

the normalized state associated with a rank-one projection P_j is

$$(\rho_9)_{P_j} = P_j.$$

Proof. One has

$$P_j \rho_9 P_j = \frac{1}{9} P_j^2 = \frac{1}{9} P_j$$

and

$$\mathrm{tr}(\rho_9 P_j) = \frac{1}{9}.$$

Consequently

$$(\rho_9)_{P_j} = \frac{(1/9)P_j}{1/9} = P_j.$$

□

163 Finite Measurement Algebra

Let

$$\{P_1, \dots, P_m\}$$

be a family of mutually orthogonal projections satisfying

$$P_i P_j = \delta_{ij} P_i$$

and

$$\sum_{i=1}^m P_i = I_{\mathcal{H}_9}.$$

Definition 163.1. The family

$$\mathcal{P} = \{P_1, \dots, P_m\}$$

is called a finite projective resolution of the identity.

For a state ρ , define

$$p_i = \text{tr}(\rho P_i).$$

Theorem 163.2. *The numbers*

$$p_1, \dots, p_m$$

define a probability distribution:

$$p_i \geq 0, \quad \sum_{i=1}^m p_i = 1.$$

Proof. Positivity follows from

$$\rho \geq 0, \quad P_i \geq 0.$$

Furthermore,

$$\sum_{i=1}^m p_i = \sum_{i=1}^m \text{tr}(\rho P_i) = \text{tr} \left(\rho \sum_{i=1}^m P_i \right) = \text{tr}(\rho) = 1.$$

□

Corollary 163.3. *For the uniform state on a nine-dimensional sector,*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$p_i = \frac{\text{rank}(P_i)}{9}.$$

Proof. Since

$$p_i = \frac{1}{9} \text{tr}(P_i)$$

and

$$\text{tr}(P_i) = \text{rank}(P_i),$$

the result follows. □

164 Compatibility of Arithmetic and Finite Spectral Observables

Let

$$q : \mathcal{L}_{\mathbb{P}} \longrightarrow \Gamma$$

be a finite arithmetic observable and

$$\Phi_F : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{H}_9$$

its realization in the stabilized sector.

Let

$$A \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be self-adjoint with spectral projections

$$\{P_\lambda\}_{\lambda \in \sigma(A)}.$$

Define the arithmetic spectral weight

$$\mu_F^\psi(\{\lambda\}) = \text{tr}(\rho_\psi P_\lambda),$$

where ρ_ψ is the density operator associated with the chosen realized state.

Theorem 164.1. *The spectral weights satisfy*

$$\sum_{\lambda \in \sigma(A)} \mu_F^\psi(\{\lambda\}) = 1.$$

Proof. The spectral projections resolve the identity:

$$\sum_{\lambda \in \sigma(A)} P_\lambda = I_{\mathcal{H}_g}.$$

Therefore

$$\sum_{\lambda \in \sigma(A)} \mu_F^\psi(\{\lambda\}) = \text{tr} \left(\rho_\psi \sum_{\lambda \in \sigma(A)} P_\lambda \right) = \text{tr}(\rho_\psi) = 1.$$

□

Remark 164.2. The formula establishes a normalized finite spectral measure once the state ρ_ψ and the observable A have been specified. The probability law is consequently derived from the pair

$$(\rho_\psi, A),$$

rather than postulated independently of the underlying operator algebra.

165 Necessary Conditions for a Born-Type Representation

Let

$$\mathcal{H}$$

be a finite-dimensional Hilbert space and let

$$\rho$$

be a density operator on $\mathcal{H}$.

Let

$$\{P_i\}_{i=1}^m$$

be a projective resolution of the identity.

Theorem 165.1. *The assignment*

$$\mathbb{P}_\rho(P_i) = \text{tr}(\rho P_i)$$

defines a finitely additive probability measure on the orthogonal projective decomposition generated by $\{P_i\}$.

Proof. Positivity and normalization have already been established. If

$$P_i P_j = 0 \quad (i \neq j),$$

then for any disjoint union of the corresponding outcomes,

$$P_I = \sum_{i \in I} P_i$$

is again a projection, and

$$\mathbb{P}_\rho(P_I) = \text{tr} \left(\rho \sum_{i \in I} P_i \right) = \sum_{i \in I} \text{tr}(\rho P_i).$$

Thus the assignment is additive on orthogonal projectors.

□

Proposition 165.2. *If*

$$\rho = \frac{1}{d} I_{\mathcal{H}},$$

then

$$\mathbb{P}_{\rho}(P) = \frac{\text{rank } P}{d}$$

for every orthogonal projection P .

Proof. One has

$$\mathbb{P}_{\rho}(P) = \frac{1}{d} \text{tr}(P) = \frac{\text{rank } P}{d}.$$

□

Remark 165.3. Thus the dimension-weighted probability law is not an additional axiom once the maximally mixed density operator has been established. It follows from the trace functional and the rank identity for orthogonal projections.

166 The Finite Trace Functional

Let

$$\mathcal{H}_9$$

be a complex Hilbert space of dimension 9, and let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Define the normalized trace

$$\tau_9 : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

by

$$\tau_9(X) = \frac{1}{9} \text{tr}(X).$$

Proposition 166.1. *The functional τ_9 satisfies*

$$\tau_9(I) = 1, \quad \tau_9(XY) = \tau_9(YX),$$

and

$$\tau_9(X^* X) \geq 0$$

for every $X, Y \in \mathcal{A}_9$.

Proof. Since

$$\text{tr}(I) = 9,$$

one has

$$\tau_9(I) = 1.$$

Cyclicity of the finite-dimensional trace gives

$$\tau_9(XY) = \frac{1}{9} \text{tr}(XY) = \frac{1}{9} \text{tr}(YX) = \tau_9(YX).$$

Finally,

$$X^* X \geq 0,$$

and the trace of a positive operator is nonnegative. Hence

$$\tau_9(X^* X) \geq 0.$$

□

Theorem 166.2. *The normalized trace defines a faithful tracial state on $\mathcal{A}_9$.*

Proof. It remains only to verify faithfulness. If

$$\tau_9(X^*X) = 0,$$

then

$$\text{tr}(X^*X) = 0.$$

The eigenvalues of X^*X are nonnegative, so their sum can vanish only if every eigenvalue is zero. Thus

$$X^*X = 0,$$

which implies

$$X = 0.$$

Therefore τ_9 is faithful. □

167 The State Associated with the Stabilized Sector

Let

$$P_9 = \Psi_9^4$$

and assume

$$\Psi_9^5 = \Psi_9^4.$$

Set

$$\mathcal{H}_9 = \text{range}(P_9)$$

and assume

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Define

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Theorem 167.1. *The operator ρ_9 is a faithful density operator on $\mathcal{H}_9$.*

Proof. Clearly

$$\rho_9 \geq 0.$$

Moreover,

$$\text{tr}(\rho_9) = \frac{1}{9} \text{tr}(I_{\mathcal{H}_9}) = \frac{1}{9} \cdot 9 = 1.$$

Since every eigenvalue of ρ_9 equals $1/9$, the operator is strictly positive and hence faithful. □

Proposition 167.2. *For every $X \in \mathcal{A}_9$,*

$$\text{tr}(\rho_9 X) = \tau_9(X).$$

Proof. Directly,

$$\text{tr}(\rho_9 X) = \text{tr}\left(\frac{1}{9} I_{\mathcal{H}_9} X\right) = \frac{1}{9} \text{tr}(X) = \tau_9(X).$$

□

Corollary 167.3. *The probability assigned to an orthogonal projection*

$$P \in \mathcal{A}_9$$

is

$$\mathbb{P}_9(P) = \text{tr}(\rho_9 P) = \frac{\text{rank } P}{9}.$$

168 The Orthomodular Projection Structure

Let

$$\mathcal{P}(\mathcal{H}_9)$$

denote the set of orthogonal projections on $\mathcal{H}_9$.

For

$$P, Q \in \mathcal{P}(\mathcal{H}_9),$$

write

$$P \leq Q$$

if

$$PQ = P.$$

Define the orthocomplement by

$$P^\perp = I_{\mathcal{H}_9} - P.$$

Proposition 168.1. *The relation $\leq$ is a partial order on $\mathcal{P}(\mathcal{H}_9)$.*

Proof. Reflexivity follows from

$$P^2 = P.$$

If

$$P \leq Q \quad \text{and} \quad Q \leq P,$$

then

$$PQ = P \quad \text{and} \quad QP = Q.$$

Taking adjoints gives

$$QP = P,$$

hence

$$P = Q.$$

Thus the relation is antisymmetric.

Finally, if

$$P \leq Q \quad \text{and} \quad Q \leq R,$$

then

$$PR = PQR = PQ = P,$$

so

$$P \leq R.$$

□

Theorem 168.2. *The projection lattice*

$$\mathcal{P}(\mathcal{H}_9)$$

is an orthomodular lattice.

Proof. The range of every projection is a closed subspace of $\mathcal{H}_9$, and orthogonal complementation corresponds to

$$P \mapsto I - P.$$

The lattice of closed subspaces of a finite-dimensional Hilbert space is orthomodular. Equivalently, if

$$P \leq Q,$$

then

$$Q = P + (Q - P)$$

as an orthogonal decomposition, and the corresponding projection identity is

$$Q = P + (I - P)Q.$$

Thus the orthomodular law holds. $\square$

169 Failure of Boolean Distributivity

The projection lattice is generally non-Boolean because orthogonal complementation does not induce a distributive lattice structure.

Theorem 169.1. *If*

$$\dim_{\mathbb{C}} \mathcal{H}_9 \geq 2,$$

then

$$\mathcal{P}(\mathcal{H}_9)$$

is not a distributive lattice.

Proof. Choose orthonormal vectors

$$e_1, e_2 \in \mathcal{H}_9$$

and define the rank-one projections

$$P = |e_1\rangle\langle e_1|,$$

$$Q = |u\rangle\langle u|, \quad u = \frac{e_1 + e_2}{\sqrt{2}}.$$

Then P and Q have distinct one-dimensional ranges, and hence

$$P \wedge Q = 0.$$

Moreover,

$$Q^\perp = |v\rangle\langle v|, \quad v = \frac{e_1 - e_2}{\sqrt{2}}.$$

The ranges of P and $Q^\perp$ are again distinct one-dimensional subspaces, so

$$P \wedge Q^\perp = 0.$$

Consequently,

$$P \vee (Q \wedge Q^\perp) = P \vee 0 = P.$$

On the other hand,

$$(P \vee Q) \wedge (P \vee Q^\perp) = I \wedge I = I,$$

because two distinct non-collinear one-dimensional subspaces in a two-dimensional subspace generate that entire two-dimensional subspace, and the corresponding common generated subspace contains the chosen two-dimensional sector. Hence the distributive identity

$$P \vee (Q \wedge Q^\perp) = (P \vee Q) \wedge (P \vee Q^\perp)$$

fails. $\square$

Remark 169.2. The non-Boolean character is therefore a structural consequence of noncommuting projections. It is not inferred from the existence of a probability measure; rather, the projection geometry is established first, and the associated trace state is subsequently evaluated on that geometry.

170 Commuting Projections and the Boolean Sector

Let

$$\mathcal{C} \subseteq \mathcal{P}(\mathcal{H}_9)$$

be a family of pairwise commuting projections.

Proposition 170.1. *The projections in $\mathcal{C}$ admit a simultaneous orthogonal decomposition of $\mathcal{H}_9$.*

Proof. A commuting family of self-adjoint operators on a finite-dimensional complex Hilbert space is simultaneously diagonalizable. Since every projection is self-adjoint, there exists an orthonormal basis in which every member of $\mathcal{C}$ is diagonal with diagonal entries in $\{0, 1\}$. $\square$

Theorem 170.2. *The lattice generated by a finite commuting family of projections is Boolean.*

Proof. In a common eigenbasis, every projection in the generated algebra is represented by a 0-1 valued diagonal function. Meet, join and orthocomplement therefore become, respectively, pointwise conjunction, disjunction and complementation. These operations form a Boolean algebra. $\square$

Corollary 170.3. *The obstruction to Booleanity of $\mathcal{P}(\mathcal{H}_9)$ is precisely the presence of noncommuting projections.*

171 Orthogonal Additivity of the Trace State

Let

$$P_1, \dots, P_m$$

be pairwise orthogonal projections:

$$P_i P_j = 0 \quad (i \neq j).$$

Theorem 171.1. *The normalized trace satisfies*

$$\tau_9 \left(\sum_{i=1}^m P_i \right) = \sum_{i=1}^m \tau_9(P_i).$$

Proof. Linearity of the trace gives

$$\tau_9 \left(\sum_{i=1}^m P_i \right) = \frac{1}{9} \operatorname{tr} \left(\sum_{i=1}^m P_i \right) = \frac{1}{9} \sum_{i=1}^m \operatorname{tr}(P_i) = \sum_{i=1}^m \tau_9(P_i).$$

$\square$

Proposition 171.2. *If*

$$\sum_{i=1}^m P_i = I_{\mathcal{H}_9},$$

then

$$\sum_{i=1}^m \tau_9(P_i) = 1.$$

Proof. Apply the preceding theorem and use

$$\tau_9(I_{\mathcal{H}_9}) = 1.$$

$\square$

Corollary 171.3. *For rank-one projections*

$$P_1, \dots, P_9$$

forming an orthogonal resolution of the identity,

$$\sum_{j=1}^9 \tau_9(P_j) = 1, \quad \tau_9(P_j) = \frac{1}{9}.$$

172 State-Independent Finite-Dimensional Representation

Let

$$\mathcal{P}(\mathcal{H}_9)$$

be the projection lattice and let

$$\mu : \mathcal{P}(\mathcal{H}_9) \longrightarrow [0, 1]$$

be a function satisfying

$$\mu(0) = 0, \quad \mu(I) = 1,$$

and

$$\mu(P + Q) = \mu(P) + \mu(Q)$$

whenever

$$PQ = 0.$$

Theorem 172.1. *The normalized trace*

$$\mu_9(P) = \frac{\text{rank } P}{9}$$

is an orthogonally additive probability measure on $\mathcal{P}(\mathcal{H}_9)$.

Proof. For every projection P ,

$$0 \leq \text{rank } P \leq 9,$$

hence

$$0 \leq \mu_9(P) \leq 1.$$

Moreover,

$$\mu_9(0) = 0, \quad \mu_9(I) = 1.$$

If P, Q are orthogonal, then

$$\text{rank}(P + Q) = \text{rank } P + \text{rank } Q,$$

so

$$\mu_9(P + Q) = \mu_9(P) + \mu_9(Q).$$

□

Remark 172.2. The result supplies an explicit finite-dimensional probability functional. No appeal to a limiting process, infinitesimal parameter, or asymptotic approximation is involved.

173 Interface with the Finite-State Operator

Let

$$\Psi_9$$

act on

$$V_9 = \mathbf{1}^\perp \subseteq \mathbb{C}^{144}$$

and assume

$$\Psi_9^4 = \Psi_9^5.$$

Set

$$P_9 = \Psi_9^4.$$

Theorem 173.1. *The stabilized operator P_9 is idempotent:*

$$P_9^2 = P_9.$$

Proof. Using the stabilization identity,

$$P_9^2 = \Psi_9^8 = \Psi_9^4 = P_9.$$

Indeed, from

$$\Psi_9^5 = \Psi_9^4$$

multiplication on the right by Ψ_9^k gives

$$\Psi_9^{k+5} = \Psi_9^{k+4}$$

for every $k \geq 0$, and repeated application with $k = 3, 2, 1, 0$ yields

$$\Psi_9^8 = \Psi_9^7 = \Psi_9^6 = \Psi_9^5 = \Psi_9^4.$$

□

Corollary 173.2. *The decomposition*

$$V_9 = \text{range}(P_9) \oplus \ker(P_9)$$

holds.

Proof. Every idempotent endomorphism has the direct-sum decomposition

$$V = \text{range}(P) \oplus \ker(P).$$

Indeed,

$$v = Pv + (I - P)v,$$

with

$$Pv \in \text{range}(P)$$

and

$$P(I - P)v = Pv - P^2v = 0.$$

If v belongs to both subspaces, then

$$v = Pv = 0.$$

□

174 The Fixed Sector of the Stabilized Operator

Let

$$P_9 = \Psi_9^4$$

with

$$\Psi_9^5 = \Psi_9^4.$$

Proposition 174.1. *One has*

$$\text{range}(P_9) = \ker(\Psi_9 - I).$$

Proof. If

$$v = P_9 u = \Psi_9^4 u,$$

then

$$\Psi_9 v = \Psi_9^5 u = \Psi_9^4 u = v.$$

Hence

$$\text{range}(P_9) \subseteq \ker(\Psi_9 - I).$$

Conversely, if

$$\Psi_9 v = v,$$

then

$$P_9 v = \Psi_9^4 v = v.$$

Thus

$$v \in \text{range}(P_9).$$

□

Corollary 174.2. *The stabilized sector is exactly the fixed-point sector:*

$$\mathcal{H}_9 = \text{range}(P_9) = \ker(\Psi_9 - I).$$

Theorem 174.3. *Every vector in the stabilized sector is invariant under all positive powers of Ψ_9 :*

$$v \in \mathcal{H}_9 \implies \Psi_9^k v = v \quad (k \geq 1).$$

Proof. By the preceding corollary,

$$\Psi_9 v = v.$$

Induction on k gives

$$\Psi_9^k v = v$$

for every $k \geq 1$.

□

175 The Stable Spectral Sector

Let

$$\mathcal{H}_9 = \text{range}(P_9) = \ker(\Psi_9 - I)$$

and assume

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

The restriction of Ψ_9 to $\mathcal{H}_9$ is therefore the identity operator.

Theorem 175.1. *The spectrum of the restricted operator*

$$\Psi_9|_{\mathcal{H}_9}$$

is

$$\sigma(\Psi_9|_{\mathcal{H}_9}) = \{1\}.$$

Proof. For every $v \in \mathcal{H}_9$,

$$\Psi_9 v = v.$$

Hence

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

The spectrum of the identity operator is the singleton

$$\{1\}.$$

□

Corollary 175.2. *Every nonzero vector in $\mathcal{H}_9$ is an eigenvector of Ψ_9 with eigenvalue 1, and the geometric and algebraic multiplicities of the eigenvalue 1 on $\mathcal{H}_9$ are both 9.*

Proof. The first assertion follows from

$$\Psi_9 v = v.$$

Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9$$

and the restriction is the identity, the eigenspace has dimension 9 and the characteristic polynomial is

$$(\lambda - 1)^9.$$

□

176 The Canonical Normalized Stable Operator

Let

$$\mathcal{H}_9$$

be a nine-dimensional complex Hilbert space. Define

$$B^* = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proposition 176.1. *The operator B^* is positive, self-adjoint, and satisfies*

$$\text{tr}(B^*) = 1.$$

Proof. Since $I_{\mathcal{H}_9}$ is self-adjoint and positive,

$$B^* = \frac{1}{9} I_{\mathcal{H}_9}$$

is self-adjoint and positive. Furthermore,

$$\text{tr}(B^*) = \frac{1}{9} \text{tr}(I_{\mathcal{H}_9}) = \frac{1}{9} \cdot 9 = 1.$$

□

Theorem 176.2. *The spectrum of B^* is*

$$\sigma(B^*) = \left\{ \frac{1}{9} \right\},$$

with algebraic multiplicity 9.

Proof. For every $v \in \mathcal{H}_9$,

$$B^* v = \frac{1}{9} v.$$

Thus every nonzero vector is an eigenvector with eigenvalue $1/9$. Since the dimension is 9, the characteristic polynomial is

$$\left(\lambda - \frac{1}{9} \right)^9.$$

□

Corollary 176.3. *The normalized trace of B^* is*

$$\tau_9(B^*) = \frac{1}{9}.$$

Proof. By definition,

$$\tau_9(B^*) = \frac{1}{9} \text{tr}(B^*) = \frac{1}{9}.$$

□

Remark 176.4. The value $1/9$ is fixed here by the dimension of the stable sector and the trace normalization. Consequently, the formula

$$B^* = \frac{1}{9} I_{\mathcal{H}_9}$$

is a normalized finite-dimensional state operator once the nine-dimensional stable sector has been established. It is not obtained from an asymptotic limiting argument.

177 Uniqueness of the Maximally Symmetric Stable State

Let

$$\mathcal{H}_9$$

be a nine-dimensional complex Hilbert space and let

$$\rho \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be a density operator.

Suppose that

$$U \rho U^* = \rho$$

for every unitary operator

$$U \in \mathcal{U}(\mathcal{H}_9).$$

Theorem 177.1. *Under the preceding invariance condition,*

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. By the spectral theorem,

$$\rho = \sum_{j=1}^9 \lambda_j P_j$$

for an orthogonal decomposition into eigenspaces.

The unitary group acts transitively on the unit sphere of $\mathcal{H}_9$. Invariance under every unitary therefore implies that all one-dimensional subspaces are spectrally equivalent. Hence all eigenvalues of ρ coincide:

$$\lambda_1 = \dots = \lambda_9 = \lambda.$$

Thus

$$\rho = \lambda I_{\mathcal{H}_9}.$$

The normalization condition gives

$$1 = \text{tr}(\rho) = 9\lambda,$$

and consequently

$$\lambda = \frac{1}{9}.$$

Therefore

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Corollary 177.2. *The operator*

$$B^* = \frac{1}{9} I_{\mathcal{H}_9}$$

is the unique density operator invariant under the full unitary group of the stable sector.

178 Exact Stabilization of the State Operator

Let

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

and suppose that

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

Define the induced conjugation action on operators by

$$\text{Ad}_{\Psi_9}(X) = \Psi_9 X \Psi_9^*.$$

Proposition 178.1. *The maximally symmetric state is invariant under every unitary operator on the stable sector:*

$$U \rho_9 U^* = \rho_9.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$U \rho_9 U^* = \frac{1}{9} U I_{\mathcal{H}_9} U^* = \frac{1}{9} U U^* = \frac{1}{9} I_{\mathcal{H}_9} = \rho_9.$$

□

Theorem 178.2. *If the restriction of Ψ_9 to $\mathcal{H}_9$ is unitary, then*

$$\text{Ad}_{\Psi_9}(\rho_9) = \rho_9.$$

Moreover,

$$\text{Ad}_{\Psi_9}^k(\rho_9) = \rho_9 \quad (k \geq 1).$$

Proof. The first assertion follows from the preceding proposition with

$$U = \Psi_9|_{\mathcal{H}_9}.$$

Iteration gives

$$\text{Ad}_{\Psi_9}^k(\rho_9) = \rho_9$$

for every $k \geq 1$. □

179 Finite-Time Stabilization and Absence of Asymptotic Error

Let

$$T$$

be an endomorphism of a finite-dimensional vector space V , and suppose that for some integer

$$m \geq 1$$

one has

$$T^{m+1} = T^m.$$

Set

$$P = T^m.$$

Theorem 179.1. *For every integer $k \geq 0$,*

$$T^{m+k} = P.$$

In particular,

$$T^{m+k} = T^m \quad (k \geq 0).$$

Proof. The case $k = 0$ is the definition of P . For $k = 1$, the assertion is the hypothesis

$$T^{m+1} = T^m.$$

Suppose

$$T^{m+k} = T^m.$$

Multiplying by T gives

$$T^{m+k+1} = T^{m+1} = T^m.$$

Induction completes the proof. □

Corollary 179.2. *The sequence*

$$T^m, T^{m+1}, T^{m+2}, \dots$$

is exactly constant.

Proof. This follows directly from

$$T^{m+k} = T^m$$

for every $k \geq 0$. □

Corollary 179.3. *For every operator norm on the finite-dimensional endomorphism algebra,*

$$\|T^{m+k} - P\| = 0 \quad (k \geq 0).$$

Proof. The operator difference vanishes identically:

$$T^{m+k} - P = 0.$$

□

Remark 179.4. The stabilization statement is therefore stronger than convergence. There is no nonzero remainder whose magnitude must be estimated and no limit whose existence must be inferred from an asymptotic argument. The stabilized operator is reached exactly after finitely many iterations.

180 Minimal Polynomial Consequences of Exact Stabilization

Let

$$T \in \text{End}_{\mathbb{C}}(V)$$

satisfy

$$T^{m+1} = T^m.$$

Theorem 180.1. *The polynomial*

$$p(X) = X^m(X - 1)$$

annihilates T :

$$p(T) = 0.$$

Proof. By substitution,

$$p(T) = T^m(T - I) = T^{m+1} - T^m = 0.$$

□

Proposition 180.2. *Every eigenvalue λ of T satisfies*

$$\lambda^m(\lambda - 1) = 0.$$

Consequently,

$$\sigma(T) \subseteq \{0, 1\}.$$

Proof. If

$$Tv = \lambda v$$

for some nonzero v , then

$$0 = p(T)v = p(\lambda)v.$$

Hence

$$p(\lambda) = 0,$$

so

$$\lambda^m(\lambda - 1) = 0.$$

Therefore

$$\lambda \in \{0, 1\}.$$

□

Theorem 180.3. *The operator T is diagonalizable over $\mathbb{C}$.*

Proof. The annihilating polynomial

$$X^m(X-1)$$

has distinct irreducible factors

$$X \quad \text{and} \quad X-1.$$

Therefore the minimal polynomial of T , being a divisor of $X^m(X-1)$, is square-free. Hence T is diagonalizable. $\square$

Corollary 180.4. *There is a direct-sum decomposition*

$$V = \ker T \oplus \ker(T-I),$$

and

$$T|_{\ker T} = 0, \quad T|_{\ker(T-I)} = I.$$

181 Identification of the Stabilization Projection

Under the hypotheses of the preceding section, let

$$P = T^m.$$

Theorem 181.1. *The operator P is the spectral projection associated with the eigenvalue 1. In particular,*

$$\text{range}(P) = \ker(T-I)$$

and

$$\ker(P) = \ker T.$$

Proof. The first equality was established directly from

$$T^{m+1} = T^m.$$

If

$$v \in \ker T,$$

then

$$Pv = T^m v = 0,$$

so

$$\ker T \subseteq \ker P.$$

Conversely, if

$$Pv = 0,$$

then

$$Tv = T(I-P)v + TPv.$$

Since

$$TP = T^{m+1} = T^m = P,$$

one has

$$T(I-P) = T - TP = T - P.$$

Using

$$Pv = 0$$

and the decomposition

$$v = Pv + (I - P)v = (I - P)v,$$

we obtain

$$Tv = (T - P)v.$$

The polynomial decomposition associated with the two spectral values 0 and 1 then gives

$$v \in \ker T.$$

Hence

$$\ker P = \ker T.$$

Thus P acts as 0 on the 0-eigenspace and as 1 on the 1-eigenspace, so it is precisely the spectral projection onto $\ker(T - I)$. $\square$

Corollary 181.2. *For the finite-state operator,*

$$P_9 = \Psi_9^4$$

is the spectral projection onto

$$\mathcal{H}_9 = \ker(\Psi_9 - I),$$

provided

$$\Psi_9^5 = \Psi_9^4.$$

182 Exact Finite-State Functional Calculus

Let

$$T \in \text{End}_{\mathbb{C}}(V)$$

satisfy

$$T^{m+1} = T^m,$$

and let

$$P = T^m.$$

Theorem 182.1. *For every polynomial*

$$f \in \mathbb{C}[X],$$

one has

$$f(T) = f(0)(I - P) + f(1)P.$$

Proof. By the preceding decomposition,

$$V = \ker T \oplus \ker(T - I).$$

On $\ker T$,

$$T = 0,$$

and therefore

$$f(T) = f(0)I.$$

On $\ker(T - I)$,

$$T = I,$$

and therefore

$$f(T) = f(1)I.$$

Since

$$I - P$$

is the projection onto $\ker T$ and P is the projection onto $\ker(T - I)$, the two restrictions combine to give

$$f(T) = f(0)(I - P) + f(1)P.$$

□

Corollary 182.2. *For every integer $k \geq 1$,*

$$T^k = P.$$

Proof. Apply the theorem to

$$f(X) = X^k.$$

Since

$$f(0) = 0, \quad f(1) = 1,$$

one obtains

$$T^k = P.$$

□

Corollary 182.3. *Every polynomial observable of the stabilized operator is determined exactly by its two spectral values at 0 and 1.*

183 Specialization to the Four-Step Constraint

Assume

$$\Psi_9^5 = \Psi_9^4$$

and define

$$P_9 = \Psi_9^4.$$

Theorem 183.1. *For every integer*

$$k \geq 4,$$

one has

$$\Psi_9^k = P_9.$$

Proof. Apply the finite stabilization theorem with

$$T = \Psi_9, \quad m = 4.$$

□

Corollary 183.2. *The complete finite-time evolution after the fourth step is*

$$\Psi_9^4 = \Psi_9^5 = \Psi_9^6 = \dots = P_9.$$

Corollary 183.3. *The characteristic polynomial of Ψ_9 divides*

$$X^4(X - 1),$$

and hence

$$\sigma(\Psi_9) \subseteq \{0, 1\}.$$

Proof. The stabilization identity gives

$$\Psi_9^4(\Psi_9 - I) = 0.$$

Thus

$$X^4(X - 1)$$

annihilates Ψ_9 . The spectral conclusion follows from the polynomial spectral mapping argument established above. $\square$

184 The Four-Step Projection and Its Rank

Let

$$P_9 = \Psi_9^4$$

and assume

$$\Psi_9^5 = \Psi_9^4.$$

Let

$$\mathcal{H}_9 = \text{range}(P_9).$$

Proposition 184.1. *The rank of P_9 is equal to the dimension of the stabilized sector:*

$$\text{rank}(P_9) = \dim_{\mathbb{C}} \mathcal{H}_9.$$

Proof. By definition,

$$\mathcal{H}_9 = \text{range}(P_9).$$

The rank of a linear operator is the dimension of its range. $\square$

Theorem 184.2. *If*

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

then

$$\text{rank}(P_9) = 9.$$

Proof. Immediate from the preceding proposition. $\square$

Corollary 184.3. *If*

$$V_9 = \mathbf{1}^\perp \subseteq \mathbb{C}^{144}$$

has dimension 143, then

$$\dim_{\mathbb{C}} \ker(P_9) = 134.$$

Proof. By rank-nullity,

$$\dim V_9 = \text{rank}(P_9) + \dim \ker(P_9).$$

Since

$$143 = 9 + \dim \ker(P_9),$$

the assertion follows. $\square$

185 Trace Characterization of the Stable Projection

Let

$$P_9 = \Psi_9^4$$

be idempotent and suppose

$$\text{rank}(P_9) = 9.$$

Proposition 185.1. *The trace of P_9 is*

$$\text{tr}(P_9) = 9.$$

Proof. An idempotent operator has spectrum contained in

$$\{0, 1\}.$$

Its trace is therefore the sum of the eigenvalues counted with algebraic multiplicity. Since the multiplicity of the eigenvalue 1 is the rank of the idempotent,

$$\text{tr}(P_9) = \text{rank}(P_9) = 9.$$

□

Corollary 185.2. *The normalized trace of the stabilized projection in $\text{End}(V_9)$, with $\dim V_9 = 143$, is*

$$\frac{\text{tr}(P_9)}{143} = \frac{9}{143}.$$

Remark 185.3. The normalization of a state on the stabilized sector and the normalization of a trace on the ambient 143-dimensional space are distinct constructions. The former gives

$$\frac{1}{9} I_{\mathcal{H}_9},$$

whereas the latter assigns the projection P_9 the normalized trace $9/143$. No identification between these two normalizations is made.

186 The Compression to the Stable Sector

Let

$$V_9 = \mathbf{1}^\perp \subseteq \mathbb{C}^{144}$$

and let

$$P_9 : V_9 \longrightarrow V_9$$

be the idempotent stabilized operator.

Define

$$\mathcal{H}_9 = \text{range}(P_9).$$

For every

$$X \in \text{End}_{\mathbb{C}}(V_9),$$

define its compression to the stable sector by

$$X_9 = P_9 X P_9|_{\mathcal{H}_9}.$$

Proposition 186.1. *The compressed operator X_9 is an endomorphism of $\mathcal{H}_9$.*

Proof. If

$$v \in \mathcal{H}_9,$$

then

$$P_9 v = v.$$

Hence

$$X_9 v = P_9 X P_9 v = P_9 X v,$$

which belongs to

$$\text{range}(P_9) = \mathcal{H}_9.$$

□

Proposition 186.2. *The compression map*

$$X \longmapsto X_9$$

is linear.

Proof. For

$$X, Y \in \text{End}_{\mathbb{C}}(V_9)$$

and

$$a, b \in \mathbb{C},$$

one has

$$P_9(aX + bY)P_9 = aP_9XP_9 + bP_9YP_9.$$

□

Theorem 186.3. *If*

$$X = P_9XP_9,$$

then X preserves the stable sector and

$$X|_{\mathcal{H}_9} = X_9.$$

Proof. For

$$v \in \mathcal{H}_9,$$

one has

$$P_9 v = v,$$

and therefore

$$Xv = P_9XP_9v = P_9Xv \in \mathcal{H}_9.$$

The equality with the compression follows directly from the definition.

□

187 The Stable Algebra

Define

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Proposition 187.1. *$\mathcal{A}_9$ is a unital finite-dimensional $\mathbb{C}$ -algebra of dimension*

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 81.$$

Proof. For a complex vector space of dimension 9, its endomorphism algebra is represented, after choosing a basis, by the algebra

$$M_9(\mathbb{C}).$$

Therefore

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 9^2 = 81.$$

□

Theorem 187.2. *The normalized trace*

$$\tau_9(X) = \frac{1}{9} \operatorname{tr}(X)$$

is the unique normalized trace on $\mathcal{A}_9$ satisfying

$$\tau_9(I) = 1$$

and

$$\tau_9(XY) = \tau_9(YX)$$

for all

$$X, Y \in \mathcal{A}_9.$$

Proof. Every trace functional on the full matrix algebra $M_9(\mathbb{C})$ is a scalar multiple of the ordinary matrix trace. The normalization

$$\tau_9(I) = 1$$

forces the scalar to be $1/9$.

□

Corollary 187.3. *The pair*

$$(\mathcal{A}_9, \tau_9)$$

provides a canonical finite-dimensional tracial algebra associated with the nine-dimensional stable sector.

188 Spectral Resolution of Observables

Let

$$A = A^* \in \mathcal{A}_9.$$

Theorem 188.1. *There exist distinct real numbers*

$$\lambda_1, \dots, \lambda_r$$

and pairwise orthogonal projections

$$E_1, \dots, E_r \in \mathcal{P}(\mathcal{H}_9)$$

such that

$$A = \sum_{j=1}^r \lambda_j E_j,$$

$$E_i E_j = 0 \quad (i \neq j),$$

and

$$\sum_{j=1}^r E_j = I_{\mathcal{H}_9}.$$

Proof. This is the finite-dimensional spectral theorem for self-adjoint operators. $\square$

Proposition 188.2. *The state induced by ρ_9 assigns to the spectral projection E_j the probability*

$$\mu_9(E_j) = \frac{\text{rank}(E_j)}{9}.$$

Proof. By definition,

$$\mu_9(E_j) = \text{tr}(\rho_9 E_j) = \frac{1}{9} \text{tr}(E_j).$$

For a projection,

$$\text{tr}(E_j) = \text{rank}(E_j).$$

Hence

$$\mu_9(E_j) = \frac{\text{rank}(E_j)}{9}.$$

$\square$

Corollary 188.3. *The expectation of A in the state ρ_9 is*

$$\langle A \rangle_{\rho_9} = \text{tr}(\rho_9 A) = \frac{1}{9} \sum_{j=1}^r \lambda_j \text{rank}(E_j).$$

Proof. Using the spectral resolution,

$$\text{tr}(\rho_9 A) = \sum_{j=1}^r \lambda_j \text{tr}(\rho_9 E_j),$$

and the preceding proposition gives the result. $\square$

189 Finite Orthogonal Resolution and Uniform Weights

Let

$$E_1, \dots, E_9$$

be rank-one mutually orthogonal projections satisfying

$$\sum_{j=1}^9 E_j = I_{\mathcal{H}_9}.$$

Theorem 189.1. *The state ρ_9 assigns equal weight to every member of the resolution:*

$$\text{tr}(\rho_9 E_j) = \frac{1}{9} \quad (1 \leq j \leq 9).$$

Proof. Since E_j has rank one,

$$\text{tr}(E_j) = 1.$$

Therefore

$$\text{tr}(\rho_9 E_j) = \frac{1}{9} \text{tr}(E_j) = \frac{1}{9}.$$

$\square$

Corollary 189.2. *For every orthonormal basis*

$$\{e_1, \dots, e_9\}$$

of $\mathcal{H}_9$, the associated rank-one projections

$$E_j = |e_j\rangle\langle e_j|$$

satisfy

$$\sum_{j=1}^9 \text{tr}(\rho_9 E_j) = 1.$$

Proof. Each term equals 1/9, hence

$$\sum_{j=1}^9 \frac{1}{9} = 1.$$

□

190 The Arithmetic Character Representation

Return now to the arithmetic lattice

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

and its logarithmic homomorphism

$$\Lambda_{\log}(x) = \sum_{p \in \mathbb{P}} n_p \log p.$$

For

$$t \in \mathbb{R},$$

let

$$\chi_t(x) = e^{it\Lambda_{\log}(x)}.$$

Proposition 190.1. *For every $x \in \mathcal{L}_{\mathbb{P}}$,*

$$\chi_t(x) = \mathfrak{a}(x)^{it}.$$

Proof. Since

$$\Lambda_{\log}(x) = \log \mathfrak{a}(x),$$

one obtains

$$\chi_t(x) = e^{it \log \mathfrak{a}(x)} = \mathfrak{a}(x)^{it}.$$

□

Theorem 190.2. *The character representation separates points of $\mathcal{L}_{\mathbb{P}}$:*

$$x \neq y \implies \exists t \in \mathbb{R} \quad \chi_t(x) \neq \chi_t(y).$$

Proof. If

$$x \neq y,$$

then

$$z = x - y \neq 0.$$

Injectivity of $\Lambda_{\log}$ gives

$$\Lambda_{\log}(z) \neq 0.$$

Choose

$$t = \frac{\pi}{\Lambda_{\log}(z)}.$$

Then

$$\chi_t(z) = e^{i\pi} = -1.$$

Since

$$\chi_t(z) = \frac{\chi_t(x)}{\chi_t(y)},$$

it follows that

$$\chi_t(x) \neq \chi_t(y).$$

□

191 The Exact Arithmetic–Spectral Interface

Let

$$\mathcal{L}_{\mathbb{P}}$$

be the arithmetic lattice and let

$$\chi_t : \mathcal{L}_{\mathbb{P}} \rightarrow \mathbb{C}^{\times}$$

be the character family.

Let

$$\pi : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be a unitary representation satisfying

$$\pi(x + y) = \pi(x)\pi(y).$$

Proposition 191.1. *For every $t \in \mathbb{R}$, the scalar character χ_t defines a one-dimensional unitary representation of $\mathcal{L}_{\mathbb{P}}$.*

Proof. The character identities

$$\chi_t(x + y) = \chi_t(x)\chi_t(y)$$

and

$$|\chi_t(x)| = 1$$

show precisely that χ_t is a homomorphism into the unitary group

$$U(1) \subset \mathbb{C}^{\times}.$$

□

Theorem 191.2. *Every finite-dimensional unitary representation*

$$\pi : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

decomposes into simultaneous character sectors.

Proof. The group $\mathcal{L}_{\mathbb{P}}$ is abelian, so the operators $\pi(x)$ commute pairwise. Each $\pi(x)$ is unitary and hence normal. A commuting family of normal operators on a finite-dimensional complex Hilbert space is simultaneously diagonalizable. Therefore $\mathcal{H}_9$ decomposes as an orthogonal direct sum of common eigenspaces, on each of which π acts by a scalar character. □

Corollary 191.3. *The arithmetic lattice acts on the finite stable sector through a finite collection of unitary characters.*

Remark 191.4. This statement supplies the precise finite-dimensional interface between the arithmetic lattice and spectral theory. It does not assert that a particular representation π is generated by the logarithmic characters χ_t ; that identification requires an additional construction specifying the representation.

192 The Arithmetic Character Measure

Let

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

be the arithmetic lattice, and let

$$\alpha : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{Q}_{>0}$$

be the multiplicative arithmetic map determined by

$$\alpha \left(\sum_p n_p e_p \right) = \prod_p p^{n_p}.$$

Let

$$\Lambda_{\log} = \log \circ \alpha.$$

For $t \in \mathbb{R}$, define

$$\chi_t(x) = e^{it\Lambda_{\log}(x)}.$$

The preceding character-separation result implies that the family $\{\chi_t\}_{t \in \mathbb{R}}$ separates the points of $\mathcal{L}_{\mathbb{P}}$.

For a finite subset

$$F = \{x_1, \dots, x_r\} \subset \mathcal{L}_{\mathbb{P}},$$

define the associated character Gram matrix

$$G_F(t_1, \dots, t_s) = \left(\chi_{t_\alpha}(x_i) \overline{\chi_{t_\alpha}(x_j)} \right)_{i,j}.$$

Proposition 192.1. *For every finite F and every finite collection of parameters $t_1, \dots, t_s$, the matrix $G_F(t_1, \dots, t_s)$ is positive semidefinite.*

Proof. For each α , define

$$v_\alpha = \left(\chi_{t_\alpha}(x_1), \dots, \chi_{t_\alpha}(x_r) \right)^\top.$$

Then

$$G_F(t_1, \dots, t_s) = \sum_{\alpha=1}^s v_\alpha v_\alpha^*.$$

Each summand is positive semidefinite, and therefore so is their sum. $\square$

Theorem 192.2. *The character construction defines a positive semidefinite kernel on $\mathcal{L}_{\mathbb{P}}$:*

$$K(x, y) = \int_{\mathbb{R}} \chi_t(x) \overline{\chi_t(y)} d\nu(t)$$

for every finite positive measure ν on $\mathbb{R}$.

Proof. For any finite family $x_1, \dots, x_r$ and complex coefficients $c_1, \dots, c_r$,

$$\sum_{i,j=1}^r \overline{c_i} c_j K(x_i, x_j) = \int_{\mathbb{R}} \left| \sum_{j=1}^r c_j \chi_t(x_j) \right|^2 d\nu(t) \geq 0.$$

Hence K is positive semidefinite. □

Corollary 192.3. *The kernel depends only on the arithmetic difference:*

$$K(x, y) = \kappa(x - y),$$

where

$$\kappa(z) = \int_{\mathbb{R}} e^{it\Lambda_{\log}(z)} d\nu(t).$$

Proof. Since

$$\chi_t(x) \overline{\chi_t(y)} = \chi_t(x - y) = e^{it\Lambda_{\log}(x-y)},$$

the assertion follows. □

193 Finite Relational Forms

Let

$$F = \{x_1, \dots, x_r\} \subset \mathcal{L}_{\mathbb{P}}$$

be finite, and let

$$K_F = (K(x_i, x_j))_{i,j=1}^r$$

be the corresponding Gram matrix.

Proposition 193.1. *The matrix K_F is Hermitian positive semidefinite.*

Proof. Hermiticity follows from

$$K(y, x) = \overline{K(x, y)}.$$

Positive semidefiniteness follows from the preceding theorem. □

Let

$$\mathcal{N}_F = \ker K_F.$$

Define the quotient space

$$\mathcal{Q}_F = \mathbb{C}^r / \mathcal{N}_F$$

and equip it with the sesquilinear form

$$\langle [u], [v] \rangle_F = u^* K_F v.$$

Theorem 193.2. *The form*

$$\langle [u], [v] \rangle_F = u^* K_F v$$

is a well-defined inner product on $\mathcal{Q}_F$.

Proof. If

$$u' = u + n, \quad n \in \mathcal{N}_F,$$

then

$$u'^* K_F v = u^* K_F v + n^* K_F v = u^* K_F v,$$

because

$$K_F n = 0$$

and K_F is Hermitian. The same argument applies to the second variable.

Moreover,

$$\langle [u], [u] \rangle_F = u^* K_F u \geq 0.$$

If this quantity vanishes, positive semidefiniteness implies

$$K_F u = 0,$$

so

$$u \in \mathcal{N}_F.$$

Thus

$$[u] = 0.$$

The form is therefore positive definite on the quotient. $\square$

Corollary 193.3. *Every finite relational kernel K_F determines canonically a finite Hilbert space*

$$\mathcal{Q}_F.$$

194 Relational Orthogonality

Let

$$\mathcal{Q}_F$$

be the Hilbert space constructed above.

For classes

$$[u], [v] \in \mathcal{Q}_F,$$

define

$$[u] \perp_F [v] \iff \langle [u], [v] \rangle_F = 0.$$

Proposition 194.1. *The relation $\perp_F$ is symmetric:*

$$[u] \perp_F [v] \implies [v] \perp_F [u].$$

Proof. Since the inner product is Hermitian,

$$\langle [v], [u] \rangle_F = \overline{\langle [u], [v] \rangle_F}.$$

Hence vanishing in one order is equivalent to vanishing in the other. $\square$

Proposition 194.2. *For every*

$$[u] \in \mathcal{Q}_F,$$

one has

$$[u] \perp_F [u] \iff [u] = 0.$$

Proof. Positive definiteness gives

$$\langle [u], [u] \rangle_F = 0$$

if and only if

$$[u] = 0.$$

□

Theorem 194.3. *For every finite family of pairwise orthogonal nonzero vectors*

$$[u_1], \dots, [u_r] \in \mathcal{Q}_F,$$

the vectors are linearly independent.

Proof. Suppose

$$\sum_{j=1}^r c_j [u_j] = 0.$$

Taking the inner product with $[u_k]$ gives

$$c_k \langle [u_k], [u_k] \rangle_F = 0,$$

because the vectors are pairwise orthogonal. Since each vector is nonzero,

$$\langle [u_k], [u_k] \rangle_F > 0,$$

and therefore

$$c_k = 0.$$

This holds for every k .

□

195 The Relational Measure

Let

$$\mathcal{Q}_F$$

be a finite-dimensional relational Hilbert space and let

$$\mathcal{P}(\mathcal{Q}_F)$$

denote its lattice of orthogonal projections.

Fix a positive trace-class operator

$$\rho_F \geq 0$$

with

$$\text{tr}(\rho_F) = 1.$$

Define

$$\mu_F(E) = \text{tr}(\rho_F E), \quad E \in \mathcal{P}(\mathcal{Q}_F).$$

Theorem 195.1. *The map*

$$\mu_F : \mathcal{P}(\mathcal{Q}_F) \longrightarrow [0, 1]$$

is a finitely additive probability measure on the projection lattice.

Proof. Positivity follows from

$$\rho_F \geq 0, \quad E \geq 0.$$

Thus

$$\mu_F(E) \geq 0.$$

Since

$$0 \leq E \leq I,$$

one has

$$0 \leq \mu_F(E) \leq \text{tr}(\rho_F) = 1.$$

Moreover,

$$\mu_F(I) = \text{tr}(\rho_F) = 1.$$

Let

$$E_1, \dots, E_r$$

be pairwise orthogonal projections. Then

$$E_i E_j = 0 \quad (i \neq j),$$

and

$$E = \sum_{j=1}^r E_j$$

is again a projection. By linearity of the trace,

$$\mu_F(E) = \text{tr} \left(\rho_F \sum_{j=1}^r E_j \right) = \sum_{j=1}^r \text{tr}(\rho_F E_j) = \sum_{j=1}^r \mu_F(E_j).$$

Hence μ_F is finitely additive on mutually orthogonal projections. □

Corollary 195.2. *For the maximally symmetric state on the nine-dimensional stable sector,*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

the induced measure is

$$\mu_9(E) = \frac{1}{9} \text{rank}(E)$$

for every projection

$$E \in \mathcal{P}(\mathcal{H}_9).$$

Proof. Substitution into the definition gives

$$\mu_9(E) = \text{tr} \left(\frac{1}{9} I E \right) = \frac{1}{9} \text{tr}(E).$$

For a projection,

$$\text{tr}(E) = \text{rank}(E).$$

□

196 Additivity on Orthogonal Resolutions

Let

$$E_1, \dots, E_r \in \mathcal{P}(\mathcal{H}_9)$$

be pairwise orthogonal projections satisfying

$$\sum_{j=1}^r E_j = I_{\mathcal{H}_9}.$$

Theorem 196.1. *The relational measure satisfies*

$$\sum_{j=1}^r \mu_9(E_j) = 1.$$

Proof. Finite additivity gives

$$\sum_{j=1}^r \mu_9(E_j) = \mu_9\left(\sum_{j=1}^r E_j\right) = \mu_9(I_{\mathcal{H}_9}) = 1.$$

□

If every E_j has rank one, then

$$r = 9$$

and

$$\mu_9(E_j) = \frac{1}{9}.$$

Thus the uniform value on rank-one resolutions is a consequence of the dimension and the normalized state, rather than an independent probability postulate.

197 Booleanity and Orthomodularity

Let

$$\mathcal{P}(\mathcal{H}_9)$$

denote the orthogonal projection lattice of the stable Hilbert space.

For

$$E, F \in \mathcal{P}(\mathcal{H}_9),$$

write

$$E \leq F$$

when

$$\text{range}(E) \subseteq \text{range}(F).$$

Proposition 197.1. *The relation $\leq$ defines the usual partial order on $\mathcal{P}(\mathcal{H}_9)$, and*

$$E \wedge F$$

is the orthogonal projection onto

$$\text{range}(E) \cap \text{range}(F).$$

Proof. These are standard properties of the projection lattice of a Hilbert space. □

Theorem 197.2. *The projection lattice*

$$\mathcal{P}(\mathcal{H}_9)$$

is orthomodular.

Proof. If

$$E \leq F,$$

then

$$\text{range}(F) = \text{range}(E) \oplus (\text{range}(F) \cap \text{range}(E)^\perp).$$

The corresponding orthogonal projections therefore satisfy

$$F = E + (F \wedge E^\perp),$$

which is the orthomodular identity. □

Theorem 197.3. *The lattice*

$$\mathcal{P}(\mathcal{H}_9)$$

is not Boolean.

Proof. Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9 \geq 2,$$

choose linearly independent nonorthogonal unit vectors

$$u, v \in \mathcal{H}_9.$$

Let

$$E = |u\rangle\langle u|, \quad F = |v\rangle\langle v|.$$

Then

$$E \neq F, \quad E \not\leq F, \quad F \not\leq E,$$

and

$$EF \neq FE.$$

In a Boolean algebra the meet and join operations are distributive and all elements commute under the corresponding projection representation. The noncommutativity above therefore excludes a Boolean realization of $\mathcal{P}(\mathcal{H}_9)$. □

Corollary 197.4. *The relational probability structure determined by*

$$(\mathcal{P}(\mathcal{H}_9), \mu_9)$$

is non-Boolean.

198 The Finite Relational Lattice

Let

$$\mathcal{Q}_F$$

be the finite relational Hilbert space associated with the positive semidefinite kernel K_F , and let

$$\mathcal{P}(\mathcal{Q}_F)$$

denote its projection lattice.

For a subspace

$$W \subseteq \mathcal{Q}_F,$$

write

$$P_W$$

for the orthogonal projection onto W .

Define the relational lattice

$$\mathcal{L}_F = \mathcal{P}(\mathcal{Q}_F)$$

with order

$$E \leq F \iff EF = E.$$

Proposition 198.1. *The order relation on $\mathcal{L}_F$ is equivalent to inclusion of ranges:*

$$E \leq F \iff \text{range}(E) \subseteq \text{range}(F).$$

Proof. If $EF = E$, then for every v ,

$$Ev = EFv,$$

so Ev belongs to $\text{range}(F)$. Hence

$$\text{range}(E) \subseteq \text{range}(F).$$

Conversely, suppose

$$\text{range}(E) \subseteq \text{range}(F).$$

Since F acts as the identity on its range,

$$FE = E.$$

Taking adjoints and using

$$E^* = E, \quad F^* = F,$$

gives

$$EF = E.$$

□

Theorem 198.2. *The lattice operations on $\mathcal{L}_F$ are determined by the corresponding subspace operations:*

$$E \wedge F = P_{\text{range}(E) \cap \text{range}(F)}$$

and

$$E^\perp = I - E.$$

The join is

$$E \vee F = P_{\text{range}(E) + \text{range}(F)}.$$

Proof. The orthogonal complement of $\text{range}(E)$ is the kernel of E , and the corresponding orthogonal projection is $I - E$. The meet is the projection onto the intersection of the two ranges, while the join is the projection onto their linear span. $\square$

Corollary 198.3. *For every*

$$E \in \mathcal{L}_F,$$

one has

$$E \wedge E^\perp = 0, \quad E \vee E^\perp = I.$$

199 Orthogonal Additivity of the Relational Measure

Let

$$\mu_F(E) = \text{tr}(\rho_F E), \quad E \in \mathcal{L}_F,$$

where

$$\rho_F \geq 0, \quad \text{tr}(\rho_F) = 1.$$

Theorem 199.1. *If*

$$E_1, \dots, E_r \in \mathcal{L}_F$$

are pairwise orthogonal, then

$$\mu_F\left(\bigvee_{j=1}^r E_j\right) = \sum_{j=1}^r \mu_F(E_j).$$

Proof. Pairwise orthogonality gives

$$E_i E_j = 0 \quad (i \neq j),$$

and consequently

$$\bigvee_{j=1}^r E_j = \sum_{j=1}^r E_j.$$

Linearity of the trace yields

$$\mu_F\left(\bigvee_{j=1}^r E_j\right) = \text{tr}\left(\rho_F \sum_{j=1}^r E_j\right) = \sum_{j=1}^r \text{tr}(\rho_F E_j).$$

Hence

$$\mu_F\left(\bigvee_{j=1}^r E_j\right) = \sum_{j=1}^r \mu_F(E_j).$$

$\square$

Corollary 199.2. *For every orthogonal decomposition of the identity,*

$$I = \bigvee_{j=1}^r E_j,$$

one has

$$\sum_{j=1}^r \mu_F(E_j) = 1.$$

200 Pure Relational States

Let

$$\mathcal{Q}_F$$

be a finite-dimensional relational Hilbert space and let

$$u \in \mathcal{Q}_F, \quad \|u\| = 1.$$

Define the rank-one projection

$$P_u = |u\rangle\langle u|.$$

Proposition 200.1. *The operator P_u is a projection of rank one:*

$$P_u^2 = P_u, \quad P_u^* = P_u, \quad \text{rank}(P_u) = 1.$$

Proof. For every $v \in \mathcal{Q}_F$,

$$P_u v = \langle u, v \rangle u.$$

Hence

$$P_u^2 v = \langle u, v \rangle P_u u = \langle u, v \rangle u = P_u v.$$

Self-adjointness follows from the definition of the rank-one orthogonal projection. Its range is

$$\mathbb{C}u,$$

which is one-dimensional. □

Define the pure state associated with u by

$$\rho_u = P_u.$$

Theorem 200.2. *For every projection*

$$E \in \mathcal{L}_F,$$

the corresponding probability is

$$\mu_u(E) = \text{tr}(P_u E) = \langle u, Eu \rangle.$$

Proof. Choose an orthonormal basis beginning with u . Since

$$P_u = |u\rangle\langle u|,$$

the trace identity

$$\text{tr}(P_u E) = \langle u, Eu \rangle$$

follows directly from the definition of the trace. □

Corollary 200.3. *For every projection E ,*

$$0 \leq \mu_u(E) \leq 1,$$

and

$$\mu_u(E) = 1 \iff u \in \text{range}(E).$$

Proof. Since E is an orthogonal projection,

$$\langle u, Eu \rangle = \|Eu\|^2,$$

so the first assertion follows from

$$\|Eu\| \leq \|u\| = 1.$$

Moreover,

$$\mu_u(E) = 1 \iff \|Eu\| = 1.$$

For an orthogonal projection and a unit vector this is equivalent to

$$Eu = u,$$

which is equivalent to

$$u \in \text{range}(E).$$

□

201 The State Representation of the Relational Measure

Let

$$\mathcal{L}_F = \mathcal{P}(\mathcal{Q}_F)$$

and let

$$\mu_F : \mathcal{L}_F \rightarrow [0, 1]$$

be a finitely additive probability measure on mutually orthogonal projections.

Theorem 201.1. *There exists a unique positive trace-one operator*

$$\rho_F \in \text{End}_{\mathbb{C}}(\mathcal{Q}_F)$$

such that

$$\mu_F(E) = \text{tr}(\rho_F E)$$

for every

$$E \in \mathcal{L}_F.$$

Proof. Choose an orthonormal basis

$$e_1, \dots, e_d$$

of $\mathcal{Q}_F$, and let

$$E_j = |e_j\rangle\langle e_j|.$$

The finite additivity of μ_F gives

$$\sum_{j=1}^d \mu_F(E_j) = 1.$$

Define a matrix ρ_F by its diagonal entries in this basis and extend the functional from diagonal projections to the full matrix algebra by linearity. The resulting functional is positive on positive operators. In finite dimension, every positive linear functional on

$$M_d(\mathbb{C})$$

has a unique density-matrix representation

$$X \longmapsto \text{tr}(\rho_F X)$$

with

$$\rho_F \geq 0.$$

Normalization gives

$$\text{tr}(\rho_F) = 1.$$

Uniqueness follows because rank-one projections span $\text{End}_{\mathbb{C}}(\mathcal{Q}_F)$. $\square$

Corollary 201.2. *The relational probability measure is represented uniquely by a density operator:*

$$\mu_F(E) = \text{tr}(\rho_F E).$$

202 Compatibility of the Stable State with the Relational Measure

Assume now that

$$\mathcal{Q}_F = \mathcal{H}_9$$

and that the relational state is invariant under the full unitary group of the stable sector.

Theorem 202.1. *The density operator representing the invariant relational measure is*

$$\rho_F = \rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. The preceding representation theorem gives a unique density operator ρ_F representing the measure. Full unitary invariance implies

$$U \rho_F U^* = \rho_F$$

for every

$$U \in \mathcal{U}(\mathcal{H}_9).$$

By the uniqueness theorem for the maximally symmetric stable state,

$$\rho_F = \frac{1}{9} I_{\mathcal{H}_9}.$$

$\square$

Corollary 202.2. *The relational measure on the stable sector is*

$$\mu_F(E) = \frac{1}{9} \text{tr}(E) = \frac{\text{rank}(E)}{9}.$$

203 The Spectral Probability Law

Let

$$A = A^* \in \mathcal{A}_9$$

and let

$$A = \sum_{j=1}^r \lambda_j E_j$$

be its spectral resolution.

Define the probability measure associated with A and ρ_9 by

$$\mathbb{P}_A(\{\lambda_j\}) = \mu_9(E_j).$$

Theorem 203.1. *The spectral probability law is*

$$\mathbb{P}_A(\{\lambda_j\}) = \text{tr}(\rho_9 E_j) = \frac{\text{rank}(E_j)}{9}.$$

Proof. The first equality is the definition of the spectral probability measure. The second is the representation of the relational measure by the state ρ_9 , and the final equality follows from

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Corollary 203.2. *The expectation of every self-adjoint observable is determined by its spectral measure:*

$$\mathbb{E}_{\rho_9}[A] = \sum_{j=1}^r \lambda_j \mathbb{P}_A(\{\lambda_j\}) = \text{tr}(\rho_9 A).$$

Proof. Substitution of the spectral resolution into the trace gives

$$\text{tr}(\rho_9 A) = \sum_{j=1}^r \lambda_j \text{tr}(\rho_9 E_j).$$

□

204 Born Representation

Let

$$\mathcal{H}$$

be a finite-dimensional complex Hilbert space and let

$$\rho$$

be a density operator on $\mathcal{H}$. For every orthogonal projection

$$E \in \mathcal{P}(\mathcal{H}),$$

define

$$\mu_\rho(E) = \text{tr}(\rho E).$$

Theorem 204.1. *The probability assigned to a projection E by the relational state is given by*

$$\mu_\rho(E) = \text{tr}(\rho E).$$

For a rank-one projection

$$E_\psi = |\psi\rangle\langle\psi|, \quad \|\psi\| = 1,$$

this becomes

$$\mu_\rho(E_\psi) = \langle\psi, \rho\psi\rangle.$$

Proof. The first identity is the state representation of the projection measure. For the rank-one projection,

$$\text{tr}(\rho |\psi\rangle\langle\psi|) = \langle\psi, \rho\psi\rangle.$$

□

Corollary 204.2. *For the pure state*

$$\rho = |\phi\rangle\langle\phi|,$$

one obtains

$$\mu_\rho(E_\psi) = |\langle\psi, \phi\rangle|^2.$$

Proof. Substituting

$$\rho = |\phi\rangle\langle\phi|$$

gives

$$\langle\psi, \rho\psi\rangle = \langle\psi, \phi\rangle\langle\phi, \psi\rangle = |\langle\psi, \phi\rangle|^2.$$

□

205 Reconstruction of the Born Rule from the Relational Measure

Let

$$\mathcal{L}_F = \mathcal{P}(\mathcal{H})$$

be the projection lattice generated by the finite relational construction, and let

$$\mu_F$$

be its normalized finitely additive probability measure.

Assume that the representation theorem identifies μ_F with the unique density operator ρ_F through

$$\mu_F(E) = \text{tr}(\rho_F E).$$

Theorem 205.1. *The relational measure induces the Born probability law on every orthogonal projection:*

$$\mu_F(E) = \text{tr}(\rho_F E), \quad E \in \mathcal{P}(\mathcal{H}).$$

For every unit vector $\psi \in \mathcal{H}$,

$$\mu_F(P_\psi) = \langle\psi, \rho_F \psi\rangle.$$

Proof. The first identity is the state representation established for μ_F . For

$$P_\psi = |\psi\rangle\langle\psi|,$$

the trace identity gives

$$\text{tr}(\rho_F P_\psi) = \langle\psi, \rho_F \psi\rangle.$$

□

Corollary 205.2. *If the relational state is pure,*

$$\rho_F = |\phi\rangle\langle\phi|,$$

then

$$\mu_F(P_\psi) = |\langle\psi, \phi\rangle|^2.$$

206 Representation of the Relational Measure

Let

$$\mathcal{H}_F$$

be a finite-dimensional complex Hilbert space with

$$\dim_{\mathbb{C}} \mathcal{H}_F \geq 3,$$

and let

$$\mu_F : \mathcal{P}(\mathcal{H}_F) \longrightarrow [0, 1]$$

be a finitely additive probability measure on the orthogonal projection lattice.

Theorem 206.1 (Gleason). *There exists a unique positive operator*

$$\rho_F \in \text{End}_{\mathbb{C}}(\mathcal{H}_F)$$

with

$$\text{tr}(\rho_F) = 1$$

such that

$$\mu_F(E) = \text{tr}(\rho_F E)$$

for every

$$E \in \mathcal{P}(\mathcal{H}_F).$$

Proof. This is the finite-dimensional form of Gleason's theorem applied to the finitely additive probability measure μ_F on the lattice of orthogonal projections of a complex Hilbert space of dimension at least three. The representing positive operator is uniquely determined by the values of μ_F on the rank-one projections, and the normalization

$$\mu_F(I) = 1$$

gives

$$\text{tr}(\rho_F) = 1.$$

□

Corollary 206.2. *The relational probability measure admits the unique density-operator representation*

$$\mu_F(E) = \text{tr}(\rho_F E).$$

Remark 206.3. The representation theorem is the point at which the relational measure acquires its operator-state representation. No additional probability axiom is introduced at this stage.

207 Rank-One Evaluation

Let

$$\rho_F$$

be the density operator representing μ_F , and let

$$\psi \in \mathcal{H}_F, \quad \|\psi\| = 1.$$

Define

$$P_\psi = |\psi\rangle\langle\psi|.$$

Proposition 207.1. *The probability of the rank-one projection P_ψ is*

$$\mu_F(P_\psi) = \langle \psi, \rho_F \psi \rangle.$$

Proof. By the representation theorem,

$$\mu_F(P_\psi) = \text{tr}(\rho_F P_\psi).$$

The rank-one trace identity gives

$$\text{tr}(\rho_F |\psi\rangle\langle\psi|) = \langle \psi, \rho_F \psi \rangle.$$

□

Corollary 207.2. *If*

$$\rho_F = P_\phi = |\phi\rangle\langle\phi|$$

for a unit vector ϕ , then

$$\mu_F(P_\psi) = |\langle \psi, \phi \rangle|^2.$$

Proof. Using

$$\rho_F = |\phi\rangle\langle\phi|$$

in the preceding proposition gives

$$\mu_F(P_\psi) = \langle \psi, \phi \rangle \langle \phi, \psi \rangle = |\langle \psi, \phi \rangle|^2.$$

□

208 Spectral Measures

Let

$$A = A^* \in \text{End}_{\mathbb{C}}(\mathcal{H}_F)$$

and write its spectral decomposition as

$$A = \sum_{j=1}^r \lambda_j E_j,$$

where

$$\lambda_1, \dots, \lambda_r \in \mathbb{R}$$

are distinct and

$$E_i E_j = 0 \quad (i \neq j),$$

with

$$\sum_{j=1}^r E_j = I.$$

Theorem 208.1. *The relational measure determines a probability distribution on the spectrum of A by*

$$\mathbb{P}_{\rho_F}^A(\{\lambda_j\}) = \text{tr}(\rho_F E_j).$$

Moreover,

$$\sum_{j=1}^r \mathbb{P}_{\rho_F}^A(\{\lambda_j\}) = 1.$$

Proof. Since the E_j are pairwise orthogonal and resolve the identity, finite additivity gives

$$\sum_{j=1}^r \operatorname{tr}(\rho_F E_j) = \operatorname{tr}\left(\rho_F \sum_{j=1}^r E_j\right) = \operatorname{tr}(\rho_F) = 1.$$

Thus the displayed values define a probability distribution on the finite spectrum of A . □

Corollary 208.2. *The expectation of A in the relational state is*

$$\mathbb{E}_{\rho_F}[A] = \sum_{j=1}^r \lambda_j \operatorname{tr}(\rho_F E_j) = \operatorname{tr}(\rho_F A).$$

Proof. By the spectral decomposition,

$$\operatorname{tr}(\rho_F A) = \operatorname{tr}\left(\rho_F \sum_{j=1}^r \lambda_j E_j\right) = \sum_{j=1}^r \lambda_j \operatorname{tr}(\rho_F E_j).$$

□

209 The Stable State as the Uniform Relational State

Let

$$\mathcal{H}_9$$

be the nine-dimensional stable sector and let

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Theorem 209.1. *For every orthogonal projection*

$$E \in \mathcal{P}(\mathcal{H}_9),$$

the corresponding relational probability is

$$\mu_9(E) = \frac{\operatorname{rank}(E)}{9}.$$

Proof. The relational measure is represented by ρ_9 , so

$$\mu_9(E) = \operatorname{tr}(\rho_9 E) = \frac{1}{9} \operatorname{tr}(E).$$

Since E is a projection,

$$\operatorname{tr}(E) = \operatorname{rank}(E).$$

□

Corollary 209.2. *For every orthonormal basis*

$$(e_1, \dots, e_9)$$

of $\mathcal{H}_9$, the associated rank-one projections

$$P_j = |e_j\rangle\langle e_j|$$

satisfy

$$\mu_9(P_j) = \frac{1}{9}$$

and

$$\sum_{j=1}^9 \mu_9(P_j) = 1.$$

Proof. Each P_j has rank one. The result follows from the theorem. □

210 The Projection-Valued Functional Calculus

Let

$$A = A^* \in \text{End}_{\mathbb{C}}(\mathcal{H}_F)$$

with spectral resolution

$$A = \sum_{j=1}^r \lambda_j E_j.$$

For every bounded function

$$f : \sigma(A) \longrightarrow \mathbb{C},$$

define

$$f(A) = \sum_{j=1}^r f(\lambda_j) E_j.$$

Proposition 210.1. *The relational state satisfies*

$$\text{tr}(\rho_F f(A)) = \sum_{j=1}^r f(\lambda_j) \text{tr}(\rho_F E_j).$$

Proof. Substitution of the spectral functional calculus gives

$$\text{tr}(\rho_F f(A)) = \text{tr} \left(\rho_F \sum_{j=1}^r f(\lambda_j) E_j \right),$$

and linearity of the trace yields the stated identity. $\square$

Corollary 210.2. *The relational measure determines the complete finite-dimensional spectral expectation functional*

$$f \longmapsto \text{tr}(\rho_F f(A)).$$

211 The Arithmetic-to-Relational State Map

Let

$$N_F$$

denote the normalized arithmetic state functional associated with the finite construction. Assume that its restriction to the relational projection algebra is positive and normalized:

$$N_F(E) \geq 0, \quad N_F(I) = 1, \quad E \in \mathcal{P}(\mathcal{H}_F).$$

Define

$$\mu_F(E) = N_F(E), \quad E \in \mathcal{P}(\mathcal{H}_F).$$

Proposition 211.1. *The functional N_F induces a finitely additive probability measure on the relational projection lattice:*

$$\mu_F : \mathcal{P}(\mathcal{H}_F) \longrightarrow [0, 1].$$

Proof. Positivity and normalization give

$$0 \leq \mu_F(E) \leq 1$$

for every projection E . If

$$E_1, \dots, E_r$$

are pairwise orthogonal, then

$$\mu_F \left(\bigvee_{j=1}^r E_j \right) = N_F \left(\sum_{j=1}^r E_j \right) = \sum_{j=1}^r N_F(E_j) = \sum_{j=1}^r \mu_F(E_j).$$

Thus μ_F is finitely additive. □

Theorem 211.2. *If*

$$\dim_{\mathbb{C}} \mathcal{H}_F \geq 3,$$

then there exists a unique density operator

$$\rho_F$$

such that

$$\boxed{N_F(E) = \mu_F(E) = \text{tr}(\rho_F E)}$$

for every

$$E \in \mathcal{P}(\mathcal{H}_F).$$

Proof. The preceding proposition provides a finitely additive probability measure on the projection lattice. The representation theorem then gives a unique density operator ρ_F satisfying

$$\mu_F(E) = \text{tr}(\rho_F E).$$

Since

$$\mu_F(E) = N_F(E),$$

the stated identity follows. □

212 The Measure Gate

The preceding construction yields the finite-state chain

$$N_F \longrightarrow \mu_F \longrightarrow \rho_F.$$

Theorem 212.1 (Measure Gate). *Assume that the normalized arithmetic functional N_F is positive on the relational projection lattice and that*

$$\dim_{\mathbb{C}} \mathcal{H}_F \geq 3.$$

Then the restriction of N_F to the relational projections has a unique operator representation

$$\boxed{N_F(E) = \text{tr}(\rho_F E)}$$

with

$$\rho_F \geq 0, \quad \text{tr}(\rho_F) = 1.$$

Consequently, for every rank-one projection

$$P_\psi = |\psi\rangle\langle\psi|,$$

one has

$$\boxed{N_F(P_\psi) = \langle\psi, \rho_F \psi\rangle.}$$

Proof. Positivity and normalization of N_F induce the finitely additive probability measure

$$\mu_F(E) = N_F(E).$$

Gleason's theorem provides the unique density operator ρ_F representing this measure. Evaluation on a rank-one projection gives

$$N_F(P_\psi) = \text{tr}(\rho_F P_\psi) = \langle \psi, \rho_F \psi \rangle.$$

□

Remark 212.2. The measure gate contains no independent probabilistic postulate. The probability functional is the restriction of the normalized arithmetic state to the relational projection lattice, and its operator representation is uniquely determined by that restriction.

213 Non-Booleanity of the Relational Event Structure

Let

$$\mathcal{L}_F = \mathcal{P}(\mathcal{H}_F)$$

with

$$\dim_{\mathbb{C}} \mathcal{H}_F \geq 2.$$

Theorem 213.1. *The relational event lattice $\mathcal{L}_F$ is non-Boolean.*

Proof. Choose linearly independent unit vectors

$$u, v \in \mathcal{H}_F$$

such that

$$0 < |\langle u, v \rangle| < 1.$$

Let

$$E = P_u, \quad F = P_v.$$

Then

$$EF \neq FE.$$

If $\mathcal{L}_F$ were Boolean, its elements would form a distributive complemented lattice. The projection representation of a Boolean family is necessarily commuting. The projections E and F violate this condition. Hence

$$\mathcal{L}_F$$

is not Boolean. □

Corollary 213.2. *The normalized functional*

$$\mu_F(E) = \text{tr}(\rho_F E)$$

is a probability measure on a non-Boolean event structure.

214 Separation of Classical and Relational Composition

Let

$$E, F \in \mathcal{P}(\mathcal{H}_F).$$

Proposition 214.1. *If E and F commute, then*

$$EF$$

is a projection and

$$EF = E \wedge F.$$

Proof. Since E and F are self-adjoint and commute,

$$(EF)^* = FE = EF.$$

Furthermore,

$$(EF)^2 = EFEF = E^2 F^2 = EF.$$

Thus EF is a projection. Its range consists precisely of vectors belonging to both ranges, so it represents the meet. $\square$

Proposition 214.2. *For arbitrary projections E and F , the product EF need not be a projection.*

Proof. If EF were a projection, then

$$(EF)^* = EF,$$

and hence

$$FE = EF.$$

Thus any pair of noncommuting projections provides an instance for which EF is not a projection. $\square$

Corollary 214.3. *The relational event composition cannot in general be represented by ordinary set intersection of events.*

215 Orthogonal Contexts

Let

$$\mathcal{C} = \{E_1, \dots, E_r\} \subseteq \mathcal{P}(\mathcal{H}_F)$$

be an orthogonal resolution of the identity:

$$E_i E_j = 0 \quad (i \neq j), \quad \sum_{j=1}^r E_j = I.$$

Theorem 215.1. *Every orthogonal context carries a normalized probability distribution*

$$p_j = \mu_F(E_j)$$

satisfying

$$p_j \geq 0, \quad \sum_{j=1}^r p_j = 1.$$

Proof. Positivity follows from the positivity of ρ_F , while normalization follows from

$$\sum_{j=1}^r p_j = \text{tr} \left(\rho_F \sum_{j=1}^r E_j \right) = \text{tr}(\rho_F) = 1.$$

□

Proposition 215.2. *The probability assigned to an event depends only on the projection representing that event:*

$$E = F \implies \mu_F(E) = \mu_F(F).$$

Proof. This follows immediately from

$$\mu_F(E) = \text{tr}(\rho_F E).$$

□

Corollary 215.3. *The measure is context-independent on projections: if the same projection belongs to two orthogonal resolutions of the identity, its assigned probability is the same in both resolutions.*

216 Finite-State Born Reconstruction

Let

$$\mathcal{H}_F$$

be the relational Hilbert space obtained from the finite construction, and let

$$N_F$$

be the normalized arithmetic state functional satisfying the hypotheses of the Measure Gate.

Let

$$\psi \in \mathcal{H}_F, \quad \|\psi\| = 1,$$

and let

$$P_\psi = |\psi\rangle\langle\psi|.$$

Theorem 216.1. *The probability assigned by the arithmetic-relational construction to the rank-one event P_ψ is*

$$\boxed{N_F(P_\psi) = \langle\psi, \rho_F \psi\rangle.}$$

If the state is pure,

$$\rho_F = P_\phi,$$

then

$$\boxed{N_F(P_\psi) = |\langle\psi, \phi\rangle|^2.}$$

Proof. The first identity is the rank-one evaluation of the unique density operator representing the restriction of N_F to the relational projection lattice. The second follows by substituting

$$\rho_F = P_\phi$$

and evaluating the resulting rank-one matrix coefficient.

□

Corollary 216.2. *For every finite orthogonal resolution*

$$I = \sum_{j=1}^r P_{\psi_j},$$

with

$$\{\psi_1, \dots, \psi_r\}$$

an orthonormal basis, one has

$$\sum_{j=1}^r N_F(P_{\psi_j}) = 1.$$

Proof. By the Born representation,

$$\sum_{j=1}^r N_F(P_{\psi_j}) = \text{tr} \left(\rho_F \sum_{j=1}^r P_{\psi_j} \right) = \text{tr}(\rho_F) = 1.$$

□

217 Specialization to the Nine-Dimensional Stable Sector

Assume that the finite relational construction identifies

$$\mathcal{H}_F = \mathcal{H}_9$$

and that its normalized state is the unique maximally symmetric state

$$\rho_F = \rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Theorem 217.1. *For every unit vector*

$$\psi \in \mathcal{H}_9,$$

the relational probability of the corresponding rank-one event is

$$N_F(P_\psi) = \frac{1}{9}.$$

Proof. The finite-state Born representation gives

$$N_F(P_\psi) = \langle \psi, \rho_9 \psi \rangle.$$

Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

and

$$\|\psi\| = 1,$$

one obtains

$$N_F(P_\psi) = \frac{1}{9}.$$

□

Corollary 217.2. *Every orthonormal basis of the stable nine-dimensional sector defines a uniform probability distribution:*

$$N_F(P_{\psi_j}) = \frac{1}{9}, \quad j = 1, \dots, 9.$$

Proof. Apply the theorem to each basis vector.

□

218 Orthogonal Additivity of the Relational Measure

Let

$$\mathcal{P}(\mathcal{H}_F)$$

denote the projection lattice of the relational Hilbert space, and let

$$\mu_F(E) = \text{tr}(\rho_F E), \quad E \in \mathcal{P}(\mathcal{H}_F).$$

Theorem 218.1. *Let*

$$E_1, \dots, E_r \in \mathcal{P}(\mathcal{H}_F)$$

be pairwise orthogonal projections. Then

$$\mu_F\left(\sum_{j=1}^r E_j\right) = \sum_{j=1}^r \mu_F(E_j).$$

Proof. Since the projections are pairwise orthogonal,

$$\sum_{j=1}^r E_j$$

is again a projection. By linearity of the trace,

$$\mu_F\left(\sum_{j=1}^r E_j\right) = \text{tr}\left(\rho_F \sum_{j=1}^r E_j\right) = \sum_{j=1}^r \text{tr}(\rho_F E_j).$$

Therefore

$$\mu_F\left(\sum_{j=1}^r E_j\right) = \sum_{j=1}^r \mu_F(E_j).$$

□

Corollary 218.2. *If*

$$E \perp F,$$

then

$$\mu_F(E + F) = \mu_F(E) + \mu_F(F).$$

Proof. Apply the theorem with

$$r = 2.$$

□

Corollary 218.3. *For every projection*

$$E \in \mathcal{P}(\mathcal{H}_F),$$

one has

$$\mu_F(E) + \mu_F(I - E) = 1.$$

Proof. The projections

$$E \quad \text{and} \quad I - E$$

are orthogonal and satisfy

$$E + (I - E) = I.$$

Hence

$$\mu_F(E) + \mu_F(I - E) = \mu_F(I) = \text{tr}(\rho_F) = 1.$$

□

219 The Projection Lattice and Orthomodularity

Let

$$\mathcal{P}(\mathcal{H}_F)$$

be the set of orthogonal projections on $\mathcal{H}_F$, ordered by

$$E \leq F \iff EF = E.$$

For

$$E, F \in \mathcal{P}(\mathcal{H}_F),$$

define

$$E^\perp = I - E.$$

Theorem 219.1. *The projection lattice*

$$\mathcal{P}(\mathcal{H}_F)$$

is orthomodular.

Proof. The order relation on projections is equivalent to inclusion of their ranges:

$$E \leq F \iff \text{range}(E) \subseteq \text{range}(F).$$

The orthogonal complement of a projection corresponds to the orthogonal complement of its range. If

$$E \leq F,$$

then

$$\text{range}(F) = \text{range}(E) \oplus (\text{range}(F) \cap \text{range}(E)^\perp).$$

The projection onto the second summand is

$$F(I - E),$$

and hence

$$F = E + F(I - E).$$

Moreover,

$$E \perp F(I - E).$$

Thus

$$F = E \vee (F \wedge E^\perp),$$

which is the orthomodular identity. □

Corollary 219.2. *For every*

$$E, F \in \mathcal{P}(\mathcal{H}_F)$$

with

$$E \leq F,$$

one has

$$\mu_F(F) = \mu_F(E) + \mu_F(F \wedge E^\perp).$$

Proof. The orthomodular decomposition gives

$$F = E + (F \wedge E^\perp)$$

as an orthogonal sum. Orthogonal additivity of μ_F therefore yields

$$\mu_F(F) = \mu_F(E) + \mu_F(F \wedge E^\perp).$$

□

220 Non-Booleanity of the Relational Projection Structure

Assume

$$\dim_{\mathbb{C}} \mathcal{H}_F \geq 2.$$

Theorem 220.1. *The projection lattice*

$$\mathcal{P}(\mathcal{H}_F)$$

is not Boolean.

Proof. Choose orthonormal vectors

$$e_1, e_2 \in \mathcal{H}_F$$

and define the rank-one projections

$$E = |e_1\rangle\langle e_1|$$

and

$$F = |v\rangle\langle v|, \quad v = \frac{e_1 + e_2}{\sqrt{2}}.$$

Then

$$EF \neq FE.$$

Indeed,

$$EF e_2 = \frac{1}{2} e_1,$$

whereas

$$FE e_2 = 0.$$

Hence

$$EF \neq FE.$$

In a Boolean algebra the meet and join operations are distributive and, in particular, all elements commute under the corresponding orthogonal-projection representation. The projection lattice therefore cannot be Boolean. $\square$

Corollary 220.2. *For the nine-dimensional stable sector,*

$$\mathcal{P}(\mathcal{H}_9)$$

is non-Boolean.

Proof. Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9 \geq 2,$$

the theorem applies. $\square$

221 Failure of Distributivity

Let

$$\mathcal{H}_F$$

have dimension at least 2. Choose orthonormal vectors

$$e_1, e_2 \in \mathcal{H}_F$$

and define

$$u = \frac{e_1 + e_2}{\sqrt{2}}, \quad v = \frac{e_1 - e_2}{\sqrt{2}}.$$

Let

$$E = P_{e_1}, \quad F = P_u, \quad G = P_v.$$

Theorem 221.1. *The projection lattice*

$$\mathcal{P}(\mathcal{H}_F)$$

fails the distributive law.

Proof. The vectors u and v are orthogonal and span the same two-dimensional subspace as e_1, e_2 . Hence

$$F \vee G = I_{\text{span}\{e_1, e_2\}},$$

and consequently

$$E \wedge (F \vee G) = E.$$

On the other hand,

$$E \wedge F = 0$$

and

$$E \wedge G = 0,$$

because the one-dimensional subspaces generated by e_1 , u , and v are pairwise distinct. Therefore

$$(E \wedge F) \vee (E \wedge G) = 0.$$

Thus

$$E \wedge (F \vee G) \neq (E \wedge F) \vee (E \wedge G),$$

which is precisely the failure of distributivity. $\square$

Corollary 221.2. *The non-Boolean character of the relational event structure is witnessed by a finite-dimensional configuration entirely contained in any two-dimensional subspace of $\mathcal{H}_F$.*

222 Dimension-Nine Specialization of the Projection Measure

Assume now

$$\mathcal{H}_F = \mathcal{H}_9$$

and

$$\rho_F = \frac{1}{9} I_{\mathcal{H}_9}.$$

Theorem 222.1. *For every projection*

$$E \in \mathcal{P}(\mathcal{H}_9),$$

the relational measure is

$$\mu_F(E) = \frac{\text{rank}(E)}{9}.$$

Proof. By definition,

$$\mu_F(E) = \text{tr}(\rho_F E).$$

Substituting

$$\rho_F = \frac{1}{9} I_{\mathcal{H}_9}$$

gives

$$\mu_F(E) = \frac{1}{9} \text{tr}(E).$$

Since E is a projection on a finite-dimensional space,

$$\text{tr}(E) = \text{rank}(E).$$

Therefore

$$\mu_F(E) = \frac{\text{rank}(E)}{9}.$$

□

Corollary 222.2. *For every pair of orthogonal projections*

$$E \perp F$$

on $\mathcal{H}_9$,

$$\mu_F(E + F) = \frac{\text{rank}(E) + \text{rank}(F)}{9}.$$

Proof. Orthogonal projections have additive rank:

$$\text{rank}(E + F) = \text{rank}(E) + \text{rank}(F).$$

Apply the theorem. □

Corollary 222.3. *Every rank-one projection in*

$$\mathcal{P}(\mathcal{H}_9)$$

has measure

$$\frac{1}{9}.$$

Proof. Set

$$\text{rank}(E) = 1$$

in the preceding theorem. □

223 Contextual Consistency of the Finite Relational Measure

Let

$$\mathcal{C}_1 = \{E_1, \dots, E_r\}$$

and

$$\mathcal{C}_2 = \{F_1, \dots, F_s\}$$

be two orthogonal resolutions of the identity in $\mathcal{P}(\mathcal{H}_F)$.

Theorem 223.1. *The measure μ_F assigns a context-independent value to every projection occurring in either resolution. In particular, if*

$$E_i = F_j,$$

then

$$\mu_F(E_i) = \mu_F(F_j).$$

Proof. Both quantities are evaluations of the same state functional:

$$\mu_F(E_i) = \text{tr}(\rho_F E_i)$$

and

$$\mu_F(F_j) = \text{tr}(\rho_F F_j).$$

If

$$E_i = F_j,$$

the two expressions coincide. □

Corollary 223.2. *The probability assignments obtained from different orthogonal resolutions are restrictions of a single measure*

$$\mu_F : \mathcal{P}(\mathcal{H}_F) \longrightarrow [0, 1].$$

Proof. The preceding theorem shows that the value assigned to a projection does not depend on the orthogonal resolution in which that projection is represented. $\square$

224 The Measure Gate

Let

$$N_F$$

be the normalized state functional associated with the finite arithmetic-relational construction. Let

$$\mathcal{P}(\mathcal{H}_F)$$

denote the projection lattice of the associated relational Hilbert space.

Assume that the restriction of N_F to $\mathcal{P}(\mathcal{H}_F)$ is normalized and finitely additive on pairwise orthogonal projections.

Theorem 224.1. *There exists a unique positive trace-class operator*

$$\rho_F \in \text{End}_{\mathbb{C}}(\mathcal{H}_F)$$

with

$$\text{tr}(\rho_F) = 1$$

such that

$$N_F(E) = \text{tr}(\rho_F E)$$

for every

$$E \in \mathcal{P}(\mathcal{H}_F).$$

Proof. The projection-valued state N_F is a normalized finitely additive probability measure on the projection lattice of the finite-dimensional Hilbert space $\mathcal{H}_F$. By the finite-dimensional representation theorem for states on a Hilbert-space projection lattice, there exists a unique positive operator ρ_F satisfying

$$N_F(E) = \text{tr}(\rho_F E)$$

for every projection E . Normalization gives

$$1 = N_F(I) = \text{tr}(\rho_F).$$

Since $\mathcal{H}_F$ is finite-dimensional, every operator is trace-class. $\square$

Corollary 224.2. *The restriction of N_F to the projection lattice determines ρ_F uniquely.*

Proof. If

$$\text{tr}(\rho_F E) = \text{tr}(\sigma_F E)$$

for every projection E , then the two state functionals coincide on all rank-one projections. Hence they have identical quadratic forms, and therefore

$$\rho_F = \sigma_F.$$

$\square$

Proposition 224.3. *The functional*

$$\mu_F(E) = \text{tr}(\rho_F E)$$

extends N_F from projections to the full matrix algebra

$$\mathcal{A}_F = \text{End}_{\mathbb{C}}(\mathcal{H}_F)$$

by

$$\mu_F(X) = \text{tr}(\rho_F X).$$

Proof. Linearity of the trace gives linearity of

$$X \longmapsto \text{tr}(\rho_F X).$$

For every projection E ,

$$\mu_F(E) = \text{tr}(\rho_F E) = N_F(E).$$

Thus the extension agrees with the original state functional on the projection lattice. $\square$

225 Extension from Projections to Observables

Let

$$\mathcal{A}_F = \text{End}_{\mathbb{C}}(\mathcal{H}_F)$$

and let

$$A = A^* \in \mathcal{A}_F.$$

Let

$$A = \sum_{j=1}^r \lambda_j E_j$$

be its spectral decomposition, where the

$$E_j$$

are pairwise orthogonal spectral projections.

Theorem 225.1. *The expectation associated with the state N_F is*

$$N_F(A) = \sum_{j=1}^r \lambda_j \mu_F(E_j) = \text{tr}(\rho_F A).$$

Proof. By the spectral decomposition,

$$A = \sum_{j=1}^r \lambda_j E_j.$$

The state functional extends linearly from projections, so

$$N_F(A) = \sum_{j=1}^r \lambda_j N_F(E_j) = \sum_{j=1}^r \lambda_j \mu_F(E_j).$$

Using

$$\mu_F(E_j) = \text{tr}(\rho_F E_j)$$

and linearity of the trace,

$$\sum_{j=1}^r \lambda_j \text{tr}(\rho_F E_j) = \text{tr} \left(\rho_F \sum_{j=1}^r \lambda_j E_j \right) = \text{tr}(\rho_F A).$$

$\square$

Corollary 225.2. *If*

$$A = \sum_{j=1}^r \lambda_j E_j$$

and

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

is any function defined on the spectrum of A , then

$$N_F(f(A)) = \sum_{j=1}^r f(\lambda_j) \mu_F(E_j).$$

Proof. Functional calculus gives

$$f(A) = \sum_{j=1}^r f(\lambda_j) E_j.$$

Apply the preceding theorem. □

226 Faithfulness and Null Events

Let

$$\rho_F$$

be the density operator representing N_F .

Definition 226.1. The state μ_F is called faithful if

$$\mu_F(E) = 0 \implies E = 0$$

for every

$$E \in \mathcal{P}(\mathcal{H}_F).$$

Theorem 226.2. *The state μ_F is faithful if and only if*

$$\rho_F > 0$$

as an operator on $\mathcal{H}_F$.

Proof. Suppose first that

$$\rho_F > 0.$$

Let E be a nonzero projection. Its range contains a nonzero vector v . Since ρ_F is positive definite,

$$\langle v, \rho_F v \rangle > 0.$$

Choose an orthonormal basis of $\text{range}(E)$. Then

$$\mu_F(E) = \text{tr}(\rho_F E) = \sum_j \langle e_j, \rho_F e_j \rangle > 0.$$

Hence

$$\mu_F(E) = 0$$

implies

$$E = 0.$$

Conversely, suppose that ρ_F is not positive definite. Then there exists a nonzero vector v such that

$$\rho_F v = 0.$$

Let

$$E = P_v$$

be the rank-one projection onto $\mathbb{C}v$. Then

$$\mu_F(E) = \text{tr}(\rho_F P_v) = \langle v, \rho_F v \rangle = 0,$$

while

$$E \neq 0.$$

Thus μ_F is not faithful. □

Corollary 226.3. *For the maximally symmetric nine-dimensional state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

the relational measure is faithful.

Proof. Every nonzero vector $v \in \mathcal{H}_9$ satisfies

$$\langle v, \rho_9 v \rangle = \frac{1}{9} \|v\|^2 > 0.$$

Hence

$$\rho_9 > 0.$$

The theorem applies. □

227 Unitary Covariance of the Relational Measure

Let

$$U \in \mathcal{U}(\mathcal{H}_F)$$

be unitary, and define the transformed projection

$$E^U = U E U^*.$$

Let

$$\rho_F^U = U \rho_F U^*.$$

Theorem 227.1. *The measure is covariant under unitary conjugation:*

$$\mu_F^U(E^U) = \mu_F(E),$$

where

$$\mu_F^U(F) = \text{tr}(\rho_F^U F).$$

Proof. By definition,

$$\mu_F^U(E^U) = \text{tr}(U\rho_F U^* U E U^*).$$

Using

$$U^* U = I$$

and cyclicity of the trace,

$$\mu_F^U(E^U) = \text{tr}(U\rho_F E U^*) = \text{tr}(\rho_F E).$$

Therefore

$$\mu_F^U(E^U) = \mu_F(E).$$

□

Corollary 227.2. *If*

$$U\rho_F U^* = \rho_F,$$

then

$$\mu_F(U E U^*) = \mu_F(E)$$

for every projection E.

Proof. Under the invariance hypothesis,

$$\rho_F^U = \rho_F.$$

Apply the theorem.

□

Corollary 227.3. *For*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\mu_9(U E U^*) = \mu_9(E)$$

for every

$$U \in \mathcal{U}(\mathcal{H}_9)$$

and every

$$E \in \mathcal{P}(\mathcal{H}_9).$$

Proof. Since

$$U\rho_9 U^* = \frac{1}{9} U I_{\mathcal{H}_9} U^* = \frac{1}{9} I_{\mathcal{H}_9},$$

the preceding corollary applies.

□

228 Rank Determination in the Maximally Symmetric Sector

Let

$$E \in \mathcal{P}(\mathcal{H}_9)$$

and

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Theorem 228.1. *The measure of a projection is determined uniquely by its rank:*

$$\mu_9(E) = \frac{\text{rank}(E)}{9}.$$

In particular, two projections of equal rank have equal measure.

Proof. The first assertion follows from

$$\mu_9(E) = \text{tr}(\rho_9 E) = \frac{1}{9} \text{tr}(E) = \frac{\text{rank}(E)}{9}.$$

If

$$\text{rank}(E) = \text{rank}(F),$$

then

$$\mu_9(E) = \mu_9(F).$$

□

Corollary 228.2. *For every r -dimensional subspace*

$$W \subseteq \mathcal{H}_9,$$

the orthogonal projection P_W satisfies

$$\mu_9(P_W) = \frac{r}{9}.$$

Proof. Since

$$\text{rank}(P_W) = \dim_{\mathbb{C}} W = r,$$

the theorem gives

$$\mu_9(P_W) = \frac{r}{9}.$$

□

Corollary 228.3. *If*

$$\mathcal{H}_9 = W_1 \oplus \cdots \oplus W_r$$

is an orthogonal decomposition, then

$$\sum_{j=1}^r \frac{\dim_{\mathbb{C}} W_j}{9} = 1.$$

Proof. Taking dimensions in the orthogonal direct sum gives

$$\sum_{j=1}^r \dim_{\mathbb{C}} W_j = 9.$$

Divide by 9.

□

229 Gleason Reconstruction on the Stable Sector

Let

$$\mathcal{H}_9$$

be a complex Hilbert space of dimension 9, and let

$$\mathcal{P}(\mathcal{H}_9)$$

denote its lattice of orthogonal projections. Suppose that

$$\mu_F : \mathcal{P}(\mathcal{H}_9) \longrightarrow [0, 1]$$

satisfies

$$\mu_F(I) = 1$$

and, whenever

$$E_i E_j = 0 \quad (i \neq j),$$

one has

$$\mu_F\left(\sum_{j=1}^r E_j\right) = \sum_{j=1}^r \mu_F(E_j).$$

Theorem 229.1 (Gleason). *There exists a unique positive trace-class operator*

$$\rho_F \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

with

$$\text{tr}(\rho_F) = 1$$

such that

$$\boxed{\mu_F(E) = \text{tr}(\rho_F E)}$$

for every

$$E \in \mathcal{P}(\mathcal{H}_9).$$

Proof. Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9 \geq 3,$$

the hypotheses place μ_F within the finite-dimensional complex case of Gleason's theorem. Hence there exists a unique positive operator ρ_F representing the measure on the projection lattice:

$$\mu_F(E) = \text{tr}(\rho_F E).$$

Evaluating at the identity gives

$$1 = \mu_F(I) = \text{tr}(\rho_F),$$

which proves the normalization. $\square$

Corollary 229.2. *The restriction of the arithmetic state functional to the projection lattice of the stable sector determines a unique density operator*

$$\rho_F.$$

In particular, the operator representation is not an additional choice once the normalized finitely additive projection measure has been fixed.

Proof. Uniqueness follows from the preceding theorem. $\square$

Proposition 229.3. *The density operator ρ_F is uniquely determined by its values on rank-one projections:*

$$\mu_F(P_\psi) = \langle \psi, \rho_F \psi \rangle \quad (\|\psi\| = 1).$$

Proof. For a unit vector ψ ,

$$P_\psi = |\psi\rangle\langle\psi|.$$

Therefore

$$\mu_F(P_\psi) = \text{tr}(\rho_F P_\psi) = \langle \psi, \rho_F \psi \rangle.$$

If two density operators have the same values on all rank-one projections, their quadratic forms coincide on every vector and hence the operators themselves coincide. $\square$

230 Invariance of the Reconstructed State

Let

$$G \subseteq \mathcal{U}(\mathcal{H}_9)$$

be a group of unitary operators. Assume that the projection measure μ_F is G -invariant:

$$\mu_F(UEU^*) = \mu_F(E)$$

for every

$$U \in G, \quad E \in \mathcal{P}(\mathcal{H}_9).$$

Theorem 230.1. *The density operator reconstructed from μ_F commutes with every element of G :*

$$U\rho_F U^* = \rho_F \quad (U \in G).$$

Equivalently,

$$U\rho_F = \rho_F U \quad (U \in G).$$

Proof. For every projection E ,

$$\mathrm{tr}(\rho_F UEU^*) = \mu_F(UEU^*) = \mu_F(E) = \mathrm{tr}(\rho_F E).$$

By cyclicity of the trace,

$$\mathrm{tr}(U^* \rho_F UE) = \mathrm{tr}(\rho_F E).$$

Thus the two positive operators

$$U^* \rho_F U \quad \text{and} \quad \rho_F$$

define the same normalized measure on $\mathcal{P}(\mathcal{H}_9)$. By the uniqueness in Gleason reconstruction,

$$U^* \rho_F U = \rho_F.$$

Conjugating by U gives

$$U\rho_F U^* = \rho_F.$$

□

Corollary 230.2. *If the unitary representation of G on $\mathcal{H}_9$ is irreducible, then*

$$\rho_F$$

is a scalar multiple of the identity.

Proof. The preceding theorem shows that

$$\rho_F$$

belongs to the commutant of the representation of G . Irreducibility and Schur's lemma imply

$$\rho_F = cI_{\mathcal{H}_9}$$

for some

$$c \geq 0.$$

□

Theorem 230.3 (Unitary rigidity). *Suppose that the action of G on $\mathcal{H}_9$ is irreducible and that μ_F is G -invariant. Then*

$$\rho_F = \frac{1}{9} I_{\mathcal{H}_9}.$$

Consequently,

$$\mu_F(E) = \frac{1}{9} \operatorname{tr}(E)$$

for every projection

$$E \in \mathcal{P}(\mathcal{H}_9).$$

Proof. By the preceding corollary,

$$\rho_F = c I_{\mathcal{H}_9}.$$

Since ρ_F is normalized,

$$1 = \operatorname{tr}(\rho_F) = c \operatorname{tr}(I_{\mathcal{H}_9}) = 9c.$$

Hence

$$c = \frac{1}{9}.$$

Substitution into the Gleason representation yields

$$\mu_F(E) = \operatorname{tr}\left(\frac{1}{9} I_{\mathcal{H}_9} E\right) = \frac{1}{9} \operatorname{tr}(E).$$

□

231 Uniformity as a Consequence of Symmetry

Let

$$E \in \mathcal{P}(\mathcal{H}_9)$$

have rank

$$\operatorname{rank}(E) = r.$$

Proposition 231.1. *Under the hypotheses of unitary rigidity,*

$$\mu_F(E) = \frac{r}{9}.$$

Proof. Since E is a projection,

$$\operatorname{tr}(E) = \operatorname{rank}(E) = r.$$

The unitary-rigidity formula therefore gives

$$\mu_F(E) = \frac{1}{9} \operatorname{tr}(E) = \frac{r}{9}.$$

□

Corollary 231.2. *All rank- r projections have the same probability:*

$$\operatorname{rank}(E) = \operatorname{rank}(F) \implies \mu_F(E) = \mu_F(F).$$

Proof. Both probabilities equal

$$\frac{r}{9}.$$

□

Corollary 231.3. *For every orthogonal family*

$$E_1, \dots, E_9$$

of rank-one projections satisfying

$$\sum_{j=1}^9 E_j = I_{\mathcal{H}_9},$$

one has

$$\mu_F(E_j) = \frac{1}{9} \quad (1 \leq j \leq 9).$$

Proof. Each E_j has rank one, so the preceding proposition gives

$$\mu_F(E_j) = \frac{1}{9}.$$

□

232 The Commutant Criterion

The full unitary invariance used above is stronger than necessary. The essential condition is that the reconstructed state lie in the commutant of the relevant symmetry representation.

Let

$$\pi : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be a unitary representation and define its commutant by

$$\pi(G)' = \{X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9) : X\pi(g) = \pi(g)X \text{ for every } g \in G\}.$$

Theorem 232.1. *If the projection measure satisfies*

$$\mu_F(\pi(g)E\pi(g)^*) = \mu_F(E)$$

for every

$$g \in G, \quad E \in \mathcal{P}(\mathcal{H}_9),$$

then

$$\rho_F \in \pi(G)'.$$

Proof. By unitary invariance of the reconstructed measure,

$$\pi(g)\rho_F\pi(g)^* = \rho_F$$

for every $g \in G$. Hence

$$\pi(g)\rho_F = \rho_F\pi(g),$$

which is precisely

$$\rho_F \in \pi(G)'.$$

□

Corollary 232.2. *If*

$$\pi(G)' = \mathbb{C}I_{\mathcal{H}_9},$$

then the normalized invariant state is necessarily

$$\rho_F = \frac{1}{9}I_{\mathcal{H}_9}.$$

Proof. The commutant criterion gives

$$\rho_F \in \mathbb{C}I_{\mathcal{H}_9}.$$

Hence

$$\rho_F = cI_{\mathcal{H}_9}.$$

The trace normalization gives

$$9c = 1.$$

□

Remark 232.3. The conclusion

$$\rho_F = \frac{1}{9}I_{\mathcal{H}_9}$$

therefore has two logically distinct possible origins. Full unitary invariance is sufficient, but not necessary. It is enough that the symmetry representation have scalar commutant. This distinction will be relevant when the arithmetic action is introduced.

233 Compatibility with the Arithmetic Representation

Let

$$\pi : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be a unitary representation of the arithmetic lattice. Assume that the projection measure is invariant under this action:

$$\mu_F(\pi(x)E\pi(x)^*) = \mu_F(E)$$

for every

$$x \in \mathcal{L}_{\mathbb{P}}, \quad E \in \mathcal{P}(\mathcal{H}_9).$$

Theorem 233.1. *The reconstructed density operator belongs to the commutant of the arithmetic representation:*

$$\rho_F \in \pi(\mathcal{L}_{\mathbb{P}})'.$$

Proof. Apply the commutant criterion with

$$G = \mathcal{L}_{\mathbb{P}}.$$

□

Proposition 233.2. *Suppose that*

$$\pi(\mathcal{L}_{\mathbb{P}})' = \mathbb{C}I_{\mathcal{H}_9}.$$

Then the arithmetic-invariant normalized state is uniquely determined:

$$\boxed{\rho_F = \frac{1}{9}I_{\mathcal{H}_9}.}$$

Proof. The preceding theorem implies

$$\rho_F \in \pi(\mathcal{L}_{\mathbb{P}})' = \mathbb{C}I_{\mathcal{H}_9}.$$

Hence

$$\rho_F = cI_{\mathcal{H}_9}.$$

The trace normalization gives

$$1 = \text{tr}(\rho_F) = 9c,$$

so

$$c = \frac{1}{9}.$$

□

Remark 233.3. The scalar character family

$$\chi_t(x) = e^{it\Lambda_{\log}(x)}$$

and the finite-dimensional representation π should not be identified without an explicit intertwining construction. What is established here is weaker and precise: whenever the arithmetic action is represented unitarily on $\mathcal{H}_9$ and the associated projection measure is invariant, the reconstructed state lies in the commutant of that action. Scalarity of the commutant then forces the maximally mixed state.

234 Character Decomposition of the Arithmetic Representation

Let

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z}e_p$$

and let

$$\pi : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be a unitary representation.

Since

$$\mathcal{L}_{\mathbb{P}}$$

is abelian, its image consists of pairwise commuting unitary operators.

Theorem 234.1. *There exists an orthogonal decomposition*

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^s \mathcal{H}_{\alpha}$$

such that each $\mathcal{H}_{\alpha}$ is invariant under $\pi(\mathcal{L}_{\mathbb{P}})$, and there exists a unitary character

$$\chi_{\alpha} : \mathcal{L}_{\mathbb{P}} \longrightarrow U(1)$$

satisfying

$$\pi(x)v = \chi_{\alpha}(x)v$$

for every

$$x \in \mathcal{L}_{\mathbb{P}}, \quad v \in \mathcal{H}_{\alpha}.$$

Proof. For every

$$x \in \mathcal{L}_{\mathbb{P}},$$

the operator

$$\pi(x)$$

is unitary and therefore normal. Moreover,

$$\pi(x)\pi(y) = \pi(x+y) = \pi(y+x) = \pi(y)\pi(x)$$

for all

$$x, y \in \mathcal{L}_{\mathbb{P}}.$$

Thus the family

$$\{\pi(x) : x \in \mathcal{L}_{\mathbb{P}}\}$$

is a commuting family of normal operators on the finite-dimensional space $\mathcal{H}_9$. Consequently it is simultaneously diagonalizable.

Hence $\mathcal{H}_9$ admits an orthogonal decomposition into common eigenspaces

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^s \mathcal{H}_{\alpha}.$$

On each common eigenspace, every $\pi(x)$ acts by a scalar of modulus one. Denote this scalar by

$$\chi_{\alpha}(x).$$

The representation identity gives

$$\chi_{\alpha}(x + y) = \chi_{\alpha}(x)\chi_{\alpha}(y),$$

so χ_{α} is a unitary character of $\mathcal{L}_{\mathbb{P}}$. □

Corollary 234.2. *The arithmetic representation is completely reducible, and every irreducible constituent is one-dimensional.*

Proof. Each common eigenspace admits a decomposition into one-dimensional common eigenspaces if necessary. Since every element of the abelian group acts by a scalar on each such line, every irreducible constituent is one-dimensional. □

235 Multiplicity Spaces and the Commutant

Let

$$\widehat{\mathcal{L}_{\mathbb{P}}}$$

denote the set of unitary characters of $\mathcal{L}_{\mathbb{P}}$. For each character

$$\chi \in \widehat{\mathcal{L}_{\mathbb{P}}},$$

define its multiplicity space by

$$\mathcal{H}_{\chi} = \{v \in \mathcal{H}_9 : \pi(x)v = \chi(x)v \text{ for every } x \in \mathcal{L}_{\mathbb{P}}\}.$$

Only finitely many such spaces are nonzero because

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Proposition 235.1. *The stable sector decomposes as*

$$\mathcal{H}_9 = \bigoplus_{\chi} \mathcal{H}_{\chi},$$

where the sum is taken over the finitely many characters occurring in the representation.

Proof. This is the simultaneous eigenspace decomposition established above. Distinct characters correspond to orthogonal common eigenspaces because the operators $\pi(x)$ are unitary. □

Theorem 235.2. *The commutant of the arithmetic representation is*

$$\pi(\mathcal{L}_{\mathbb{P}})' = \bigoplus_{\chi} \text{End}_{\mathbb{C}}(\mathcal{H}_{\chi}).$$

Proof. Let

$$X \in \pi(\mathcal{L}_{\mathbb{P}})'.$$

If

$$v \in \mathcal{H}_{\chi},$$

then for every

$$x \in \mathcal{L}_{\mathbb{P}}$$

one has

$$\pi(x)Xv = X\pi(x)v = X(\chi(x)v) = \chi(x)Xv.$$

Therefore

$$Xv \in \mathcal{H}_{\chi}.$$

Hence X preserves every multiplicity space.

Conversely, any operator which acts independently on the spaces $\mathcal{H}_{\chi}$ commutes with every $\pi(x)$, because $\pi(x)$ acts on $\mathcal{H}_{\chi}$ by the scalar $\chi(x)$.

Thus

$$\pi(\mathcal{L}_{\mathbb{P}})' = \bigoplus_{\chi} \text{End}_{\mathbb{C}}(\mathcal{H}_{\chi}).$$

□

Corollary 235.3. *The arithmetic commutant is scalar if and only if every character occurs with multiplicity one and exactly one character occurs.*

Proof. If

$$\pi(\mathcal{L}_{\mathbb{P}})' = \mathbb{C}I_{\mathcal{H}_9},$$

then the preceding decomposition can contain only one nonzero summand, and that summand must have dimension one. Conversely, if exactly one one-dimensional character space occurs, then the representation is scalar and its commutant is the full matrix algebra rather than the scalar algebra.

Therefore the stated equivalence requires the additional condition that the representation algebra generated by the arithmetic action be irreducible. For an abelian group, this is possible only in dimension one. Hence, on a nine-dimensional space, the arithmetic representation alone cannot have scalar commutant. □

Remark 235.4. The preceding observation is structurally decisive. Since $\mathcal{L}_{\mathbb{P}}$ is abelian, no nine-dimensional unitary representation of $\mathcal{L}_{\mathbb{P}}$ can be irreducible. Thus the condition

$$\pi(\mathcal{L}_{\mathbb{P}})' = \mathbb{C}I_{\mathcal{H}_9}$$

cannot be obtained from the arithmetic lattice alone.

Consequently, if the state

$$\rho_F = \frac{1}{9}I_{\mathcal{H}_9}$$

is to be derived from a commutant argument, the relevant symmetry algebra must contain additional noncommuting operators beyond the image of $\mathcal{L}_{\mathbb{P}}$.

236 The Logarithmic Characters and Their Separation

For

$$t \in \mathbb{R},$$

consider the unitary character

$$\chi_t : \mathcal{L}_{\mathbb{P}} \longrightarrow U(1)$$

defined by

$$\chi_t(x) = \exp(it\Lambda_{\log}(x)),$$

where

$$\Lambda_{\log}(x) = \sum_{p \in \mathbb{P}} n_p \log p$$

for

$$x = \sum_{p \in \mathbb{P}} n_p e_p.$$

Proposition 236.1. *The map*

$$t \longmapsto \chi_t$$

is a one-parameter family of unitary characters of $\mathcal{L}_{\mathbb{P}}$.

Proof. For every

$$x, y \in \mathcal{L}_{\mathbb{P}},$$

one has

$$\Lambda_{\log}(x + y) = \Lambda_{\log}(x) + \Lambda_{\log}(y).$$

Therefore

$$\chi_t(x + y) = e^{it\Lambda_{\log}(x+y)} = e^{it\Lambda_{\log}(x)} e^{it\Lambda_{\log}(y)} = \chi_t(x) \chi_t(y).$$

Moreover,

$$|\chi_t(x)| = 1.$$

□

Theorem 236.2. *The family*

$$\{\chi_t : t \in \mathbb{R}\}$$

separates points of $\mathcal{L}_{\mathbb{P}}$.

Proof. Let

$$x \neq y.$$

Then

$$z = x - y \neq 0.$$

Unique factorization implies

$$\Lambda_{\log}(z) \neq 0.$$

Choose

$$t = \frac{\pi}{\Lambda_{\log}(z)}.$$

Then

$$\chi_t(z) = e^{i\pi} = -1.$$

If

$$\chi_t(x) = \chi_t(y),$$

then

$$\chi_t(z) = \chi_t(x - y) = \chi_t(x)\chi_t(y)^{-1} = 1,$$

which contradicts

$$\chi_t(z) = -1.$$

Hence

$$\chi_t(x) \neq \chi_t(y).$$

□

237 The Character Group Generated by the Logarithmic Embedding

Let

$$\Gamma_{\log} = \Lambda_{\log}(\mathcal{L}_{\mathbb{P}}) \subseteq \mathbb{R}.$$

Proposition 237.1. *The subgroup*

$$\Gamma_{\log}$$

is the free abelian subgroup of $\mathbb{R}$ generated by

$$\{\log p : p \in \mathbb{P}\}.$$

Proof. By definition,

$$\Gamma_{\log} = \left\{ \sum_{p \in \mathbb{P}} n_p \log p : n_p \in \mathbb{Z}, n_p = 0 \text{ for all but finitely many } p \right\}.$$

Unique factorization shows that the logarithms of distinct primes are linearly independent over $\mathbb{Q}$. Indeed, if

$$\sum_{j=1}^r n_j \log p_j = 0$$

with

$$n_j \in \mathbb{Z},$$

then

$$\prod_{j=1}^r p_j^{n_j} = 1,$$

and hence

$$n_1 = \cdots = n_r = 0.$$

Thus $\Gamma_{\log}$ is a free abelian group on the generators $\log p$. □

Theorem 237.2. *The logarithmic character family factors through the additive group $\Gamma_{\log}$:*

$$\chi_t(x) = e^{it\gamma}, \quad \gamma = \Lambda_{\log}(x) \in \Gamma_{\log}.$$

Moreover, the induced characters

$$\tilde{\chi}_t : \Gamma_{\log} \longrightarrow U(1)$$

are given by

$$\tilde{\chi}_t(\gamma) = e^{it\gamma}.$$

Proof. The definition of χ_t depends on x only through $\Lambda_{\log}(x)$. Since $\Lambda_{\log}$ is injective, this defines an equivalent character on its image $\Gamma_{\log}$. □

238 Finite-Dimensional Arithmetic Spectra

Let

$$\pi : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be a unitary representation. Let

$$\chi_1, \dots, \chi_s$$

be the distinct characters occurring in its simultaneous spectral decomposition, and let

$$m_j = \dim_{\mathbb{C}} \mathcal{H}_{\chi_j}.$$

Then

$$\sum_{j=1}^s m_j = 9.$$

Theorem 238.1. *For every*

$$x \in \mathcal{L}_{\mathbb{P}},$$

the characteristic polynomial of $\pi(x)$ is

$$\det(\lambda I - \pi(x)) = \prod_{j=1}^s (\lambda - \chi_j(x))^{m_j}.$$

Proof. On the multiplicity space

$$\mathcal{H}_{\chi_j},$$

the operator $\pi(x)$ acts as the scalar

$$\chi_j(x).$$

Thus the characteristic polynomial is the product of the corresponding linear factors, each repeated according to the dimension m_j of its multiplicity space. $\square$

Corollary 238.2. *The determinant and trace of the arithmetic operator are*

$$\det \pi(x) = \prod_{j=1}^s \chi_j(x)^{m_j}$$

and

$$\mathrm{tr} \pi(x) = \sum_{j=1}^s m_j \chi_j(x).$$

Proof. Both identities follow immediately from the characteristic polynomial and the eigenvalue decomposition. $\square$

Proposition 238.3. *If the representation is faithful, then*

$$\bigcap_{j=1}^s \ker \chi_j = \{0\}.$$

Proof. An element

$$x \in \mathcal{L}_{\mathbb{P}}$$

belongs to the kernel of π precisely when

$$\pi(x) = I_{\mathcal{H}_9}.$$

Under the simultaneous character decomposition this is equivalent to

$$\chi_j(x) = 1 \quad (1 \leq j \leq s).$$

Therefore

$$\ker \pi = \bigcap_{j=1}^s \ker \chi_j.$$

Faithfulness gives

$$\ker \pi = \{0\}.$$

□

239 Necessary Extension of the Arithmetic Symmetry

The preceding analysis shows that the arithmetic lattice by itself does not provide an irreducible nine-dimensional unitary symmetry. A nine-dimensional scalar commutant must therefore arise from a larger operator algebra.

Let

$$\mathfrak{A}_F \subseteq \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be a unital $*$ -algebra containing the image of the arithmetic representation:

$$\pi(\mathcal{L}_{\mathbb{P}}) \subseteq \mathfrak{A}_F.$$

Define its commutant by

$$\mathfrak{A}'_F = \{X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9) : XA = AX \text{ for every } A \in \mathfrak{A}_F\}.$$

Theorem 239.1. *Suppose that*

$$\mathfrak{A}'_F = \mathbb{C}I_{\mathcal{H}_9}.$$

If the normalized projection measure μ_F is invariant under unitary elements of $\mathfrak{A}_F$, then

$$\rho_F = \frac{1}{9}I_{\mathcal{H}_9}.$$

Proof. Invariance of the projection measure implies, by the commutant criterion,

$$\rho_F \in \mathfrak{A}'_F.$$

By hypothesis,

$$\mathfrak{A}'_F = \mathbb{C}I_{\mathcal{H}_9}.$$

Therefore

$$\rho_F = cI_{\mathcal{H}_9}.$$

The normalization condition gives

$$1 = \text{tr}(\rho_F) = 9c,$$

hence

$$c = \frac{1}{9}.$$

□

Remark 239.2. The role of the arithmetic lattice is therefore sharply separated from that of the full finite-state symmetry algebra. The lattice supplies the arithmetic character structure, while scalarity of the commutant requires a sufficiently large noncommutative algebra acting on the stable sector. Any identification of $\mathfrak{A}_F$ with a concrete algebra must consequently be proved from the finite relational construction rather than inferred from the abelian arithmetic action alone.

240 Conditional States on Finite Subspaces

Let

$$\mathcal{H}_9$$

be the nine-dimensional stable sector and let

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Let

$$E \in \mathcal{P}(\mathcal{H}_9)$$

be a nonzero orthogonal projection, and write

$$r_E = \text{rank}(E).$$

Define the conditional state associated with the event E by

$$\rho_{9,E} = \frac{E\rho_9 E}{\text{tr}(\rho_9 E)}.$$

Proposition 240.1. *For every nonzero projection E ,*

$$\text{tr}(\rho_9 E) = \frac{r_E}{9}.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\text{tr}(\rho_9 E) = \frac{1}{9} \text{tr}(E).$$

For an orthogonal projection,

$$\text{tr}(E) = \text{rank}(E) = r_E.$$

Therefore

$$\text{tr}(\rho_9 E) = \frac{r_E}{9}.$$

□

Theorem 240.2. *The conditional state associated with E is the normalized identity on the range of E :*

$$\boxed{\rho_{9,E} = \frac{1}{r_E} E.}$$

Proof. Using

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

we obtain

$$E\rho_9 E = \frac{1}{9} E^2 = \frac{1}{9} E.$$

By the preceding proposition,

$$\text{tr}(\rho_9 E) = \frac{r_E}{9}.$$

Hence

$$\rho_{9,E} = \frac{\frac{1}{9} E}{r_E/9} = \frac{1}{r_E} E.$$

□

Corollary 240.3. *The conditional state is supported precisely on*

$$\text{range}(E)$$

and satisfies

$$\text{tr}(\rho_{9,E}) = 1.$$

Proof. Since

$$\rho_{9,E} = \frac{1}{r_E} E,$$

its range is

$$\text{range}(\rho_{9,E}) = \text{range}(E).$$

Moreover,

$$\text{tr}(\rho_{9,E}) = \frac{1}{r_E} \text{tr}(E) = 1.$$

□

Proposition 240.4. *If E has rank one, then*

$$\rho_{9,E} = E.$$

Thus conditioning the maximally symmetric state on a one-dimensional event produces the corresponding pure state.

Proof. For

$$\text{rank}(E) = 1,$$

the preceding theorem gives

$$\rho_{9,E} = E.$$

A rank-one orthogonal projection is a pure density operator. □

241 Conditional Born Probabilities

Let

$$E, F \in \mathcal{P}(\mathcal{H}_9)$$

be nonzero projections. With respect to the conditional state

$$\rho_{9,E} = \frac{1}{\text{rank}(E)} E,$$

define the conditional probability of F by

$$\mu_{9,E}(F) = \text{tr}(\rho_{9,E} F).$$

Theorem 241.1. *The conditional probability is given by*

$$\mu_{9,E}(F) = \frac{\text{tr}(EF)}{\text{rank}(E)}.$$

Proof. Substituting

$$\rho_{9,E} = \frac{1}{\text{rank}(E)} E$$

into the definition yields

$$\mu_{9,E}(F) = \text{tr}\left(\frac{1}{\text{rank}(E)} EF\right) = \frac{\text{tr}(EF)}{\text{rank}(E)}.$$

□

Proposition 241.2. *If $F \leq E$, then*

$$\mu_{9,E}(F) = \frac{\text{rank}(F)}{\text{rank}(E)}.$$

Proof. The relation

$$F \leq E$$

for orthogonal projections implies

$$EF = FE = F.$$

Consequently,

$$\mu_{9,E}(F) = \frac{\text{tr}(F)}{\text{rank}(E)} = \frac{\text{rank}(F)}{\text{rank}(E)}.$$

□

Corollary 241.3. *If*

$$E = \sum_{j=1}^r E_j$$

is an orthogonal decomposition of E into nonzero projections, then

$$\sum_{j=1}^r \mu_{9,E}(E_j) = 1.$$

Proof. Since

$$E_i E_j = 0 \quad (i \neq j)$$

and

$$\sum_{j=1}^r E_j = E,$$

one has

$$\sum_{j=1}^r \mu_{9,E}(E_j) = \frac{1}{\text{rank}(E)} \sum_{j=1}^r \text{tr}(E_j).$$

The trace is additive on finite orthogonal sums, so

$$\sum_{j=1}^r \text{tr}(E_j) = \text{tr}(E) = \text{rank}(E).$$

Hence the sum equals 1.

□

242 Orthogonal Decompositions of the Stable State

Let

$$\{E_1, \dots, E_r\}$$

be an orthogonal resolution of the identity on $\mathcal{H}_9$:

$$E_i E_j = 0 \quad (i \neq j), \quad \sum_{j=1}^r E_j = I_{\mathcal{H}_9}.$$

Set

$$r_j = \text{rank}(E_j).$$

Theorem 242.1. *The maximally symmetric state admits the decomposition*

$$\rho_9 = \sum_{j=1}^r \frac{r_j}{9} \rho_{9,E_j},$$

where

$$\rho_{9,E_j} = \frac{1}{r_j} E_j.$$

Proof. By the conditional-state formula,

$$\rho_{9,E_j} = \frac{1}{r_j} E_j.$$

Therefore

$$\frac{r_j}{9} \rho_{9,E_j} = \frac{1}{9} E_j.$$

Summing over j gives

$$\sum_{j=1}^r \frac{r_j}{9} \rho_{9,E_j} = \frac{1}{9} \sum_{j=1}^r E_j = \frac{1}{9} I_{\mathcal{H}_9} = \rho_9.$$

□

Corollary 242.2. *The weights in the orthogonal decomposition are determined solely by the dimensions of the corresponding subspaces:*

$$p_j = \frac{r_j}{9}, \quad \sum_{j=1}^r p_j = 1.$$

Proof. The expression for p_j follows from

$$\text{tr}(\rho_9 E_j) = \frac{\text{rank}(E_j)}{9}.$$

Since the projections form a resolution of the identity,

$$\sum_{j=1}^r r_j = \text{rank}(I_{\mathcal{H}_9}) = 9.$$

Hence

$$\sum_{j=1}^r p_j = \frac{1}{9} \sum_{j=1}^r r_j = 1.$$

□

243 Unitary Covariance of the Conditional Construction

Let

$$U \in \mathcal{U}(\mathcal{H}_9)$$

and let

$$E \in \mathcal{P}(\mathcal{H}_9)$$

be nonzero.

Define

$$E^U = U E U^*.$$

Proposition 243.1. *The ranks of E and E^U coincide:*

$$\text{rank}(E^U) = \text{rank}(E).$$

Proof. Since U is invertible,

$$\text{range}(E^U) = U(\text{range}(E)).$$

Unitary maps preserve dimension, and therefore

$$\text{rank}(E^U) = \dim U(\text{range}(E)) = \dim \text{range}(E) = \text{rank}(E).$$

□

Theorem 243.2. *The conditional-state assignment is equivariant under the unitary group:*

$$\boxed{\rho_{9,E^U} = U\rho_{9,E}U^*}.$$

Proof. By the conditional-state formula and the rank invariance,

$$\rho_{9,E^U} = \frac{1}{\text{rank}(E^U)} E^U = \frac{1}{\text{rank}(E)} U E U^*.$$

Thus

$$\rho_{9,E^U} = U \left(\frac{1}{\text{rank}(E)} E \right) U^* = U \rho_{9,E} U^*.$$

□

Corollary 243.3. *For every pair of projections E, F ,*

$$\mu_{9,E^U}(F^U) = \mu_{9,E}(F).$$

Proof. Using unitary covariance,

$$\mu_{9,E^U}(F^U) = \text{tr}(\rho_{9,E^U} F^U) = \text{tr}(U \rho_{9,E} U^* U F U^*).$$

Cyclicity of the trace gives

$$\mu_{9,E^U}(F^U) = \text{tr}(\rho_{9,E} F) = \mu_{9,E}(F).$$

□

244 Moments of Self-Adjoint Observables in the Stable State

Let

$$A = A^* \in \mathcal{A}_9.$$

Theorem 244.1. *For every polynomial*

$$f \in \mathbb{C}[X],$$

the expectation of $f(A)$ in the state ρ_9 is

$$\boxed{\text{tr}(\rho_9 f(A)) = \frac{1}{9} \text{tr}(f(A)).}$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\mathrm{tr}(\rho_9 f(A)) = \mathrm{tr}\left(\frac{1}{9} f(A)\right) = \frac{1}{9} \mathrm{tr}(f(A)).$$

□

Corollary 244.2. *If*

$$A = \sum_{j=1}^r \lambda_j E_j$$

is the spectral resolution of A , then

$$\mathrm{tr}(\rho_9 f(A)) = \frac{1}{9} \sum_{j=1}^r f(\lambda_j) \mathrm{rank}(E_j).$$

Proof. The functional calculus gives

$$f(A) = \sum_{j=1}^r f(\lambda_j) E_j.$$

Taking the normalized trace yields

$$\mathrm{tr}(\rho_9 f(A)) = \frac{1}{9} \sum_{j=1}^r f(\lambda_j) \mathrm{tr}(E_j),$$

and

$$\mathrm{tr}(E_j) = \mathrm{rank}(E_j).$$

□

Proposition 244.3. *The variance of a self-adjoint observable A in the state ρ_9 is*

$$\mathrm{Var}_{\rho_9}(A) = \frac{1}{9} \mathrm{tr}(A^2) - \left(\frac{1}{9} \mathrm{tr}(A)\right)^2.$$

Proof. By definition,

$$\mathrm{Var}_{\rho_9}(A) = \mathrm{tr}(\rho_9 A^2) - \mathrm{tr}(\rho_9 A)^2.$$

Applying

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

to both terms gives

$$\mathrm{Var}_{\rho_9}(A) = \frac{1}{9} \mathrm{tr}(A^2) - \left(\frac{1}{9} \mathrm{tr}(A)\right)^2.$$

□

245 Characterization by Vanishing Variance

Let

$$A = A^* \in \mathcal{A}_9.$$

Theorem 245.1. *The following conditions are equivalent:*

$$\text{Var}_{\rho_9}(A) = 0,$$

$$A = \lambda I_{\mathcal{H}_9} \quad \text{for some } \lambda \in \mathbb{R}.$$

Proof. If

$$A = \lambda I_{\mathcal{H}_9},$$

then

$$\text{tr}(\rho_9 A) = \lambda$$

and

$$\text{tr}(\rho_9 A^2) = \lambda^2,$$

so the variance vanishes.

Conversely, suppose

$$\text{Var}_{\rho_9}(A) = 0.$$

Set

$$\lambda = \text{tr}(\rho_9 A).$$

Then

$$0 = \text{tr}(\rho_9 (A - \lambda I)^2).$$

Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

this becomes

$$0 = \frac{1}{9} \text{tr}((A - \lambda I)^2).$$

The operator

$$(A - \lambda I)^2$$

is positive. A positive operator with zero trace in finite dimension is zero. Hence

$$(A - \lambda I)^2 = 0.$$

Because $A - \lambda I$ is self-adjoint, this implies

$$A - \lambda I = 0.$$

Therefore

$$A = \lambda I_{\mathcal{H}_9}.$$

□

Corollary 245.2. *The maximally symmetric state is faithful on $\mathcal{A}_9$: for every positive operator*

$$X \geq 0,$$

$$\text{tr}(\rho_9 X) = 0 \implies X = 0.$$

Proof. One has

$$\text{tr}(\rho_9 X) = \frac{1}{9} \text{tr}(X).$$

If this quantity vanishes, positivity of X implies

$$X = 0.$$

□

246 The GNS Realization of the Stable State

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

and let

$$\tau_9(X) = \frac{1}{9} \text{tr}(X)$$

be the normalized trace.

Define

$$\langle X, Y \rangle_{\tau_9} = \tau_9(Y^* X), \quad X, Y \in \mathcal{A}_9.$$

Proposition 246.1. *The form*

$$\langle \cdot, \cdot \rangle_{\tau_9}$$

is an inner product on $\mathcal{A}_9$.

Proof. Sesquilinearity and conjugate symmetry follow from the corresponding properties of the Hilbert–Schmidt inner product. Moreover,

$$\langle X, X \rangle_{\tau_9} = \frac{1}{9} \text{tr}(X^* X).$$

Since $X^* X$ is positive,

$$\text{tr}(X^* X) \geq 0.$$

Equality holds if and only if

$$X^* X = 0,$$

which is equivalent to

$$X = 0.$$

Thus the form is positive definite. □

Theorem 246.2. *The GNS Hilbert space associated with*

$$(\mathcal{A}_9, \tau_9)$$

is canonically identified, as a Hilbert space, with

$$\mathcal{A}_9$$

equipped with the normalized Hilbert–Schmidt inner product

$$\langle X, Y \rangle_{\tau_9} = \frac{1}{9} \text{tr}(Y^* X).$$

Its complex dimension is

$$81.$$

Proof. Because τ_9 is faithful, its null space

$$\mathcal{N}_{\tau_9} = \{X \in \mathcal{A}_9 : \tau_9(X^* X) = 0\}$$

is trivial. Hence the GNS quotient is already

$$\mathcal{A}_9.$$

Since

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 9^2 = 81,$$

the resulting Hilbert space has dimension 81. □

Proposition 246.3. *The left regular representation*

$$\pi_{\tau_9}(A)X = AX$$

defines a faithful representation of $\mathcal{A}_9$ on its GNS Hilbert space.

Proof. For

$$A, X \in \mathcal{A}_9,$$

one has

$$\pi_{\tau_9}(A)X = AX.$$

If

$$\pi_{\tau_9}(A) = 0,$$

then in particular

$$AI_{\mathcal{H}_9} = 0,$$

so

$$A = 0.$$

Thus the representation is faithful. □

247 The Commutant and the Standard Finite Representation

Let

$$L_A(X) = AX, \quad R_A(X) = XA$$

denote respectively left and right multiplication on $\mathcal{A}_9$.

Proposition 247.1. *For all*

$$A, B, X \in \mathcal{A}_9,$$

one has

$$L_AR_B(X) = R_B L_A(X).$$

Proof. Direct computation gives

$$L_AR_B(X) = A(XB) = (AX)B = R_B L_A(X).$$

□

Theorem 247.2. *The commutant of the left multiplication representation of $\mathcal{A}_9$ on itself is the right multiplication algebra:*

$$\boxed{\{L_A : A \in \mathcal{A}_9\}' = \{R_B : B \in \mathcal{A}_9\}.$$

Proof. The inclusion

$$\{R_B : B \in \mathcal{A}_9\} \subseteq \{L_A : A \in \mathcal{A}_9\}'$$

follows from the preceding proposition.

Conversely, let

$$S \in \mathcal{B}(\mathcal{A}_9)$$

commute with every left multiplication operator. Set

$$B = S(I_{\mathcal{H}_9}).$$

For every

$$X \in \mathcal{A}_9,$$

one has

$$S(X) = S(L_X I_{\mathcal{H}_9}) = L_X S(I_{\mathcal{H}_9}) = XB = R_B(X).$$

Hence

$$S = R_B.$$

□

248 The Modular Operator of the Maximally Symmetric State

Let

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

On the GNS Hilbert space $\mathcal{A}_9$, define the modular operator by

$$\Delta_{\rho_9}(X) = \rho_9 X \rho_9^{-1}.$$

Proposition 248.1. *The modular operator of the maximally symmetric state is the identity:*

$$\Delta_{\rho_9} = I_{\mathcal{A}_9}.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}, \quad \rho_9^{-1} = 9 I_{\mathcal{H}_9},$$

one obtains

$$\Delta_{\rho_9}(X) = \frac{1}{9} I_{\mathcal{H}_9} X 9 I_{\mathcal{H}_9} = X.$$

Thus

$$\Delta_{\rho_9} = I_{\mathcal{A}_9}.$$

□

Corollary 248.2. *The modular automorphism group associated with ρ_9 is trivial:*

$$\sigma_t^{\rho_9}(X) = X \quad (t \in \mathbb{R}, X \in \mathcal{A}_9).$$

Proof. The modular automorphism group is given by

$$\sigma_t^{\rho_9}(X) = \rho_9^{it} X \rho_9^{-it}.$$

Because ρ_9 is a scalar multiple of the identity,

$$\rho_9^{it} = 9^{-it} I_{\mathcal{H}_9},$$

and therefore

$$\sigma_t^{\rho_9}(X) = X.$$

□

249 Entropy of the Stable State

Let

$$S(\rho) = -\operatorname{tr}(\rho \log \rho)$$

denote the von Neumann entropy of a density operator.

Theorem 249.1. *The entropy of the maximally symmetric state on the nine-dimensional stable sector is*

$$S(\rho_9) = \log 9.$$

Proof. The only eigenvalue of ρ_9 is

$$\frac{1}{9}$$

with multiplicity 9. Hence

$$S(\rho_9) = -9 \left(\frac{1}{9} \right) \log \left(\frac{1}{9} \right) = \log 9.$$

□

Theorem 249.2. *Among all density operators on $\mathcal{H}_9$, the entropy satisfies*

$$S(\rho) \leq \log 9,$$

with equality if and only if

$$\rho = \rho_9.$$

Proof. Let

$$\lambda_1, \dots, \lambda_9$$

be the eigenvalues of ρ . They satisfy

$$\lambda_j \geq 0, \quad \sum_{j=1}^9 \lambda_j = 1.$$

The entropy is

$$S(\rho) = - \sum_{j=1}^9 \lambda_j \log \lambda_j.$$

The strict concavity of

$$-x \log x$$

on $(0, \infty)$, together with the normalization constraint, implies

$$- \sum_{j=1}^9 \lambda_j \log \lambda_j \leq \log 9.$$

Equality occurs precisely when

$$\lambda_1 = \dots = \lambda_9 = \frac{1}{9}.$$

Hence

$$\rho = \frac{1}{9} I_{\mathcal{H}_9} = \rho_9.$$

□

Corollary 249.3. *The state ρ_9 is simultaneously characterized by any of the following equivalent properties:*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

$$U \rho_9 U^* = \rho_9 \quad \forall U \in \mathcal{U}(\mathcal{H}_9),$$

$$S(\rho_9) = \max_{\rho} S(\rho),$$

and

$$\Delta_{\rho_9} = I_{\mathcal{A}_9}.$$

250 Relative Entropy from the Stable State

For density operators ρ, σ on $\mathcal{H}_9$, with $\text{supp}(\rho) \subseteq \text{supp}(\sigma)$, define the relative entropy by

$$D(\rho \parallel \sigma) = \text{tr}(\rho(\log \rho - \log \sigma)).$$

Theorem 250.1. *For every density operator ρ on $\mathcal{H}_9$,*

$$D(\rho \parallel \rho_9) = \log 9 - S(\rho).$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\log \rho_9 = -\log 9 I_{\mathcal{H}_9}.$$

Therefore

$$D(\rho \parallel \rho_9) = \text{tr}(\rho \log \rho) + \log 9 \text{tr}(\rho).$$

Because

$$\text{tr}(\rho) = 1$$

and

$$S(\rho) = -\text{tr}(\rho \log \rho),$$

it follows that

$$D(\rho \parallel \rho_9) = \log 9 - S(\rho).$$

□

Corollary 250.2. *The entropy bound*

$$S(\rho) \leq \log 9$$

is equivalent to the nonnegativity of relative entropy:

$$D(\rho \parallel \rho_9) \geq 0.$$

Equality holds if and only if

$$\rho = \rho_9.$$

251 Finite-State Symmetry and Ergodic Averaging

Let

$$G$$

be a finite group with a unitary representation

$$U : G \longrightarrow \mathcal{U}(\mathcal{H}_g).$$

For

$$X \in \mathcal{A}_g,$$

define the group average

$$\mathcal{E}_G(X) = \frac{1}{|G|} \sum_{g \in G} U(g) X U(g)^*.$$

Proposition 251.1. *The averaging map $\mathcal{E}_G$ is unital, positive, and trace-preserving.*

Proof. Unitality follows from

$$\mathcal{E}_G(I) = \frac{1}{|G|} \sum_{g \in G} U(g) U(g)^* = I.$$

Each map

$$X \mapsto U(g) X U(g)^*$$

is positive, so their average is positive. Finally,

$$\mathrm{tr}(U(g) X U(g)^*) = \mathrm{tr}(X)$$

by cyclicity of the trace. Averaging over g preserves the trace. □

Theorem 251.2. *The averaging map satisfies*

$$\mathcal{E}_G^2 = \mathcal{E}_G$$

and

$$\mathcal{E}_G(X)$$

is invariant under the representation of G .

Proof. For $h \in G$,

$$U(h) \mathcal{E}_G(X) U(h)^* = \frac{1}{|G|} \sum_{g \in G} U(hg) X U(hg)^*.$$

Since

$$g \mapsto hg$$

is a permutation of G ,

$$U(h) \mathcal{E}_G(X) U(h)^* = \mathcal{E}_G(X).$$

Thus the range of $\mathcal{E}_G$ consists of invariant operators. Applying the averaging map to an invariant operator leaves it unchanged, and hence

$$\mathcal{E}_G^2 = \mathcal{E}_G.$$

□

Corollary 251.3. *The maximally symmetric state satisfies*

$$\mathcal{E}_G(\rho_9) = \rho_9$$

for every finite unitary symmetry group G .

Proof. Each conjugation fixes the scalar operator ρ_9 :

$$U(g)\rho_9U(g)^* = \rho_9.$$

Therefore their average is also ρ_9 . □

252 The Stable Sector as a Representation Space

Let

$$\pi : G \longrightarrow \mathcal{U}(V_9)$$

be a unitary representation satisfying

$$\pi(g)P_9 = P_9\pi(g) \quad (g \in G).$$

Proposition 252.1. *The stable sector*

$$\mathcal{H}_9 = \text{range}(P_9)$$

is invariant under $\pi(G)$.

Proof. Let

$$v \in \mathcal{H}_9.$$

Then

$$P_9 v = v.$$

Consequently,

$$P_9\pi(g)v = \pi(g)P_9v = \pi(g)v.$$

Hence

$$\pi(g)v \in \text{range}(P_9) = \mathcal{H}_9.$$

□

Theorem 252.2. *The restriction*

$$\pi_9(g) = \pi(g)|_{\mathcal{H}_9}$$

defines a unitary representation of G on $\mathcal{H}_9$.

Proof. By the preceding proposition, each $\pi(g)$ preserves $\mathcal{H}_9$. Since $\pi(g)$ is unitary on V_9 , its restriction to the invariant subspace $\mathcal{H}_9$ is unitary. Moreover,

$$\pi_9(g_1g_2) = \pi(g_1g_2)|_{\mathcal{H}_9} = \pi(g_1)\pi(g_2)|_{\mathcal{H}_9} = \pi_9(g_1)\pi_9(g_2).$$

Thus π_9 is a unitary representation. □

Corollary 252.3. *The stable state*

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

is invariant under the restricted representation:

$$\pi_9(g)\rho_9\pi_9(g)^* = \rho_9 \quad (g \in G).$$

Proof. The operator ρ_9 is a scalar multiple of the identity on $\mathcal{H}_9$, so it is invariant under every unitary conjugation. □

253 The Commutant of the Stable Representation

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

and define the commutant of the restricted representation by

$$\mathcal{A}_9^G = \{A \in \mathcal{A}_9 : A\pi_9(g) = \pi_9(g)A \text{ for all } g \in G\}.$$

Proposition 253.1. *The space*

$$\mathcal{A}_9^G$$

is a unital subalgebra of $\mathcal{A}_9$.

Proof. Let

$$A, B \in \mathcal{A}_9^G$$

and

$$\alpha, \beta \in \mathbb{C}.$$

For every $g \in G$,

$$(\alpha A + \beta B)\pi_9(g) = \alpha A\pi_9(g) + \beta B\pi_9(g) = \alpha\pi_9(g)A + \beta\pi_9(g)B,$$

hence

$$(\alpha A + \beta B)\pi_9(g) = \pi_9(g)(\alpha A + \beta B).$$

Similarly,

$$(AB)\pi_9(g) = A\pi_9(g)B = \pi_9(g)AB.$$

Finally,

$$I_{\mathcal{H}_9}\pi_9(g) = \pi_9(g) = \pi_9(g)I_{\mathcal{H}_9}.$$

Thus $\mathcal{A}_9^G$ is a unital subalgebra. □

Proposition 253.2. *The density operator*

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

belongs to $\mathcal{A}_9^G$.

Proof. For every $g \in G$,

$$\rho_9\pi_9(g) = \frac{1}{9}\pi_9(g) = \pi_9(g)\rho_9.$$

Hence

$$\rho_9 \in \mathcal{A}_9^G.$$

□

254 Isotypic Decomposition of the Stable Representation

Assume that G is compact. Let

$$\widehat{G}$$

denote the set of equivalence classes of irreducible finite-dimensional unitary representations of G .

Theorem 254.1. *The finite-dimensional unitary representation*

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

admits an orthogonal decomposition

$$\mathcal{H}_9 = \bigoplus_{\alpha \in \widehat{G}} \mathcal{H}_\alpha,$$

where each $\mathcal{H}_\alpha$ is the isotypic component associated with an irreducible representation class α , and only finitely many components are nonzero.

Proof. Since G is compact and π_9 is finite-dimensional and unitary, complete reducibility applies. Thus $\mathcal{H}_9$ decomposes as an orthogonal direct sum of irreducible invariant subspaces. Grouping equivalent irreducible summands yields the stated isotypic decomposition. $\square$

Corollary 254.2. *There exist pairwise inequivalent irreducible unitary representations*

$$\sigma_1, \dots, \sigma_s$$

of G , with dimensions

$$d_1, \dots, d_s,$$

and multiplicities

$$m_1, \dots, m_s,$$

such that

$$\mathcal{H}_9 \cong \bigoplus_{\alpha=1}^s \left(\mathbb{C}^{m_\alpha} \otimes \mathbb{C}^{d_\alpha} \right),$$

and

$$\sum_{\alpha=1}^s m_\alpha d_\alpha = 9.$$

Proof. This is the standard multiplicity-space form of the finite-dimensional isotypic decomposition. $\square$

255 Structure of the Representation Commutant

With respect to the decomposition

$$\mathcal{H}_9 \cong \bigoplus_{\alpha=1}^s \left(\mathbb{C}^{m_\alpha} \otimes \mathbb{C}^{d_\alpha} \right),$$

the representation takes the form

$$\pi_9(g) = \bigoplus_{\alpha=1}^s \left(I_{m_\alpha} \otimes \sigma_\alpha(g) \right).$$

Theorem 255.1. *The commutant of the stable representation is*

$$\mathcal{A}_9^G \cong \bigoplus_{\alpha=1}^s M_{m_\alpha}(\mathbb{C}) \otimes I_{d_\alpha}.$$

Proof. Let

$$A \in \mathcal{A}_9^G.$$

Write A in block form with respect to the isotypic decomposition. If $\alpha \neq \beta$, an off-diagonal block

$$A_{\alpha\beta} : \mathbb{C}^{m_\beta} \otimes \mathbb{C}^{d_\beta} \longrightarrow \mathbb{C}^{m_\alpha} \otimes \mathbb{C}^{d_\alpha}$$

would intertwine inequivalent irreducible representations σ_β and σ_α . By Schur's lemma,

$$A_{\alpha\beta} = 0.$$

Thus A is block diagonal.

On a fixed isotypic component,

$$\mathbb{C}^{m_\alpha} \otimes \mathbb{C}^{d_\alpha},$$

the commutation relation is

$$A_\alpha(I_{m_\alpha} \otimes \sigma_\alpha(g)) = (I_{m_\alpha} \otimes \sigma_\alpha(g))A_\alpha.$$

Again by Schur's lemma, the commutant of the irreducible factor $\sigma_\alpha(G)$ is scalar on the irreducible space. Hence

$$A_\alpha = B_\alpha \otimes I_{d_\alpha}$$

for some

$$B_\alpha \in M_{m_\alpha}(\mathbb{C}).$$

Therefore

$$\mathcal{A}_9^G \cong \bigoplus_{\alpha=1}^s M_{m_\alpha}(\mathbb{C}) \otimes I_{d_\alpha}.$$

□

Corollary 255.2. *The dimension of the commutant is*

$$\dim_{\mathbb{C}} \mathcal{A}_9^G = \sum_{\alpha=1}^s m_\alpha^2.$$

Proof. Since

$$\dim_{\mathbb{C}} M_{m_\alpha}(\mathbb{C}) = m_\alpha^2,$$

and the direct sum is algebraic, the asserted formula follows. □

256 The Multiplicity-Free Case

Definition 256.1. The representation π_9 is called multiplicity-free if

$$m_\alpha = 1$$

for every irreducible representation occurring in its decomposition.

Proposition 256.2. *If π_9 is multiplicity-free, then*

$$\mathcal{A}_9^G$$

is commutative.

Proof. If every multiplicity equals one, then

$$M_{m_\alpha}(\mathbb{C}) = \mathbb{C}$$

for every α . Hence

$$\mathcal{A}_9^G \cong \bigoplus_{\alpha=1}^s \mathbb{C} I_{d_\alpha},$$

which is commutative. □

Corollary 256.3. *If π_9 is irreducible, then*

$$\mathcal{A}_9^G = \mathbb{C} I_{\mathcal{H}_9}.$$

Proof. Irreducibility means

$$s = 1, \quad m_1 = 1.$$

The commutant theorem therefore gives

$$\mathcal{A}_9^G = \mathbb{C} I_{\mathcal{H}_9}.$$

□

257 Invariant Operators and Schur's Lemma

Assume from now on that π_9 is irreducible.

Theorem 257.1. *Every operator*

$$A \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

satisfying

$$A\pi_9(g) = \pi_9(g)A \quad (g \in G)$$

is a scalar multiple of the identity:

$$A = \lambda I_{\mathcal{H}_9}$$

for some

$$\lambda \in \mathbb{C}.$$

Proof. This is Schur's lemma applied to the irreducible complex representation π_9 . □

Corollary 257.2. *Every self-adjoint operator in the commutant has the form*

$$A = \lambda I_{\mathcal{H}_9}, \quad \lambda \in \mathbb{R}.$$

Corollary 257.3. *Every positive operator in the commutant has the form*

$$A = \lambda I_{\mathcal{H}_9}, \quad \lambda \geq 0.$$

Theorem 257.4. *The only G -invariant density operator on the irreducible stable sector is*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. Let ρ be a G -invariant density operator. Its invariance means

$$\pi_9(g)\rho\pi_9(g)^* = \rho.$$

Multiplying on the right by $\pi_9(g)$ gives

$$\pi_9(g)\rho = \rho\pi_9(g).$$

Thus

$$\rho \in \mathcal{A}_9^G.$$

By irreducibility,

$$\rho = \lambda I_{\mathcal{H}_9}.$$

Since

$$\text{tr}(\rho) = 1$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

one obtains

$$1 = 9\lambda.$$

Hence

$$\lambda = \frac{1}{9}.$$

Therefore

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

258 Invariant Vectors

Let

$$\mathcal{H}_9^G = \{v \in \mathcal{H}_9 : \pi_9(g)v = v \text{ for all } g \in G\}.$$

Proposition 258.1. *The invariant subspace*

$$\mathcal{H}_9^G$$

is a closed linear subspace of $\mathcal{H}_9$.

Proof. For every $g \in G$,

$$\ker(\pi_9(g) - I)$$

is a linear subspace. Since

$$\mathcal{H}_9^G = \bigcap_{g \in G} \ker(\pi_9(g) - I),$$

it is a linear subspace. Because $\mathcal{H}_9$ is finite-dimensional, every linear subspace is closed. □

Theorem 258.2. *If π_9 is irreducible, then*

$$\mathcal{H}_9^G = \{0\}$$

unless π_9 is the trivial representation.

Proof. The invariant subspace $\mathcal{H}_9^G$ is invariant under $\pi_9(G)$. Irreducibility therefore implies

$$\mathcal{H}_9^G = \{0\}$$

or

$$\mathcal{H}_9^G = \mathcal{H}_9.$$

In the latter case,

$$\pi_9(g) = I_{\mathcal{H}_9}$$

for every $g \in G$, so the representation is trivial. □

259 Averaging over the Group

Assume that G is compact and let

$$dg$$

denote its normalized Haar measure.

Define the averaging operator

$$\mathfrak{P}_G = \int_G \pi_9(g) dg.$$

Proposition 259.1. *The operator $\mathfrak{P}_G$ is self-adjoint.*

Proof. Since $\pi_9(g)$ is unitary,

$$\pi_9(g)^* = \pi_9(g^{-1}).$$

Therefore

$$\mathfrak{P}_G^* = \int_G \pi_9(g)^* dg = \int_G \pi_9(g^{-1}) dg.$$

Haar measure is invariant under inversion, hence

$$\mathfrak{P}_G^* = \int_G \pi_9(g) dg = \mathfrak{P}_G.$$

□

Proposition 259.2. *The averaging operator is idempotent:*

$$\mathfrak{P}_G^2 = \mathfrak{P}_G.$$

Proof. Using Fubini's theorem and the representation law,

$$\mathfrak{P}_G^2 = \int_G \int_G \pi_9(g) \pi_9(h) dg dh = \int_G \int_G \pi_9(gh) dg dh.$$

For fixed h , left invariance of Haar measure gives

$$\int_G \pi_9(gh) dg = \int_G \pi_9(k) dk = \mathfrak{P}_G.$$

Since the Haar measure is normalized,

$$\mathfrak{P}_G^2 = \mathfrak{P}_G.$$

□

Theorem 259.3. *The operator $\mathfrak{P}_G$ is the orthogonal projection onto $\mathcal{H}_9^G$:*

$$\text{range}(\mathfrak{P}_G) = \mathcal{H}_9^G.$$

Proof. For $v \in \mathcal{H}_9$ and $h \in G$,

$$\pi_9(h)\mathfrak{P}_G v = \int_G \pi_9(hg) v dg.$$

By left invariance of Haar measure,

$$\int_G \pi_9(hg) v dg = \int_G \pi_9(g) v dg = \mathfrak{P}_G v.$$

Thus

$$\mathfrak{P}_G v \in \mathcal{H}_9^G.$$

Hence

$$\text{range}(\mathfrak{P}_G) \subseteq \mathcal{H}_9^G.$$

Conversely, if

$$v \in \mathcal{H}_9^G,$$

then

$$\mathfrak{P}_G v = \int_G v dg = v.$$

Thus every invariant vector belongs to the range of $\mathfrak{P}_G$. Therefore

$$\text{range}(\mathfrak{P}_G) = \mathcal{H}_9^G.$$

Since $\mathfrak{P}_G$ is self-adjoint and idempotent, it is the orthogonal projection onto this subspace. $\square$

Corollary 259.4. *If π_9 is irreducible and nontrivial, then*

$$\mathfrak{P}_G = 0.$$

If π_9 is trivial, then

$$\mathfrak{P}_G = I_{\mathcal{H}_9}.$$

260 Character of the Stable Representation

Define the character of π_9 by

$$\chi_{\pi_9}(g) = \text{tr}(\pi_9(g)).$$

Proposition 260.1. *The character satisfies*

$$\chi_{\pi_9}(e) = 9$$

and

$$|\chi_{\pi_9}(g)| \leq 9 \quad (g \in G).$$

Proof. Since

$$\pi_9(e) = I_{\mathcal{H}_9},$$

one has

$$\chi_{\pi_9}(e) = \text{tr}(I_{\mathcal{H}_9}) = 9.$$

Every $\pi_9(g)$ is unitary, so all its eigenvalues have modulus one. Writing them as

$$\lambda_1, \dots, \lambda_9,$$

one obtains

$$\chi_{\pi_9}(g) = \sum_{j=1}^9 \lambda_j.$$

Hence

$$|\chi_{\pi_9}(g)| \leq \sum_{j=1}^9 |\lambda_j| = 9.$$

□

Theorem 260.2. *The multiplicity of the trivial representation in π_9 is*

$$\dim_{\mathbb{C}} \mathcal{H}_9^G = \int_G \chi_{\pi_9}(g) dg.$$

Proof. By linearity of the trace,

$$\int_G \chi_{\pi_9}(g) dg = \text{tr} \left(\int_G \pi_9(g) dg \right) = \text{tr}(\mathfrak{P}_G).$$

Since $\mathfrak{P}_G$ is the orthogonal projection onto $\mathcal{H}_9^G$,

$$\text{tr}(\mathfrak{P}_G) = \dim_{\mathbb{C}} \mathcal{H}_9^G.$$

□

261 Character Orthogonality and Multiplicity

Assume again that G is compact. Let

$$\sigma_\alpha, \sigma_\beta$$

be irreducible unitary representations with characters

$$\chi_\alpha, \chi_\beta.$$

Theorem 261.1. *The irreducible characters satisfy*

$$\int_G \chi_\alpha(g) \overline{\chi_\beta(g)} dg = \delta_{\alpha\beta}.$$

Proof. This is the orthogonality relation for irreducible characters of a compact group with normalized Haar measure. □

Corollary 261.2. *If*

$$\pi_9 \cong \bigoplus_{\alpha=1}^s m_\alpha \sigma_\alpha,$$

then

$$\chi_{\pi_9} = \sum_{\alpha=1}^s m_\alpha \chi_\alpha,$$

and

$$m_\alpha = \int_G \chi_{\pi_9}(g) \overline{\chi_\alpha(g)} dg.$$

Proof. The character is additive under direct sums, so

$$\chi_{\pi_9} = \sum_{\beta=1}^s m_\beta \chi_\beta.$$

Taking the inner product with χ_α and using character orthogonality gives

$$\int_G \chi_{\pi_9}(g) \overline{\chi_\alpha(g)} dg = \sum_{\beta=1}^s m_\beta \delta_{\alpha\beta} = m_\alpha.$$

□

262 Dimension Constraint on the Stable Representation

Let

$$\pi_9 \cong \bigoplus_{\alpha=1}^s m_\alpha \sigma_\alpha$$

be the irreducible decomposition of the stable representation.

Theorem 262.1. *The dimensions and multiplicities satisfy the exact identity*

$$\sum_{\alpha=1}^s m_\alpha d_\alpha = 9,$$

where

$$d_\alpha = \dim_{\mathbb{C}} \sigma_\alpha.$$

Proof. Dimension is additive under direct sums. Since each copy of σ_α has dimension d_α and occurs m_α times,

$$\dim_{\mathbb{C}} \mathcal{H}_9 = \sum_{\alpha=1}^s m_\alpha d_\alpha.$$

Because

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

the result follows. □

Corollary 262.2. *Only finitely many irreducible representation types can occur in the stable sector, and their multiplicity data form a finite partition of the dimension 9.*

Remark 262.3. The dimension identity alone does not determine the representation type. In particular, the conclusion that the stable sector is nine-dimensional does not by itself imply irreducibility. Any such identification requires an additional hypothesis or an independent argument from the finite relational construction.

263 The Complementary Representation Sector

Let

$$K_9 = \ker(P_9) \subseteq V_9.$$

Proposition 263.1. *The subspace K_9 is invariant under the representation π .*

Proof. Let

$$v \in K_9.$$

Then

$$P_9 v = 0.$$

Using the commutation relation

$$P_9 \pi(g) = \pi(g) P_9,$$

we obtain

$$P_9 \pi(g) v = \pi(g) P_9 v = 0.$$

Hence

$$\pi(g) v \in \ker(P_9) = K_9.$$

□

Theorem 263.2. *There is a $\pi(G)$ -invariant direct-sum decomposition*

$$V_9 = \mathcal{H}_9 \oplus K_9.$$

With respect to this decomposition,

$$\pi(g) = \pi_9(g) \oplus \pi_{K_9}(g),$$

where

$$\pi_{K_9}(g) = \pi(g)|_{K_9}.$$

Proof. Since

$$P_9^2 = P_9,$$

every vector $v \in V_9$ admits the decomposition

$$v = P_9 v + (I - P_9) v.$$

The first summand belongs to

$$\text{range}(P_9) = \mathcal{H}_9,$$

while

$$P_9(I - P_9)v = (P_9 - P_9^2)v = 0,$$

so the second summand belongs to

$$\ker(P_9) = K_9.$$

If

$$v \in \mathcal{H}_9 \cap K_9,$$

then $P_9 v = v$ and $P_9 v = 0$, hence $v = 0$. Therefore

$$V_9 = \mathcal{H}_9 \oplus K_9.$$

Both summands are invariant by the preceding results. Consequently, $\pi(g)$ is block diagonal with respect to this decomposition, giving the stated formula. □

Remark 263.3. The decomposition above is algebraic and does not require P_9 to be self-adjoint. If, in addition,

$$P_9 = P_9^*,$$

then

$$K_9 = \mathcal{H}_9^\perp$$

and the decomposition becomes orthogonal:

$$V_9 = \mathcal{H}_9 \oplus \mathcal{H}_9^\perp.$$

264 Dimension of the Complementary Sector

Assume

$$\dim_{\mathbb{C}} V_9 = 143$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Theorem 264.1. *The complementary sector has dimension*

$$\dim_{\mathbb{C}} K_9 = 134.$$

Proof. From

$$V_9 = \mathcal{H}_9 \oplus K_9$$

one obtains

$$\dim_{\mathbb{C}} V_9 = \dim_{\mathbb{C}} \mathcal{H}_9 + \dim_{\mathbb{C}} K_9.$$

Therefore

$$143 = 9 + \dim_{\mathbb{C}} K_9,$$

and hence

$$\dim_{\mathbb{C}} K_9 = 134.$$

□

Corollary 264.2. *The representation π decomposes into invariant sectors of dimensions 9 and 134:*

$$\pi = \pi_9 \oplus \pi_{K_9}.$$

265 Complete Reducibility of the Stable Representation

Let

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be the unitary representation obtained by restriction.

Theorem 265.1. *The representation π_9 admits a decomposition*

$$\mathcal{H}_9 = \mathcal{H}^{(1)} \oplus \cdots \oplus \mathcal{H}^{(s)}$$

into irreducible G -invariant subspaces.

Proof. If π_9 is irreducible, there is nothing to prove. Otherwise there exists a nonzero proper invariant subspace

$$\mathcal{H}^{(1)} \subsetneq \mathcal{H}_9.$$

Because $\pi_9(g)$ is unitary for every $g \in G$, the orthogonal complement

$$(\mathcal{H}^{(1)})^\perp$$

is also invariant. Indeed, for

$$v \in (\mathcal{H}^{(1)})^\perp, \quad w \in \mathcal{H}^{(1)},$$

one has

$$\langle \pi_9(g)v, w \rangle = \langle v, \pi_9(g)^{-1}w \rangle = 0,$$

because

$$\pi_9(g)^{-1}w \in \mathcal{H}^{(1)}.$$

Thus

$$\mathcal{H}_9 = \mathcal{H}^{(1)} \oplus (\mathcal{H}^{(1)})^\perp$$

as an orthogonal direct sum of invariant subspaces.

Repeating the argument on the remaining finite-dimensional invariant subspace yields, after finitely many steps, a decomposition into irreducible invariant subspaces. $\square$

Corollary 265.2. *There exist irreducible unitary representations*

$$\pi^{(1)}, \dots, \pi^{(s)}$$

such that

$$\pi_9 \cong \pi^{(1)} \oplus \dots \oplus \pi^{(s)}$$

and

$$\sum_{a=1}^s \dim_{\mathbb{C}} \mathcal{H}^{(a)} = 9.$$

266 The Commutant of the Stable Representation

Define the commutant

$$\pi_9(G)' = \{X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9) : X\pi_9(g) = \pi_9(g)X \text{ for all } g \in G\}.$$

Proposition 266.1. *The commutant $\pi_9(G)'$ is a unital subalgebra of $\text{End}_{\mathbb{C}}(\mathcal{H}_9)$.*

Proof. The identity operator commutes with every $\pi_9(g)$. If $X, Y \in \pi_9(G)'$ and $a, b \in \mathbb{C}$, then

$$(aX + bY)\pi_9(g) = \pi_9(g)(aX + bY).$$

Moreover,

$$(XY)\pi_9(g) = X\pi_9(g)Y = \pi_9(g)XY.$$

Thus the commutant is closed under linear combinations and products. $\square$

Proposition 266.2. *The density operator*

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

belongs to the commutant:

$$\rho_9 \in \pi_9(G)'.$$

Proof. For every $g \in G$,

$$\rho_9 \pi_9(g) = \frac{1}{9} \pi_9(g) = \pi_9(g) \rho_9.$$

Hence

$$\rho_9 \in \pi_9(G)'. \quad \square$$

267 Invariant Density Operators and the Commutant

Let

$$\rho \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be a density operator satisfying

$$\pi_9(g) \rho \pi_9(g)^* = \rho \quad (g \in G).$$

Proposition 267.1. *The invariance condition is equivalent to*

$$\rho \pi_9(g) = \pi_9(g) \rho \quad (g \in G).$$

Consequently,

$$\rho \in \pi_9(G)'.$$

Proof. Since $\pi_9(g)$ is unitary,

$$\pi_9(g)^* = \pi_9(g)^{-1}.$$

Multiplying

$$\pi_9(g) \rho \pi_9(g)^* = \rho$$

on the right by $\pi_9(g)$ gives

$$\pi_9(g) \rho = \rho \pi_9(g).$$

Thus ρ commutes with every representation operator. $\square$

Theorem 267.2. *If the stable representation π_9 is irreducible, then every density operator invariant under $\pi_9(G)$ is equal to*

$$\frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. Let ρ be invariant under $\pi_9(G)$. By the preceding proposition,

$$\rho \in \pi_9(G)'.$$

Since π_9 is irreducible over $\mathbb{C}$, Schur's lemma gives

$$\pi_9(G)' = \mathbb{C} I_{\mathcal{H}_9}.$$

Therefore

$$\rho = \lambda I_{\mathcal{H}_9}$$

for some

$$\lambda \geq 0.$$

The trace normalization gives

$$1 = \text{tr}(\rho) = 9\lambda.$$

Hence

$$\lambda = \frac{1}{9},$$

and consequently

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Corollary 267.3. *If the stable representation is irreducible, the maximally symmetric state ρ_9 is the unique G -invariant density operator on $\mathcal{H}_9$.*

268 Invariant Observables in the Irreducible Case

Assume that

$$\pi_9$$

is irreducible.

Theorem 268.1. *Every operator*

$$A \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

satisfying

$$A\pi_9(g) = \pi_9(g)A \quad (g \in G)$$

is a scalar multiple of the identity:

$$A = \lambda I_{\mathcal{H}_9}$$

for some

$$\lambda \in \mathbb{C}.$$

Proof. The commutation relation states that

$$A \in \pi_9(G)'$$

Irreducibility and Schur's lemma imply

$$\pi_9(G)' = \mathbb{C} I_{\mathcal{H}_9}.$$

Therefore

$$A = \lambda I_{\mathcal{H}_9}.$$

□

Corollary 268.2. *Every self-adjoint operator invariant under the irreducible stable representation has the form*

$$A = \lambda I_{\mathcal{H}_9}, \quad \lambda \in \mathbb{R}.$$

269 The Stable State as a Central State

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Proposition 269.1. *The density operator*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

lies in the center of $\mathcal{A}_9$.

Proof. For every

$$X \in \mathcal{A}_9,$$

one has

$$\rho_9 X = \frac{1}{9} X = X \rho_9.$$

Hence

$$\rho_9 \in Z(\mathcal{A}_9).$$

□

Theorem 269.2. *The state*

$$\omega_9(X) = \text{tr}(\rho_9 X)$$

is the normalized trace state on $\mathcal{A}_9$, namely

$$\omega_9(X) = \frac{1}{9} \text{tr}(X).$$

Moreover,

$$\omega_9(XY) = \omega_9(YX)$$

for all

$$X, Y \in \mathcal{A}_9.$$

Proof. Substitution of

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

gives

$$\omega_9(X) = \frac{1}{9} \text{tr}(X).$$

Cyclicity of the finite-dimensional trace yields

$$\omega_9(XY) = \frac{1}{9} \text{tr}(XY) = \frac{1}{9} \text{tr}(YX) = \omega_9(YX).$$

□

270 Unitary Covariance of the Stable State

Let

$$U \in \mathcal{U}(\mathcal{H}_9).$$

Proposition 270.1. *For every*

$$X \in \mathcal{A}_9,$$

the state ω_9 satisfies

$$\omega_9(UXU^*) = \omega_9(X).$$

Proof. Using the trace representation,

$$\omega_9(UXU^*) = \frac{1}{9} \operatorname{tr}(UXU^*).$$

By cyclicity of the trace,

$$\operatorname{tr}(UXU^*) = \operatorname{tr}(XU^*U) = \operatorname{tr}(X).$$

Therefore

$$\omega_9(UXU^*) = \omega_9(X).$$

□

Corollary 270.2. *The state ω_9 is invariant not only under the restricted representation $\pi_9(G)$, but under the full unitary group $\mathcal{U}(\mathcal{H}_9)$.*

271 State Evaluation of Spectral Projections

Let

$$E \in \mathcal{A}_9$$

be an orthogonal projection of rank

$$r.$$

Proposition 271.1. *The stable state evaluates E according to*

$$\omega_9(E) = \frac{r}{9}.$$

Proof. Since

$$E^2 = E = E^*,$$

the eigenvalues of E are 0 and 1, and the multiplicity of 1 is its rank. Hence

$$\operatorname{tr}(E) = r.$$

Therefore

$$\omega_9(E) = \frac{1}{9} \operatorname{tr}(E) = \frac{r}{9}.$$

□

Corollary 271.2. *For any pair of orthogonal projections*

$$E, F \in \mathcal{A}_9$$

with

$$EF = 0,$$

one has

$$\omega_9(E + F) = \omega_9(E) + \omega_9(F).$$

Proof. Linearity of the trace gives

$$\omega_9(E + F) = \omega_9(E) + \omega_9(F).$$

□

Theorem 271.3. *If*

$$E_1, \dots, E_r$$

are pairwise orthogonal projections satisfying

$$\sum_{j=1}^r E_j = I_{\mathcal{H}_9},$$

then

$$\sum_{j=1}^r \text{rank}(E_j) = 9$$

and

$$\sum_{j=1}^r \omega_9(E_j) = 1.$$

Proof. Taking traces of

$$\sum_{j=1}^r E_j = I_{\mathcal{H}_9}$$

gives

$$\sum_{j=1}^r \text{tr}(E_j) = \text{tr}(I_{\mathcal{H}_9}) = 9.$$

Since

$$\text{tr}(E_j) = \text{rank}(E_j),$$

the first identity follows. Dividing by 9 yields

$$\sum_{j=1}^r \omega_9(E_j) = \frac{1}{9} \sum_{j=1}^r \text{rank}(E_j) = 1.$$

□

272 The Stable Sector and Finite Spectral Resolution

Let

$$A = A^* \in \mathcal{A}_9.$$

Let

$$\sigma(A) = \{\lambda_1, \dots, \lambda_r\}$$

be the set of distinct eigenvalues of A , and let

$$E_1, \dots, E_r$$

be the corresponding spectral projections.

Theorem 272.1. *The expectation of A in the stable state is*

$$\omega_9(A) = \frac{1}{9} \sum_{j=1}^r \lambda_j \text{rank}(E_j).$$

Proof. The finite-dimensional spectral theorem gives

$$A = \sum_{j=1}^r \lambda_j E_j.$$

Therefore

$$\omega_9(A) = \sum_{j=1}^r \lambda_j \omega_9(E_j).$$

By the preceding proposition,

$$\omega_9(E_j) = \frac{\text{rank}(E_j)}{9}.$$

Substitution gives

$$\omega_9(A) = \frac{1}{9} \sum_{j=1}^r \lambda_j \text{rank}(E_j).$$

□

Corollary 272.2. *The variance of a self-adjoint observable A in the stable state is*

$$\text{Var}_{\omega_9}(A) = \frac{1}{9} \sum_{j=1}^r (\lambda_j - \omega_9(A))^2 \text{rank}(E_j).$$

Proof. By definition,

$$\text{Var}_{\omega_9}(A) = \omega_9(A^2) - \omega_9(A)^2.$$

Since

$$A^2 = \sum_{j=1}^r \lambda_j^2 E_j,$$

one has

$$\omega_9(A^2) = \frac{1}{9} \sum_{j=1}^r \lambda_j^2 \text{rank}(E_j).$$

Using

$$\sum_{j=1}^r \text{rank}(E_j) = 9$$

and expanding the square

$$(\lambda_j - \omega_9(A))^2$$

gives the stated expression. □

273 Faithfulness of the Stable Trace

Let

$$\omega_9(X) = \frac{1}{9} \text{tr}(X), \quad X \in \mathcal{A}_9.$$

Proposition 273.1. *The state ω_9 is faithful on the positive cone of $\mathcal{A}_9$. Namely,*

$$X \geq 0, \quad \omega_9(X) = 0 \implies X = 0.$$

Proof. Since $X \geq 0$, the spectral theorem gives

$$X = \sum_{j=1}^r \lambda_j E_j, \quad \lambda_j \geq 0,$$

where the E_j are the spectral projections corresponding to the nonzero eigenvalues. Hence

$$\omega_9(X) = \frac{1}{9} \sum_{j=1}^r \lambda_j \operatorname{rank}(E_j).$$

Every term in the sum is nonnegative. If

$$\omega_9(X) = 0,$$

then

$$\lambda_j = 0$$

for every j , so $X = 0$. □

Corollary 273.2. *For every*

$$X \in \mathcal{A}_9,$$

one has

$$\omega_9(X^* X) = 0 \implies X = 0.$$

Proof. The operator $X^* X$ is positive. The preceding proposition therefore implies

$$X^* X = 0.$$

Hence

$$\|Xv\|^2 = \langle v, X^* X v \rangle = 0$$

for every $v \in \mathcal{H}_9$, and consequently $X = 0$. □

274 The Hilbert–Schmidt Structure Induced by the Stable State

Define

$$\langle X, Y \rangle_9 = \omega_9(X^* Y) = \frac{1}{9} \operatorname{tr}(X^* Y)$$

for

$$X, Y \in \mathcal{A}_9.$$

Theorem 274.1. *The form*

$$\langle \cdot, \cdot \rangle_9$$

is an inner product on $\mathcal{A}_9$.

Proof. Sesquilinearity follows from the linearity of the trace. Moreover,

$$\langle Y, X \rangle_9 = \frac{1}{9} \operatorname{tr}(Y^* X) = \frac{1}{9} \overline{\operatorname{tr}(X^* Y)} = \overline{\langle X, Y \rangle_9}.$$

Finally,

$$\langle X, X \rangle_9 = \frac{1}{9} \operatorname{tr}(X^* X) \geq 0.$$

If this quantity vanishes, faithfulness of ω_9 gives

$$X = 0.$$

Thus the form is positive definite. □

Corollary 274.2. *The algebra $\mathcal{A}_9$ becomes a 81-dimensional Hilbert space with respect to the normalized Hilbert–Schmidt inner product*

$$\langle X, Y \rangle_9 = \frac{1}{9} \operatorname{tr}(X^* Y).$$

275 Orthogonal Invariance of the Operator-Space Structure

For

$$U \in \mathcal{U}(\mathcal{H}_9),$$

define

$$\operatorname{Ad}_U(X) = UXU^*.$$

Proposition 275.1. *The adjoint action*

$$\operatorname{Ad}_U : \mathcal{A}_9 \longrightarrow \mathcal{A}_9$$

is unitary with respect to $\langle \cdot, \cdot \rangle_9$.

Proof. For

$$X, Y \in \mathcal{A}_9,$$

one has

$$\begin{aligned} \langle \operatorname{Ad}_U(X), \operatorname{Ad}_U(Y) \rangle_9 &= \frac{1}{9} \operatorname{tr}((UXU^*)^* UYU^*) \\ &= \frac{1}{9} \operatorname{tr}(UX^* YU^*) \\ &= \frac{1}{9} \operatorname{tr}(X^* Y) \\ &= \langle X, Y \rangle_9. \end{aligned}$$

Thus Ad_U preserves the inner product. □

Corollary 275.2. *For every $U \in \mathcal{U}(\mathcal{H}_9)$,*

$$\|\operatorname{Ad}_U(X)\|_9 = \|X\|_9.$$

276 The Representation-Induced Operator Action

Let

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be the stable representation. Define

$$\alpha_g = \operatorname{Ad}_{\pi_9(g)}$$

on $\mathcal{A}_9$.

Theorem 276.1. *The map*

$$\alpha : G \longrightarrow \mathcal{U}(\mathcal{A}_9)$$

defined by

$$\alpha_g(X) = \pi_9(g)X\pi_9(g)^*$$

is a unitary representation of G on the Hilbert space $\mathcal{A}_9$.

Proof. For $g, h \in G$,

$$\begin{aligned}\alpha_g(\alpha_h(X)) &= \pi_g(g) (\pi_g(h) X \pi_g(h)^*) \pi_g(g)^* \\ &= \pi_g(gh) X \pi_g(gh)^* \\ &= \alpha_{gh}(X).\end{aligned}$$

Furthermore,

$$\alpha_e(X) = X$$

because

$$\pi_g(e) = I_{\mathcal{H}_g}.$$

Each α_g is unitary by the preceding proposition. Therefore α is a unitary representation. $\square$

Proposition 276.2. *The stable state density operator is a fixed vector of the induced operator representation:*

$$\alpha_g(\rho_g) = \rho_g \quad (g \in G).$$

Proof. Since

$$\rho_g = \frac{1}{g} I_{\mathcal{H}_g},$$

one has

$$\alpha_g(\rho_g) = \frac{1}{g} \pi_g(g) I_{\mathcal{H}_g} \pi_g(g)^* = \frac{1}{g} I_{\mathcal{H}_g} = \rho_g.$$

$\square$

277 Fixed Operators of the Induced Action

Define

$$\mathcal{A}_g^G = \{X \in \mathcal{A}_g : \alpha_g(X) = X \text{ for all } g \in G\}.$$

Proposition 277.1. *The fixed-point space $\mathcal{A}_g^G$ is precisely the commutant of the stable representation:*

$$\mathcal{A}_g^G = \pi_g(G)'$$

Proof. The equality

$$\alpha_g(X) = X$$

is equivalent to

$$\pi_g(g) X \pi_g(g)^* = X.$$

Multiplication on the right by $\pi_g(g)$ gives

$$\pi_g(g) X = X \pi_g(g).$$

Thus X commutes with every $\pi_g(g)$, which is exactly the definition of the commutant. $\square$

Theorem 277.2. *If π_g is irreducible, then*

$$\mathcal{A}_g^G = \mathbb{C} I_{\mathcal{H}_g}.$$

Proof. By the preceding proposition,

$$\mathcal{A}_g^G = \pi_g(G)'.$$

Irreducibility of π_g and Schur's lemma imply

$$\pi_g(G)' = \mathbb{C} I_{\mathcal{H}_g}.$$

$\square$

278 Invariant Self-Adjoint Observables

Assume again that π_g is irreducible.

Corollary 278.1. *If*

$$A = A^*$$

and

$$\alpha_g(A) = A \quad (g \in G),$$

then

$$A = \lambda I_{\mathcal{H}_g}$$

for some

$$\lambda \in \mathbb{R}.$$

Proof. The invariance condition gives

$$A \in \mathcal{A}_g^G.$$

Hence

$$A = \lambda I_{\mathcal{H}_g}$$

for some $\lambda \in \mathbb{C}$. Since A is self-adjoint,

$$A^* = \bar{\lambda} I_{\mathcal{H}_g} = A,$$

and therefore

$$\lambda \in \mathbb{R}.$$

□

279 The Averaging Projection for a Compact Symmetry Group

Assume now that G is a compact group and let

$$dg$$

denote its normalized Haar measure:

$$\int_G dg = 1.$$

Define

$$\mathcal{E}_G(X) = \int_G \alpha_g(X) dg = \int_G \pi_g(g) X \pi_g(g)^* dg.$$

Proposition 279.1. *The map*

$$\mathcal{E}_G : \mathcal{A}_g \longrightarrow \mathcal{A}_g$$

is linear and satisfies

$$\mathcal{E}_G(X) \in \mathcal{A}_g^G$$

for every $X \in \mathcal{A}_g$.

Proof. Linearity follows from linearity of the integral. Let $h \in G$. By left invariance of Haar measure,

$$\begin{aligned}\alpha_h(\mathcal{E}_G(X)) &= \int_G \alpha_h(\alpha_g(X)) dg \\ &= \int_G \alpha_{hg}(X) dg \\ &= \int_G \alpha_g(X) dg \\ &= \mathcal{E}_G(X).\end{aligned}$$

Thus

$$\mathcal{E}_G(X) \in \mathcal{A}_9^G.$$

□

Proposition 279.2. *The averaging map is idempotent:*

$$\mathcal{E}_G^2 = \mathcal{E}_G.$$

Proof. By the preceding proposition,

$$\mathcal{E}_G(X) \in \mathcal{A}_9^G.$$

Every fixed element is unchanged by the averaging operation, since

$$\alpha_g(Y) = Y$$

for all $g \in G$ whenever $Y \in \mathcal{A}_9^G$. Therefore

$$\mathcal{E}_G(\mathcal{E}_G(X)) = \mathcal{E}_G(X).$$

□

Theorem 279.3. *If π_9 is irreducible, then*

$$\mathcal{E}_G(X) = \omega_9(X) I_{\mathcal{H}_9}$$

for every

$$X \in \mathcal{A}_9.$$

Proof. Irreducibility gives

$$\mathcal{A}_9^G = \mathbb{C} I_{\mathcal{H}_9}.$$

Hence

$$\mathcal{E}_G(X) = \lambda(X) I_{\mathcal{H}_9}$$

for some scalar $\lambda(X)$.

The trace is invariant under unitary conjugation, so

$$\begin{aligned}\mathrm{tr}(\mathcal{E}_G(X)) &= \int_G \mathrm{tr}(\pi_9(g) X \pi_9(g)^*) dg \\ &= \int_G \mathrm{tr}(X) dg \\ &= \mathrm{tr}(X).\end{aligned}$$

On the other hand,

$$\mathrm{tr}(\mathcal{E}_G(X)) = 9\lambda(X).$$

Consequently,

$$\lambda(X) = \frac{1}{9} \operatorname{tr}(X) = \omega_9(X).$$

Thus

$$\mathcal{E}_G(X) = \omega_9(X) I_{\mathcal{H}_9}.$$

□

280 Rigidity of the Maximally Symmetric State

Theorem 280.1. *Assume that π_9 is irreducible. Let*

$$\rho$$

be a density operator on $\mathcal{H}_9$ satisfying

$$\pi_9(g) \rho \pi_9(g)^* = \rho \quad (g \in G).$$

Then

$$\rho = \rho_9.$$

Proof. The invariance condition implies

$$\rho \in \mathcal{A}_9^G.$$

By irreducibility,

$$\mathcal{A}_9^G = \mathbb{C} I_{\mathcal{H}_9}.$$

Hence

$$\rho = \lambda I_{\mathcal{H}_9}$$

for some $\lambda \geq 0$. Since ρ is a density operator,

$$1 = \operatorname{tr}(\rho) = 9\lambda.$$

Thus

$$\lambda = \frac{1}{9},$$

and therefore

$$\rho = \frac{1}{9} I_{\mathcal{H}_9} = \rho_9.$$

□

Corollary 280.2. *Under irreducibility, group invariance and trace normalization determine the stable state uniquely.*

281 Separation of States by Stable Observables

Let

$$\rho, \sigma$$

be density operators on $\mathcal{H}_9$.

Proposition 281.1. *If*

$$\mathrm{tr}(\rho X) = \mathrm{tr}(\sigma X)$$

for every

$$X \in \mathcal{A}_9,$$

then

$$\rho = \sigma.$$

Proof. Set

$$D = \rho - \sigma.$$

The hypothesis gives

$$\mathrm{tr}(DX) = 0$$

for every $X \in \mathcal{A}_9$. Choosing

$$X = D^*$$

yields

$$\mathrm{tr}(DD^*) = 0.$$

Since $DD^* \geq 0$, faithfulness of the trace implies

$$DD^* = 0.$$

Hence

$$D = 0,$$

and therefore

$$\rho = \sigma.$$

□

Corollary 281.2. *The finite-dimensional observable algebra*

$$\mathcal{A}_9 = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9)$$

separates density operators on the stable sector.

282 The Stable State as the Unique Tracial State

Theorem 282.1. *Let*

$$\varphi : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a positive normalized linear functional satisfying

$$\varphi(XY) = \varphi(YX) \quad (X, Y \in \mathcal{A}_9).$$

Then

$$\varphi = \omega_9.$$

Proof. Every positive linear functional on the finite-dimensional matrix algebra $\mathcal{A}_9$ has the form

$$\varphi(X) = \mathrm{tr}(\rho X)$$

for a unique positive operator ρ satisfying

$$\mathrm{tr}(\rho) = 1.$$

The tracial property gives

$$\operatorname{tr}(\rho XY) = \operatorname{tr}(\rho YX)$$

for all $X, Y \in \mathcal{A}_9$. By cyclicity,

$$\operatorname{tr}(\rho XY) = \operatorname{tr}(Y\rho X),$$

whereas

$$\operatorname{tr}(\rho YX) = \operatorname{tr}(X\rho Y).$$

Equivalently,

$$\operatorname{tr}((\rho X - X\rho)Y) = 0$$

for every Y . Nondegeneracy of the trace pairing implies

$$\rho X = X\rho$$

for every $X \in \mathcal{A}_9$. Thus

$$\rho \in Z(\mathcal{A}_9).$$

Since

$$Z(\mathcal{A}_9) = \mathbb{C}I_{\mathcal{H}_9},$$

we have

$$\rho = \lambda I_{\mathcal{H}_9}.$$

Normalization gives

$$9\lambda = 1,$$

hence

$$\rho = \frac{1}{9}I_{\mathcal{H}_9}.$$

Therefore

$$\varphi(X) = \frac{1}{9} \operatorname{tr}(X) = \omega_9(X).$$

□

Corollary 282.2. *The normalized trace*

$$\omega_9(X) = \frac{1}{9} \operatorname{tr}(X)$$

is the unique tracial state on $\mathcal{A}_9$.

283 The Commutant of the Stable Projection

Let

$$V_9$$

be a finite-dimensional complex Hilbert space and let

$$P_9 \in \operatorname{End}_{\mathbb{C}}(V_9)$$

be an idempotent operator:

$$P_9^2 = P_9.$$

Set

$$\mathcal{H}_9 = \operatorname{range}(P_9).$$

Define the commutant of P_9 by

$$\mathcal{C}(P_9) = \{X \in \operatorname{End}_{\mathbb{C}}(V_9) : XP_9 = P_9X\}.$$

Proposition 283.1. *If*

$$X \in \mathcal{C}(P_9),$$

then both

$$\text{range}(P_9) \quad \text{and} \quad \ker(P_9)$$

are invariant under X .

Proof. Let

$$v \in \text{range}(P_9).$$

Then

$$P_9 v = v.$$

Since

$$XP_9 = P_9X,$$

we obtain

$$P_9Xv = XP_9v = Xv.$$

Hence

$$Xv \in \text{range}(P_9).$$

Now let

$$v \in \ker(P_9).$$

Then

$$P_9 v = 0,$$

and therefore

$$P_9Xv = XP_9v = 0.$$

Thus

$$Xv \in \ker(P_9).$$

□

Theorem 283.2. *There is a canonical vector-space decomposition*

$$V_9 = \text{range}(P_9) \oplus \ker(P_9).$$

Moreover, every

$$v \in V_9$$

has the unique decomposition

$$v = P_9 v + (I - P_9) v,$$

with

$$P_9 v \in \text{range}(P_9)$$

and

$$(I - P_9) v \in \ker(P_9).$$

Proof. For every

$$v \in V_9,$$

write

$$v = P_9 v + (I - P_9) v.$$

The first term belongs to

$$\text{range}(P_9).$$

Furthermore,

$$P_9(I - P_9) = P_9 - P_9^2 = 0,$$

so

$$(I - P_9)v \in \ker(P_9).$$

It remains to prove that the intersection is trivial. If

$$v \in \text{range}(P_9) \cap \ker(P_9),$$

then

$$P_9 v = v$$

and simultaneously

$$P_9 v = 0.$$

Hence

$$v = 0.$$

Therefore the sum is direct. □

Corollary 283.3. *Every operator*

$$X \in \mathcal{C}(P_9)$$

is block diagonal with respect to the decomposition

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9).$$

Proof. By the preceding proposition, both summands are invariant under X . Therefore the matrix of X relative to this decomposition has vanishing off-diagonal blocks. □

Theorem 283.4. *The restriction map*

$$\mathcal{C}(P_9) \longrightarrow \text{End}_{\mathbb{C}}(\mathcal{H}_9) \oplus \text{End}_{\mathbb{C}}(\ker P_9)$$

given by

$$X \longmapsto (X|_{\mathcal{H}_9}, X|_{\ker P_9})$$

is an algebra isomorphism.

Proof. The map is an algebra homomorphism by construction.

Let

$$A \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

and

$$B \in \text{End}_{\mathbb{C}}(\ker P_9).$$

Using the direct-sum decomposition

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9),$$

define

$$X = A \oplus B.$$

Then

$$XP_9 = P_9X,$$

because P_9 acts as the identity on $\mathcal{H}_9$ and as zero on $\ker(P_9)$. Thus every pair (A, B) lies in the image.

If X restricts to zero on both summands, then $X = 0$. Hence the map is injective. $\square$

284 The Stable Representation and the Commutant

Let

$$\pi : G \longrightarrow \mathcal{U}(V_9)$$

be a unitary representation satisfying

$$\pi(g)P_9 = P_9\pi(g) \quad (g \in G).$$

Proposition 284.1. *For every*

$$g \in G,$$

the operator

$$\pi_9(g) = \pi(g)|_{\mathcal{H}_9}$$

belongs to

$$\mathcal{U}(\mathcal{H}_9).$$

Proof. The invariance of $\mathcal{H}_9$ was established above. Since $\pi(g)$ is unitary on V_9 , its restriction preserves the inner product on $\mathcal{H}_9$. The inverse restriction is

$$\pi(g^{-1})|_{\mathcal{H}_9}.$$

Thus

$$\pi_9(g) \in \mathcal{U}(\mathcal{H}_9).$$

$\square$

Theorem 284.2. *The restriction of the commutant representation to the stable sector defines a homomorphism*

$$\mathcal{C}(P_9) \longrightarrow \text{End}_{\mathbb{C}}(\mathcal{H}_9), \quad X \longmapsto X|_{\mathcal{H}_9}.$$

Its image contains the algebra generated by

$$\{\pi_9(g) : g \in G\}.$$

Proof. The first assertion follows from the invariance of $\mathcal{H}_9$ under every element of $\mathcal{C}(P_9)$.

For every $g \in G$, the relation

$$\pi(g)P_9 = P_9\pi(g)$$

shows that

$$\pi(g) \in \mathcal{C}(P_9).$$

Therefore

$$\pi_9(g) = \pi(g)|_{\mathcal{H}_9}$$

belongs to the image of the restriction map. Since the image is an algebra, it contains the algebra generated by these restrictions. $\square$

285 Irreducible Stable Sectors

Let

$$\pi_g : G \longrightarrow \mathcal{U}(\mathcal{H}_g)$$

be the restricted representation.

Definition 285.1. A nonzero subspace

$$W \subseteq \mathcal{H}_g$$

is called G -invariant if

$$\pi_g(g)W \subseteq W \quad (g \in G).$$

The representation π_g is called irreducible if its only G -invariant subspaces are

$$\{0\} \quad \text{and} \quad \mathcal{H}_g.$$

Proposition 285.2. *Every finite-dimensional unitary representation admits an orthogonal decomposition into irreducible invariant subspaces.*

Proof. If π_g is irreducible, there is nothing to prove. Otherwise choose a proper nonzero invariant subspace $W \subsetneq \mathcal{H}_g$.

Because every $\pi_g(g)$ is unitary, the orthogonal complement $W^\perp$ is also invariant. Indeed, if

$$v \in W^\perp, \quad w \in W,$$

then

$$\langle \pi_g(g)v, w \rangle = \langle v, \pi_g(g)^{-1}w \rangle = 0,$$

since

$$\pi_g(g)^{-1}w \in W.$$

Hence

$$\mathcal{H}_g = W \oplus W^\perp.$$

Repeating the argument on the proper invariant summands terminates after finitely many steps because

$$\dim_{\mathbb{C}} \mathcal{H}_g < \infty.$$

□

Corollary 285.3. *There exist irreducible unitary representations*

$$\pi_1, \dots, \pi_s$$

and positive integers

$$d_1, \dots, d_s$$

such that

$$\mathcal{H}_g \cong \bigoplus_{j=1}^s \mathbb{C}^{d_j}$$

as a G -representation, with

$$\sum_{j=1}^s d_j = 9.$$

286 The Isotypic Decomposition

Let

$$\widehat{G}_{\mathcal{H}_9}$$

denote the finite set of equivalence classes of irreducible representations occurring in π_9 .

For each

$$\alpha \in \widehat{G}_{\mathcal{H}_9},$$

let

$$W_\alpha$$

be the corresponding isotypic component.

Theorem 286.1. *The stable representation has an orthogonal isotypic decomposition*

$$\mathcal{H}_9 = \bigoplus_{\alpha \in \widehat{G}_{\mathcal{H}_9}} W_\alpha,$$

where each

$$W_\alpha$$

is G -invariant and is a direct sum of copies of one irreducible representation.

Proof. The complete reducibility established above gives a decomposition into irreducible unitary representations. Grouping equivalent irreducible summands produces the isotypic components. Orthogonality follows from unitarity and the finite-dimensional orthogonal decomposition. $\square$

Proposition 286.2. *For every*

$$g \in G,$$

the operator $\pi_9(g)$ preserves each isotypic component W_α .

Proof. By construction each W_α is a sum of G -invariant irreducible summands. Hence it is invariant under every $\pi_9(g)$. $\square$

287 Invariant Projections of the Stable Representation

Let

$$Q \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be an orthogonal projection.

Proposition 287.1. *The following conditions are equivalent:*

$$Q\pi_9(g) = \pi_9(g)Q \quad (g \in G),$$

and

$$\text{range}(Q)$$

is invariant under $\pi_9(G)$.

Proof. Suppose first that

$$Q\pi_9(g) = \pi_9(g)Q.$$

If

$$v \in \text{range}(Q),$$

then

$$Qv = v,$$

and therefore

$$Q\pi_9(g)v = \pi_9(g)Qv = \pi_9(g)v.$$

Thus

$$\pi_9(g)v \in \text{range}(Q).$$

Conversely, suppose

$$W = \text{range}(Q)$$

is invariant under $\pi_9(G)$. Since Q is orthogonal and $\pi_9(g)$ is unitary, $W^\perp$ is also invariant. Hence $\pi_9(g)$ is block diagonal with respect to

$$\mathcal{H}_9 = W \oplus W^\perp,$$

and therefore commutes with the orthogonal projection Q . □

Corollary 287.2. *The G -invariant orthogonal projections on $\mathcal{H}_9$ are precisely the projections in the commutant*

$$\pi_9(G)'.$$

288 The Maximally Symmetric State and Representation Theory

Recall

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Theorem 288.1. *The state ρ_9 belongs to the commutant of every unitary representation of G on $\mathcal{H}_9$:*

$$\rho_9\pi_9(g) = \pi_9(g)\rho_9 \quad (g \in G).$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\rho_9\pi_9(g) = \frac{1}{9}\pi_9(g) = \pi_9(g)\rho_9.$$

□

Corollary 288.2. *The state ρ_9 is invariant under the conjugation action of $\pi_9(G)$:*

$$\pi_9(g)\rho_9\pi_9(g)^* = \rho_9 \quad (g \in G).$$

Proposition 288.3. *Let*

$$Q \in \pi_9(G)'$$

be an orthogonal projection. Then

$$\text{tr}(\rho_9 Q) = \frac{\text{rank}(Q)}{9}.$$

Proof. Using

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

we obtain

$$\mathrm{tr}(\rho_9 Q) = \frac{1}{9} \mathrm{tr}(Q).$$

Since Q is an orthogonal projection,

$$\mathrm{tr}(Q) = \mathrm{rank}(Q).$$

Therefore

$$\mathrm{tr}(\rho_9 Q) = \frac{\mathrm{rank}(Q)}{9}.$$

□

289 Dimension Formula for the Stable Representation

Suppose the isotypic decomposition has the form

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^s (\mathbb{C}^{m_\alpha} \otimes W_\alpha),$$

where the

$$W_\alpha$$

are pairwise inequivalent irreducible G -representations and

$$d_\alpha = \dim_{\mathbb{C}} W_\alpha.$$

Theorem 289.1. *The dimensions satisfy*

$$\sum_{\alpha=1}^s m_\alpha d_\alpha = 9.$$

Proof. Taking dimensions in the orthogonal direct sum gives

$$\dim_{\mathbb{C}} \mathcal{H}_9 = \sum_{\alpha=1}^s \dim_{\mathbb{C}} (\mathbb{C}^{m_\alpha} \otimes W_\alpha).$$

Since

$$\dim_{\mathbb{C}} (\mathbb{C}^{m_\alpha} \otimes W_\alpha) = m_\alpha d_\alpha$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

the result follows. □

Corollary 289.2. *The possible irreducible dimensions occurring in the stable sector satisfy*

$$d_\alpha \leq 9.$$

Proof. Each summand contributes the positive integer

$$m_\alpha d_\alpha$$

to a sum equal to 9. Hence

$$d_\alpha \leq 9.$$

□

Remark 289.3. No irreducibility of the nine-dimensional stable representation is assumed here. Such an assertion would require information about the specific group G and the specific representation π . The dimension 9 alone does not determine the representation-theoretic decomposition.

290 Intertwining of the Stable Projection and the Representation

Let

$$P_9 : V_9 \longrightarrow V_9$$

be the stabilized projection and let

$$\pi : G \longrightarrow \mathcal{U}(V_9)$$

satisfy

$$\pi(g)P_9 = P_9\pi(g) \quad (g \in G).$$

Proposition 290.1. *For every*

$$g \in G,$$

the operator $\pi(g)$ preserves both

$$\mathcal{H}_9 = \text{range}(P_9)$$

and

$$\ker(P_9).$$

Proof. The invariance of $\mathcal{H}_9$ was established above. Let

$$v \in \ker(P_9).$$

Then

$$P_9 v = 0.$$

Using the commutation relation,

$$P_9\pi(g)v = \pi(g)P_9v = 0.$$

Therefore

$$\pi(g)v \in \ker(P_9).$$

□

Theorem 290.2. *There is a G -invariant direct-sum decomposition*

$$V_9 = \ker(P_9) \oplus \mathcal{H}_9.$$

Moreover, with respect to this decomposition,

$$P_9 = 0 \oplus I_{\mathcal{H}_9},$$

and

$$\pi(g) = \pi_0(g) \oplus \pi_9(g),$$

where

$$\pi_0(g) = \pi(g)|_{\ker(P_9)}$$

and

$$\pi_9(g) = \pi(g)|_{\mathcal{H}_9}.$$

Proof. Since P_9 is idempotent,

$$P_9^2 = P_9.$$

For every

$$v \in V_9,$$

one has

$$v = (v - P_9 v) + P_9 v.$$

Furthermore,

$$P_9(v - P_9 v) = P_9 v - P_9^2 v = 0,$$

so

$$v - P_9 v \in \ker(P_9),$$

whereas

$$P_9 v \in \text{range}(P_9) = \mathcal{H}_9.$$

Thus

$$V_9 = \ker(P_9) + \mathcal{H}_9.$$

If

$$v \in \ker(P_9) \cap \mathcal{H}_9,$$

then

$$P_9 v = 0$$

and, since $v \in \mathcal{H}_9$,

$$P_9 v = v.$$

Hence

$$v = 0,$$

and the sum is direct.

The invariance of both summands follows from the preceding proposition. On $\ker(P_9)$, the projection P_9 vanishes, while on $\mathcal{H}_9$ it acts as the identity. The block decomposition of $\pi(g)$ follows from the invariance of the two summands. $\square$

Corollary 290.3. *The stabilized projection P_9 commutes with the entire representation:*

$$P_9 \pi(g) = \pi(g) P_9 \quad (g \in G),$$

and is the projection onto the G -invariant summand $\mathcal{H}_9$.

Proposition 290.4. *If*

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

then

$$\dim_{\mathbb{C}} \ker(P_9) = \dim_{\mathbb{C}} V_9 - 9.$$

In particular, if

$$\dim_{\mathbb{C}} V_9 = 143,$$

then

$$\dim_{\mathbb{C}} \ker(P_9) = 134.$$

Proof. The assertion follows from the direct-sum decomposition

$$V_9 = \ker(P_9) \oplus \mathcal{H}_9$$

and rank-nullity. $\square$

291 The Commutant of the Stable Representation

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

and define the commutant of the restricted representation by

$$\mathcal{A}_9^G = \{X \in \mathcal{A}_9 : X\pi_9(g) = \pi_9(g)X \text{ for all } g \in G\}.$$

Proposition 291.1. *The set*

$$\mathcal{A}_9^G$$

is a unital subalgebra of $\mathcal{A}_9$.

Proof. The identity operator commutes with every $\pi_9(g)$. If

$$X, Y \in \mathcal{A}_9^G,$$

then

$$(X + Y)\pi_9(g) = \pi_9(g)(X + Y)$$

and

$$(XY)\pi_9(g) = X\pi_9(g)Y = \pi_9(g)XY.$$

Scalar multiples are handled similarly. Hence $\mathcal{A}_9^G$ is a unital subalgebra. □

Proposition 291.2. *The state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

belongs to the commutant $\mathcal{A}_9^G$.

Proof. For every

$$g \in G,$$

one has

$$\rho_9\pi_9(g) = \frac{1}{9}\pi_9(g) = \pi_9(g)\rho_9.$$

□

Theorem 291.3. *For every*

$$X \in \mathcal{A}_9,$$

the group average

$$\frac{1}{|G|} \sum_{g \in G} \pi_9(g)X\pi_9(g)^*$$

is defined whenever G is finite and belongs to $\mathcal{A}_9^G$.

Proof. Let

$$\mathcal{E}_G(X) = \frac{1}{|G|} \sum_{g \in G} \pi_9(g)X\pi_9(g)^*.$$

For $h \in G$,

$$\pi_9(h)\mathcal{E}_G(X)\pi_9(h)^* = \frac{1}{|G|} \sum_{g \in G} \pi_9(hg)X\pi_9(hg)^*.$$

Since multiplication by h permutes the elements of the finite group, the last sum equals

$$\frac{1}{|G|} \sum_{g \in G} \pi_9(g)X\pi_9(g)^* = \mathcal{E}_G(X).$$

Thus

$$\mathcal{E}_G(X) \in \mathcal{A}_9^G.$$

□

Remark 291.4. The preceding averaging construction requires finiteness of G . For a general compact group, the corresponding statement requires normalized Haar measure. No averaging assertion for an arbitrary noncompact group is used here.

292 Trace Preservation Under the Stable Representation

Let

$$X \in \mathcal{A}_9$$

and

$$g \in G.$$

Proposition 292.1. *Unitary conjugation preserves the ordinary trace:*

$$\mathrm{tr}(\pi_9(g)X\pi_9(g)^*) = \mathrm{tr}(X).$$

Proof. By cyclicity of the trace,

$$\mathrm{tr}(\pi_9(g)X\pi_9(g)^*) = \mathrm{tr}(X\pi_9(g)^*\pi_9(g)).$$

Since $\pi_9(g)$ is unitary,

$$\pi_9(g)^*\pi_9(g) = I_{\mathcal{H}_9}.$$

Consequently,

$$\mathrm{tr}(\pi_9(g)X\pi_9(g)^*) = \mathrm{tr}(X).$$

□

Corollary 292.2. *The normalized trace*

$$\tau_9(X) = \frac{1}{9} \mathrm{tr}(X)$$

is invariant under the adjoint action of G :

$$\tau_9(\pi_9(g)X\pi_9(g)^*) = \tau_9(X).$$

Theorem 292.3. *The state ρ_9 induces the normalized trace through*

$$\mathrm{tr}(\rho_9 X) = \tau_9(X) \quad (X \in \mathcal{A}_9).$$

Proof. Using

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one obtains

$$\mathrm{tr}(\rho_9 X) = \frac{1}{9} \mathrm{tr}(X) = \tau_9(X).$$

□

Corollary 292.4. *For every*

$$X \in \mathcal{A}_9$$

and every

$$g \in G,$$

the expectation determined by ρ_9 is representation-invariant:

$$\mathrm{tr}(\rho_9 \pi_9(g) X \pi_9(g)^*) = \mathrm{tr}(\rho_9 X).$$

Proof. By

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

and trace invariance under unitary conjugation,

$$\mathrm{tr}(\rho_9 \pi_9(g) X \pi_9(g)^*) = \frac{1}{9} \mathrm{tr}(\pi_9(g) X \pi_9(g)^*) = \frac{1}{9} \mathrm{tr}(X).$$

This equals

$$\mathrm{tr}(\rho_9 X).$$

□

293 Faithfulness and the Effective Symmetry Group

Let

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be the restricted representation introduced above. Define its kernel by

$$\ker(\pi_9) = \{g \in G : \pi_9(g) = I_{\mathcal{H}_9}\}.$$

Proposition 293.1. *The kernel*

$$\ker(\pi_9)$$

is a normal subgroup of G .

Proof. Let

$$g \in \ker(\pi_9)$$

and let

$$h \in G.$$

Since π_9 is a representation,

$$\pi_9(hgh^{-1}) = \pi_9(h)\pi_9(g)\pi_9(h)^{-1}.$$

Because

$$\pi_9(g) = I_{\mathcal{H}_9},$$

it follows that

$$\pi_9(hgh^{-1}) = I_{\mathcal{H}_9}.$$

Hence

$$hgh^{-1} \in \ker(\pi_9),$$

which proves normality.

□

Theorem 293.2. *The representation π_9 factors uniquely through the quotient*

$$G_{\text{eff}} = G / \ker(\pi_9).$$

More precisely, there exists a unique faithful unitary representation

$$\bar{\pi}_9 : G_{\text{eff}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

such that

$$\pi_9 = \bar{\pi}_9 \circ q,$$

where

$$q : G \longrightarrow G_{\text{eff}}$$

is the quotient homomorphism.

Proof. For

$$g \in G,$$

write

$$[g] = g \ker(\pi_9).$$

Define

$$\bar{\pi}_9([g]) = \pi_9(g).$$

If

$$[g] = [h],$$

then

$$h^{-1}g \in \ker(\pi_9),$$

and therefore

$$\pi_9(h^{-1}g) = I_{\mathcal{H}_9}.$$

Hence

$$\pi_9(g) = \pi_9(h),$$

so the definition is independent of the representative.

The homomorphism property follows from

$$\bar{\pi}_9([g][h]) = \bar{\pi}_9([gh]) = \pi_9(gh) = \pi_9(g)\pi_9(h).$$

Its kernel is trivial by construction. Thus $\bar{\pi}_9$ is faithful. Uniqueness follows from the surjectivity of q . $\square$

Corollary 293.3. *All symmetry information carried by the stable sector is determined by the effective group*

$$G_{\text{eff}} = G / \ker(\pi_9).$$

294 Character Decomposition of the Stable Representation

Assume now that G is abelian. Let

$$\widehat{G}$$

denote its unitary character group.

Theorem 294.1. *There exist finitely many characters*

$$\chi_1, \dots, \chi_s \in \widehat{G}$$

and an orthogonal decomposition

$$\mathcal{H}_g = \bigoplus_{j=1}^s \mathcal{H}_{\chi_j}$$

such that

$$\pi_g(g)v = \chi_j(g)v$$

for every

$$v \in \mathcal{H}_{\chi_j}, \quad g \in G.$$

Proof. The family

$$\{\pi_g(g) : g \in G\}$$

consists of pairwise commuting unitary operators. Hence it is a commuting family of normal operators on the finite-dimensional complex Hilbert space $\mathcal{H}_g$. Simultaneous diagonalization yields an orthonormal basis consisting of common eigenvectors.

If v is a common eigenvector, define

$$\chi_v(g)$$

by

$$\pi_g(g)v = \chi_v(g)v.$$

Since $\pi_g(g)$ is unitary,

$$|\chi_v(g)| = 1.$$

Moreover,

$$\pi_g(gh)v = \pi_g(g)\pi_g(h)v = \chi_v(h)\chi_v(g)v,$$

so

$$\chi_v(gh) = \chi_v(g)\chi_v(h).$$

Thus χ_v is a unitary character. Grouping together common eigenvectors carrying the same character gives the asserted orthogonal decomposition. $\square$

Corollary 294.2. *The dimension of the stable sector satisfies*

$$9 = \sum_{j=1}^s \dim_{\mathbb{C}} \mathcal{H}_{\chi_j}.$$

Proof. This is the dimension formula associated with the direct-sum decomposition of $\mathcal{H}_g$. $\square$

Proposition 294.3. *For every $g \in G$, the trace of the restricted representation is*

$$\mathrm{tr}(\pi_g(g)) = \sum_{j=1}^s \dim_{\mathbb{C}}(\mathcal{H}_{\chi_j}) \chi_j(g).$$

Proof. On the subspace

$$\mathcal{H}_{\chi_j},$$

the operator $\pi_9(g)$ is multiplication by $\chi_j(g)$. Its trace on that subspace is therefore

$$\dim_{\mathbb{C}}(\mathcal{H}_{\chi_j})\chi_j(g).$$

The trace of the direct sum is the sum of these contributions. □

295 Compatibility with the Arithmetic Character Family

Let

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

and let

$$\chi_t(x) = e^{it\Lambda_{\log}(x)}$$

be the logarithmic character family.

Suppose that the stable representation is defined on the arithmetic lattice:

$$\pi_9 : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9).$$

Theorem 295.1. *The representation π_9 admits a finite character decomposition*

$$\mathcal{H}_9 = \bigoplus_{j=1}^s \mathcal{H}_j$$

such that each $\mathcal{H}_j$ is invariant under $\pi_9(\mathcal{L}_{\mathbb{P}})$ and there exists a unitary character

$$\eta_j : \mathcal{L}_{\mathbb{P}} \longrightarrow U(1)$$

with

$$\pi_9(x)|_{\mathcal{H}_j} = \eta_j(x) I_{\mathcal{H}_j}.$$

Proof. The arithmetic lattice is abelian under addition. Therefore its image under π_9 is a commuting family of unitary operators. The character decomposition theorem applies directly. □

Proposition 295.2. *If, for some real numbers*

$$t_1, \dots, t_s,$$

the characters of the stable representation satisfy

$$\eta_j = \chi_{t_j},$$

then

$$\pi_9(x) = \sum_{j=1}^s \chi_{t_j}(x) Q_j,$$

where Q_j denotes the orthogonal projection onto $\mathcal{H}_j$.

Proof. The projections Q_j form the orthogonal resolution

$$\sum_{j=1}^s Q_j = I_{\mathcal{H}_9}.$$

On $\mathcal{H}_j$, the operator $\pi_9(x)$ acts by the scalar $\chi_{t_j}(x)$. Hence its spectral decomposition is

$$\pi_9(x) = \sum_{j=1}^s \chi_{t_j}(x) Q_j.$$

□

Remark 295.3. The preceding proposition is conditional on the identification

$$\eta_j = \chi_{t_j}.$$

The logarithmic character family by itself does not imply that every unitary representation of $\mathcal{L}_{\mathbb{P}}$ is of this form. Such an identification is an additional structural statement and must be established from the construction of the representation.

296 Spectral Data of Arithmetic Translations

Assume the hypotheses of the preceding proposition and let

$$x \in \mathcal{L}_{\mathbb{P}}.$$

Then

$$\pi_9(x) = \sum_{j=1}^s e^{it_j \Lambda_{\log}(x)} Q_j.$$

Theorem 296.1. *The spectrum of $\pi_9(x)$ is contained in the finite set*

$$\left\{ e^{it_1 \Lambda_{\log}(x)}, \dots, e^{it_s \Lambda_{\log}(x)} \right\}.$$

Proof. Each subspace $\mathcal{H}_j$ is an eigenspace for $\pi_9(x)$ with eigenvalue

$$e^{it_j \Lambda_{\log}(x)}.$$

Since

$$\mathcal{H}_9 = \bigoplus_{j=1}^s \mathcal{H}_j,$$

there are no other spectral values. □

Corollary 296.2. *The normalized trace of an arithmetic translation is*

$$\tau_9(\pi_9(x)) = \frac{1}{9} \sum_{j=1}^s \dim_{\mathbb{C}}(\mathcal{H}_j) e^{it_j \Lambda_{\log}(x)}.$$

Proof. Using the spectral decomposition,

$$\mathrm{tr}(\pi_9(x)) = \sum_{j=1}^s \dim_{\mathbb{C}}(\mathcal{H}_j) e^{it_j \Lambda_{\log}(x)}.$$

Division by 9 gives the result. □

297 Positive Definiteness of the Stable Character

Define

$$\varphi_9(x) = \tau_9(\pi_9(x)) \quad (x \in \mathcal{L}_{\mathbb{P}}).$$

Theorem 297.1. *The function*

$$\varphi_9 : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{C}$$

is normalized and positive definite:

$$\varphi_9(0) = 1$$

and, for every finite collection

$$x_1, \dots, x_n \in \mathcal{L}_{\mathbb{P}}$$

and

$$c_1, \dots, c_n \in \mathbb{C},$$

one has

$$\sum_{i,j=1}^n \overline{c_i} c_j \varphi_9(x_i - x_j) \geq 0.$$

Proof. Since

$$\pi_9(0) = I_{\mathcal{H}_9},$$

one has

$$\varphi_9(0) = \tau_9(I_{\mathcal{H}_9}) = 1.$$

For the positivity assertion, set

$$A = \sum_{j=1}^n c_j \pi_9(x_j).$$

Since the representation is unitary,

$$\pi_9(x_i - x_j) = \pi_9(x_i)^* \pi_9(x_j).$$

Therefore

$$\begin{aligned} \sum_{i,j=1}^n \overline{c_i} c_j \varphi_9(x_i - x_j) &= \tau_9 \left(\sum_{i,j=1}^n \overline{c_i} c_j \pi_9(x_i)^* \pi_9(x_j) \right) \\ &= \tau_9(A^* A). \end{aligned}$$

The operator $A^* A$ is positive, and the normalized trace is positive on positive operators. Hence

$$\tau_9(A^* A) \geq 0.$$

□

Corollary 297.2. *The function*

$$\varphi_9$$

satisfies

$$|\varphi_9(x)| \leq 1 \quad (x \in \mathcal{L}_{\mathbb{P}}).$$

Proof. Since $\pi_9(x)$ is unitary, all its eigenvalues have modulus one. Equivalently,

$$\varphi_9(x) = \frac{1}{9} \operatorname{tr}(\pi_9(x))$$

is the arithmetic mean of nine complex numbers of modulus one, counted with multiplicity. The triangle inequality gives

$$|\varphi_9(x)| \leq \frac{1}{9} \operatorname{tr}(I_{\mathcal{H}_9}) = 1.$$

□

298 Decomposition of the Stable Representation

Let

$$\pi_g : G \longrightarrow \mathcal{U}(\mathcal{H}_g)$$

be the restricted representation obtained above. Assume that G is abelian.

Proposition 298.1. *The family*

$$\{\pi_g(g) : g \in G\}$$

is a commuting family of unitary operators on $\mathcal{H}_g$.

Proof. For $g, h \in G$, the representation identity gives

$$\pi_g(g)\pi_g(h) = \pi_g(gh).$$

Since G is abelian,

$$gh = hg,$$

and hence

$$\pi_g(g)\pi_g(h) = \pi_g(h)\pi_g(g).$$

Each $\pi_g(g)$ is unitary by construction. □

Theorem 298.2. *There exists an orthogonal decomposition*

$$\mathcal{H}_g = \bigoplus_{\alpha=1}^s \mathcal{H}_\alpha$$

into nonzero subspaces invariant under $\pi_g(G)$, such that for every α there is a unitary character

$$\chi_\alpha : G \longrightarrow \mathbb{T}$$

satisfying

$$\pi_g(g)v = \chi_\alpha(g)v \quad (g \in G, v \in \mathcal{H}_\alpha).$$

Here

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}.$$

Proof. Because $\mathcal{H}_g$ is finite-dimensional and the operators $\pi_g(g)$ are commuting normal operators, they admit a common orthonormal eigenbasis. Grouping the common eigenspaces according to their joint eigenvalue functions gives

$$\mathcal{H}_g = \bigoplus_{\alpha=1}^s \mathcal{H}_\alpha.$$

For $v \in \mathcal{H}_\alpha$, define

$$\chi_\alpha(g)$$

by

$$\pi_g(g)v = \chi_\alpha(g)v.$$

The representation law implies

$$\chi_\alpha(gh) = \chi_\alpha(g)\chi_\alpha(h),$$

and unitarity gives

$$|\chi_\alpha(g)| = 1.$$

Thus χ_α is a unitary character of G . □

Corollary 298.3. *The character decomposition satisfies*

$$\sum_{\alpha=1}^s \dim_{\mathbb{C}} \mathcal{H}_{\alpha} = 9.$$

Proof. The decomposition is an orthogonal direct sum of $\mathcal{H}_9$. Taking dimensions gives

$$\dim_{\mathbb{C}} \mathcal{H}_9 = \sum_{\alpha=1}^s \dim_{\mathbb{C}} \mathcal{H}_{\alpha} = 9.$$

□

299 Character Projectors

For each character sector $\mathcal{H}_{\alpha}$, let

$$Q_{\alpha}$$

denote the orthogonal projection onto $\mathcal{H}_{\alpha}$.

Proposition 299.1. *The projections Q_{α} satisfy*

$$Q_{\alpha}^* = Q_{\alpha}, \quad Q_{\alpha} Q_{\beta} = 0 \quad (\alpha \neq \beta), \quad \sum_{\alpha=1}^s Q_{\alpha} = I_{\mathcal{H}_9}.$$

Proof. These are the defining properties of the orthogonal projections associated with the orthogonal direct-sum decomposition

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^s \mathcal{H}_{\alpha}.$$

□

Theorem 299.2. *Each character projector commutes with the restricted representation:*

$$Q_{\alpha} \pi_9(g) = \pi_9(g) Q_{\alpha} \quad (g \in G).$$

Moreover,

$$\pi_9(g) Q_{\alpha} = \chi_{\alpha}(g) Q_{\alpha}.$$

Proof. If $v \in \mathcal{H}_{\beta}$, then

$$Q_{\alpha} v = \begin{cases} v, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases}$$

while

$$\pi_9(g) v = \chi_{\beta}(g) v.$$

Therefore

$$Q_{\alpha} \pi_9(g) v = \pi_9(g) Q_{\alpha} v$$

for every vector in each summand. Hence the operators commute.

For $v \in \mathcal{H}_{\alpha}$,

$$\pi_9(g) Q_{\alpha} v = \pi_9(g) v = \chi_{\alpha}(g) v = \chi_{\alpha}(g) Q_{\alpha} v.$$

For $v \in \mathcal{H}_{\beta}$, $\beta \neq \alpha$, both sides vanish. Thus

$$\pi_9(g) Q_{\alpha} = \chi_{\alpha}(g) Q_{\alpha}.$$

□

300 Trace Weights of the Character Sectors

Let

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

and define

$$w_\alpha = \text{tr}(\rho_9 Q_\alpha).$$

Proposition 300.1. *For every character sector,*

$$w_\alpha = \frac{\dim_{\mathbb{C}} \mathcal{H}_\alpha}{9}.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$w_\alpha = \frac{1}{9} \text{tr}(Q_\alpha).$$

The trace of an orthogonal projection equals its rank, hence

$$\text{tr}(Q_\alpha) = \dim_{\mathbb{C}} \mathcal{H}_\alpha.$$

Therefore

$$w_\alpha = \frac{\dim_{\mathbb{C}} \mathcal{H}_\alpha}{9}.$$

□

Theorem 300.2. *The character-sector weights form a probability distribution:*

$$w_\alpha \geq 0, \quad \sum_{\alpha=1}^s w_\alpha = 1.$$

Proof. Nonnegativity follows from

$$\dim_{\mathbb{C}} \mathcal{H}_\alpha \geq 0.$$

Furthermore,

$$\sum_{\alpha=1}^s w_\alpha = \frac{1}{9} \sum_{\alpha=1}^s \dim_{\mathbb{C}} \mathcal{H}_\alpha = \frac{9}{9} = 1.$$

□

301 Expectation of the Representation

For $g \in G$, define

$$M_g = \pi_9(g).$$

Theorem 301.1. *The normalized trace of M_g is*

$$\tau_9(M_g) = \frac{1}{9} \sum_{\alpha=1}^s \dim_{\mathbb{C}} \mathcal{H}_\alpha \chi_\alpha(g).$$

Proof. From the character decomposition,

$$M_g = \sum_{\alpha=1}^s \chi_{\alpha}(g) Q_{\alpha}.$$

Taking the normalized trace gives

$$\tau_9(M_g) = \frac{1}{9} \sum_{\alpha=1}^s \chi_{\alpha}(g) \operatorname{tr}(Q_{\alpha}).$$

Since

$$\operatorname{tr}(Q_{\alpha}) = \dim_{\mathbb{C}} \mathcal{H}_{\alpha},$$

the stated formula follows. $\square$

Corollary 301.2. *The function*

$$\chi_{\rho_9}(g) = \tau_9(\pi_9(g))$$

is the normalized character of the finite-dimensional representation π_9 .

Proof. The preceding theorem expresses χ_{ρ_9} as the normalized trace of the representation. $\square$

302 Arithmetic Characters on the Prime Lattice

Let

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

be the free abelian group on the primes. For

$$x = \sum_{p \in \mathbb{P}} n_p e_p$$

with finite support, define

$$\Lambda_{\log}(x) = \sum_{p \in \mathbb{P}} n_p \log p.$$

For $t \in \mathbb{R}$, define

$$\chi_t(x) = \exp(it \Lambda_{\log}(x)).$$

Proposition 302.1. *Each χ_t is a unitary character of $\mathcal{L}_{\mathbb{P}}$.*

Proof. For $x, y \in \mathcal{L}_{\mathbb{P}}$,

$$\Lambda_{\log}(x + y) = \Lambda_{\log}(x) + \Lambda_{\log}(y).$$

Consequently,

$$\chi_t(x + y) = e^{it \Lambda_{\log}(x)} e^{it \Lambda_{\log}(y)} = \chi_t(x) \chi_t(y).$$

Moreover,

$$|\chi_t(x)| = 1.$$

Thus χ_t is a homomorphism into $\mathbb{T}$. $\square$

Theorem 302.2. *The family*

$$\{\chi_t : t \in \mathbb{R}\}$$

separates the points of $\mathcal{L}_{\mathbb{P}}$.

Proof. Let

$$x \neq y.$$

Then

$$z = x - y \neq 0.$$

Unique factorization implies that the logarithms of distinct positive rational numbers are distinct, and therefore

$$\Lambda_{\log}(z) \neq 0.$$

Choose

$$t = \frac{\pi}{\Lambda_{\log}(z)}.$$

Then

$$\chi_t(z) = e^{i\pi} = -1.$$

Hence

$$\chi_t(x) = -\chi_t(y),$$

and in particular

$$\chi_t(x) \neq \chi_t(y).$$

□

303 Representation-Theoretic Separation

Suppose now that

$$\pi : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

is a finite-dimensional unitary representation.

Theorem 303.1. *There exist finitely many unitary characters*

$$\eta_1, \dots, \eta_s$$

of $\mathcal{L}_{\mathbb{P}}$, together with mutually orthogonal nonzero subspaces

$$\mathcal{H}_1, \dots, \mathcal{H}_s,$$

such that

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^s \mathcal{H}_{\alpha}$$

and

$$\pi(x)|_{\mathcal{H}_{\alpha}} = \eta_{\alpha}(x) I_{\mathcal{H}_{\alpha}} \quad (x \in \mathcal{L}_{\mathbb{P}}).$$

Proof. The group $\mathcal{L}_{\mathbb{P}}$ is abelian. Hence its image under π is a commuting family of unitary operators. Simultaneous diagonalization therefore yields the asserted orthogonal decomposition, and the joint eigenvalue functions are characters by the representation law. □

Remark 303.2. The logarithmic characters χ_t and the characters η_{α} arising from an arbitrary finite-dimensional representation are distinct objects. No equality

$$\eta_{\alpha} = \chi_t$$

is implied without an additional hypothesis specifying the representation. This distinction is necessary for the arithmetic construction to remain logically non-circular.

304 Criterion for a Logarithmic Character Realization

Let

$$\pi : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be as above.

Proposition 304.1. *If there exist real numbers*

$$t_1, \dots, t_s$$

such that

$$\eta_{\alpha}(x) = \chi_{t_{\alpha}}(x) \quad (x \in \mathcal{L}_{\mathbb{P}}),$$

then

$$\pi(x) = \sum_{\alpha=1}^s \chi_{t_{\alpha}}(x) Q_{\alpha}.$$

Proof. On $\mathcal{H}_{\alpha}$,

$$\pi(x) = \eta_{\alpha}(x) I_{\mathcal{H}_{\alpha}} = \chi_{t_{\alpha}}(x) I_{\mathcal{H}_{\alpha}}.$$

Using

$$I_{\mathcal{H}_9} = \sum_{\alpha=1}^s Q_{\alpha},$$

the operator is therefore

$$\pi(x) = \sum_{\alpha=1}^s \chi_{t_{\alpha}}(x) Q_{\alpha}.$$

□

Corollary 304.2. *Under the logarithmic-character realization,*

$$\tau_9(\pi(x)) = \frac{1}{9} \sum_{\alpha=1}^s \dim_{\mathbb{C}} \mathcal{H}_{\alpha} \mathfrak{a}(x)^{it_{\alpha}},$$

where

$$\mathfrak{a}(x) = \prod_{p \in \mathbb{P}} p^{n_p}$$

for

$$x = \sum_{p \in \mathbb{P}} n_p e_p.$$

Proof. By the preceding proposition,

$$\pi(x) = \sum_{\alpha=1}^s \chi_{t_{\alpha}}(x) Q_{\alpha}.$$

Taking the normalized trace and using

$$\chi_{t_{\alpha}}(x) = \mathfrak{a}(x)^{it_{\alpha}}$$

gives the formula.

□

305 Compatibility with the Stable Projection

Let

$$V_9$$

be the ambient finite-dimensional space and

$$P_9 : V_9 \longrightarrow V_9$$

the stabilized projection satisfying

$$P_9^2 = P_9.$$

Assume that

$$\dim_{\mathbb{C}} \text{range}(P_9) = 9.$$

Theorem 305.1. *If*

$$\pi(g)P_9 = P_9\pi(g) \quad (g \in G),$$

then the ambient representation decomposes as

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9)$$

into G -invariant subspaces, provided P_9 is an orthogonal projection.

Proof. For an orthogonal projection,

$$V_9 = \text{range}(P_9) \oplus \ker(P_9).$$

The range is G -invariant by the argument already established.

Let

$$v \in \ker(P_9).$$

Then

$$P_9 v = 0.$$

Using commutation,

$$P_9 \pi(g) v = \pi(g) P_9 v = 0.$$

Hence

$$\pi(g) v \in \ker(P_9).$$

Thus both summands are invariant. □

Corollary 305.2. *Under the preceding hypotheses,*

$$\pi = \pi_9 \oplus \pi_{\ker P_9},$$

where

$$\pi_9 = \pi|_{\mathcal{H}_9}.$$

Proof. The invariant orthogonal decomposition of V_9 yields the direct-sum decomposition of the representation. □

306 The Stable Sector as the Fixed Sector of the Stabilization

Assume in addition that

$$\Psi_9^5 = \Psi_9^4$$

and

$$P_9 = \Psi_9^4.$$

Theorem 306.1. *The stable sector satisfies*

$$\mathcal{H}_9 = \text{range}(P_9) = \ker(\Psi_9 - I).$$

Furthermore,

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

Proof. The equality between the range of P_9 and the fixed-point space was established from

$$\Psi_9 P_9 = \Psi_9^5 = \Psi_9^4 = P_9.$$

If

$$v = P_9 u,$$

then

$$\Psi_9 v = v.$$

Conversely, if

$$\Psi_9 v = v,$$

then

$$P_9 v = \Psi_9^4 v = v.$$

Hence $v \in \text{range}(P_9)$.

The second assertion is precisely the defining fixed-point property on this range. $\square$

Corollary 306.2. *The restriction of the stabilized operator to the stable sector carries no nontrivial dynamics:*

$$\Psi_9^k|_{\mathcal{H}_9} = I_{\mathcal{H}_9} \quad (k \geq 1).$$

Proof. Since

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9},$$

raising the restriction to any positive integer power leaves it unchanged. $\square$

307 Finite-State Trace Identity

Let

$$A \in \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Theorem 307.1. *For the maximally symmetric state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\text{tr}(\rho_9 A) = \frac{1}{9} \text{tr}(A).$$

Proof. By linearity of the trace,

$$\mathrm{tr}(\rho_9 A) = \mathrm{tr}\left(\frac{1}{9} A\right) = \frac{1}{9} \mathrm{tr}(A).$$

□

Corollary 307.2. *For every orthogonal projection*

$$Q \in \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the state probability is

$$\mathrm{tr}(\rho_9 Q) = \frac{\mathrm{rank}(Q)}{9}.$$

Proof. Apply the preceding theorem and use

$$\mathrm{tr}(Q) = \mathrm{rank}(Q).$$

□

Corollary 307.3. *For every unit vector*

$$\psi \in \mathcal{H}_9,$$

the rank-one projection

$$P_\psi = |\psi\rangle\langle\psi|$$

satisfies

$$\mathrm{tr}(\rho_9 P_\psi) = \frac{1}{9}.$$

Proof. Since

$$\mathrm{rank}(P_\psi) = 1,$$

the preceding corollary gives

$$\mathrm{tr}(\rho_9 P_\psi) = \frac{1}{9}.$$

□

308 State Determination by Rank-One Projections

Let

$$\omega : \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9) \longrightarrow \mathbb{C}$$

be a positive normalized linear functional, so that

$$\omega(I_{\mathcal{H}_9}) = 1.$$

Theorem 308.1. *There exists a unique positive operator*

$$\rho_\omega \in \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9)$$

such that

$$\mathrm{tr}(\rho_\omega) = 1$$

and

$$\omega(A) = \mathrm{tr}(\rho_\omega A) \quad (A \in \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9)).$$

Proof. Choose an orthonormal basis

$$\{e_1, \dots, e_9\}$$

of $\mathcal{H}_9$, and let

$$E_{ij} = |e_i\rangle\langle e_j|$$

be the corresponding matrix units. Define

$$(\rho_\omega)_{ji} = \omega(E_{ij}).$$

Linearity then gives

$$\omega(A) = \text{tr}(\rho_\omega A)$$

for every A .

It remains to verify positivity. If

$$A = B^* B,$$

then positivity of ω gives

$$\text{tr}(\rho_\omega B^* B) = \omega(B^* B) \geq 0.$$

Taking B to be arbitrary rank-one operators shows that

$$\langle v, \rho_\omega v \rangle \geq 0$$

for every $v \in \mathcal{H}_9$. Hence

$$\rho_\omega \geq 0.$$

Finally,

$$\text{tr}(\rho_\omega) = \omega(I_{\mathcal{H}_9}) = 1.$$

If ρ and $\tilde{\rho}$ both represent ω , then

$$\text{tr}((\rho - \tilde{\rho})A) = 0$$

for every A . Taking $A = E_{ji}$ for all i, j shows that every matrix coefficient of $\rho - \tilde{\rho}$ vanishes. Therefore

$$\rho = \tilde{\rho}.$$

□

Corollary 308.2. *A normalized positive state on*

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

is completely determined by its values on the rank-one projections.

Proof. The representing density operator is uniquely determined by the state. Its matrix coefficients are recovered from the associated linear functional, and hence the state is determined on the whole matrix algebra. □

309 Tomographic Determination of the Finite State

Let

$$\{e_1, \dots, e_9\}$$

be an orthonormal basis of $\mathcal{H}_9$, and let

$$E_j = |e_j\rangle\langle e_j|.$$

For a unit vector

$$\psi \in \mathcal{H}_9,$$

write

$$P_\psi = |\psi\rangle\langle\psi|.$$

Proposition 309.1. *For every density operator ρ ,*

$$\text{tr}(\rho P_\psi) = \langle\psi, \rho\psi\rangle.$$

Proof. The operator P_ψ has matrix coefficients

$$(P_\psi)_{ij} = \langle e_i, \psi\rangle\langle\psi, e_j\rangle.$$

Therefore

$$\text{tr}(\rho P_\psi) = \sum_{i,j} \langle e_i, \rho e_j\rangle\langle e_j, \psi\rangle\langle\psi, e_i\rangle,$$

which is precisely

$$\langle\psi, \rho\psi\rangle.$$

□

Theorem 309.2. *The values*

$$\text{tr}(\rho P_\psi)$$

for all unit vectors $\psi \in \mathcal{H}_9$ determine the density operator ρ uniquely.

Proof. Suppose that two density operators ρ and σ satisfy

$$\text{tr}(\rho P_\psi) = \text{tr}(\sigma P_\psi)$$

for every unit vector ψ . Set

$$A = \rho - \sigma.$$

Then $A = A^*$ and

$$\langle\psi, A\psi\rangle = 0$$

for every unit vector ψ , hence for every $\psi \in \mathcal{H}_9$.

For arbitrary $u, v \in \mathcal{H}_9$, the polarization identity gives

$$\langle u, Av\rangle = \frac{1}{4} \sum_{k=0}^3 i^k \left\langle u + i^k v, A(u + i^k v) \right\rangle.$$

Every term on the right-hand side vanishes. Thus

$$\langle u, Av\rangle = 0$$

for all u, v , and consequently

$$A = 0.$$

Therefore

$$\rho = \sigma.$$

□

310 Characterization of the Maximally Symmetric State

Let

$$\rho$$

be a density operator on $\mathcal{H}_9$.

Theorem 310.1. *The following conditions are equivalent:*

$$(i) \quad \rho = \frac{1}{9} I_{\mathcal{H}_9};$$

$$(ii) \quad \text{tr}(\rho P_\psi) = \frac{1}{9} \quad \text{for every unit vector } \psi \in \mathcal{H}_9;$$

$$(iii) \quad U\rho U^* = \rho \quad \text{for every } U \in \mathcal{U}(\mathcal{H}_9).$$

Proof. (i) implies (ii). For every unit vector ψ ,

$$\text{tr}(\rho P_\psi) = \frac{1}{9} \text{tr}(P_\psi) = \frac{1}{9}.$$

(ii) implies (i). The operator

$$\frac{1}{9} I_{\mathcal{H}_9}$$

has the same rank-one probabilities as ρ . By the tomographic determination theorem,

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

(i) implies (iii). For every unitary U ,

$$U\rho U^* = \frac{1}{9} U I_{\mathcal{H}_9} U^* = \frac{1}{9} I_{\mathcal{H}_9} = \rho.$$

(iii) implies (i). The invariance condition implies that every one-dimensional subspace is spectrally equivalent under the unitary group. Hence ρ has only one eigenvalue. Since

$$\text{tr}(\rho) = 1$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

that eigenvalue is

$$\frac{1}{9}.$$

Therefore

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

311 Purity and the Uniform Stable State

For a density operator ρ , define its purity by

$$\mathcal{P}(\rho) = \text{tr}(\rho^2).$$

Proposition 311.1. *For every density operator on the nine-dimensional stable sector,*

$$\frac{1}{9} \leq \mathcal{P}(\rho) \leq 1.$$

Proof. Let

$$\lambda_1, \dots, \lambda_9$$

be the eigenvalues of ρ . Positivity and normalization imply

$$\lambda_j \geq 0, \quad \sum_{j=1}^9 \lambda_j = 1.$$

Hence

$$\mathcal{P}(\rho) = \sum_{j=1}^9 \lambda_j^2.$$

The upper bound follows from

$$\lambda_j^2 \leq \lambda_j$$

for $0 \leq \lambda_j \leq 1$, giving

$$\mathcal{P}(\rho) \leq 1.$$

For the lower bound, Cauchy–Schwarz gives

$$\left(\sum_{j=1}^9 \lambda_j \right)^2 \leq 9 \sum_{j=1}^9 \lambda_j^2.$$

Since the left-hand side equals 1,

$$\mathcal{P}(\rho) \geq \frac{1}{9}.$$

□

Theorem 311.2. *The maximally symmetric state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is the unique density operator attaining the minimum purity:

$$\text{tr}(\rho_9^2) = \frac{1}{9}.$$

Proof. Direct computation gives

$$\rho_9^2 = \frac{1}{81} I_{\mathcal{H}_9},$$

and therefore

$$\text{tr}(\rho_9^2) = \frac{1}{81} \text{tr}(I_{\mathcal{H}_9}) = \frac{9}{81} = \frac{1}{9}.$$

Equality in the Cauchy–Schwarz inequality used above holds precisely when

$$\lambda_1 = \dots = \lambda_9.$$

Normalization then forces

$$\lambda_j = \frac{1}{9}$$

for every j . Hence

$$\rho = \rho_9.$$

□

312 Entropy Characterization

For a density operator ρ , define the von Neumann entropy by

$$S(\rho) = -\operatorname{tr}(\rho \log \rho).$$

Theorem 312.1. *For every density operator on $\mathcal{H}_9$,*

$$S(\rho) \leq \log 9.$$

Equality holds if and only if

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. Let

$$\lambda_1, \dots, \lambda_9$$

be the eigenvalues of ρ . Then

$$S(\rho) = -\sum_{j=1}^9 \lambda_j \log \lambda_j.$$

The function

$$x \mapsto -x \log x$$

is strictly concave on $(0, 1]$. Under the constraint

$$\sum_{j=1}^9 \lambda_j = 1,$$

Jensen's inequality gives

$$S(\rho) \leq -9 \left(\frac{1}{9} \right) \log \left(\frac{1}{9} \right) = \log 9.$$

Strict concavity shows that equality is possible only when

$$\lambda_1 = \dots = \lambda_9 = \frac{1}{9}.$$

Thus

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Corollary 312.2. *The state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is characterized equivalently as the unique state that is invariant under the full unitary group, the unique state assigning probability $1/9$ to every rank-one projection, the unique state of minimal purity, and the unique state of maximal von Neumann entropy.

Proof. The four characterizations were established respectively in the preceding results. □

313 Compatibility of Stabilization and State Invariance

Let

$$P_9 = \Psi_9^4$$

with

$$\Psi_9^5 = \Psi_9^4,$$

and let

$$\mathcal{H}_9 = \text{range}(P_9).$$

Theorem 313.1. *Suppose that*

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9$$

and that

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Then

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}$$

and

$$\Psi_9^k \rho_9 (\Psi_9^*)^k = \rho_9 \quad (k \geq 1),$$

provided the restriction of Ψ_9 to $\mathcal{H}_9$ is unitary.

Proof. The fixed-sector identity gives

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

Consequently,

$$\Psi_9^k|_{\mathcal{H}_9} = I_{\mathcal{H}_9}$$

for every $k \geq 1$. Therefore

$$\Psi_9^k \rho_9 (\Psi_9^*)^k = \rho_9.$$

The same conclusion follows from the unitary invariance of the scalar operator ρ_9 . $\square$

Corollary 313.2. *Once the finite stabilization identity and the nine-dimensionality of the stable sector have been established, the normalized maximally symmetric state is stationary under the stabilized dynamics.*

Proof. This is the preceding identity restricted to $\mathcal{H}_9$. $\square$

314 Separation of Structural and Representation-Theoretic Inputs

The preceding construction depends on three logically distinct ingredients.

Proposition 314.1. *The following assertions are independent in the sense that each requires its own hypothesis:*

$$\Psi_9^5 = \Psi_9^4,$$

$$\dim_{\mathbb{C}} \text{range}(P_9) = 9,$$

and

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. The first assertion is an algebraic stabilization identity for Ψ_9 . The second is a rank statement concerning the resulting projection P_9 . The third is a statement about a normalized positive functional on the resulting nine-dimensional space. Neither the stabilization identity alone determines the rank, nor does the rank alone determine a distinguished density operator without an additional symmetry or normalization principle. $\square$

Remark 314.2. This separation prevents the finite-dimensional state from being used to prove the dimension of the stable sector, and prevents the stabilization identity from being used to infer properties of a state that have not been imposed or derived independently. In particular, the value $1/9$ enters the state construction only after the nine-dimensional stable sector has been established.

315 Finite-Dimensional Closure of the Stable State Construction

Theorem 315.1. *Assume*

$$\Psi_9^5 = \Psi_9^4, \quad P_9 = \Psi_9^4, \quad \dim_{\mathbb{C}} \text{range}(P_9) = 9.$$

Let

$$\mathcal{H}_9 = \text{range}(P_9).$$

Then the following conclusions hold:

$$P_9^2 = P_9,$$

$$\mathcal{H}_9 = \ker(\Psi_9 - I),$$

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9},$$

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

and the unique density operator invariant under

$$\mathcal{U}(\mathcal{H}_9)$$

is

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Moreover, for every rank-one projection

$$P_\psi = |\psi\rangle\langle\psi|, \quad \|\psi\| = 1,$$

one has

$$\text{tr}(\rho_9 P_\psi) = \frac{1}{9}.$$

Proof. From

$$\Psi_9^5 = \Psi_9^4$$

one obtains

$$P_9^2 = \Psi_9^8 = \Psi_9^4 = P_9,$$

using the exact stabilization of all powers from the fourth step onward.

The fixed-point identity follows from

$$\Psi_9 P_9 = P_9$$

and the converse implication

$$\Psi_9 v = v \implies P_9 v = v.$$

Thus

$$\mathcal{H}_9 = \ker(\Psi_9 - I).$$

The restriction of Ψ_9 to this space is therefore the identity. The dimension is the stated rank hypothesis.

Finally, the unitary-invariance characterization gives

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

and the rank-one evaluation yields

$$\mathrm{tr}(\rho_9 P_\psi) = \frac{1}{9} \mathrm{tr}(P_\psi) = \frac{1}{9}.$$

□

316 Equivariance of the Stabilization Projection

Let

$$P_9 = \Psi_9^4$$

and let

$$\pi : G \longrightarrow \mathcal{U}(V_9)$$

be a unitary representation satisfying

$$\pi(g) P_9 = P_9 \pi(g) \quad (g \in G).$$

Proposition 316.1. *The stabilization projection P_9 is G -equivariant:*

$$P_9 \pi(g) = \pi(g) P_9 \quad (g \in G).$$

Proof. This is precisely the assumed commutation relation. □

Theorem 316.2. *For every*

$$g \in G,$$

the operator $\pi(g)$ preserves both

$$\mathcal{H}_9 = \mathrm{range}(P_9)$$

and

$$\ker(P_9).$$

Consequently,

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9)$$

is a decomposition into G -invariant subspaces.

Proof. The invariance of $\mathcal{H}_9$ was established previously. Let

$$v \in \ker(P_9).$$

Then

$$P_9 v = 0.$$

Hence

$$P_9 \pi(g) v = \pi(g) P_9 v = 0,$$

so

$$\pi(g) v \in \ker(P_9).$$

Since P_9 is idempotent, its range and kernel give the direct-sum decomposition

$$V_9 = \text{range}(P_9) \oplus \ker(P_9).$$

Both summands are invariant under $\pi(G)$. □

Corollary 316.3. *Relative to the decomposition*

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9),$$

every representation operator has block-diagonal form

$$\pi(g) = \begin{pmatrix} \pi_9(g) & 0 \\ 0 & \pi_0(g) \end{pmatrix},$$

where

$$\pi_9(g) = \pi(g)|_{\mathcal{H}_9}$$

and

$$\pi_0(g) = \pi(g)|_{\ker(P_9)}.$$

Proof. Both summands are invariant under $\pi(g)$. Hence the off-diagonal blocks vanish, giving the stated decomposition. □

317 Restriction and Quotient Description of the Stable Sector

Let

$$W_9 = \ker(P_9).$$

Proposition 317.1. *The quotient space*

$$V_9 / W_9$$

is canonically isomorphic to

$$\mathcal{H}_9.$$

Proof. Define

$$\Phi : V_9 / W_9 \longrightarrow \mathcal{H}_9$$

by

$$\Phi(v + W_9) = P_9 v.$$

If

$$v + W_9 = v' + W_9,$$

then

$$v - v' \in W_9,$$

and therefore

$$P_9(v - v') = 0.$$

Thus Φ is well-defined.

It is surjective because every

$$u \in \mathcal{H}_9$$

satisfies

$$u = P_9 u.$$

If

$$\Phi(v + W_9) = 0,$$

then

$$P_9 v = 0,$$

so

$$v \in W_9.$$

Hence

$$v + W_9 = W_9,$$

and Φ is injective. □

Theorem 317.2. *The canonical isomorphism*

$$\Phi : V_9 / \ker(P_9) \longrightarrow \mathcal{H}_9$$

is G -equivariant.

Proof. The representation preserves $\ker(P_9)$, so it induces a quotient representation

$$\bar{\pi}(g)(v + \ker(P_9)) = \pi(g)v + \ker(P_9).$$

For every $v \in V_9$,

$$\Phi(\bar{\pi}(g)(v + \ker(P_9))) = P_9 \pi(g)v.$$

Using the commutation relation,

$$P_9 \pi(g)v = \pi(g)P_9 v.$$

Therefore

$$\Phi(\bar{\pi}(g)(v + \ker(P_9))) = \pi_9(g)\Phi(v + \ker(P_9)).$$

Thus Φ intertwines the quotient representation with π_9 . □

318 Dimension of the Stable Representation

Assume

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Theorem 318.1. *The stable representation*

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

is a nine-dimensional unitary representation.

Proof. The preceding construction shows that π_9 acts unitarily on $\mathcal{H}_9$. By assumption,

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Hence π_9 has dimension 9. □

Corollary 318.2. *If*

$$\dim_{\mathbb{C}} V_9 = 143,$$

then

$$\dim_{\mathbb{C}} \ker(P_9) = 134.$$

Proof. Since

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9),$$

one has

$$143 = 9 + \dim_{\mathbb{C}} \ker(P_9).$$

Therefore

$$\dim_{\mathbb{C}} \ker(P_9) = 134. \quad \square$$

319 Intertwining of the Stabilized Dynamics

Assume

$$\Psi_9^5 = \Psi_9^4$$

and

$$P_9 = \Psi_9^4.$$

Proposition 319.1. *The stabilization projection commutes with Ψ_9 :*

$$\Psi_9 P_9 = P_9 \Psi_9 = P_9.$$

Proof. By definition,

$$\Psi_9 P_9 = \Psi_9^5 = \Psi_9^4 = P_9.$$

Since powers of the same operator commute,

$$P_9 \Psi_9 = \Psi_9^4 \Psi_9 = \Psi_9^5 = P_9. \quad \square$$

Theorem 319.2. *The stabilized sector is pointwise fixed by the dynamics, while the kernel of the stabilization projection is annihilated after four iterations:*

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9},$$

and

$$\Psi_9^4|_{\ker(P_9)} = 0.$$

Proof. If

$$\nu \in \mathcal{H}_9,$$

then

$$P_9 \nu = \nu.$$

Since

$$\Psi_9 P_9 = P_9,$$

it follows that

$$\Psi_9 \nu = \Psi_9 P_9 \nu = P_9 \nu = \nu.$$

If

$$\nu \in \ker(P_9),$$

then

$$\Psi_9^4 \nu = P_9 \nu = 0.$$

□

Corollary 319.3. *The finite-time dynamics admits the decomposition*

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9)$$

such that

$$\Psi_9 = \begin{pmatrix} I_{\mathcal{H}_9} & 0 \\ 0 & N_9 \end{pmatrix},$$

where

$$N_9^4 = 0.$$

Proof. The two summands are invariant under Ψ_9 . Its restriction to $\mathcal{H}_9$ is the identity, while its restriction

$$N_9 = \Psi_9|_{\ker(P_9)}$$

satisfies

$$N_9^4 = 0.$$

□

320 Nilpotent Transient Sector

Let

$$N_9 = \Psi_9|_{\ker(P_9)}.$$

Theorem 320.1. *The transient operator N_9 is nilpotent of index at most four:*

$$N_9^4 = 0.$$

Consequently,

$$\sigma(N_9) \subseteq \{0\}.$$

Proof. The identity

$$N_9^4 = 0$$

follows from the preceding theorem. If

$$N_9 v = \lambda v$$

for some nonzero v , then

$$0 = N_9^4 v = \lambda^4 v.$$

Thus

$$\lambda = 0.$$

Hence

$$\sigma(N_9) \subseteq \{0\}.$$

□

Corollary 320.2. *The characteristic polynomial of N_9 is*

$$\chi_{N_9}(\lambda) = \lambda^{134}$$

when

$$\dim_{\mathbb{C}} \ker(P_9) = 134.$$

Proof. A nilpotent operator has only the eigenvalue 0, and the degree of its characteristic polynomial equals the dimension of the underlying space. Here that dimension is 134. □

321 Characteristic Polynomial of the Full Stabilized Operator

Assume

$$\dim_{\mathbb{C}} V_9 = 143$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Theorem 321.1. *The characteristic polynomial of Ψ_9 has the form*

$$\chi_{\Psi_9}(\lambda) = (\lambda - 1)^9 \lambda^{134}.$$

Proof. Relative to

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9),$$

the operator has block form

$$\Psi_9 = \begin{pmatrix} I_9 & 0 \\ 0 & N_9 \end{pmatrix},$$

with

$$N_9^4 = 0.$$

Therefore

$$\chi_{\Psi_9}(\lambda) = \chi_{I_9}(\lambda) \chi_{N_9}(\lambda).$$

The first factor is

$$(\lambda - 1)^9,$$

while the second is

$$\lambda^{134}.$$

Hence

$$\chi_{\Psi_9}(\lambda) = (\lambda - 1)^9 \lambda^{134}.$$

□

Corollary 321.2. *The eigenvalue 1 has algebraic multiplicity 9, whereas the eigenvalue 0 has algebraic multiplicity 134.*

Proof. These multiplicities are read directly from

$$\chi_{\Psi_9}(\lambda) = (\lambda - 1)^9 \lambda^{134}.$$

□

322 Jordan Structure of the Transient Sector

Let

$$W_9 = \ker(P_9)$$

and

$$N_9 = \Psi_9|_{W_9}.$$

Then

$$N_9^4 = 0.$$

Proposition 322.1. *Every Jordan block of N_9 has size at most four.*

Proof. The minimal polynomial of N_9 divides

$$X^4.$$

Hence the nilpotency index of N_9 is at most four. The size of the largest Jordan block of a nilpotent operator equals its nilpotency index, and therefore no Jordan block can have size greater than four. □

Theorem 322.2. *There exist nonnegative integers*

$$a_1, a_2, a_3, a_4$$

such that

$$\dim_{\mathbb{C}} W_9 = a_1 + 2a_2 + 3a_3 + 4a_4,$$

and N_9 is similar to a direct sum of

$$a_1$$

Jordan blocks of size 1,

$$a_2$$

Jordan blocks of size 2,

$$a_3$$

Jordan blocks of size 3, and

$$a_4$$

Jordan blocks of size 4.

Proof. The Jordan canonical form exists over $\mathbb{C}$. Since N_9 is nilpotent, every Jordan block corresponds to the eigenvalue 0. The preceding proposition restricts the possible block sizes to 1, 2, 3, 4. Counting dimensions gives the stated identity. □

Corollary 322.3. *If*

$$\dim_{\mathbb{C}} W_9 = 134,$$

then

$$134 = a_1 + 2a_2 + 3a_3 + 4a_4.$$

323 Ranks of the Transient Powers

Theorem 323.1. *For*

$$1 \leq k \leq 4,$$

the rank of N_9^k is

$$\text{rank}(N_9^k) = \sum_{j=k+1}^4 (j-k)a_j.$$

In particular,

$$\text{rank}(N_9) = a_2 + 2a_3 + 3a_4,$$

$$\text{rank}(N_9^2) = a_3 + 2a_4,$$

and

$$\text{rank}(N_9^3) = a_4.$$

Proof. A nilpotent Jordan block of size j has

$$\max(j-k, 0)$$

as the rank of its k -th power. Summing these contributions over all Jordan blocks yields

$$\text{rank}(N_9^k) = \sum_{j=k+1}^4 (j-k)a_j.$$

The displayed formulas follow by taking $k = 1, 2, 3$. □

Corollary 323.2. *The dimensions of the kernels of the transient powers satisfy*

$$\dim \ker(N_9) = a_1 + a_2 + a_3 + a_4,$$

$$\dim \ker(N_9^2) = a_1 + 2a_2 + 2a_3 + 2a_4,$$

and

$$\dim \ker(N_9^3) = a_1 + 2a_2 + 3a_3 + 3a_4.$$

Proof. Apply the rank-nullity theorem to N_9^k and use

$$\dim W_9 = a_1 + 2a_2 + 3a_3 + 4a_4.$$

□

324 Minimal Polynomial of the Stabilized Dynamics

Theorem 324.1. *The minimal polynomial of Ψ_9 divides*

$$X^4(X-1).$$

Proof. The stabilization identity gives

$$\Psi_9^5 = \Psi_9^4,$$

and therefore

$$\Psi_9^4(\Psi_9 - I) = 0.$$

Thus

$$X^4(X-1)$$

is an annihilating polynomial for Ψ_9 . By definition, the minimal polynomial divides every annihilating polynomial. □

Proposition 324.2. *The minimal polynomial has the form*

$$m_{\Psi_9}(X) = X^s(X - 1),$$

for some

$$0 \leq s \leq 4,$$

unless the transient sector is trivial, in which case

$$m_{\Psi_9}(X) = X - 1.$$

Proof. The restriction of Ψ_9 to $\mathcal{H}_9$ is the identity, so the factor $X - 1$ occurs. On W_9 , the operator is nilpotent, hence its minimal polynomial is X^s for some $0 \leq s \leq 4$. Since the two factors are coprime, the minimal polynomial of the direct sum is their least common multiple, giving the stated form. $\square$

Corollary 324.3. *The exact stabilization relation*

$$\Psi_9^5 = \Psi_9^4$$

does not by itself imply that the nilpotency index of the transient sector is exactly four.

Proof. The minimal polynomial may contain

$$X, \quad X^2, \quad X^3, \quad \text{or} \quad X^4.$$

The stabilization identity supplies only the upper bound 4. $\square$

325 Polynomial Reduction at the Stabilized Operator

Theorem 325.1. *For every polynomial*

$$f(X) \in \mathbb{C}[X],$$

there exist uniquely determined scalars

$$f(1), \quad c_0, c_1, c_2, c_3 \in \mathbb{C}$$

such that

$$f(\Psi_9) = f(1)P_9 + c_0(I - P_9) + c_1N_9 + c_2N_9^2 + c_3N_9^3$$

relative to the decomposition

$$V_9 = \mathcal{H}_9 \oplus W_9.$$

Proof. On $\mathcal{H}_9$,

$$\Psi_9 = I,$$

so

$$f(\Psi_9)|_{\mathcal{H}_9} = f(1)I_{\mathcal{H}_9}.$$

On W_9 ,

$$\Psi_9 = N_9$$

and

$$N_9^4 = 0.$$

Writing

$$f(X) = c_0 + c_1 X + c_2 X^2 + c_3 X^3 + X^4 q(X)$$

for a polynomial q , one obtains

$$f(N_9) = c_0 I_{W_9} + c_1 N_9 + c_2 N_9^2 + c_3 N_9^3.$$

Combining the two invariant summands gives the asserted expression. $\square$

Corollary 325.2. *For every polynomial f ,*

$$f(\Psi_9)P_9 = f(1)P_9.$$

Proof. Since

$$\Psi_9 P_9 = P_9,$$

one has

$$f(\Psi_9)P_9 = f(1)P_9.$$

$\square$

326 Exact Functional Calculus on the Stable Sector

Let

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Proposition 326.1. *The restriction map*

$$\text{End}_{\mathbb{C}}(V_9) \longrightarrow \text{End}_{\mathbb{C}}(\mathcal{H}_9), \quad A \longmapsto A|_{\mathcal{H}_9},$$

is surjective.

Proof. Let

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Using

$$V_9 = \mathcal{H}_9 \oplus W_9,$$

define

$$\tilde{X}(v + w) = Xv \quad (v \in \mathcal{H}_9, w \in W_9).$$

Then

$$\tilde{X} \in \text{End}_{\mathbb{C}}(V_9)$$

and

$$\tilde{X}|_{\mathcal{H}_9} = X.$$

$\square$

Theorem 326.2. *The map*

$$\iota : \text{End}_{\mathbb{C}}(\mathcal{H}_9) \longrightarrow \text{End}_{\mathbb{C}}(V_9)$$

defined by

$$\iota(X) = P_9 X P_9$$

is an injective algebra homomorphism onto

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

Proof. For

$$X, Y \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the operators $P_9 X P_9$ and $P_9 Y P_9$ act as X and Y on $\mathcal{H}_9$ and vanish on W_9 . Hence

$$\iota(XY) = P_9 XY P_9 = (P_9 X P_9)(P_9 Y P_9) = \iota(X)\iota(Y).$$

Linearity is immediate.

If

$$\iota(X) = 0,$$

then restricting to $\mathcal{H}_9$ gives

$$X = 0,$$

so ι is injective.

Finally, every element of

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9$$

is of the form $P_9 A P_9$ and its restriction to $\mathcal{H}_9$ defines an element of $\text{End}_{\mathbb{C}}(\mathcal{H}_9)$. Thus the map is onto the stated corner algebra. $\square$

Corollary 326.3. *There is a canonical algebra isomorphism*

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9) \cong P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

327 The Stable Corner and Its Identity

Define

$$\mathcal{A}_9 = P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

Proposition 327.1. *The element P_9 is the identity of the algebra $\mathcal{A}_9$.*

Proof. For every

$$A = P_9 X P_9 \in \mathcal{A}_9,$$

one has

$$P_9 A = P_9^2 X P_9 = P_9 X P_9 = A,$$

and similarly

$$A P_9 = P_9 X P_9^2 = A.$$

Thus P_9 is the multiplicative identity of $\mathcal{A}_9$. $\square$

Theorem 327.2. *If*

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

then

$$\mathcal{A}_9 \cong M_9(\mathbb{C})$$

and

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 81.$$

Proof. By the canonical algebra isomorphism above,

$$\mathcal{A}_9 \cong \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

choosing a basis identifies the latter algebra with

$$M_9(\mathbb{C}),$$

whose complex dimension is

$$9^2 = 81.$$

□

328 The Trace on the Stable Corner

Let

$$\mathcal{A}_9 = P_9 \text{End}_{\mathbb{C}}(V_9) P_9$$

and assume

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Definition 328.1. For

$$A \in \mathcal{A}_9,$$

define

$$\tau_9(A) = \frac{1}{9} \text{Tr}_{V_9}(A).$$

Proposition 328.2. *The functional τ_9 is a normalized trace on $\mathcal{A}_9$:*

$$\tau_9(AB) = \tau_9(BA), \quad \tau_9(P_9) = 1.$$

Proof. Since A, B are endomorphisms of the finite-dimensional space V_9 , cyclicity of the ordinary trace gives

$$\text{Tr}_{V_9}(AB) = \text{Tr}_{V_9}(BA).$$

Hence

$$\tau_9(AB) = \tau_9(BA).$$

Furthermore,

$$\text{Tr}_{V_9}(P_9) = \text{rank}(P_9) = 9,$$

and therefore

$$\tau_9(P_9) = \frac{1}{9} \text{Tr}_{V_9}(P_9) = 1.$$

□

Theorem 328.3. *For every*

$$A \in \mathcal{A}_9,$$

one has

$$\tau_9(A) = \frac{1}{9} \text{Tr}_{\mathcal{H}_9}(A|_{\mathcal{H}_9}).$$

Proof. Every $A \in \mathcal{A}_9$ satisfies

$$A = P_9 A P_9.$$

Consequently,

$$A|_{\ker(P_9)} = 0.$$

Relative to

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9),$$

the operator A has the form

$$A = \begin{pmatrix} A_9 & 0 \\ 0 & 0 \end{pmatrix},$$

where

$$A_9 = A|_{\mathcal{H}_9}.$$

Therefore

$$\mathrm{Tr}_{V_9}(A) = \mathrm{Tr}_{\mathcal{H}_9}(A_9),$$

which gives the result. □

Corollary 328.4. *Under the canonical identification*

$$\mathcal{A}_9 \cong M_9(\mathbb{C}),$$

the trace τ_9 is the normalized matrix trace

$$\tau_9(A) = \frac{1}{9} \mathrm{tr}(A).$$

329 Positivity on the Stable Corner

Let

$$\mathcal{A}_9 = P_9 \mathrm{End}_{\mathbb{C}}(V_9) P_9.$$

Definition 329.1. An element

$$A \in \mathcal{A}_9$$

is called positive if

$$A = A^*$$

and

$$\langle Av, v \rangle \geq 0 \quad (v \in \mathcal{H}_9).$$

Proposition 329.2. *If*

$$A \in \mathcal{A}_9$$

is positive, then

$$\tau_9(A) \geq 0.$$

Proof. The restriction

$$A_9 = A|_{\mathcal{H}_9}$$

is a positive operator on the finite-dimensional Hilbert space $\mathcal{H}_9$. Its eigenvalues are therefore nonnegative. Hence

$$\mathrm{Tr}_{\mathcal{H}_9}(A_9) \geq 0.$$

The preceding theorem gives

$$\tau_9(A) = \frac{1}{9} \mathrm{Tr}_{\mathcal{H}_9}(A_9) \geq 0.$$

□

Theorem 329.3. *The functional*

$$\tau_9 : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

is faithful:

$$A \geq 0, \quad \tau_9(A) = 0 \implies A = 0.$$

Proof. Let $A \geq 0$. Its eigenvalues

$$\lambda_1, \dots, \lambda_9$$

are nonnegative. Since

$$\tau_9(A) = \frac{1}{9} \sum_{j=1}^9 \lambda_j,$$

the equality

$$\tau_9(A) = 0$$

forces

$$\lambda_j = 0 \quad (1 \leq j \leq 9).$$

Thus $A = 0$. □

330 The Stable State as a Trace State

Define

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Theorem 330.1. *For every*

$$A \in \mathcal{A}_9,$$

the state determined by ρ_9 coincides with the normalized trace:

$$\text{Tr}_{\mathcal{H}_9}(\rho_9 A) = \tau_9(A).$$

Proof. By definition,

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Hence

$$\text{Tr}_{\mathcal{H}_9}(\rho_9 A) = \frac{1}{9} \text{Tr}_{\mathcal{H}_9}(A) = \tau_9(A).$$

□

Corollary 330.2. *For every positive*

$$A \in \mathcal{A}_9,$$

$$\tau_9(A) \geq 0,$$

and

$$\tau_9(P_9) = 1.$$

Thus τ_9 is a state on the unital finite-dimensional $$ -algebra $\mathcal{A}_9$.*

331 Unitary Invariance of the Stable Trace

Let

$$U \in \mathcal{U}(\mathcal{H}_9).$$

Proposition 331.1. *For every*

$$A \in \mathcal{A}_9,$$

one has

$$\tau_9(UAU^*) = \tau_9(A).$$

Proof. By cyclicity of the trace,

$$\tau_9(UAU^*) = \frac{1}{9} \operatorname{Tr}_{\mathcal{H}_9}(UAU^*) = \frac{1}{9} \operatorname{Tr}_{\mathcal{H}_9}(AU^*U).$$

Since

$$U^*U = I_{\mathcal{H}_9},$$

it follows that

$$\tau_9(UAU^*) = \frac{1}{9} \operatorname{Tr}_{\mathcal{H}_9}(A) = \tau_9(A).$$

□

Theorem 331.2. *The stable state ρ_9 is invariant under every unitary automorphism of $\mathcal{A}_9$ induced by conjugation:*

$$U\rho_9U^* = \rho_9 \quad (U \in \mathcal{U}(\mathcal{H}_9)).$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$U\rho_9U^* = \frac{1}{9} UI_{\mathcal{H}_9}U^* = \frac{1}{9} UU^* = \frac{1}{9} I_{\mathcal{H}_9} = \rho_9.$$

□

332 Restriction of the Arithmetic Representation

Let

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be the restricted representation constructed above.

Definition 332.1. The induced action of G on the stable algebra is

$$\alpha_g(A) = \pi_9(g)A\pi_9(g)^* \quad (A \in \mathcal{A}_9).$$

Proposition 332.2. *For every*

$$g \in G,$$

the map

$$\alpha_g : \mathcal{A}_9 \longrightarrow \mathcal{A}_9$$

is a unital $$ -automorphism.*

Proof. Since $\pi_9(g)$ is unitary,

$$\alpha_g(AB) = \pi_9(g)AB\pi_9(g)^* = \alpha_g(A)\alpha_g(B).$$

Moreover,

$$\alpha_g(A^*) = \pi_9(g)A^*\pi_9(g)^* = \alpha_g(A)^*.$$

Finally,

$$\alpha_g(P_9) = P_9,$$

because P_9 acts as the identity on $\mathcal{H}_9$. The inverse of α_g is

$$\alpha_{g^{-1}}.$$

Thus α_g is a unital $*$ -automorphism. □

Theorem 332.3. *The family*

$$\{\alpha_g\}_{g \in G}$$

defines an action of G by trace-preserving $$ -automorphisms of $\mathcal{A}_9$:*

$$\alpha_{g_1 g_2} = \alpha_{g_1} \circ \alpha_{g_2},$$

and

$$\tau_9 \circ \alpha_g = \tau_9.$$

Proof. The representation identity

$$\pi_9(g_1 g_2) = \pi_9(g_1)\pi_9(g_2)$$

implies

$$\alpha_{g_1 g_2}(A) = \pi_9(g_1)\pi_9(g_2)A\pi_9(g_2)^*\pi_9(g_1)^* = \alpha_{g_1}(\alpha_{g_2}(A)).$$

Trace preservation follows from the unitary invariance of τ_9 . □

Corollary 332.4. *The stable finite-dimensional algebra carries a canonical G -invariant state:*

$$\tau_9(\alpha_g(A)) = \tau_9(A) \quad (g \in G, A \in \mathcal{A}_9).$$

333 Compatibility of Stabilization with the Group Action

Theorem 333.1. *For every*

$$g \in G$$

and every

$$X \in \text{End}_{\mathbb{C}}(V_9),$$

one has

$$P_9\pi(g)X\pi(g)^*P_9 = \pi(g)P_9XP_9\pi(g)^*.$$

Proof. Using

$$P_9\pi(g) = \pi(g)P_9$$

and, by taking adjoints,

$$\pi(g)^*P_9 = P_9\pi(g)^*,$$

one obtains

$$P_9\pi(g)X\pi(g)^*P_9 = \pi(g)P_9XP_9\pi(g)^*.$$

□

Corollary 333.2. *Compression to the stable sector commutes with the induced group action:*

$$(\pi(g)X\pi(g)^*)_9 = \pi_9(g)X\pi_9(g)^*.$$

Proof. This is the preceding identity restricted to $\mathcal{H}_9$. □

334 The Commutant of the Stable Projection

Let

$$P_9 = \Psi_9^4$$

and suppose

$$\Psi_9^5 = \Psi_9^4.$$

Let

$$\mathcal{H}_9 = \text{range}(P_9).$$

Proposition 334.1. *The operator P_9 is idempotent:*

$$P_9^2 = P_9.$$

Proof. Using

$$P_9 = \Psi_9^4$$

and

$$\Psi_9^5 = \Psi_9^4,$$

one obtains

$$P_9^2 = \Psi_9^8 = \Psi_9^4 = P_9,$$

where the equality

$$\Psi_9^k = \Psi_9^4 \quad (k \geq 4)$$

follows by repeated multiplication of the stabilization identity by Ψ_9 . □

Proposition 334.2. *For every*

$$X \in \text{End}_{\mathbb{C}}(V_9)$$

satisfying

$$XP_9 = P_9X,$$

the subspaces

$$\text{range}(P_9) \quad \text{and} \quad \ker(P_9)$$

are invariant under X .

Proof. Let

$$v \in \text{range}(P_9).$$

Then

$$v = P_9 u$$

for some $u \in V_9$. Hence

$$Xv = XP_9 u = P_9 Xu \in \text{range}(P_9).$$

Thus $\text{range}(P_9)$ is X -invariant.

If

$$v \in \ker(P_9),$$

then

$$P_9 v = 0.$$

Consequently,

$$P_9 X v = X P_9 v = 0,$$

and therefore

$$X v \in \ker(P_9).$$

□

Theorem 334.3. *If X commutes with P_9 , then with respect to the direct-sum decomposition*

$$V_9 = \text{range}(P_9) \oplus \ker(P_9),$$

the operator X is block diagonal.

Proof. Since P_9 is idempotent,

$$V_9 = \text{range}(P_9) \oplus \ker(P_9).$$

The preceding proposition shows that both summands are invariant under X . Hence X has no off-diagonal components relative to this decomposition and therefore has the form

$$X = \begin{pmatrix} X_{\text{st}} & 0 \\ 0 & X_{\text{tr}} \end{pmatrix},$$

where

$$X_{\text{st}} \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

and

$$X_{\text{tr}} \in \text{End}_{\mathbb{C}}(\ker P_9).$$

□

Corollary 334.4. *The commutant of P_9 ,*

$$\{X \in \text{End}_{\mathbb{C}}(V_9) : XP_9 = P_9X\},$$

preserves the stable and transient sectors separately.

335 The Stable Restriction of Commuting Observables

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

For every

$$X \in \text{End}_{\mathbb{C}}(V_9)$$

commuting with P_9 , define

$$\text{res}_9(X) = X|_{\mathcal{H}_9}.$$

Proposition 335.1. *The map*

$$\text{res}_9 : \{X : XP_9 = P_9X\} \longrightarrow \mathcal{A}_9$$

is an algebra homomorphism.

Proof. For commuting operators X, Y and scalars a, b ,

$$\text{res}_9(aX + bY) = a\text{res}_9(X) + b\text{res}_9(Y).$$

Moreover,

$$\text{res}_9(XY) = (XY)|_{\mathcal{H}_9} = X|_{\mathcal{H}_9} Y|_{\mathcal{H}_9} = \text{res}_9(X) \text{res}_9(Y).$$

Finally,

$$\text{res}_9(I_{V_9}) = I_{\mathcal{H}_9}.$$

Thus res_9 is a unital algebra homomorphism. □

Theorem 335.2. *Every operator*

$$A \in \mathcal{A}_9$$

admits a commuting extension to V_9 of the form

$$\tilde{A} = A \oplus 0$$

relative to

$$V_9 = \mathcal{H}_9 \oplus \ker(P_9).$$

This extension satisfies

$$\tilde{A}P_9 = P_9\tilde{A}$$

and

$$\text{res}_9(\tilde{A}) = A.$$

Proof. Define

$$\tilde{A}(\nu_{\text{st}} + \nu_{\text{tr}}) = A\nu_{\text{st}}$$

for

$$\nu_{\text{st}} \in \mathcal{H}_9, \quad \nu_{\text{tr}} \in \ker(P_9).$$

Then

$$P_9(\nu_{\text{st}} + \nu_{\text{tr}}) = \nu_{\text{st}}.$$

Consequently,

$$\tilde{A}P_9(\nu_{\text{st}} + \nu_{\text{tr}}) = A\nu_{\text{st}},$$

while

$$P_9\tilde{A}(\nu_{\text{st}} + \nu_{\text{tr}}) = P_9A\nu_{\text{st}} = A\nu_{\text{st}},$$

because

$$A\nu_{\text{st}} \in \mathcal{H}_9.$$

Hence

$$\tilde{A}P_9 = P_9\tilde{A}.$$

The restriction to $\mathcal{H}_9$ is evidently A . □

336 The Stable Algebra as a Full Matrix Algebra

Assume now that

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Theorem 336.1. *The stable algebra*

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

is isomorphic, as a unital complex algebra, to

$$M_9(\mathbb{C}).$$

Proof. Choose an ordered basis

$$(e_1, \dots, e_9)$$

of $\mathcal{H}_9$. Every endomorphism A is represented uniquely by a 9×9 complex matrix

$$[A]_e.$$

Composition of endomorphisms corresponds to matrix multiplication and the identity endomorphism corresponds to the identity matrix. Hence

$$\mathcal{A}_9 \cong M_9(\mathbb{C}).$$

□

Corollary 336.2. *The stable algebra has complex dimension*

$$81.$$

Proof. Since

$$\dim_{\mathbb{C}} M_9(\mathbb{C}) = 9^2,$$

one obtains

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 81.$$

□

Proposition 336.3. *The normalized trace*

$$\tau_9(A) = \frac{1}{9} \operatorname{tr}(A)$$

is faithful on positive elements:

$$A \geq 0, \quad \tau_9(A) = 0 \implies A = 0.$$

Proof. If $A \geq 0$, all eigenvalues of A are nonnegative. The condition

$$\tau_9(A) = 0$$

implies

$$\operatorname{tr}(A) = 0.$$

Hence every eigenvalue is zero, and therefore

$$A = 0.$$

□

337 The Maximally Mixed State on the Stable Algebra

Define

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proposition 337.1. *The functional*

$$\omega_9(A) = \operatorname{tr}(\rho_9 A)$$

coincides with the normalized trace:

$$\omega_9 = \tau_9.$$

Proof. For every

$$A \in \mathcal{A}_9,$$

one has

$$\omega_9(A) = \operatorname{tr}\left(\frac{1}{9} I_{\mathcal{H}_9} A\right) = \frac{1}{9} \operatorname{tr}(A) = \tau_9(A).$$

□

Theorem 337.2. *The functional*

$$\omega_9 : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

is a normalized positive tracial state:

$$\omega_9(I) = 1,$$

$$A \geq 0 \implies \omega_9(A) \geq 0,$$

and

$$\omega_9(AB) = \omega_9(BA)$$

for all

$$A, B \in \mathcal{A}_9.$$

Proof. Normalization follows from

$$\omega_9(I) = \frac{1}{9} \operatorname{tr}(I) = \frac{1}{9} \cdot 9 = 1.$$

If $A \geq 0$, then

$$\operatorname{tr}(A) \geq 0,$$

and hence

$$\omega_9(A) \geq 0.$$

Finally,

$$\omega_9(AB) = \frac{1}{9} \operatorname{tr}(AB) = \frac{1}{9} \operatorname{tr}(BA) = \omega_9(BA).$$

□

338 Unitary Invariance and Uniqueness

Let

$$\mathcal{U}(\mathcal{H}_9)$$

denote the unitary group of the stable sector.

Proposition 338.1. *For every*

$$U \in \mathcal{U}(\mathcal{H}_9)$$

and every

$$A \in \mathcal{A}_9,$$

one has

$$\omega_9(UAU^*) = \omega_9(A).$$

Proof. Using cyclicity of the trace,

$$\omega_9(UAU^*) = \frac{1}{9} \operatorname{tr}(UAU^*) = \frac{1}{9} \operatorname{tr}(AU^*U) = \frac{1}{9} \operatorname{tr}(A) = \omega_9(A).$$

□

Theorem 338.2. *The normalized tracial state on $\mathcal{A}_9$ is unique.*

Proof. Let

$$\varphi : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a normalized tracial state. Choose matrix units

$$E_{ij}, \quad 1 \leq i, j \leq 9,$$

relative to an orthonormal basis of $\mathcal{H}_9$.

For $i \neq j$,

$$E_{ij} = E_{ii}E_{ij},$$

and the tracial property gives

$$\varphi(E_{ij}) = \varphi(E_{ii}E_{ij}) = \varphi(E_{ij}E_{ii}) = 0.$$

For the diagonal matrix units, if $i \neq j$, then

$$E_{ij}E_{ji} = E_{ii}, \quad E_{ji}E_{ij} = E_{jj}.$$

Therefore

$$\varphi(E_{ii}) = \varphi(E_{ij}E_{ji}) = \varphi(E_{ji}E_{ij}) = \varphi(E_{jj}).$$

Thus all diagonal values coincide with some constant c . Since

$$I = \sum_{i=1}^9 E_{ii},$$

normalization gives

$$1 = \varphi(I) = 9c,$$

hence

$$c = \frac{1}{9}.$$

For

$$A = \sum_{i,j=1}^9 a_{ij}E_{ij},$$

it follows that

$$\varphi(A) = \frac{1}{9} \sum_{i=1}^9 a_{ii} = \frac{1}{9} \operatorname{tr}(A) = \tau_9(A).$$

Therefore

$$\varphi = \tau_9 = \omega_9.$$

□

339 Stable Expectations

Let

$$A = A^* \in \mathcal{A}_9.$$

Write its spectral decomposition as

$$A = \sum_{j=1}^r \lambda_j E_j,$$

where the λ_j are distinct real numbers and the E_j are pairwise orthogonal spectral projections.

Theorem 339.1. *The expectation of A in the stable state is*

$$\omega_9(A) = \frac{1}{9} \sum_{j=1}^r \lambda_j \text{rank}(E_j).$$

Proof. By linearity,

$$\omega_9(A) = \sum_{j=1}^r \lambda_j \omega_9(E_j).$$

Since

$$\omega_9(E_j) = \frac{1}{9} \text{tr}(E_j) = \frac{1}{9} \text{rank}(E_j),$$

the stated formula follows. $\square$

Corollary 339.2. *The stable state assigns to each spectral subspace $E_j \mathcal{H}_9$ the probability*

$$\omega_9(E_j) = \frac{\dim_{\mathbb{C}}(E_j \mathcal{H}_9)}{9}.$$

Corollary 339.3. *If A has nine distinct eigenvalues, then every eigenspace has dimension one and the corresponding spectral projections all have probability*

$$\frac{1}{9}.$$

340 Compatibility with the Stabilization Dynamics

Let

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

Proposition 340.1. *Every stable observable*

$$A \in \mathcal{A}_9$$

commutes with the restricted stabilization operator:

$$A \Psi_9|_{\mathcal{H}_9} = \Psi_9|_{\mathcal{H}_9} A.$$

Proof. Since

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9},$$

both sides are equal to A . $\square$

Theorem 340.2. *The Heisenberg evolution induced by the stabilized operator is trivial on the stable algebra:*

$$\text{Ad}_{\Psi_9|_{\mathcal{H}_9}}(A) = A \quad (A \in \mathcal{A}_9).$$

Consequently,

$$\text{Ad}_{\Psi_9|_{\mathcal{H}_9}}^k(A) = A \quad (k \geq 1).$$

Proof. For every

$$A \in \mathcal{A}_9,$$

one has

$$\text{Ad}_{\Psi_9|_{\mathcal{H}_9}}(A) = \Psi_9|_{\mathcal{H}_9} A (\Psi_9|_{\mathcal{H}_9})^* = I_{\mathcal{H}_9} A I_{\mathcal{H}_9} = A.$$

Iteration gives the second assertion. $\square$

Corollary 340.3. *For every stable observable A ,*

$$\omega_9(\text{Ad}_{\Psi_9|_{\mathcal{H}_9}}^k(A)) = \omega_9(A) \quad (k \geq 1).$$

Proof. The preceding theorem gives equality of the operators themselves. $\square$

341 The Fixed-Point Algebra of the Stabilized Dynamics

Let

$$\alpha_9 : \mathcal{A}_9 \longrightarrow \mathcal{A}_9$$

be the automorphism induced by the stabilized dynamics:

$$\alpha_9(A) = \Psi_9|_{\mathcal{H}_9} A (\Psi_9|_{\mathcal{H}_9})^*.$$

Proposition 341.1. *The automorphism α_9 is the identity:*

$$\alpha_9 = \text{id}_{\mathcal{A}_9}.$$

Proof. The restriction of Ψ_9 to $\mathcal{H}_9$ is $I_{\mathcal{H}_9}$. Hence

$$\alpha_9(A) = I_{\mathcal{H}_9} A I_{\mathcal{H}_9} = A$$

for every

$$A \in \mathcal{A}_9.$$

□

Theorem 341.2. *The fixed-point algebra of the stabilized dynamics is the whole stable algebra:*

$$\mathcal{A}_9^{\alpha_9} = \{A \in \mathcal{A}_9 : \alpha_9(A) = A\} = \mathcal{A}_9.$$

Proof. By the preceding proposition,

$$\alpha_9(A) = A$$

for every $A \in \mathcal{A}_9$. Therefore every element of $\mathcal{A}_9$ is fixed. □

342 The Spectral Projection as a Conditional Expectation

Let

$$\mathcal{B}_9 = \{P_9 X P_9 : X \in \text{End}_{\mathbb{C}}(V_9)\},$$

identified with

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

through restriction to $\mathcal{H}_9$.

Define

$$\mathbb{E}_9(X) = P_9 X P_9|_{\mathcal{H}_9}.$$

Proposition 342.1. *The map*

$$\mathbb{E}_9 : \text{End}_{\mathbb{C}}(V_9) \longrightarrow \mathcal{A}_9$$

is linear and satisfies

$$\mathbb{E}_9(X) = X$$

whenever

$$X = P_9 X P_9.$$

Proof. Linearity follows from the linearity of multiplication by P_9 on both sides. If

$$X = P_9 X P_9,$$

then

$$\mathbb{E}_9(X) = P_9 X P_9|_{\mathcal{H}_9} = X|_{\mathcal{H}_9}.$$

□

Proposition 342.2. *The compression map satisfies*

$$\mathbb{E}_9^2 = \mathbb{E}_9$$

on the compressed operator algebra.

Proof. For every X ,

$$P_9(P_9XP_9)P_9 = P_9^2XP_9^2 = P_9XP_9.$$

Thus a second application does not change the compressed operator. $\square$

Remark 342.3. The map $\mathbb{E}_9$ is a genuine conditional expectation onto the corner $P_9 \text{End}(V_9)P_9$ when that corner is viewed as an operator algebra with unit P_9 . No assertion that the compression is multiplicative on the full ambient algebra is required.

343 The Stable Corner

Define

$$P_9 \text{End}_{\mathbb{C}}(V_9)P_9 = \{P_9XP_9 : X \in \text{End}_{\mathbb{C}}(V_9)\}.$$

Theorem 343.1. *The map*

$$\Phi_9 : P_9 \text{End}_{\mathbb{C}}(V_9)P_9 \longrightarrow \text{End}_{\mathbb{C}}(\mathcal{H}_9), \quad \Phi_9(X) = X|_{\mathcal{H}_9},$$

is a unital algebra isomorphism, where the unit of the corner is P_9 .

Proof. If $X = P_9XP_9$, then X preserves $\mathcal{H}_9$, so Φ_9 is well-defined.

For

$$X, Y \in P_9 \text{End}(V_9)P_9,$$

one has

$$(XY)|_{\mathcal{H}_9} = X|_{\mathcal{H}_9}Y|_{\mathcal{H}_9},$$

so Φ_9 is multiplicative.

It is linear by construction. Its unit is

$$\Phi_9(P_9) = P_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

To prove surjectivity, let

$$A \in \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Extend A by zero on $\ker(P_9)$, obtaining

$$\tilde{A} = A \oplus 0.$$

Then

$$P_9\tilde{A}P_9 = \tilde{A}$$

and

$$\Phi_9(\tilde{A}) = A.$$

Finally, if

$$\Phi_9(X) = 0,$$

then X vanishes on $\mathcal{H}_9$. Since

$$X = P_9XP_9,$$

it also vanishes on $\ker(P_9)$, and hence $X = 0$. Thus Φ_9 is injective. $\square$

Corollary 343.2. *The stable corner has dimension*

$$\dim_{\mathbb{C}}(P_9 \text{End}_{\mathbb{C}}(V_9)P_9) = 81.$$

344 The Stable State on the Corner

Define

$$\tilde{\tau}_9(P_9 X P_9) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(P_9 X P_9|_{\mathcal{H}_9}).$$

Proposition 344.1. *The functional $\tilde{\tau}_9$ is well-defined on the stable corner.*

Proof. If

$$P_9 X P_9 = P_9 Y P_9,$$

then their restrictions to $\mathcal{H}_9$ coincide. Hence their traces on $\mathcal{H}_9$ coincide, and therefore

$$\tilde{\tau}_9(P_9 X P_9) = \tilde{\tau}_9(P_9 Y P_9).$$

□

Theorem 344.2. *The functional $\tilde{\tau}_9$ is the unique normalized trace on the stable corner.*

Proof. The isomorphism Φ_9 identifies the stable corner with

$$M_9(\mathbb{C}).$$

Under this identification,

$$\tilde{\tau}_9 = \frac{1}{9} \operatorname{tr}.$$

The uniqueness of the normalized trace on $M_9(\mathbb{C})$ therefore implies the assertion. □

345 Stable and Transient Spectral Subspaces

Let

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9, \quad \mathcal{K}_9 = \ker(P_9).$$

Theorem 345.1. *The stabilized operator P_9 acts as*

$$P_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}$$

and

$$P_9|_{\mathcal{K}_9} = 0.$$

Proof. The first identity follows from

$$\mathcal{H}_9 = \operatorname{range}(P_9)$$

and idempotence:

$$P_9^2 = P_9.$$

If

$$v \in \mathcal{K}_9,$$

then

$$P_9 v = 0$$

by definition. □

Proposition 345.2. *The decomposition*

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9$$

is invariant under Ψ_9 .

Proof. The stable sector is

$$\mathcal{H}_9 = \ker(\Psi_9 - I),$$

hence it is invariant under Ψ_9 .

For

$$v \in \ker(P_9),$$

one has

$$P_9 \Psi_9 v = \Psi_9 P_9 v = 0,$$

because

$$P_9 = \Psi_9^4$$

commutes with Ψ_9 . Thus

$$\Psi_9 v \in \ker(P_9).$$

□

Theorem 345.3. *Relative to*

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9,$$

the operator Ψ_9 has the block form

$$\Psi_9 = I_{\mathcal{H}_9} \oplus T_0,$$

where

$$T_0 : \mathcal{K}_9 \longrightarrow \mathcal{K}_9$$

satisfies

$$T_0^4 = 0.$$

Proof. The invariance of the two summands gives the block decomposition. On $\mathcal{H}_9$,

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

Let

$$v \in \mathcal{K}_9.$$

Since

$$P_9 = \Psi_9^4$$

and $P_9 v = 0$,

$$\Psi_9^4 v = 0.$$

Thus

$$T_0^4 v = 0$$

for every $v \in \mathcal{K}_9$, proving

$$T_0^4 = 0.$$

□

Corollary 345.4. *The transient sector is nilpotent of index at most 4:*

$$\Psi_9^4|_{\mathcal{K}_9} = 0.$$

Corollary 345.5. *The entire finite-state operator admits the canonical decomposition*

$$\Psi_9 = I_{\mathcal{H}_9} \oplus T_0, \quad T_0^4 = 0.$$

Thus the stabilized dynamics consists of an identity component on the stable sector and a nilpotent component on its complementary invariant sector.

346 Jordan Structure of the Transient Sector

Theorem 346.1. *All Jordan blocks of Ψ_9 associated with the eigenvalue 1 have size 1, while all Jordan blocks associated with the eigenvalue 0 have size at most 4.*

Proof. The block decomposition

$$\Psi_9 = I_{\mathcal{H}_9} \oplus T_0$$

shows that the restriction to the stable sector is already diagonal, with eigenvalue 1.

On $\mathcal{K}_9$,

$$T_0^4 = 0.$$

Hence the minimal polynomial of T_0 divides X^4 , so every Jordan block of T_0 has size at most 4. Since the stable and transient spectra are respectively $\{1\}$ and $\{0\}$, the assertion follows. $\square$

Corollary 346.2. *The minimal polynomial of Ψ_9 divides*

$$X^4(X - 1).$$

Proof. The minimal polynomial of the identity block is $X - 1$, while that of the nilpotent block divides X^4 . Since these factors are coprime, their product annihilates the direct sum. $\square$

347 Rank and Nullity in the 143-Dimensional Sector

Assume

$$V_9 = \mathbf{1}^\perp \subseteq \mathbb{C}^{144}, \quad \dim_{\mathbb{C}} V_9 = 143,$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Theorem 347.1. *The transient sector has dimension*

$$\dim_{\mathbb{C}} \mathcal{K}_9 = 134.$$

Proof. Since

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9,$$

one has

$$143 = \dim_{\mathbb{C}} \mathcal{H}_9 + \dim_{\mathbb{C}} \mathcal{K}_9.$$

Therefore

$$\dim_{\mathbb{C}} \mathcal{K}_9 = 143 - 9 = 134.$$

$\square$

Corollary 347.2. *The stabilized projection satisfies*

$$\text{rank}(P_9) = 9, \quad \text{nullity}(P_9) = 134.$$

Proof. The rank is the dimension of the range, while the nullity is the dimension of the kernel. $\square$

348 Characteristic Polynomial Under the Exact Stabilization

Theorem 348.1. *Let*

$$d_0 = \dim_{\mathbb{C}} \ker(P_9), \quad d_1 = \dim_{\mathbb{C}} \operatorname{range}(P_9).$$

Then

$$\chi_{\Psi_9}(X) = X^{d_0}(X-1)^{d_1}.$$

In particular, if

$$\dim_{\mathbb{C}} V_9 = 143$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

then

$$\chi_{\Psi_9}(X) = X^{134}(X-1)^9.$$

Proof. The block decomposition gives

$$\Psi_9 = I_{\mathcal{H}_9} \oplus T_0, \quad T_0^4 = 0.$$

Thus the characteristic polynomial of T_0 is

$$X^{d_0},$$

while that of the identity on $\mathcal{H}_9$ is

$$(X-1)^{d_1}.$$

The characteristic polynomial of a direct sum is the product of the characteristic polynomials of its blocks. Hence

$$\chi_{\Psi_9}(X) = X^{d_0}(X-1)^{d_1}.$$

Substituting

$$d_0 = 134, \quad d_1 = 9$$

gives the stated formula. □

349 The Commutant of the Stable Representation

Let

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be the restricted unitary representation obtained above. Define its commutant by

$$\pi_9(G)' = \{X \in \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9) : X\pi_9(g) = \pi_9(g)X \text{ for all } g \in G\}.$$

Proposition 349.1. *The commutant $\pi_9(G)'$ is a unital subalgebra of $\operatorname{End}_{\mathbb{C}}(\mathcal{H}_9)$.*

Proof. Let

$$X, Y \in \pi_9(G)'$$

and let

$$a, b \in \mathbb{C}.$$

For every $g \in G$,

$$(aX + bY)\pi_9(g) = aX\pi_9(g) + bY\pi_9(g) = a\pi_9(g)X + b\pi_9(g)Y = \pi_9(g)(aX + bY).$$

Thus

$$aX + bY \in \pi_9(G)'.$$

Moreover,

$$(XY)\pi_9(g) = X\pi_9(g)Y = \pi_9(g)XY,$$

so

$$XY \in \pi_9(G)'.$$

Finally,

$$I_{\mathcal{H}_9}\pi_9(g) = \pi_9(g) = \pi_9(g)I_{\mathcal{H}_9},$$

hence

$$I_{\mathcal{H}_9} \in \pi_9(G)'.$$

Therefore $\pi_9(G)'$ is a unital subalgebra. $\square$

Proposition 349.2. *The adjoint of every element of the commutant again belongs to the commutant:*

$$X \in \pi_9(G)' \implies X^* \in \pi_9(G)'.$$

Proof. Let

$$X\pi_9(g) = \pi_9(g)X.$$

Taking adjoints gives

$$\pi_9(g)^* X^* = X^* \pi_9(g)^*.$$

Replacing g by g^{-1} and using

$$\pi_9(g^{-1}) = \pi_9(g)^*,$$

we obtain

$$\pi_9(g)X^* = X^*\pi_9(g),$$

which is equivalent to

$$X^*\pi_9(g) = \pi_9(g)X^*.$$

Hence

$$X^* \in \pi_9(G)'.$$

$\square$

Theorem 349.3. *The commutant $\pi_9(G)'$ is a finite-dimensional $*$ -subalgebra of*

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Proof. The preceding propositions show that $\pi_9(G)'$ is closed under addition, scalar multiplication, multiplication, adjoints, and contains the identity. Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

one has

$$\dim_{\mathbb{C}} \text{End}_{\mathbb{C}}(\mathcal{H}_9) = 81.$$

Consequently the commutant is finite-dimensional. $\square$

350 Irreducible Decomposition of the Stable Representation

Theorem 350.1. *There exist pairwise inequivalent irreducible unitary representations*

$$\sigma_1, \dots, \sigma_s$$

of G , finite-dimensional Hilbert spaces

$$M_1, \dots, M_s,$$

and a unitary isomorphism

$$\mathcal{H}_9 \cong \bigoplus_{j=1}^s (M_j \otimes W_j),$$

where W_j carries σ_j , such that

$$\pi_9(g) \cong \bigoplus_{j=1}^s (I_{M_j} \otimes \sigma_j(g)).$$

Proof. Since G acts unitarily on the finite-dimensional complex Hilbert space $\mathcal{H}_9$, every invariant subspace admits an invariant orthogonal complement. Indeed, if $W \subseteq \mathcal{H}_9$ is invariant under $\pi_9(G)$, then for

$$v \in W^\perp, \quad w \in W,$$

one has

$$\langle \pi_9(g)v, w \rangle = \langle v, \pi_9(g)^{-1}w \rangle = 0,$$

because

$$\pi_9(g)^{-1}w \in W.$$

Thus $W^\perp$ is invariant.

Starting from any nonzero invariant subspace, choose an irreducible invariant subspace and split it off orthogonally. Finite dimensionality allows this procedure to terminate after finitely many steps. Grouping isomorphic irreducible summands yields the asserted multiplicity-space decomposition. $\square$

Corollary 350.2. *The dimension constraint is*

$$9 = \sum_{j=1}^s (\dim_{\mathbb{C}} M_j)(\dim_{\mathbb{C}} W_j).$$

Proof. Taking dimensions in the decomposition of $\mathcal{H}_9$ gives

$$\dim_{\mathbb{C}} \mathcal{H}_9 = \sum_{j=1}^s \dim_{\mathbb{C}} M_j \dim_{\mathbb{C}} W_j.$$

Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

the stated identity follows. $\square$

Proposition 350.3. *The commutant of the stable representation has the corresponding decomposition*

$$\pi_9(G)' \cong \bigoplus_{j=1}^s (\text{End}_{\mathbb{C}}(M_j) \otimes I_{W_j}).$$

Proof. An operator commuting with every

$$I_{M_j} \otimes \sigma_j(g)$$

preserves the isotypic components. Schur's lemma implies that on each irreducible factor W_j , the intertwining operators are scalar. Consequently the operators in the commutant act arbitrarily on the multiplicity space M_j and trivially on W_j , giving

$$\text{End}_{\mathbb{C}}(M_j) \otimes I_{W_j}.$$

Inequivalent isotypic components cannot be mixed by an intertwiner, again by Schur's lemma. This yields the asserted direct sum. $\square$

351 Characterization of the Maximally Symmetric State

Let

$$\rho_g = \frac{1}{9} I_{\mathcal{H}_g}.$$

Proposition 351.1. *The density operator ρ_g belongs to the commutant of every unitary representation of G on $\mathcal{H}_g$.*

Proof. For every

$$g \in G,$$

one has

$$\rho_g \pi_g(g) = \frac{1}{9} \pi_g(g) = \pi_g(g) \rho_g.$$

Hence

$$\rho_g \in \pi_g(G)'. \quad \square$$

Theorem 351.2. *Let*

$$\rho \in \text{End}_{\mathbb{C}}(\mathcal{H}_g)$$

be a density operator invariant under $\pi_g(G)$. If the stable representation is irreducible, then

$$\rho = \frac{1}{9} I_{\mathcal{H}_g}.$$

Proof. Invariance means

$$\pi_g(g) \rho \pi_g(g)^* = \rho \quad (g \in G).$$

Multiplication on the right by $\pi_g(g)$ gives

$$\pi_g(g) \rho = \rho \pi_g(g).$$

Thus

$$\rho \in \pi_g(G)'.$$

If π_g is irreducible, Schur's lemma implies

$$\pi_g(G)' = \mathbb{C} I_{\mathcal{H}_g}.$$

Therefore

$$\rho = \lambda I_{\mathcal{H}_g}$$

for some

$$\lambda \geq 0.$$

Since ρ is normalized,

$$1 = \text{tr}(\rho) = 9\lambda.$$

Hence

$$\lambda = \frac{1}{9},$$

and consequently

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Remark 351.3. Irreducibility is essential for the uniqueness assertion in the preceding theorem. Without it, the commutant may contain non-scalar operators, and invariance under G alone does not force a density operator to be proportional to the identity.

352 The Arithmetic Character and the Stable Representation

Let

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

and let

$$\Lambda_{\log} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{R}$$

be the homomorphism defined by

$$\Lambda_{\log} \left(\sum_{p \in \mathbb{P}} n_p e_p \right) = \sum_{p \in \mathbb{P}} n_p \log p.$$

For

$$t \in \mathbb{R},$$

define

$$\chi_t(x) = \exp(it\Lambda_{\log}(x)).$$

Proposition 352.1. *Each χ_t is a unitary character of $\mathcal{L}_{\mathbb{P}}$.*

Proof. For

$$x, y \in \mathcal{L}_{\mathbb{P}},$$

additivity of $\Lambda_{\log}$ gives

$$\Lambda_{\log}(x + y) = \Lambda_{\log}(x) + \Lambda_{\log}(y).$$

Therefore

$$\chi_t(x + y) = e^{it\Lambda_{\log}(x)} e^{it\Lambda_{\log}(y)} = \chi_t(x) \chi_t(y).$$

Moreover,

$$|\chi_t(x)| = 1.$$

Thus χ_t is a homomorphism from $\mathcal{L}_{\mathbb{P}}$ into $U(1)$. □

Theorem 352.2. *The family*

$$\{\chi_t : t \in \mathbb{R}\}$$

separates the points of $\mathcal{L}_{\mathbb{P}}$.

Proof. Let

$$x \neq y.$$

Then

$$z = x - y \neq 0.$$

Unique factorization implies that

$$\Lambda_{\log}(z) \neq 0.$$

Choose

$$t = \frac{\pi}{\Lambda_{\log}(z)}.$$

Then

$$\chi_t(z) = e^{i\pi} = -1.$$

If

$$\chi_t(x) = \chi_t(y),$$

then

$$\chi_t(z) = \chi_t(x - y) = \chi_t(x)\chi_t(y)^{-1} = 1,$$

a contradiction. Hence

$$\chi_t(x) \neq \chi_t(y).$$

□

353 Finite-Dimensional Representations of the Arithmetic Lattice

Let

$$\pi_{\mathbb{P}} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_{\mathfrak{g}})$$

be a unitary representation.

Theorem 353.1. *There exist unitary characters*

$$\eta_1, \dots, \eta_s : \mathcal{L}_{\mathbb{P}} \longrightarrow U(1)$$

and an orthogonal decomposition

$$\mathcal{H}_{\mathfrak{g}} = \bigoplus_{j=1}^s \mathcal{K}_j$$

such that

$$\pi_{\mathbb{P}}(x)|_{\mathcal{K}_j} = \eta_j(x)I_{\mathcal{K}_j} \quad (x \in \mathcal{L}_{\mathbb{P}}).$$

Proof. The image of $\mathcal{L}_{\mathbb{P}}$ is a commuting family of unitary operators because

$$\mathcal{L}_{\mathbb{P}}$$

is abelian. Hence the family consists of commuting normal operators. The simultaneous spectral theorem in finite dimension provides an orthogonal decomposition into common eigenspaces. On each common eigenspace $\mathcal{K}_j$, every $\pi_{\mathbb{P}}(x)$ acts by a scalar

$$\eta_j(x) \in U(1).$$

The representation identity implies

$$\eta_j(x + y) = \eta_j(x)\eta_j(y),$$

so η_j is a unitary character.

□

Corollary 353.2. *The arithmetic action on the nine-dimensional stable sector is completely reducible and is determined by at most nine character sectors, counted with multiplicity.*

Proof. Each nonzero common eigenspace has dimension at least one, while

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Therefore there can be at most nine distinct nonzero summands in the orthogonal decomposition. $\square$

Remark 353.3. The characters η_j need not coincide with the particular one-parameter family

$$\chi_t(x) = e^{it\Lambda_{\log}(x)}.$$

Such an identification requires an additional hypothesis on $\pi_{\mathbb{P}}$. The finite-dimensional spectral decomposition alone establishes only the existence of unitary characters η_j .

354 Reduction Induced by the Stabilization Projection

Let

$$V_9$$

be the finite-dimensional relational space, let

$$P_9^2 = P_9,$$

and set

$$\mathcal{H}_9 = \text{range}(P_9), \quad \mathcal{K}_9 = \ker(P_9).$$

Proposition 354.1. *One has the direct-sum decomposition*

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9.$$

Proof. For every

$$v \in V_9,$$

write

$$v = P_9 v + (I - P_9) v.$$

Since

$$P_9(P_9 v) = P_9 v,$$

the first term belongs to

$$\text{range}(P_9) = \mathcal{H}_9.$$

Moreover,

$$P_9(I - P_9)v = P_9 v - P_9^2 v = 0,$$

so

$$(I - P_9)v \in \ker(P_9) = \mathcal{K}_9.$$

Thus

$$V_9 = \mathcal{H}_9 + \mathcal{K}_9.$$

If

$$v \in \mathcal{H}_9 \cap \mathcal{K}_9,$$

then

$$P_9 v = v$$

because $v \in \mathcal{H}_9$, whereas

$$P_9 v = 0$$

because $v \in \mathcal{K}_9$. Hence $v = 0$, proving that the sum is direct. $\square$

Proposition 354.2. *If*

$$\pi(g)P_9 = P_9\pi(g) \quad (g \in G),$$

then both subspaces

$$\mathcal{H}_9 \quad \text{and} \quad \mathcal{K}_9$$

are invariant under $\pi(g)$.

Proof. The invariance of $\mathcal{H}_9$ was established above.

Let

$$v \in \mathcal{K}_9.$$

Then

$$P_9 v = 0.$$

Consequently,

$$P_9 \pi(g) v = \pi(g) P_9 v = 0.$$

Therefore

$$\pi(g) v \in \ker(P_9) = \mathcal{K}_9.$$

$\square$

Theorem 354.3. *With respect to the direct-sum decomposition*

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9,$$

every operator $\pi(g)$ has block-diagonal form

$$\pi(g) = \begin{pmatrix} \pi_9(g) & 0 \\ 0 & \pi_0(g) \end{pmatrix},$$

where

$$\pi_9(g) = \pi(g)|_{\mathcal{H}_9}, \quad \pi_0(g) = \pi(g)|_{\mathcal{K}_9}.$$

Proof. By the preceding proposition, both summands are invariant under $\pi(g)$. Hence the image of a vector in either summand remains in that same summand, so the off-diagonal blocks vanish. The diagonal blocks are precisely the corresponding restrictions. $\square$

Corollary 354.4. *The representation decomposes algebraically as*

$$\pi = \pi_9 \oplus \pi_0.$$

Proof. This is the block decomposition obtained in the preceding theorem. $\square$

Proposition 354.5. *The stabilization projection commutes with the representation:*

$$P_9\pi(g) = \pi(g)P_9,$$

and, in the decomposition

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9,$$

it is represented by

$$P_9 = \begin{pmatrix} I_{\mathcal{H}_9} & 0 \\ 0 & 0 \end{pmatrix}.$$

Proof. The commutation relation is part of the hypothesis. Since P_9 acts as the identity on its range and vanishes on its kernel, its matrix relative to

$$V_9 = \text{range}(P_9) \oplus \ker(P_9)$$

has the stated form. □

355 The Stable Sector as the Nonzero Spectral Component

Assume now that

$$\Psi_9^5 = \Psi_9^4$$

and

$$P_9 = \Psi_9^4.$$

Proposition 355.1. *The stabilization projection satisfies*

$$P_9^2 = P_9$$

and

$$\Psi_9 P_9 = P_9 = P_9 \Psi_9.$$

Proof. Using

$$\Psi_9^5 = \Psi_9^4,$$

one obtains

$$P_9^2 = \Psi_9^8 = \Psi_9^4 = P_9,$$

where the equality

$$\Psi_9^k = \Psi_9^4 \quad (k \geq 4)$$

follows by repeated multiplication of the stabilization identity by Ψ_9 .

Similarly,

$$\Psi_9 P_9 = \Psi_9^5 = \Psi_9^4 = P_9,$$

and, since powers of the same operator commute,

$$P_9 \Psi_9 = \Psi_9^5 = P_9.$$

□

Theorem 355.2. *The decomposition*

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9$$

satisfies

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}$$

and

$$\Psi_9|_{\mathcal{K}_9}$$

is nilpotent of index at most 4:

$$(\Psi_9|_{\mathcal{K}_9})^4 = 0.$$

Proof. If

$$v \in \mathcal{H}_9,$$

then

$$v = P_9 v.$$

Therefore

$$\Psi_9 v = \Psi_9 P_9 v = P_9 v = v.$$

Thus

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

If

$$v \in \mathcal{K}_9,$$

then

$$P_9 v = 0.$$

Since

$$P_9 = \Psi_9^4,$$

one obtains

$$\Psi_9^4 v = 0.$$

Hence

$$(\Psi_9|_{\mathcal{K}_9})^4 = 0.$$

□

Corollary 355.3. *The minimal polynomial of the restriction of Ψ_9 to $\mathcal{K}_9$ divides*

$$X^4.$$

Proof. The fourth power of the restricted operator vanishes identically. Therefore its minimal polynomial divides X^4 . □

Corollary 355.4. *The minimal polynomial of Ψ_9 on V_9 divides*

$$X^4(X - 1).$$

Proof. On

$$\mathcal{H}_9$$

the operator satisfies

$$\Psi_9 - I = 0,$$

while on

$$\mathcal{K}_9$$

it satisfies

$$\Psi_9^4 = 0.$$

Since

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9,$$

the product polynomial

$$X^4(X-1)$$

annihilates the full operator. □

356 Rank and Trace of the Stabilization Projection

Assume

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9$$

and

$$\dim_{\mathbb{C}} V_9 = 143.$$

Theorem 356.1. *The stabilization projection has*

$$\text{rank}(P_9) = 9$$

and

$$\dim_{\mathbb{C}} \ker(P_9) = 134.$$

Proof. By definition,

$$\text{rank}(P_9) = \dim_{\mathbb{C}} \text{range}(P_9) = \dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Rank-nullity gives

$$143 = 9 + \dim_{\mathbb{C}} \ker(P_9),$$

hence

$$\dim_{\mathbb{C}} \ker(P_9) = 134. \quad \square$$

Proposition 356.2. *The trace of P_9 on V_9 is*

$$\text{tr}_{V_9}(P_9) = 9.$$

Proof. Relative to

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9,$$

the operator P_9 has the block form

$$P_9 = \begin{pmatrix} I_{\mathcal{H}_9} & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore

$$\text{tr}_{V_9}(P_9) = \text{tr}(I_{\mathcal{H}_9}) = \dim_{\mathbb{C}} \mathcal{H}_9 = 9. \quad \square$$

Corollary 356.3. *The normalized ambient trace of the stabilized projection is*

$$\frac{1}{143} \text{tr}_{V_9}(P_9) = \frac{9}{143}.$$

357 The Stable Compression Algebra

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

For

$$X \in \text{End}_{\mathbb{C}}(V_9),$$

define

$$C_9(X) = P_9 X P_9|_{\mathcal{H}_9}.$$

Proposition 357.1. *The map*

$$C_9 : \text{End}_{\mathbb{C}}(V_9) \longrightarrow \mathcal{A}_9$$

is linear.

Proof. For

$$X, Y \in \text{End}_{\mathbb{C}}(V_9)$$

and

$$a, b \in \mathbb{C},$$

one has

$$P_9(aX + bY)P_9 = aP_9XP_9 + bP_9YP_9.$$

Restriction to $\mathcal{H}_9$ preserves this equality. □

Theorem 357.2. *The compression map is surjective.*

Proof. Let

$$A \in \mathcal{A}_9.$$

Define

$$\tilde{A} = A \oplus 0$$

with respect to

$$V_9 = \mathcal{H}_9 \oplus \mathcal{K}_9.$$

Then

$$P_9 \tilde{A} P_9 = A \oplus 0.$$

Consequently,

$$C_9(\tilde{A}) = A.$$

Thus every element of $\mathcal{A}_9$ is obtained by compression. □

Corollary 357.3. *The stable algebra is canonically identified with the corner algebra*

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

Proof. The map

$$A \longmapsto A \oplus 0$$

identifies

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

with the operators on V_9 whose only nonzero block is the $\mathcal{H}_9$ -to- $\mathcal{H}_9$ block. These are exactly the operators of the form

$$P_9 X P_9.$$

□

Proposition 357.4. *Under the identification*

$$\mathcal{A}_9 \simeq P_9 \operatorname{End}_{\mathbb{C}}(V_9) P_9,$$

multiplication is preserved:

$$C_9(XY) = C_9(X)C_9(Y)$$

whenever

$$X = P_9 X P_9, \quad Y = P_9 Y P_9.$$

Proof. Using

$$P_9^2 = P_9,$$

one obtains

$$P_9 X Y P_9 = P_9 X P_9 Y P_9.$$

Restricting to $\mathcal{H}_9$ gives

$$C_9(XY) = C_9(X)C_9(Y).$$

□

Corollary 357.5. *If*

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

then

$$\mathcal{A}_9 \simeq M_9(\mathbb{C})$$

and

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 81.$$

Proof. Choosing a basis of the nine-dimensional space $\mathcal{H}_9$ identifies its endomorphism algebra with $M_9(\mathbb{C})$. Hence

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 9^2 = 81.$$

□

358 The Trace Structure of the Stable Algebra

Let

$$\mathcal{A}_9 = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9)$$

with

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Proposition 358.1. *The ordinary trace on $\mathcal{A}_9$ is cyclic:*

$$\operatorname{tr}(XY) = \operatorname{tr}(YX) \quad (X, Y \in \mathcal{A}_9).$$

Proof. Choose a basis of $\mathcal{H}_9$ and identify

$$\mathcal{A}_9 \simeq M_9(\mathbb{C}).$$

For matrices $X = (x_{ij})$ and $Y = (y_{ij})$,

$$\operatorname{tr}(XY) = \sum_{i,j} x_{ij} y_{ji} = \sum_{i,j} y_{ij} x_{ji} = \operatorname{tr}(YX).$$

□

Theorem 358.2. *Let*

$$\tau : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a linear functional satisfying

$$\tau(XY) = \tau(YX) \quad (X, Y \in \mathcal{A}_9).$$

Then there exists a unique scalar $c \in \mathbb{C}$ such that

$$\tau(X) = c \operatorname{tr}(X) \quad (X \in \mathcal{A}_9).$$

Proof. Choose matrix units

$$e_{ij}, \quad 1 \leq i, j \leq 9.$$

For $i \neq j$,

$$e_{ij} = e_{ii}e_{ij},$$

whereas

$$e_{ij}e_{ii} = 0.$$

Cyclicity therefore gives

$$\tau(e_{ij}) = \tau(e_{ii}e_{ij}) = \tau(e_{ij}e_{ii}) = 0.$$

For diagonal matrix units,

$$e_{ii} = e_{i1}e_{1i}$$

and

$$e_{11} = e_{1i}e_{i1}.$$

Hence

$$\tau(e_{ii}) = \tau(e_{i1}e_{1i}) = \tau(e_{1i}e_{i1}) = \tau(e_{11}).$$

Thus all diagonal matrix units have the same value. Setting

$$c = \tau(e_{11}),$$

we obtain, for

$$X = \sum_{i,j} x_{ij} e_{ij},$$

the identity

$$\tau(X) = c \sum_{i=1}^9 x_{ii} = c \operatorname{tr}(X).$$

The scalar c is uniquely determined by

$$c = \tau(e_{11}).$$

□

Corollary 358.3. *There is a unique normalized trace on $\mathcal{A}_9$, namely*

$$\tau_9(X) = \frac{1}{9} \operatorname{tr}(X),$$

characterized by

$$\tau_9(I_{\mathcal{H}_9}) = 1$$

and

$$\tau_9(XY) = \tau_9(YX).$$

Proof. The theorem gives

$$\tau = c \operatorname{tr}.$$

Since

$$1 = \tau(I_{\mathcal{H}_9}) = c \operatorname{tr}(I_{\mathcal{H}_9}) = 9c,$$

one has

$$c = \frac{1}{9}.$$

□

359 Positive Functionals and Density Operators

Let

$$\varphi : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a linear functional satisfying

$$\varphi(X^* X) \geq 0 \quad (X \in \mathcal{A}_9)$$

and

$$\varphi(I_{\mathcal{H}_9}) = 1.$$

Theorem 359.1. *There exists a unique positive operator*

$$\rho \in \mathcal{A}_9$$

such that

$$\operatorname{tr}(\rho) = 1$$

and

$$\varphi(X) = \operatorname{tr}(\rho X) \quad (X \in \mathcal{A}_9).$$

Proof. Choose an orthonormal basis of $\mathcal{H}_9$ and let

$$e_{ij} = |e_i\rangle\langle e_j|$$

be the associated matrix units. Define

$$\rho_{ji} = \varphi(e_{ij}).$$

Then

$$\rho = \sum_{i,j=1}^9 \rho_{ij} e_{ij}.$$

For

$$v = \sum_{i=1}^9 v_i e_i,$$

let

$$X_v = |v\rangle\langle e_1|.$$

Then

$$X_v^* X_v = \|e_1\|^2 |v\rangle\langle v| = |v\rangle\langle v|.$$

Positivity of φ yields

$$0 \leq \varphi(|v\rangle\langle v|) = \langle v, \rho v \rangle.$$

Hence ρ is positive.

Moreover,

$$\mathrm{tr}(\rho) = \sum_{i=1}^9 \rho_{ii} = \sum_{i=1}^9 \varphi(e_{ii}) = \varphi(I_{\mathcal{H}_9}) = 1.$$

Finally, if

$$X = \sum_{i,j=1}^9 x_{ij} e_{ij},$$

then

$$\mathrm{tr}(\rho X) = \sum_{i,j=1}^9 \rho_{ji} x_{ij} = \sum_{i,j=1}^9 \varphi(e_{ij}) x_{ij} = \varphi(X).$$

If ρ' is another operator with the same property, then

$$\mathrm{tr}((\rho - \rho')X) = 0$$

for every $X \in \mathcal{A}_9$. Taking $X = e_{ji}$ for all i, j gives

$$\rho = \rho'.$$

Thus the representing density operator is unique. □

Corollary 359.2. *The normalized positive functionals on*

$$\mathcal{A}_9 \simeq M_9(\mathbb{C})$$

are in one-to-one correspondence with density operators

$$\rho \geq 0, \quad \mathrm{tr}(\rho) = 1.$$

360 The Maximally Symmetric State

Define

$$\rho_{\mathrm{sym}} = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proposition 360.1. *The operator ρ_{sym} is a density operator.*

Proof. Positivity follows from

$$I_{\mathcal{H}_9} \geq 0.$$

Furthermore,

$$\mathrm{tr}(\rho_{\mathrm{sym}}) = \frac{1}{9} \mathrm{tr}(I_{\mathcal{H}_9}) = 1. \quad \square$$

Theorem 360.2. *The following conditions are equivalent for a density operator $\rho \in \mathcal{A}_9$:*

$$(i) \quad U\rho U^* = \rho \quad \text{for every } U \in \mathcal{U}(\mathcal{H}_9);$$

$$(ii) \quad \rho = \frac{1}{9} I_{\mathcal{H}_9};$$

$$(iii) \quad \rho X = X\rho \quad \text{for every } X \in \mathcal{A}_9.$$

Proof. Suppose first that (i) holds. Then ρ commutes with every unitary operator, because

$$U\rho U^* = \rho$$

is equivalent to

$$U\rho = \rho U.$$

Every operator in $\mathcal{A}_9$ is a linear combination of unitaries. Consequently ρ commutes with every element of $\mathcal{A}_9$. Thus (iii) holds.

Conversely, if (iii) holds, then in particular

$$U\rho = \rho U$$

for every unitary U , and therefore

$$U\rho U^* = \rho.$$

Hence (i) and (iii) are equivalent.

It remains to prove that (iii) implies (ii). Relative to any orthonormal basis, write

$$\rho = (\rho_{ij}).$$

Commutation with every diagonal unitary

$$D = \text{diag}(z_1, \dots, z_9), \quad |z_j| = 1,$$

gives

$$z_i \rho_{ij} = \rho_{ij} z_j.$$

Choosing $z_i \neq z_j$ yields

$$\rho_{ij} = 0 \quad (i \neq j).$$

Thus ρ is diagonal. Commutation with the permutation unitaries interchanging two basis vectors then forces all diagonal entries to be equal. Hence

$$\rho = \lambda I_{\mathcal{H}_9}.$$

The trace normalization gives

$$1 = \text{tr}(\rho) = 9\lambda,$$

so

$$\lambda = \frac{1}{9}.$$

Therefore

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Corollary 360.3. *If the symmetry group acting on the stable sector generates the full matrix algebra*

$$\mathcal{A}_9,$$

then the unique invariant density operator is

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. Invariance under a family of unitaries whose generated algebra is $\mathcal{A}_9$ implies commutation with every element of $\mathcal{A}_9$. The preceding theorem then applies. □

361 Extremal States and Pure Stable States

Let

$$\mathfrak{S}(\mathcal{H}_9) = \{\rho \in \mathcal{A}_9 : \rho \geq 0, \operatorname{tr}(\rho) = 1\}.$$

Proposition 361.1. *The set*

$$\mathfrak{S}(\mathcal{H}_9)$$

is convex.

Proof. Let

$$\rho_1, \rho_2 \in \mathfrak{S}(\mathcal{H}_9)$$

and let

$$0 \leq t \leq 1.$$

Then

$$\rho = t\rho_1 + (1-t)\rho_2$$

is positive and satisfies

$$\operatorname{tr}(\rho) = t + (1-t) = 1.$$

Hence

$$\rho \in \mathfrak{S}(\mathcal{H}_9).$$

□

Theorem 361.2. *A density operator*

$$\rho \in \mathfrak{S}(\mathcal{H}_9)$$

is an extreme point of the state space if and only if

$$\rho = P_\psi$$

for some unit vector

$$\psi \in \mathcal{H}_9.$$

Proof. Suppose first that

$$\rho = P_\psi.$$

If

$$P_\psi = t\rho_1 + (1-t)\rho_2$$

with

$$0 < t < 1$$

and

$$\rho_1, \rho_2 \in \mathfrak{S}(\mathcal{H}_9),$$

then for every

$$v \perp \psi,$$

one has

$$0 = \langle v, P_\psi v \rangle = t\langle v, \rho_1 v \rangle + (1-t)\langle v, \rho_2 v \rangle.$$

Both terms are nonnegative, so both vanish. Positivity then implies that the ranges of ρ_1 and ρ_2 are contained in $\mathbb{C}\psi$. Since both operators have trace one,

$$\rho_1 = \rho_2 = P_\psi.$$

Thus P_ψ is extreme.

Conversely, suppose that ρ has rank at least two. Its spectral decomposition contains

$$\rho = \sum_{j=1}^r \lambda_j P_j, \quad r \geq 2,$$

with

$$\lambda_j > 0$$

and

$$\sum_{j=1}^r \lambda_j \operatorname{rank}(P_j) = 1.$$

Choose two distinct spectral projections P_1, P_2 . For sufficiently small $\varepsilon > 0$, define

$$\rho_{\pm} = \rho \pm \varepsilon \left(P_1 - \frac{\operatorname{rank}(P_1)}{\operatorname{rank}(P_2)} P_2 \right).$$

The perturbation has trace zero, and for sufficiently small ε both ρ_+ and ρ_- remain positive. Moreover,

$$\rho = \frac{1}{2}\rho_+ + \frac{1}{2}\rho_-, \quad \rho_+ \neq \rho_-.$$

Thus ρ is not extreme.

Therefore the extreme points are exactly the rank-one projections. □

362 Entropy of the Maximally Symmetric Stable State

For a density operator

$$\rho \in \mathfrak{S}(\mathcal{H}_9),$$

define its von Neumann entropy by

$$S(\rho) = -\operatorname{tr}(\rho \log \rho),$$

with the convention

$$0 \log 0 = 0.$$

Theorem 362.1. *For every density operator*

$$\rho \in \mathfrak{S}(\mathcal{H}_9),$$

one has

$$S(\rho) \leq \log 9.$$

Equality holds if and only if

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. Let

$$\lambda_1, \dots, \lambda_9$$

be the eigenvalues of ρ , counted with multiplicity. Then

$$\lambda_j \geq 0, \quad \sum_{j=1}^9 \lambda_j = 1,$$

and

$$S(\rho) = - \sum_{j=1}^9 \lambda_j \log \lambda_j.$$

The function

$$x \mapsto -x \log x$$

is strictly concave on $(0, \infty)$. By the concavity inequality,

$$-\frac{1}{9} \sum_{j=1}^9 \lambda_j \log \lambda_j \leq -\left(\frac{1}{9} \sum_{j=1}^9 \lambda_j\right) \log \left(\frac{1}{9} \sum_{j=1}^9 \lambda_j\right).$$

Since

$$\sum_{j=1}^9 \lambda_j = 1,$$

this gives

$$S(\rho) \leq \log 9.$$

Strict concavity shows that equality occurs precisely when

$$\lambda_1 = \cdots = \lambda_9 = \frac{1}{9}.$$

Hence equality is equivalent to

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Corollary 362.2. *The state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is characterized intrinsically as the unique state of maximal entropy on the nine-dimensional stable sector.

Proof. This is the equality statement of the preceding theorem. □

363 The Stable-State Conditional Expectation

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

and let

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Define the linear functional

$$\tau_9 : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

by

$$\tau_9(X) = \text{tr}(\rho_9 X) = \frac{1}{9} \text{tr}(X).$$

Proposition 363.1. *The functional τ_9 is positive and normalized:*

$$X \geq 0 \implies \tau_9(X) \geq 0, \quad \tau_9(I_{\mathcal{H}_9}) = 1.$$

Proof. If $X \geq 0$, then X has nonnegative eigenvalues, and therefore

$$\mathrm{tr}(X) \geq 0.$$

Hence

$$\tau_9(X) = \frac{1}{9} \mathrm{tr}(X) \geq 0.$$

Moreover,

$$\tau_9(I_{\mathcal{H}_9}) = \frac{1}{9} \mathrm{tr}(I_{\mathcal{H}_9}) = \frac{1}{9} \cdot 9 = 1.$$

□

Theorem 363.2. *The functional τ_9 is tracial:*

$$\tau_9(XY) = \tau_9(YX) \quad (X, Y \in \mathcal{A}_9).$$

Proof. The cyclicity of the finite-dimensional trace gives

$$\mathrm{tr}(XY) = \mathrm{tr}(YX).$$

Multiplication by $1/9$ yields

$$\tau_9(XY) = \tau_9(YX).$$

□

Corollary 363.3. *For every unitary*

$$U \in \mathcal{U}(\mathcal{H}_9),$$

one has

$$\tau_9(UXU^*) = \tau_9(X) \quad (X \in \mathcal{A}_9).$$

Proof. Using $U^*U = I_{\mathcal{H}_9}$,

$$\begin{aligned} \tau_9(UXU^*) &= \frac{1}{9} \mathrm{tr}(UXU^*) \\ &= \frac{1}{9} \mathrm{tr}(XU^*U) \\ &= \frac{1}{9} \mathrm{tr}(X) \\ &= \tau_9(X). \end{aligned}$$

□

364 Uniqueness of the Normalized Trace on the Stable Algebra

Theorem 364.1. *Let*

$$\varphi : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a linear functional satisfying

$$\varphi(XY) = \varphi(YX) \quad (X, Y \in \mathcal{A}_9)$$

and

$$\varphi(I_{\mathcal{H}_9}) = 1.$$

Then

$$\varphi = \tau_9.$$

Proof. Choose an orthonormal basis of $\mathcal{H}_9$ and let

$$e_{ij}$$

denote the associated matrix units. For $i \neq j$,

$$e_{ij} = e_{ii}e_{ij}, \quad e_{ij}e_{ii} = 0.$$

Traciality therefore gives

$$\varphi(e_{ij}) = \varphi(e_{ii}e_{ij}) = \varphi(e_{ij}e_{ii}) = 0.$$

For the diagonal matrix units,

$$e_{ii} = e_{ij}e_{ji}$$

and

$$e_{jj} = e_{ji}e_{ij}.$$

Hence

$$\varphi(e_{ii}) = \varphi(e_{ij}e_{ji}) = \varphi(e_{ji}e_{ij}) = \varphi(e_{jj}).$$

Thus there exists $\lambda \in \mathbb{C}$ such that

$$\varphi(e_{ii}) = \lambda \quad (1 \leq i \leq 9).$$

Since

$$I_{\mathcal{H}_9} = \sum_{i=1}^9 e_{ii},$$

normalization gives

$$1 = \varphi(I_{\mathcal{H}_9}) = 9\lambda.$$

Therefore

$$\lambda = \frac{1}{9}.$$

For

$$X = \sum_{i,j=1}^9 x_{ij}e_{ij},$$

one consequently has

$$\varphi(X) = \frac{1}{9} \sum_{i=1}^9 x_{ii} = \frac{1}{9} \operatorname{tr}(X) = \tau_9(X).$$

Thus

$$\varphi = \tau_9.$$

□

Corollary 364.2. *The normalized trace on the stable algebra is uniquely determined by the algebraic requirements*

$$\tau_9(I_{\mathcal{H}_9}) = 1$$

and

$$\tau_9(XY) = \tau_9(YX).$$

365 The Stable State as a Faithful State

Proposition 365.1. *The state*

$$\tau_9(X) = \frac{1}{9} \operatorname{tr}(X)$$

is faithful on positive operators:

$$X \geq 0, \quad \tau_9(X) = 0 \implies X = 0.$$

Proof. A positive operator has nonnegative eigenvalues. Since

$$\tau_9(X) = \frac{1}{9} \operatorname{tr}(X) = 0,$$

the sum of its nonnegative eigenvalues is zero. Hence every eigenvalue vanishes, and therefore

$$X = 0.$$

□

Corollary 365.2. *For every nonzero vector*

$$\psi \in \mathcal{H}_9,$$

the rank-one projection

$$P_\psi = \frac{|\psi\rangle\langle\psi|}{\|\psi\|^2}$$

satisfies

$$\tau_9(P_\psi) = \frac{1}{9}.$$

Proof. The operator P_ψ has rank one, and hence

$$\operatorname{tr}(P_\psi) = 1.$$

Therefore

$$\tau_9(P_\psi) = \frac{1}{9}.$$

□

366 The Stable State and Spectral Measures

Let

$$A = A^* \in \mathcal{A}_9$$

and write its spectral decomposition as

$$A = \sum_{\lambda \in \sigma(A)} \lambda E_\lambda,$$

where the E_λ are the pairwise orthogonal spectral projections.

Theorem 366.1. *The state τ_9 induces the probability measure*

$$\mu_A(\{\lambda\}) = \tau_9(E_\lambda) = \frac{\operatorname{rank}(E_\lambda)}{9}$$

on the finite spectrum $\sigma(A)$.

Proof. Each E_λ is a projection, so

$$\tau_9(E_\lambda) = \frac{1}{9} \operatorname{tr}(E_\lambda) = \frac{\operatorname{rank}(E_\lambda)}{9}.$$

Furthermore,

$$\sum_{\lambda \in \sigma(A)} E_\lambda = I_{\mathcal{H}_9}.$$

Consequently,

$$\sum_{\lambda \in \sigma(A)} \mu_A(\{\lambda\}) = \tau_9(I_{\mathcal{H}_9}) = 1.$$

Positivity follows from positivity of the projections. □

Corollary 366.2. *For every polynomial*

$$f \in \mathbb{C}[X],$$

one has

$$\tau_9(f(A)) = \sum_{\lambda \in \sigma(A)} f(\lambda) \frac{\operatorname{rank}(E_\lambda)}{9}.$$

Proof. Functional calculus gives

$$f(A) = \sum_{\lambda \in \sigma(A)} f(\lambda) E_\lambda.$$

Applying τ_9 and using linearity yields

$$\tau_9(f(A)) = \sum_{\lambda \in \sigma(A)} f(\lambda) \tau_9(E_\lambda),$$

and the theorem gives the stated formula. □

367 The Stable Sector and Matrix Coefficients

Let

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be the restricted representation.

For

$$\psi, \varphi \in \mathcal{H}_9,$$

define the matrix coefficient

$$c_{\psi, \varphi}(g) = \langle \psi, \pi_9(g) \varphi \rangle.$$

Proposition 367.1. *For every*

$$g \in G,$$

one has

$$|c_{\psi, \varphi}(g)| \leq \|\psi\| \|\varphi\|.$$

Proof. Since $\pi_9(g)$ is unitary,

$$\|\pi_9(g) \varphi\| = \|\varphi\|.$$

The Cauchy–Schwarz inequality therefore gives

$$|c_{\psi, \varphi}(g)| = |\langle \psi, \pi_9(g) \varphi \rangle| \leq \|\psi\| \|\pi_9(g) \varphi\| = \|\psi\| \|\varphi\|.$$

□

Theorem 367.2. *If ψ is a unit vector, then*

$$g \mapsto \langle \psi, \pi_9(g)\psi \rangle$$

is a normalized positive-definite function on G .

Proof. Normalization follows from

$$\langle \psi, \pi_9(e)\psi \rangle = \langle \psi, \psi \rangle = 1.$$

For arbitrary

$$g_1, \dots, g_n \in G$$

and

$$z_1, \dots, z_n \in \mathbb{C},$$

one has

$$\begin{aligned} \sum_{i,j=1}^n \overline{z_i} z_j \langle \psi, \pi_9(g_i^{-1} g_j) \psi \rangle &= \sum_{i,j=1}^n \overline{z_i} z_j \langle \pi_9(g_i) \psi, \pi_9(g_j) \psi \rangle \\ &= \left\| \sum_{j=1}^n z_j \pi_9(g_j) \psi \right\|^2 \\ &\geq 0. \end{aligned}$$

Thus the function is positive definite. □

368 Character Decomposition of an Abelian Arithmetic Action

Let

$$\mathcal{L}_{\mathbb{P}} = \bigoplus_{p \in \mathbb{P}} \mathbb{Z} e_p$$

and suppose that

$$\pi_9 : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

is a unitary representation whose image consists of commuting operators.

Theorem 368.1. *There exist finitely many unitary characters*

$$\chi_1, \dots, \chi_s : \mathcal{L}_{\mathbb{P}} \longrightarrow U(1)$$

and an orthogonal decomposition

$$\mathcal{H}_9 = \mathcal{H}_{\chi_1} \oplus \dots \oplus \mathcal{H}_{\chi_s}$$

such that

$$\pi_9(x) v = \chi_j(x) v$$

for every

$$x \in \mathcal{L}_{\mathbb{P}}, \quad v \in \mathcal{H}_{\chi_j}.$$

Moreover,

$$\sum_{j=1}^s \dim_{\mathbb{C}} \mathcal{H}_{\chi_j} = 9.$$

Proof. The operators

$$\pi_9(x)$$

are commuting unitary operators and hence commuting normal operators. A finite-dimensional commuting family of normal operators admits a simultaneous orthogonal eigenspace decomposition. On each common eigenspace $\mathcal{H}_{\chi_j}$, every $\pi_9(x)$ acts by a scalar of modulus one. The representation identity

$$\pi_9(x+y) = \pi_9(x)\pi_9(y)$$

then implies

$$\chi_j(x+y) = \chi_j(x)\chi_j(y).$$

Thus each scalar action is a unitary character. The dimensions of the orthogonal common eigenspaces sum to

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

□

Corollary 368.2. *If*

$$\chi : \mathcal{L}_{\mathbb{P}} \longrightarrow U(1)$$

is a character occurring in the decomposition, then its multiplicity is

$$\dim_{\mathbb{C}} \mathcal{H}_{\chi},$$

and the representation is completely determined, up to unitary equivalence, by the finite multiset of occurring characters with their multiplicities.

369 The Logarithmic Character Family

For

$$t \in \mathbb{R},$$

define

$$\chi_t(x) = \exp(it\Lambda_{\log}(x)),$$

where

$$\Lambda_{\log} \left(\sum_{p \in \mathbb{P}} n_p e_p \right) = \sum_{p \in \mathbb{P}} n_p \log p.$$

Proposition 369.1. *Each χ_t is a unitary character of $\mathcal{L}_{\mathbb{P}}$.*

Proof. For

$$x, y \in \mathcal{L}_{\mathbb{P}},$$

additivity of $\Lambda_{\log}$ gives

$$\Lambda_{\log}(x+y) = \Lambda_{\log}(x) + \Lambda_{\log}(y).$$

Consequently,

$$\chi_t(x+y) = e^{it\Lambda_{\log}(x)} e^{it\Lambda_{\log}(y)} = \chi_t(x)\chi_t(y).$$

Moreover,

$$|\chi_t(x)| = 1.$$

Thus χ_t is a unitary character. □

Proposition 369.2. *The family*

$$\{\chi_t : t \in \mathbb{R}\}$$

separates points of $\mathcal{L}_{\mathbb{P}}$.

Proof. Let

$$x \neq y.$$

Then

$$z = x - y \neq 0.$$

Unique factorization implies

$$\Lambda_{\log}(z) \neq 0.$$

Choose

$$t = \frac{\pi}{\Lambda_{\log}(z)}.$$

Then

$$\chi_t(z) = e^{i\pi} = -1.$$

Since

$$\chi_t(z) = \chi_t(x)\chi_t(y)^{-1},$$

one has

$$\chi_t(x) \neq \chi_t(y).$$

□

370 Separation of Arithmetic Configurations by the Logarithmic Coordinate

Let

$$\alpha : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{Q}_{>0}$$

be defined by

$$\alpha \left(\sum_{p \in \mathbb{P}} n_p e_p \right) = \prod_{p \in \mathbb{P}} p^{n_p}.$$

Theorem 370.1. *The logarithmic map*

$$\Lambda_{\log} : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathbb{R}$$

is injective.

Proof. Suppose

$$\Lambda_{\log}(x) = 0$$

for

$$x = \sum_{p \in \mathbb{P}} n_p e_p.$$

Then

$$\sum_{p \in \mathbb{P}} n_p \log p = 0.$$

Exponentiation gives

$$\prod_{p \in \mathbb{P}} p^{n_p} = 1.$$

Separating positive and negative exponents,

$$\prod_{n_p > 0} p^{n_p} = \prod_{n_p < 0} p^{-n_p}.$$

Unique factorization in $\mathbb{Z}$ implies

$$n_p = 0 \quad \text{for every } p.$$

Hence

$$x = 0.$$

□

Corollary 370.2. *The arithmetic encoding*

$$x \longmapsto \mathfrak{a}(x)$$

is injective on $\mathcal{L}_{\mathbb{P}}$.

Proof. If

$$\mathfrak{a}(x) = \mathfrak{a}(y),$$

then

$$\mathfrak{a}(x - y) = 1.$$

Applying the logarithm gives

$$\Lambda_{\log}(x - y) = 0.$$

The injectivity of $\Lambda_{\log}$ yields

$$x - y = 0,$$

and hence

$$x = y.$$

□

371 Scope of the Arithmetic–Representation Identification

Let

$$\pi_9 : \mathcal{L}_{\mathbb{P}} \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

be a finite-dimensional unitary representation.

Proposition 371.1. *The existence of the logarithmic character family*

$$\{\chi_t\}_{t \in \mathbb{R}}$$

does not, by itself, imply that every irreducible constituent of π_9 is one of the characters χ_t .

Proof. The family $\{\chi_t\}$ is a specified collection of characters of $\mathcal{L}_{\mathbb{P}}$. The simultaneous diagonalization theorem shows only that the irreducible constituents of π_9 are unitary characters of $\mathcal{L}_{\mathbb{P}}$. An additional hypothesis is required to identify those characters with members of the particular one-parameter family $\{\chi_t\}$. □

Theorem 371.2. *Suppose that every character occurring in the simultaneous character decomposition of π_9 is of the form*

$$\chi_t$$

for some

$$t \in \mathbb{R}.$$

Then

$$\mathcal{H}_9 = \bigoplus_{j=1}^s \mathcal{H}_{t_j},$$

where

$$\pi_9(x)v = \chi_{t_j}(x)v$$

for

$$v \in \mathcal{H}_{t_j}.$$

Proof. This is precisely the simultaneous character decomposition together with the stated identification of each occurring character with a member of the logarithmic family. $\square$

Remark 371.3. The final identification in the theorem is an additional structural hypothesis, not a consequence of finite-dimensionality alone. Any proof of the arithmetic–spectral correspondence must therefore establish this identification independently from the existence of the finite stable sector.

372 Intertwining and Reduction to the Stable Sector

Let

$$V_9 = \mathbf{1}^\perp \subseteq \mathbb{C}^{144}, \quad P_9 = \Psi_9^4, \quad \mathcal{H}_9 = \text{range}(P_9),$$

and assume

$$\Psi_9^5 = \Psi_9^4.$$

Let

$$\pi : G \longrightarrow \mathcal{U}(V_9)$$

be a unitary representation satisfying

$$\pi(g)P_9 = P_9\pi(g) \quad (g \in G).$$

Proposition 372.1. *For every $g \in G$, the operator $\pi(g)$ induces a well-defined unitary operator on $\mathcal{H}_9$, and the induced representation is*

$$\pi_9(g) = \pi(g)|_{\mathcal{H}_9}.$$

Proof. The invariance of $\mathcal{H}_9$ has already been established. Since $\pi(g)$ is unitary on V_9 and $\mathcal{H}_9$ is invariant under $\pi(g)$, the restriction is an isometry of $\mathcal{H}_9$.

It remains to verify surjectivity. Since

$$\pi(g)P_9 = P_9\pi(g),$$

the same relation holds for g^{-1} . Hence $\mathcal{H}_9$ is invariant under both $\pi(g)$ and $\pi(g)^{-1}$. Thus

$$\pi_9(g) \in \mathcal{U}(\mathcal{H}_9).$$

$\square$

Theorem 372.2. *The compression of an operator commuting with P_9 is compatible with the restricted representation. More precisely, if*

$$X \in \text{End}_{\mathbb{C}}(V_9)$$

satisfies

$$XP_9 = P_9X,$$

then

$$X_9 = X|_{\mathcal{H}_9}$$

belongs to

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

and, for every $g \in G$,

$$\pi_9(g)X_9\pi_9(g)^* = (\pi(g)X\pi(g)^*)|_{\mathcal{H}_9}.$$

Proof. The commutation relation implies that X preserves $\mathcal{H}_9$. Hence its restriction X_9 is well defined.

For $v \in \mathcal{H}_9$,

$$\begin{aligned} \pi_9(g)X_9\pi_9(g)^*v &= \pi(g)X\pi(g)^*v \\ &= (\pi(g)X\pi(g)^*)v. \end{aligned}$$

Since $\mathcal{H}_9$ is invariant under $\pi(g)$ and $\pi(g)^*$, the right-hand side belongs to $\mathcal{H}_9$. This proves the asserted identity. $\square$

373 The Commutant of the Stable Representation

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Define the commutant of the restricted representation by

$$\mathcal{A}_9^G = \{X \in \mathcal{A}_9 : X\pi_9(g) = \pi_9(g)X \text{ for every } g \in G\}.$$

Proposition 373.1. *The space $\mathcal{A}_9^G$ is a unital subalgebra of $\mathcal{A}_9$.*

Proof. The identity commutes with every $\pi_9(g)$. If $X, Y \in \mathcal{A}_9^G$, then

$$(X + Y)\pi_9(g) = \pi_9(g)(X + Y)$$

and

$$XY\pi_9(g) = X\pi_9(g)Y = \pi_9(g)XY.$$

Scalar multiples are treated similarly. Hence $\mathcal{A}_9^G$ is a unital subalgebra. $\square$

Theorem 373.2. *If the restricted representation π_9 is irreducible, then*

$$\mathcal{A}_9^G = \mathbb{C}I_{\mathcal{H}_9}.$$

Proof. Let

$$X \in \mathcal{A}_9^G.$$

Since $\mathcal{H}_9$ is finite dimensional over $\mathbb{C}$, X has an eigenvalue λ . The eigenspace

$$E_\lambda = \ker(X - \lambda I)$$

is nonzero. Because X commutes with every $\pi_9(g)$, the eigenspace E_λ is invariant under $\pi_9(G)$. Irreducibility therefore gives

$$E_\lambda = \mathcal{H}_9.$$

Consequently,

$$X = \lambda I_{\mathcal{H}_9}.$$

□

Corollary 373.3. *If π_9 is irreducible, every operator on $\mathcal{H}_9$ invariant under conjugation by $\pi_9(G)$ is a scalar multiple of the identity.*

Proof. Conjugation invariance means

$$\pi_9(g)X\pi_9(g)^* = X.$$

Multiplication on the right by $\pi_9(g)$ gives

$$\pi_9(g)X = X\pi_9(g),$$

so $X \in \mathcal{A}_9^G$. The theorem applies. □

374 Uniqueness of the G -Invariant State

Assume from now on that

$$\pi_9 : G \longrightarrow \mathcal{U}(\mathcal{H}_9)$$

is irreducible.

Let

$$\rho \in \mathcal{A}_9$$

be a density operator:

$$\rho \geq 0, \quad \text{tr}(\rho) = 1.$$

Theorem 374.1. *If*

$$\pi_9(g)\rho\pi_9(g)^* = \rho \quad (g \in G),$$

then

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. The invariance relation implies

$$\pi_9(g)\rho = \rho\pi_9(g) \quad (g \in G).$$

Hence

$$\rho \in \mathcal{A}_9^G.$$

By irreducibility,

$$\rho = \lambda I_{\mathcal{H}_9}$$

for some $\lambda \in \mathbb{R}$. Taking traces yields

$$1 = \text{tr}(\rho) = 9\lambda.$$

Therefore

$$\lambda = \frac{1}{9},$$

and hence

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Corollary 374.2. *Under irreducibility, the normalized trace state*

$$\tau_9(X) = \frac{1}{9} \operatorname{tr}(X)$$

is the unique G -invariant state on $\mathcal{A}_9$.

Proof. Every density operator representing a G -invariant state is equal to

$$\frac{1}{9} I_{\mathcal{H}_9}$$

by the theorem. Thus the corresponding functional is uniquely

$$X \mapsto \operatorname{tr} \left(\frac{1}{9} I_{\mathcal{H}_9} X \right) = \frac{1}{9} \operatorname{tr}(X).$$

□

375 The Projection as an Equivariant Retraction

The idempotence of P_9 follows from the stabilization relation:

$$P_9^2 = \Psi_9^8 = \Psi_9^4 = P_9.$$

Proposition 375.1. *The map*

$$P_9 : V_9 \longrightarrow \mathcal{H}_9$$

is a retraction onto the stable sector:

$$P_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

Proof. If

$$v \in \mathcal{H}_9 = \operatorname{range}(P_9),$$

then

$$P_9 v = v$$

because P_9 is idempotent. □

Theorem 375.2. *The retraction P_9 is G -equivariant:*

$$P_9 \pi(g) = \pi_9(g) P_9 \quad (g \in G).$$

Proof. The assumed commutation relation gives

$$P_9 \pi(g) = \pi(g) P_9.$$

The range of P_9 is $\mathcal{H}_9$, so the action of $\pi(g)$ on that range is precisely $\pi_9(g)$. Therefore

$$P_9 \pi(g) = \pi_9(g) P_9.$$

□

Corollary 375.3. *The decomposition*

$$V_9 = \ker(P_9) \oplus \mathcal{H}_9$$

is G -invariant.

Proof. The space $\mathcal{H}_9$ is invariant by construction. If $v \in \ker(P_9)$, then

$$P_9 \pi(g) v = \pi(g) P_9 v = 0.$$

Thus

$$\pi(g) v \in \ker(P_9).$$

The direct-sum decomposition follows from the idempotence of P_9 . □

376 Separation of the Stable and Transient Spectra

Theorem 376.1. *The decomposition*

$$V_9 = \ker(P_9) \oplus \mathcal{H}_9$$

reduces Ψ_9 , and the restrictions satisfy

$$\Psi_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}, \quad (\Psi_9|_{\ker(P_9)})^4 = 0.$$

Proof. The first identity follows from

$$\mathcal{H}_9 = \ker(\Psi_9 - I).$$

If

$$v \in \ker(P_9),$$

then

$$\Psi_9^4 v = P_9 v = 0.$$

Thus

$$(\Psi_9|_{\ker(P_9)})^4 = 0.$$

Both subspaces are invariant under Ψ_9 , since

$$P_9 \Psi_9 = \Psi_9 P_9 = P_9.$$

Hence the decomposition reduces Ψ_9 . □

Corollary 376.2. *With respect to*

$$V_9 = \ker(P_9) \oplus \mathcal{H}_9,$$

the operator Ψ_9 has block form

$$\Psi_9 = \begin{pmatrix} N_9 & 0 \\ 0 & I_{\mathcal{H}_9} \end{pmatrix}, \quad N_9^4 = 0.$$

Proof. This is the operator matrix associated with the reducing decomposition established above. □

Theorem 376.3. *The minimal polynomial of Ψ_9 divides*

$$X^4(X - 1),$$

and the restriction to the stable sector contributes exactly the factor $X - 1$.

Proof. The stabilization relation gives

$$\Psi_9^4(\Psi_9 - I) = 0,$$

so the minimal polynomial divides

$$X^4(X - 1).$$

On $\mathcal{H}_9$,

$$\Psi_9 = I_{\mathcal{H}_9}.$$

Since $\mathcal{H}_9 \neq \{0\}$, the restriction has minimal polynomial $X - 1$. Therefore the stable sector contributes the factor $X - 1$ to the minimal polynomial of the full operator. □

377 Exact Polynomial Reduction on the Stable Sector

Let

$$f(X) = \sum_{k=0}^d a_k X^k$$

be a polynomial.

Proposition 377.1. *The restriction of $f(\Psi_9)$ to the stable sector is*

$$f(\Psi_9)|_{\mathcal{H}_9} = f(1)I_{\mathcal{H}_9}.$$

Proof. For

$$v \in \mathcal{H}_9,$$

one has

$$\Psi_9^k v = v \quad (k \geq 0).$$

Therefore

$$f(\Psi_9)v = \sum_{k=0}^d a_k \Psi_9^k v = \left(\sum_{k=0}^d a_k \right) v = f(1)v.$$

□

Corollary 377.2. *For every polynomial f ,*

$$\rho_9 f(\Psi_9)|_{\mathcal{H}_9} = f(1)\rho_9.$$

Proof. The preceding proposition gives

$$f(\Psi_9)|_{\mathcal{H}_9} = f(1)I_{\mathcal{H}_9}.$$

Multiplication by

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

yields the result. □

Corollary 377.3. *The expectation of every polynomial observable of the stabilized operator is its value at the stable spectral point:*

$$\tau_9(f(\Psi_9)|_{\mathcal{H}_9}) = f(1).$$

Proof. By the preceding proposition,

$$f(\Psi_9)|_{\mathcal{H}_9} = f(1)I_{\mathcal{H}_9}.$$

Hence

$$\tau_9(f(\Psi_9)|_{\mathcal{H}_9}) = f(1)\tau_9(I_{\mathcal{H}_9}) = f(1).$$

□

378 The Commutant of the Stable Representation

Let

$$\pi_g : G \longrightarrow \mathcal{U}(\mathcal{H}_g)$$

be the restricted unitary representation constructed above. Define its commutant by

$$\pi_g(G)' = \{X \in \text{End}_{\mathbb{C}}(\mathcal{H}_g) : X\pi_g(g) = \pi_g(g)X \text{ for all } g \in G\}.$$

Proposition 378.1. *The algebra*

$$\pi_g(G)'$$

is a unital $$ -subalgebra of $\text{End}_{\mathbb{C}}(\mathcal{H}_g)$.*

Proof. Let

$$X, Y \in \pi_g(G)'$$

and let

$$a, b \in \mathbb{C}.$$

Then, for every $g \in G$,

$$(aX + bY)\pi_g(g) = aX\pi_g(g) + bY\pi_g(g) = a\pi_g(g)X + b\pi_g(g)Y = \pi_g(g)(aX + bY).$$

Thus the commutant is a linear subspace.

Moreover,

$$(XY)\pi_g(g) = X\pi_g(g)Y = \pi_g(g)XY,$$

so it is closed under multiplication, and clearly

$$I_{\mathcal{H}_g} \in \pi_g(G)'.$$

Finally, since $\pi_g(g)$ is unitary,

$$X\pi_g(g) = \pi_g(g)X$$

implies

$$\pi_g(g)^* X^* = X^* \pi_g(g)^*.$$

Replacing g by g^{-1} gives

$$X^* \pi_g(g) = \pi_g(g) X^*.$$

Hence $X^* \in \pi_g(G)'$. □

Proposition 378.2. *The state*

$$\rho_g = \frac{1}{9} I_{\mathcal{H}_g}$$

belongs to the commutant $\pi_g(G)'$.

Proof. For every $g \in G$,

$$\rho_g \pi_g(g) = \frac{1}{9} \pi_g(g) = \pi_g(g) \rho_g.$$

Thus

$$\rho_g \in \pi_g(G)'. \quad \square$$

Theorem 378.3. *The invariance of ρ_g under the representation is equivalent to its belonging to the commutant:*

$$\pi_g(g)\rho_g\pi_g(g)^* = \rho_g \quad (g \in G)$$

if and only if

$$\rho_g\pi_g(g) = \pi_g(g)\rho_g \quad (g \in G).$$

Proof. Since $\pi_g(g)$ is unitary,

$$\pi_g(g)\rho_g\pi_g(g)^* = \rho_g$$

is equivalent, after right multiplication by $\pi_g(g)$, to

$$\pi_g(g)\rho_g = \rho_g\pi_g(g).$$

The converse follows by multiplying the latter identity on the right by $\pi_g(g)^*$. $\square$

379 Invariant Subspaces and Reducing Decompositions

Let

$$W \subseteq \mathcal{H}_g$$

be a closed subspace invariant under $\pi_g(G)$.

Proposition 379.1. *The orthogonal complement $W^\perp$ is also invariant under $\pi_g(G)$.*

Proof. Let

$$v \in W^\perp$$

and

$$w \in W.$$

Since W is invariant,

$$\pi_g(g^{-1})w \in W.$$

Therefore

$$\langle \pi_g(g)v, w \rangle = \langle v, \pi_g(g)^*w \rangle = \langle v, \pi_g(g^{-1})w \rangle = 0.$$

Hence

$$\pi_g(g)v \in W^\perp.$$

Thus $W^\perp$ is invariant. $\square$

Theorem 379.2. *Every invariant subspace of the finite-dimensional unitary representation π_g is reducing. Equivalently,*

$$\mathcal{H}_g = W \oplus W^\perp$$

is an orthogonal decomposition into G -invariant subspaces.

Proof. The orthogonal decomposition is standard for finite-dimensional Hilbert spaces, while the preceding proposition shows that both summands are invariant under every $\pi_g(g)$. $\square$

Corollary 379.3. *If Q_W denotes the orthogonal projection onto a G -invariant subspace W , then*

$$Q_W\pi_g(g) = \pi_g(g)Q_W \quad (g \in G).$$

Proof. The decomposition

$$\mathcal{H}_g = W \oplus W^\perp$$

is preserved by $\pi_g(g)$. Hence $\pi_g(g)$ is block diagonal relative to this decomposition, and therefore commutes with the orthogonal projection Q_W . $\square$

380 Complete Reducibility of the Stable Representation

Theorem 380.1. *The representation π_9 is completely reducible. More precisely, there exist irreducible G -invariant subspaces*

$$\mathcal{K}_1, \dots, \mathcal{K}_s$$

such that

$$\mathcal{H}_9 = \mathcal{K}_1 \oplus \dots \oplus \mathcal{K}_s$$

orthogonally and

$$\dim_{\mathbb{C}} \mathcal{K}_1 + \dots + \dim_{\mathbb{C}} \mathcal{K}_s = 9.$$

Proof. If π_9 is irreducible, there is nothing to prove. Otherwise there exists a nonzero proper invariant subspace $W \subsetneq \mathcal{H}_9$. By the preceding theorem,

$$\mathcal{H}_9 = W \oplus W^\perp$$

with both summands invariant.

If either summand is reducible, the same argument can be applied to it. Because

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

this process terminates after finitely many steps and yields an orthogonal direct sum of irreducible invariant subspaces. $\square$

Corollary 380.2. *The stable representation admits an orthogonal decomposition into irreducible representation sectors whose dimensions form a partition of 9.*

Remark 380.3. No particular partition of 9 follows from finite-dimensional unitarity alone. Determining the multiplicities and dimensions of the irreducible sectors requires information about the group G and the representation π_9 .

381 Schur's Lemma on Irreducible Stable Sectors

Let

$$\mathcal{K} \subseteq \mathcal{H}_9$$

be an irreducible G -invariant subspace.

Theorem 381.1. *If*

$$X \in \text{End}_{\mathbb{C}}(\mathcal{K})$$

commutes with the restricted representation

$$\pi_{\mathcal{K}}(g) = \pi_9(g)|_{\mathcal{K}},$$

then

$$X = \lambda I_{\mathcal{K}}$$

for some

$$\lambda \in \mathbb{C}.$$

Proof. Let λ be any eigenvalue of X , which exists because $\mathcal{K}$ is finite-dimensional over $\mathbb{C}$. The eigenspace

$$E_\lambda = \ker(X - \lambda I_{\mathcal{K}})$$

is nonzero.

For $v \in E_\lambda$ and $g \in G$,

$$X\pi_{\mathcal{K}}(g)v = \pi_{\mathcal{K}}(g)Xv = \lambda\pi_{\mathcal{K}}(g)v.$$

Thus

$$\pi_{\mathcal{K}}(g)v \in E_\lambda.$$

Hence E_λ is a nonzero invariant subspace of $\mathcal{K}$. Irreducibility implies

$$E_\lambda = \mathcal{K}.$$

Therefore

$$X = \lambda I_{\mathcal{K}}.$$

□

Corollary 381.2. *On an irreducible stable sector, every self-adjoint operator commuting with the representation is a real scalar multiple of the identity.*

Proof. Apply the theorem and use self-adjointness to obtain

$$\lambda = \bar{\lambda}.$$

□

382 Multiplicity Spaces

Let the decomposition of the stable representation be written in the form

$$\mathcal{H}_g \cong \bigoplus_{\alpha=1}^r (\mathbb{C}^{m_\alpha} \otimes \mathcal{K}_\alpha),$$

where the $\mathcal{K}_\alpha$ are pairwise inequivalent irreducible representations and

$$m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha$$

denotes the dimension contributed by the corresponding isotypic component.

Theorem 382.1. *The commutant of the stable representation has the decomposition*

$$\pi_g(G)' \cong \bigoplus_{\alpha=1}^r (M_{m_\alpha}(\mathbb{C}) \otimes I_{\mathcal{K}_\alpha}).$$

Proof. On each isotypic component

$$\mathbb{C}^{m_\alpha} \otimes \mathcal{K}_\alpha,$$

the representation acts trivially on the multiplicity factor and irreducibly on $\mathcal{K}_\alpha$. By Schur's lemma, an operator intertwining the irreducible factor with itself is scalar on $\mathcal{K}_\alpha$. Consequently the full commutant on that isotypic component consists precisely of arbitrary operators on the multiplicity space tensored with the identity on $\mathcal{K}_\alpha$.

If $\alpha \neq \beta$, inequivalence of the irreducible representations implies that there are no nonzero intertwiners between $\mathcal{K}_\alpha$ and $\mathcal{K}_\beta$. Hence the commutant is the direct sum of the stated matrix algebras. □

Corollary 382.2. *The structure of the commutant is determined entirely by the multiplicities of the irreducible representation sectors.*

383 The Trace State on the Representation Algebra

Define

$$\tau_9(X) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(X), \quad X \in \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Proposition 383.1. *The functional τ_9 is a tracial state:*

$$\tau_9(I_{\mathcal{H}_9}) = 1,$$

$$\tau_9(X^* X) \geq 0,$$

and

$$\tau_9(XY) = \tau_9(YX).$$

Proof. Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

one has

$$\tau_9(I_{\mathcal{H}_9}) = \frac{1}{9} \operatorname{tr}(I_{\mathcal{H}_9}) = 1.$$

Moreover,

$$X^* X$$

is positive, so its ordinary trace is nonnegative. Finally, cyclicity of the finite-dimensional trace gives

$$\operatorname{tr}(XY) = \operatorname{tr}(YX).$$

Dividing by 9 yields the result. □

Theorem 383.2. *The trace state is invariant under the representation:*

$$\tau_9(\pi_9(g) X \pi_9(g)^*) = \tau_9(X) \quad (g \in G).$$

Proof. By cyclicity of the trace and unitarity of $\pi_9(g)$,

$$\begin{aligned} \tau_9(\pi_9(g) X \pi_9(g)^*) &= \frac{1}{9} \operatorname{tr}(\pi_9(g) X \pi_9(g)^*) \\ &= \frac{1}{9} \operatorname{tr}(X \pi_9(g)^* \pi_9(g)) \\ &= \frac{1}{9} \operatorname{tr}(X) \\ &= \tau_9(X). \end{aligned}$$

□

Corollary 383.3. *For every $X \in \mathcal{A}_9$,*

$$\tau_9(\operatorname{Ad}_{\pi_9(g)}(X)) = \tau_9(X).$$

Thus the normalized trace is a G -invariant state on the stable operator algebra.

384 Invariant States and the Ergodic Criterion

Let

$$\mathfrak{S}_G = \{\rho \in \text{End}_{\mathbb{C}}(\mathcal{H}_9) : \rho \geq 0, \text{tr}(\rho) = 1, \pi_9(g)\rho\pi_9(g)^* = \rho \ \forall g \in G\}.$$

Proposition 384.1. *The set $\mathfrak{S}_G$ is a convex compact subset of the state space of $\mathcal{H}_9$.*

Proof. Convexity follows from linearity of the trace and preservation of positivity under convex combinations. The invariance equations are linear, so the set is closed. The finite-dimensional density-operator space is compact, and hence its closed subset $\mathfrak{S}_G$ is compact. $\square$

Theorem 384.2. *The state*

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

belongs to $\mathfrak{S}_G$.

Proof. Positivity and normalization follow from

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

and

$$\text{tr}(\rho_9) = 1.$$

Invariance follows from

$$\pi_9(g)\rho_9\pi_9(g)^* = \frac{1}{9}\pi_9(g)\pi_9(g)^* = \frac{1}{9}I_{\mathcal{H}_9}.$$

$\square$

Theorem 384.3. *The invariant state ρ_9 is the unique element of $\mathfrak{S}_G$ if and only if*

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

Proof. Suppose first that

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

Every G -invariant density operator commutes with $\pi_9(G)$, hence belongs to the commutant. It must therefore have the form

$$\rho = \lambda I_{\mathcal{H}_9}.$$

The trace normalization gives

$$1 = \text{tr}(\rho) = 9\lambda,$$

so

$$\rho = \frac{1}{9}I_{\mathcal{H}_9}.$$

Conversely, suppose that ρ_9 is the unique invariant density operator. If the commutant were larger than $\mathbb{C}I_{\mathcal{H}_9}$, it would contain a non-scalar self-adjoint operator A . Replacing A by

$$A - \tau_9(A)I_{\mathcal{H}_9},$$

we may assume

$$\tau_9(A) = 0$$

while retaining

$$A = A^*$$

and

$$A \neq 0.$$

For sufficiently small

$$\varepsilon > 0,$$

the operator

$$\rho_\varepsilon = \frac{1}{9}I_{\mathcal{H}_9} + \varepsilon A$$

is positive and has trace one. Since A belongs to the commutant, ρ_ε is G -invariant. This contradicts uniqueness. Hence

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

□

Corollary 384.4. *If the stable representation is irreducible, then*

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

is the unique G -invariant density operator.

Proof. For an irreducible finite-dimensional complex representation, Schur's lemma gives

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

The preceding theorem applies. □

385 The Stable Representation and Its Isotypic Decomposition

Let

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^r \mathcal{H}_\alpha$$

be the orthogonal decomposition into isotypic components of the finite-dimensional unitary representation π_9 , so that

$$\mathcal{H}_\alpha \cong \mathbb{C}^{m_\alpha} \otimes \mathcal{K}_\alpha,$$

where the $\mathcal{K}_\alpha$ are pairwise inequivalent irreducible representations and

$$\sum_{\alpha=1}^r m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha = 9.$$

Let

$$Q_\alpha$$

denote the orthogonal projection onto $\mathcal{H}_\alpha$.

Proposition 385.1. *Each Q_α commutes with the stable representation:*

$$Q_\alpha \pi_9(g) = \pi_9(g) Q_\alpha \quad (g \in G).$$

Proof. Every isotypic component is G -invariant. Since the representation is unitary, its orthogonal complement is invariant as well. Hence the decomposition

$$\mathcal{H}_9 = \mathcal{H}_\alpha \oplus \mathcal{H}_\alpha^\perp$$

is preserved by $\pi_9(g)$. The corresponding orthogonal projection therefore commutes with $\pi_9(g)$. □

Theorem 385.2. *The projections $Q_1, \dots, Q_r$ satisfy*

$$Q_\alpha Q_\beta = \delta_{\alpha\beta} Q_\alpha,$$

$$\sum_{\alpha=1}^r Q_\alpha = I_{\mathcal{H}_9},$$

and

$$\text{tr}(Q_\alpha) = m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha.$$

Proof. The first two identities follow from the orthogonal direct-sum decomposition. The range of Q_α is

$$\mathcal{H}_\alpha \cong \mathbb{C}^{m_\alpha} \otimes \mathcal{K}_\alpha,$$

whose dimension is

$$m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha.$$

Since the trace of a finite-dimensional orthogonal projection equals its rank, the final identity follows. $\square$

Corollary 385.3. *The normalized trace weights of the isotypic components are*

$$\tau_9(Q_\alpha) = \frac{m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha}{9},$$

and

$$\sum_{\alpha=1}^r \tau_9(Q_\alpha) = 1.$$

386 The Canonical Conditional State on an Isotypic Component

For every α , define the normalized restriction of the trace state to $\mathcal{H}_\alpha$ by

$$\tau_\alpha(X) = \frac{\text{tr}_{\mathcal{H}_\alpha}(X)}{m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha}, \quad X \in \text{End}_{\mathbb{C}}(\mathcal{H}_\alpha).$$

Proposition 386.1. *The functional τ_α is a normalized tracial state on $\text{End}_{\mathbb{C}}(\mathcal{H}_\alpha)$.*

Proof. The space $\mathcal{H}_\alpha$ has finite positive dimension

$$d_\alpha = m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha.$$

Therefore

$$\tau_\alpha(I_{\mathcal{H}_\alpha}) = \frac{1}{d_\alpha} \text{tr}(I_{\mathcal{H}_\alpha}) = 1.$$

Positivity follows from positivity of the ordinary trace, while the tracial identity follows from cyclicity of the trace. $\square$

Theorem 386.2. *The normalized trace state on every isotypic component is invariant under the restricted representation:*

$$\tau_\alpha(\pi_\alpha(g) X \pi_\alpha(g)^*) = \tau_\alpha(X),$$

where

$$\pi_\alpha(g) = \pi_9(g)|_{\mathcal{H}_\alpha}.$$

Proof. The operator $\pi_\alpha(g)$ is unitary on $\mathcal{H}_\alpha$. Consequently,

$$\begin{aligned}\tau_\alpha(\pi_\alpha(g)X\pi_\alpha(g)^*) &= \frac{1}{d_\alpha} \operatorname{tr}(\pi_\alpha(g)X\pi_\alpha(g)^*) \\ &= \frac{1}{d_\alpha} \operatorname{tr}(X\pi_\alpha(g)^*\pi_\alpha(g)) \\ &= \frac{1}{d_\alpha} \operatorname{tr}(X) \\ &= \tau_\alpha(X).\end{aligned}$$

□

387 Block Form of the Maximally Mixed Stable State

Relative to the isotypic decomposition,

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^r \mathcal{H}_\alpha,$$

the identity decomposes as

$$I_{\mathcal{H}_9} = \bigoplus_{\alpha=1}^r I_{\mathcal{H}_\alpha}.$$

Theorem 387.1. *The canonical stable state admits the block decomposition*

$$\rho_9 = \frac{1}{9} \bigoplus_{\alpha=1}^r I_{\mathcal{H}_\alpha}.$$

Equivalently,

$$\rho_9 = \bigoplus_{\alpha=1}^r \frac{1}{9} I_{\mathcal{H}_\alpha}.$$

Proof. This follows immediately from

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

and the orthogonal direct-sum decomposition of $\mathcal{H}_9$. □

Proposition 387.2. *The probability of the isotypic event Q_α in the state ρ_9 is*

$$\operatorname{tr}(\rho_9 Q_\alpha) = \frac{m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha}{9}.$$

Proof. Using

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one obtains

$$\operatorname{tr}(\rho_9 Q_\alpha) = \frac{1}{9} \operatorname{tr}(Q_\alpha).$$

The preceding trace formula gives

$$\operatorname{tr}(Q_\alpha) = m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha.$$

□

Corollary 387.3. *The state ρ_9 induces on the finite family of isotypic sectors the probability distribution*

$$p_\alpha = \frac{m_\alpha \dim_{\mathbb{C}} \mathcal{K}_\alpha}{9}.$$

388 The Representation-Theoretic Content of Uniform Rank-One Weights

Let

$$\psi_1, \dots, \psi_9$$

be an orthonormal basis of $\mathcal{H}_9$, and let

$$P_j = |\psi_j\rangle\langle\psi_j|.$$

Proposition 388.1. *The rank-one projections satisfy*

$$P_i P_j = \delta_{ij} P_j$$

and

$$\sum_{j=1}^9 P_j = I_{\mathcal{H}_9}.$$

Proof. Orthogonality of the basis gives

$$\langle\psi_i, \psi_j\rangle = \delta_{ij}.$$

Therefore

$$P_i P_j = |\psi_i\rangle\langle\psi_i, \psi_j\rangle\langle\psi_j| = \delta_{ij} P_j.$$

The resolution of the identity by an orthonormal basis gives the second identity. $\square$

Theorem 388.2. *For the state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\text{tr}(\rho_9 P_j) = \frac{1}{9} \quad (1 \leq j \leq 9).$$

Proof. Since P_j has rank one,

$$\text{tr}(P_j) = 1.$$

Consequently,

$$\text{tr}(\rho_9 P_j) = \frac{1}{9} \text{tr}(P_j) = \frac{1}{9}.$$

$\square$

Corollary 388.3. *The uniformity of the rank-one probabilities is independent of the choice of orthonormal basis.*

Proof. The proof of the preceding theorem uses only the facts that ρ_9 is a scalar multiple of the identity and that P_j has rank one. Neither depends on the choice of basis. $\square$

389 Characterization of the Uniform Rank-One State

Theorem 389.1. *Let*

$$\rho \geq 0, \quad \text{tr}(\rho) = 1$$

on the nine-dimensional Hilbert space $\mathcal{H}_9$. Suppose that for every unit vector $\psi \in \mathcal{H}_9$,

$$\langle\psi, \rho\psi\rangle = \frac{1}{9}.$$

Then

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. Define

$$A = \rho - \frac{1}{9} I_{\mathcal{H}_9}.$$

The hypothesis gives

$$\langle \psi, A\psi \rangle = 0$$

for every unit vector ψ , and hence, by homogeneity,

$$\langle v, Av \rangle = 0$$

for every $v \in \mathcal{H}_9$.

Since A is self-adjoint, the polarization identity implies

$$\langle u, Av \rangle = 0$$

for all

$$u, v \in \mathcal{H}_9.$$

Therefore

$$A = 0,$$

which gives

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

Corollary 389.2. *Among all density operators on $\mathcal{H}_9$, the maximally mixed state is uniquely characterized by the property that every rank-one projection has the same probability.*

390 Compatibility of the Stable Projection with the Representation

Recall that

$$P_9 = \Psi_9^4$$

and

$$\mathcal{H}_9 = \text{range}(P_9).$$

Proposition 390.1. *If*

$$\pi(g)P_9 = P_9\pi(g) \quad (g \in G),$$

then

$$\pi(g)P_9\pi(g)^* = P_9.$$

Proof. Starting from

$$\pi(g)P_9 = P_9\pi(g),$$

multiply on the right by $\pi(g)^*$. Since

$$\pi(g)\pi(g)^* = I,$$

one obtains

$$\pi(g)P_9\pi(g)^* = P_9.$$

□

Theorem 390.2. *The stabilized projection P_9 is a G -invariant event:*

$$\text{Ad}_{\pi(g)}(P_9) = P_9 \quad (g \in G).$$

Proof. This is precisely the preceding proposition. $\square$

Corollary 390.3. *The stable sector is preserved by the full representation, and the orthogonal projection onto that sector is an invariant observable.*

Proof. The invariance of the range was established from the commutation relation, while the projection itself is invariant by conjugation. $\square$

391 Restriction of the Ambient Trace to the Stable Sector

Let

$$V_9$$

be the ambient finite-dimensional space containing $\mathcal{H}_9$, and suppose

$$\dim_{\mathbb{C}} V_9 = 143, \quad \dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

Let

$$\text{tr}_{V_9}$$

denote the ordinary trace on $\text{End}_{\mathbb{C}}(V_9)$.

Proposition 391.1. *For every*

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

viewed as an operator supported on the stable sector,

$$\text{tr}_{V_9}(X) = \text{tr}_{\mathcal{H}_9}(X).$$

Proof. Relative to the orthogonal decomposition

$$V_9 = \mathcal{H}_9 \oplus \mathcal{H}_9^{\perp},$$

the extension of X by zero has block form

$$\begin{pmatrix} X & 0 \\ 0 & 0 \end{pmatrix}.$$

Its trace is therefore the trace of the first diagonal block. $\square$

Theorem 391.2. *For every $X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$,*

$$\tau_9(X) = \frac{143}{9} \left(\frac{1}{143} \text{tr}_{V_9}(X) \right).$$

Proof. By definition,

$$\tau_9(X) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(X).$$

The preceding proposition identifies this trace with the ambient trace of the zero extension of X . Hence

$$\tau_9(X) = \frac{1}{9} \text{tr}_{V_9}(X) = \frac{143}{9} \left(\frac{1}{143} \text{tr}_{V_9}(X) \right).$$

$\square$

Remark 391.3. The factor

$$\frac{143}{9}$$

records the change of normalization between the ambient trace and the intrinsic normalized trace of the nine-dimensional stable sector. The two traces should therefore not be identified.

392 The Stable Compression of the Ambient Trace

Let

$$\tau_{V_9}(X) = \frac{1}{143} \operatorname{tr}_{V_9}(X), \quad X \in \operatorname{End}_{\mathbb{C}}(V_9),$$

and let

$$P_9 = \Psi_9^4$$

be the orthogonal projection onto

$$\mathcal{H}_9 = \operatorname{range}(P_9), \quad \dim_{\mathbb{C}} \mathcal{H}_9 = 9.$$

For

$$X \in \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9),$$

define its ambient zero extension by

$$\tilde{X} = P_9 X P_9 \in \operatorname{End}_{\mathbb{C}}(V_9).$$

Proposition 392.1. *The map*

$$\iota_9 : \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9) \longrightarrow \operatorname{End}_{\mathbb{C}}(V_9), \quad X \longmapsto \tilde{X},$$

is an injective unital $$ -homomorphism onto the corner algebra*

$$P_9 \operatorname{End}_{\mathbb{C}}(V_9) P_9.$$

Proof. For

$$X, Y \in \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the operators $\tilde{X}$ and $\tilde{Y}$ vanish on

$$\mathcal{H}_9^{\perp}$$

and preserve $\mathcal{H}_9$. Hence

$$\tilde{X} \tilde{Y} = \widetilde{XY}.$$

Similarly,

$$\widetilde{X^*} = \tilde{X}^*.$$

Moreover,

$$\widetilde{I_{\mathcal{H}_9}} = P_9,$$

which is the identity of the corner algebra

$$P_9 \operatorname{End}_{\mathbb{C}}(V_9) P_9.$$

If

$$\tilde{X} = 0,$$

then its restriction to

$$\mathcal{H}_9$$

is X , and therefore $X = 0$. Thus ι_9 is injective.

Conversely, if

$$A = P_9 A P_9,$$

then A vanishes on $\mathcal{H}_9^{\perp}$ and restricts to an operator on

$$\mathcal{H}_9.$$

Hence $A = \tilde{X}$ for

$$X = A|_{\mathcal{H}_9}.$$

Therefore the image is exactly

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

□

Theorem 392.2. *The normalized trace on the stable sector is the unique normalized trace on its full matrix algebra*

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Equivalently, if

$$\varphi : \text{End}_{\mathbb{C}}(\mathcal{H}_9) \longrightarrow \mathbb{C}$$

satisfies

$$\varphi(I_{\mathcal{H}_9}) = 1$$

and

$$\varphi(XY) = \varphi(YX)$$

for all

$$X, Y \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

then

$$\varphi(X) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(X) = \tau_9(X).$$

Proof. Choose matrix units

$$E_{ij} = |\psi_i\rangle\langle\psi_j|$$

associated with any orthonormal basis

$$\{\psi_1, \dots, \psi_9\}$$

of $\mathcal{H}_9$.

For $i \neq j$,

$$E_{ij} = E_{ii}E_{ij},$$

while

$$E_{ij} = E_{ij}E_{ii}.$$

The tracial property therefore gives

$$\varphi(E_{ij}) = \varphi(E_{ii}E_{ij}) = \varphi(E_{ij}E_{ii}).$$

Since

$$E_{ij}E_{ii} = 0 \quad (i \neq j),$$

we obtain

$$\varphi(E_{ij}) = 0.$$

For the diagonal matrix units, if $i \neq j$, then

$$E_{ij}E_{ji} = E_{ii}, \quad E_{ji}E_{ij} = E_{jj}.$$

Hence

$$\varphi(E_{ii}) = \varphi(E_{ij}E_{ji}) = \varphi(E_{ji}E_{ij}) = \varphi(E_{jj}).$$

Thus all diagonal matrix units have the same value, say c . Since

$$I_{\mathcal{H}_9} = \sum_{i=1}^9 E_{ii},$$

normalization gives

$$1 = \varphi(I_{\mathcal{H}_9}) = \sum_{i=1}^9 \varphi(E_{ii}) = 9c.$$

Therefore

$$c = \frac{1}{9}.$$

For

$$X = \sum_{i,j=1}^9 x_{ij} E_{ij},$$

linearity yields

$$\varphi(X) = \sum_{i,j=1}^9 x_{ij} \varphi(E_{ij}) = \frac{1}{9} \sum_{i=1}^9 x_{ii} = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(X).$$

Hence

$$\varphi = \tau_9.$$

□

Corollary 392.3. *The normalized trace state*

$$\tau_9(X) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(X)$$

is intrinsic to the full operator algebra of the stable sector and does not depend on the choice of orthonormal basis.

Proof. The preceding theorem characterizes τ_9 solely by normalization and the tracial identity. Neither condition refers to a choice of basis. □

393 The Stable Trace as a Conditional Ambient State

Define the compression of an ambient operator

$$A \in \operatorname{End}_{\mathbb{C}}(V_9)$$

to the stable sector by

$$C_9(A) = P_9 A P_9|_{\mathcal{H}_9}.$$

Proposition 393.1. *The compression map*

$$C_9 : \operatorname{End}_{\mathbb{C}}(V_9) \longrightarrow \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9)$$

is completely positive and unital relative to the corner identity P_9 :

$$C_9(I_{V_9}) = I_{\mathcal{H}_9}.$$

Proof. For every positive operator

$$A \geq 0$$

on V_9 and every

$$v \in \mathcal{H}_9,$$

one has

$$\langle v, C_9(A)v \rangle = \langle v, Av \rangle \geq 0.$$

Thus C_9 is positive.

More generally, for every $n \geq 1$ and every positive operator

$$[A_{ij}]_{i,j=1}^n \in M_n(\text{End}_{\mathbb{C}}(V_9)),$$

compression by

$$I_n \otimes P_9$$

preserves positivity:

$$(I_n \otimes P_9)[A_{ij}](I_n \otimes P_9) \geq 0.$$

After restriction to

$$\mathbb{C}^n \otimes \mathcal{H}_9,$$

this is precisely positivity of

$$[C_9(A_{ij})].$$

Hence C_9 is completely positive.

Finally,

$$C_9(I_{V_9}) = P_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

□

Theorem 393.2. *Let*

$$\widehat{\tau}_9 = \tau_9 \circ C_9.$$

Then

$$\widehat{\tau}_9(A) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(P_9 A P_9|_{\mathcal{H}_9})$$

defines a normalized positive functional on

$$\text{End}_{\mathbb{C}}(V_9).$$

Moreover,

$$\widehat{\tau}_9(A) = \frac{1}{9} \text{tr}_{V_9}(P_9 A)$$

for every

$$A \in \text{End}_{\mathbb{C}}(V_9).$$

Proof. Since C_9 is unital and completely positive and τ_9 is a positive normalized functional,

$$\widehat{\tau}_9 = \tau_9 \circ C_9$$

is positive and satisfies

$$\widehat{\tau}_9(I_{V_9}) = \tau_9(I_{\mathcal{H}_9}) = 1.$$

It remains to identify its trace expression. Relative to

$$V_9 = \mathcal{H}_9 \oplus \mathcal{H}_9^{\perp},$$

the operator

$$P_9 A P_9$$

has block form

$$\begin{pmatrix} A_{11} & 0 \\ 0 & 0 \end{pmatrix},$$

where

$$A_{11} = P_9 A P_9|_{\mathcal{H}_9}.$$

Consequently,

$$\mathrm{tr}_{V_9}(P_9 A P_9) = \mathrm{tr}_{\mathcal{H}_9}(A_{11}).$$

By cyclicity of the finite-dimensional trace,

$$\mathrm{tr}_{V_9}(P_9 A P_9) = \mathrm{tr}_{V_9}(P_9 A).$$

Therefore

$$\widehat{\tau}_9(A) = \frac{1}{9} \mathrm{tr}_{V_9}(P_9 A).$$

□

Corollary 393.3. *The stable trace depends on an ambient observable only through its stable compression:*

$$C_9(A) = C_9(B) \implies \widehat{\tau}_9(A) = \widehat{\tau}_9(B).$$

Proof. Immediate from

$$\widehat{\tau}_9 = \tau_9 \circ C_9.$$

□

Theorem 393.4. *The functional $\widehat{\tau}_9$ is invariant under the ambient representation whenever*

$$\pi(g)P_9 = P_9\pi(g) \quad (g \in G).$$

More precisely,

$$\widehat{\tau}_9(\pi(g)A\pi(g)^*) = \widehat{\tau}_9(A)$$

for every

$$A \in \mathrm{End}_{\mathbb{C}}(V_9)$$

and every

$$g \in G.$$

Proof. The commutation relation implies

$$\pi(g)P_9 = P_9\pi(g)$$

and, taking adjoints,

$$\pi(g)^*P_9 = P_9\pi(g)^*.$$

Hence

$$P_9\pi(g)A\pi(g)^* = \pi(g)P_9A\pi(g)^*.$$

Using the trace expression established above,

$$\begin{aligned}
 \widehat{\tau}_9(\pi(g)A\pi(g)^*) &= \frac{1}{9} \operatorname{tr}_{V_9}(P_9\pi(g)A\pi(g)^*) \\
 &= \frac{1}{9} \operatorname{tr}_{V_9}(\pi(g)P_9A\pi(g)^*) \\
 &= \frac{1}{9} \operatorname{tr}_{V_9}(P_9A\pi(g)^*\pi(g)) \\
 &= \frac{1}{9} \operatorname{tr}_{V_9}(P_9A) \\
 &= \widehat{\tau}_9(A).
 \end{aligned}$$

Thus the compressed stable state is invariant under the full ambient representation. $\square$

Corollary 393.5. *For the stable projection itself,*

$$\widehat{\tau}_9(P_9) = 1,$$

whereas the normalized ambient trace satisfies

$$\tau_{V_9}(P_9) = \frac{9}{143}.$$

Thus the stable conditioning changes the normalization from the ambient dimension 143 to the intrinsic stable dimension 9.

Proof. Since

$$P_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9},$$

one has

$$\widehat{\tau}_9(P_9) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(I_{\mathcal{H}_9}) = 1.$$

On the other hand,

$$\operatorname{tr}_{V_9}(P_9) = \operatorname{rank}(P_9) = 9,$$

and therefore

$$\tau_{V_9}(P_9) = \frac{1}{143} \operatorname{tr}_{V_9}(P_9) = \frac{9}{143}.$$

$\square$

394 The Stable Corner Algebra and Its State Space

Define the stable corner algebra by

$$\mathcal{A}_9 = P_9 \operatorname{End}_{\mathbb{C}}(V_9) P_9.$$

By the preceding identification,

$$\mathcal{A}_9 \cong \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9) \cong M_9(\mathbb{C}).$$

Proposition 394.1. *The stable corner algebra $\mathcal{A}_9$ is a unital finite-dimensional C^* -algebra with identity P_9 .*

Proof. If

$$A, B \in \mathcal{A}_9,$$

then

$$A = P_9 A P_9, \quad B = P_9 B P_9.$$

Consequently,

$$AB = P_9 A B P_9 \in \mathcal{A}_9,$$

and

$$A^* = P_9 A^* P_9 \in \mathcal{A}_9.$$

Thus $\mathcal{A}_9$ is a $*$ -subalgebra of $\text{End}_{\mathbb{C}}(V_9)$.

For every $A \in \mathcal{A}_9$,

$$P_9 A = P_9(P_9 A P_9) = P_9 A P_9 = A,$$

and similarly

$$A P_9 = A.$$

Hence P_9 is the identity element of $\mathcal{A}_9$.

Finally, since

$$\mathcal{H}_9 = \text{range}(P_9)$$

has dimension 9, the restriction map

$$A \longmapsto A|_{\mathcal{H}_9}$$

identifies $\mathcal{A}_9$ with

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9) \cong M_9(\mathbb{C}).$$

The latter is a finite-dimensional C^* -algebra, and therefore so is $\mathcal{A}_9$. □

Theorem 394.2. *The functional*

$$\widehat{\tau}_9(A) = \frac{1}{9} \text{tr}_{V_9}(P_9 A), \quad A \in \mathcal{A}_9,$$

is the unique normalized tracial state on $\mathcal{A}_9$.

Proof. For $A \in \mathcal{A}_9$ one has

$$P_9 A = A,$$

and therefore

$$\widehat{\tau}_9(A) = \frac{1}{9} \text{tr}_{V_9}(A) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(A).$$

Under the canonical identification

$$\mathcal{A}_9 \cong \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

this is precisely the normalized trace τ_9 .

The uniqueness of the normalized tracial state on $\text{End}_{\mathbb{C}}(\mathcal{H}_9)$ was established above. Hence $\widehat{\tau}_9$ is the unique normalized tracial state on $\mathcal{A}_9$. □

Corollary 394.3. *If*

$$A \in \mathcal{A}_9$$

is positive, then

$$\widehat{\tau}_9(A) \geq 0,$$

and

$$\widehat{\tau}_9(A^* A) = 0$$

if and only if

$$A = 0.$$

Proof. Positivity follows from positivity of the finite-dimensional trace.

If

$$\widehat{\tau}_9(A^*A) = 0,$$

then

$$\mathrm{tr}_{\mathcal{H}_9}(A^*A) = 0.$$

The operator A^*A is positive, so all its eigenvalues are nonnegative. Their sum is zero, hence every eigenvalue is zero. Therefore

$$A^*A = 0,$$

and consequently

$$A = 0.$$

□

395 Invariant States on the Stable Corner

Let

$$\mathfrak{S}(\mathcal{A}_9) = \{\omega : \mathcal{A}_9 \longrightarrow \mathbb{C} : \omega \geq 0, \omega(P_9) = 1\}$$

denote the state space of the stable corner algebra.

The representation induces an action

$$\alpha_g(A) = \pi_9(g)A\pi_9(g)^*, \quad A \in \mathcal{A}_9.$$

Proposition 395.1. *For every $g \in G$, the map*

$$\alpha_g : \mathcal{A}_9 \longrightarrow \mathcal{A}_9$$

is a $$ -automorphism.*

Proof. Since

$$\pi_9(g)$$

is unitary on $\mathcal{H}_9$,

$$\alpha_g(AB) = \pi_9(g)AB\pi_9(g)^* = \alpha_g(A)\alpha_g(B),$$

and

$$\alpha_g(A^*) = \pi_9(g)A^*\pi_9(g)^* = \alpha_g(A)^*.$$

Moreover,

$$\alpha_g(P_9) = P_9$$

when P_9 is regarded as the identity of the corner algebra.

The inverse of α_g is

$$\alpha_{g^{-1}},$$

because

$$\pi_9(g^{-1}) = \pi_9(g)^*.$$

Thus α_g is a $*$ -automorphism.

□

Theorem 395.2. *The normalized stable trace*

$$\widehat{\tau}_9$$

is invariant under the induced action of G :

$$\widehat{\tau}_9(\alpha_g(A)) = \widehat{\tau}_9(A) \quad (A \in \mathcal{A}_9, g \in G).$$

Proof. For $A \in \mathcal{A}_g$,

$$\widehat{\tau}_g(\alpha_g(A)) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_g} (\pi_g(g) A \pi_g(g)^*).$$

Cyclicity of the finite-dimensional trace gives

$$\widehat{\tau}_g(\alpha_g(A)) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_g} (A \pi_g(g)^* \pi_g(g)).$$

Since

$$\pi_g(g)^* \pi_g(g) = I_{\mathcal{H}_g},$$

it follows that

$$\widehat{\tau}_g(\alpha_g(A)) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_g} (A) = \widehat{\tau}_g(A).$$

□

Proposition 395.3. *A density operator*

$$\rho \in \operatorname{End}_{\mathbb{C}}(\mathcal{H}_g)$$

defines a G -invariant state on $\mathcal{A}_g$ if and only if

$$\rho \geq 0, \quad \operatorname{tr}_{\mathcal{H}_g}(\rho) = 1,$$

and

$$\rho \pi_g(g) = \pi_g(g) \rho \quad (g \in G).$$

Proof. The state associated with ρ is

$$\omega_\rho(A) = \operatorname{tr}_{\mathcal{H}_g}(\rho A).$$

It is invariant precisely when

$$\omega_\rho(\alpha_g(A)) = \omega_\rho(A)$$

for every A and g . Thus

$$\operatorname{tr}(\rho \pi_g(g) A \pi_g(g)^*) = \operatorname{tr}(\rho A).$$

By cyclicity,

$$\operatorname{tr}(\pi_g(g)^* \rho \pi_g(g) A) = \operatorname{tr}(\rho A)$$

for every A . Nondegeneracy of the trace pairing on $\operatorname{End}_{\mathbb{C}}(\mathcal{H}_g)$ implies

$$\pi_g(g)^* \rho \pi_g(g) = \rho.$$

Equivalently,

$$\rho \pi_g(g) = \pi_g(g) \rho.$$

Conversely, if ρ commutes with every $\pi_g(g)$, then

$$\begin{aligned} \omega_\rho(\alpha_g(A)) &= \operatorname{tr}(\rho \pi_g(g) A \pi_g(g)^*) \\ &= \operatorname{tr}(\pi_g(g)^* \rho \pi_g(g) A) \\ &= \operatorname{tr}(\rho A) \\ &= \omega_\rho(A). \end{aligned}$$

Hence the state is G -invariant.

□

396 Irreducibility and Uniqueness of the Stable State

Theorem 396.1. *The following statements are equivalent:*

1. *The representation π_9 on $\mathcal{H}_9$ is irreducible.*
2. *The commutant satisfies*

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

3. *The normalized trace state*

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

is the unique G -invariant density operator on $\mathcal{H}_9$.

Proof. The equivalence of (1) and (2) follows from Schur's lemma for finite-dimensional complex representations.

The equivalence of (2) and (3) was established by the characterization of the invariant density operators.

For completeness, suppose that (3) holds and that π_9 is reducible. Then there exists a nonzero proper invariant subspace

$$\mathcal{W} \subsetneq \mathcal{H}_9.$$

Because π_9 is unitary,

$$\mathcal{W}^\perp$$

is also invariant. Let

$$Q_{\mathcal{W}}$$

be the orthogonal projection onto $\mathcal{W}$. Then

$$Q_{\mathcal{W}}\pi_9(g) = \pi_9(g)Q_{\mathcal{W}} \quad (g \in G).$$

The operator

$$\rho_t = \frac{1}{9}I_{\mathcal{H}_9} + t \left(Q_{\mathcal{W}} - \frac{\dim_{\mathbb{C}} \mathcal{W}}{9} I_{\mathcal{H}_9} \right)$$

has trace one for every t . Since the second term is self-adjoint and nonzero, sufficiently small nonzero t make ρ_t positive. It is then a G -invariant density operator distinct from ρ_9 , contradicting (3).

Thus π_9 is irreducible. □

Corollary 396.2. *If the nine-dimensional stable representation is irreducible, then the arithmetic-relational state on the stable operator algebra is forced to be*

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}.$$

Proof. Irreducibility implies

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}$$

by Schur's lemma. The preceding theorem then gives uniqueness of the invariant density operator, whose trace normalization forces

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}.$$

□

397 The Stable Trace as the Intrinsic State Functional

The preceding restriction identifies the normalized trace on $\mathcal{H}_9$ with the ambient trace after the corresponding normalization change. We now isolate the intrinsic statement that is independent of the particular realization of $\mathcal{H}_9$ inside V_9 .

Theorem 397.1. *Let*

$$\iota: \mathcal{H}_9 \hookrightarrow V_9$$

be the canonical inclusion and let

$$\tilde{X} = \iota X \iota^*$$

denote the extension of

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

by zero on $\mathcal{H}_9^\perp$. Then

$$\text{tr}_{V_9}(\tilde{X}) = \text{tr}_{\mathcal{H}_9}(X)$$

and consequently

$$\tau_9(X) = \frac{1}{9} \text{tr}_{V_9}(\tilde{X}).$$

In particular, the state functional on the stable sector is determined solely by the intrinsic nine-dimensional trace.

Proof. Choose an orthonormal basis

$$(e_1, \dots, e_9)$$

of $\mathcal{H}_9$, and extend it to an orthonormal basis

$$(e_1, \dots, e_9, f_1, \dots, f_{134})$$

of V_9 . Since $\tilde{X}$ vanishes on $\mathcal{H}_9^\perp$,

$$\langle f_a, \tilde{X} f_a \rangle = 0 \quad (1 \leq a \leq 134).$$

Therefore

$$\begin{aligned} \text{tr}_{V_9}(\tilde{X}) &= \sum_{j=1}^9 \langle e_j, \tilde{X} e_j \rangle + \sum_{a=1}^{134} \langle f_a, \tilde{X} f_a \rangle \\ &= \sum_{j=1}^9 \langle e_j, X e_j \rangle \\ &= \text{tr}_{\mathcal{H}_9}(X). \end{aligned}$$

The second identity follows immediately from the definition

$$\tau_9(X) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(X).$$

□

Proposition 397.2. *The intrinsic state τ_9 is faithful: if*

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9), \quad X \geq 0,$$

and

$$\tau_9(X) = 0,$$

then

$$X = 0.$$

Proof. The condition

$$\tau_9(X) = 0$$

is equivalent to

$$\mathrm{tr}_{\mathcal{H}_9}(X) = 0.$$

Since X is positive, all of its eigenvalues are nonnegative. Their sum is its trace, hence every eigenvalue is zero. Therefore

$$X = 0.$$

□

Theorem 397.3. *For every*

$$X \in \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the state τ_9 admits the density-operator representation

$$\tau_9(X) = \mathrm{tr}_{\mathcal{H}_9}(\rho_9 X), \quad \rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Moreover, ρ_9 is the unique density operator with this property.

Proof. Direct substitution gives

$$\mathrm{tr}_{\mathcal{H}_9}(\rho_9 X) = \mathrm{tr}_{\mathcal{H}_9}\left(\frac{1}{9} I_{\mathcal{H}_9} X\right) = \frac{1}{9} \mathrm{tr}_{\mathcal{H}_9}(X) = \tau_9(X).$$

Suppose that another density operator

$$\sigma$$

satisfies

$$\mathrm{tr}_{\mathcal{H}_9}(\sigma X) = \tau_9(X)$$

for every X . Taking

$$X = |\psi\rangle\langle\phi|$$

for arbitrary

$$\psi, \phi \in \mathcal{H}_9$$

gives

$$\langle\phi, \sigma\psi\rangle = \frac{1}{9} \langle\phi, \psi\rangle.$$

Hence

$$\langle\phi, (\sigma - \frac{1}{9} I_{\mathcal{H}_9})\psi\rangle = 0$$

for all ϕ, ψ , and therefore

$$\sigma = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

398 Compression of Ambient Observables

Let

$$P_9$$

be the orthogonal projection onto $\mathcal{H}_9$. For an arbitrary ambient observable

$$A \in \text{End}_{\mathbb{C}}(V_9),$$

define its stable compression by

$$A_9 = P_9 A P_9|_{\mathcal{H}_9}.$$

Proposition 398.1. *The compression map*

$$\mathfrak{C}_9 : \text{End}_{\mathbb{C}}(V_9) \longrightarrow \text{End}_{\mathbb{C}}(\mathcal{H}_9), \quad \mathfrak{C}_9(A) = P_9 A P_9|_{\mathcal{H}_9},$$

is linear and unital.

Proof. Linearity follows from linearity of operator multiplication:

$$\mathfrak{C}_9(aA + bB) = P_9(aA + bB)P_9|_{\mathcal{H}_9} = a\mathfrak{C}_9(A) + b\mathfrak{C}_9(B).$$

For the identity operator,

$$\mathfrak{C}_9(I_{V_9}) = P_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9},$$

because $P_9 v = v$ for every

$$v \in \mathcal{H}_9.$$

□

Proposition 398.2. *The compression map is positive. More precisely,*

$$A \geq 0 \implies \mathfrak{C}_9(A) \geq 0.$$

Proof. Let

$$v \in \mathcal{H}_9.$$

Since

$$P_9 v = v,$$

one has

$$\langle v, \mathfrak{C}_9(A) v \rangle = \langle v, P_9 A P_9 v \rangle = \langle v, A v \rangle.$$

If $A \geq 0$, the last quantity is nonnegative. Hence

$$\mathfrak{C}_9(A) \geq 0.$$

□

Theorem 398.3. *The stable state applied to an ambient observable depends only on its compression to the stable sector:*

$$\text{tr}_{V_9}(\widetilde{\rho}_9 A) = \text{tr}_{\mathcal{H}_9}(\rho_9 \mathfrak{C}_9(A)),$$

where

$$\widetilde{\rho}_9 = \frac{1}{9} P_9$$

is the extension of ρ_9 by zero to V_9 .

Proof. Using

$$\widetilde{\rho}_9 = \frac{1}{9} P_9$$

and cyclicity of the finite-dimensional trace,

$$\begin{aligned} \mathrm{tr}_{V_9}(\widetilde{\rho}_9 A) &= \frac{1}{9} \mathrm{tr}_{V_9}(P_9 A) \\ &= \frac{1}{9} \mathrm{tr}_{V_9}(P_9 A P_9) \\ &= \frac{1}{9} \mathrm{tr}_{\mathcal{H}_9}(P_9 A P_9|_{\mathcal{H}_9}) \\ &= \mathrm{tr}_{\mathcal{H}_9}(\rho_9 \mathfrak{C}_9(A)). \end{aligned}$$

The second equality follows from

$$P_9 = P_9^2$$

and cyclicity:

$$\mathrm{tr}(P_9 A) = \mathrm{tr}(P_9 A P_9).$$

□

Corollary 398.4. *If*

$$A P_9 = P_9 A,$$

then

$$\mathfrak{C}_9(A) = A|_{\mathcal{H}_9},$$

and hence

$$\mathrm{tr}_{V_9}(\widetilde{\rho}_9 A) = \frac{1}{9} \mathrm{tr}_{\mathcal{H}_9}(A|_{\mathcal{H}_9}).$$

Proof. Commutation with P_9 implies

$$P_9 A P_9 = A P_9.$$

For

$$v \in \mathcal{H}_9,$$

one has

$$P_9 v = v,$$

and therefore

$$P_9 A P_9 v = A v.$$

Thus

$$\mathfrak{C}_9(A) = A|_{\mathcal{H}_9}.$$

The trace identity follows from the preceding theorem. □

399 Induced Dynamics on the Stable Sector

Let

$$U : G \longrightarrow \mathcal{U}(V_9)$$

be a unitary representation of the dynamical symmetry group on the ambient finite-dimensional space V_9 . Assume that the stable projection satisfies

$$U(g) P_9 = P_9 U(g) \quad (g \in G).$$

For each $g \in G$, define

$$U_9(g) = U(g)|_{\mathcal{H}_9}.$$

Proposition 399.1. *The family*

$$\{U_9(g) : g \in G\}$$

is a well-defined unitary representation of G on $\mathcal{H}_9$.

Proof. Since

$$U(g)P_9 = P_9U(g),$$

the range of P_9 is invariant under $U(g)$. Indeed, if

$$v \in \mathcal{H}_9,$$

then

$$P_9v = v,$$

and hence

$$P_9U(g)v = U(g)P_9v = U(g)v.$$

Thus

$$U(g)v \in \mathcal{H}_9.$$

The restriction $U_9(g)$ is therefore well defined. Since $U(g)$ is unitary on V_9 ,

$$\langle U_9(g)v, U_9(g)w \rangle = \langle v, w \rangle \quad (v, w \in \mathcal{H}_9),$$

so $U_9(g)$ is unitary on $\mathcal{H}_9$.

Finally, for $g, h \in G$,

$$\begin{aligned} U_9(gh) &= U(gh)|_{\mathcal{H}_9} \\ &= U(g)U(h)|_{\mathcal{H}_9} \\ &= U_9(g)U_9(h). \end{aligned}$$

Hence U_9 is a unitary representation of G on $\mathcal{H}_9$. □

Theorem 399.2. *The stable sector is dynamically closed: if*

$$v \in \mathcal{H}_9,$$

then

$$U(g)v \in \mathcal{H}_9 \quad (g \in G),$$

and the induced evolution is completely determined by

$$U_9(g).$$

In particular, no component in $\mathcal{H}_9^\perp$ is generated by the action of G on a stable state.

Proof. The invariance relation established above gives

$$P_9U(g)v = U(g)v$$

for every

$$v \in \mathcal{H}_9.$$

Therefore

$$(I - P_9)U(g)v = 0,$$

which is equivalent to

$$U(g)v \in \text{range}(P_9) = \mathcal{H}_9.$$

The action on the stable sector is consequently exactly the restriction

$$U_9(g) = U(g)|_{\mathcal{H}_9}.$$

□

Proposition 399.3. *The complementary subspace*

$$\mathcal{H}_9^\perp$$

is also invariant under $U(g)$.

Proof. Let

$$w \in \mathcal{H}_9^\perp$$

and

$$v \in \mathcal{H}_9.$$

Using unitarity,

$$\langle U(g)w, v \rangle = \langle w, U(g)^* v \rangle.$$

Since

$$U(g)^* = U(g^{-1})$$

and $\mathcal{H}_9$ is invariant under the representation,

$$U(g)^* v \in \mathcal{H}_9.$$

Because

$$w \in \mathcal{H}_9^\perp,$$

it follows that

$$\langle w, U(g)^* v \rangle = 0.$$

Hence

$$\langle U(g)w, v \rangle = 0$$

for every $v \in \mathcal{H}_9$, and therefore

$$U(g)w \in \mathcal{H}_9^\perp.$$

□

Theorem 399.4. *Relative to the orthogonal decomposition*

$$V_9 = \mathcal{H}_9 \oplus \mathcal{H}_9^\perp,$$

every operator $U(g)$ has block-diagonal form

$$U(g) = \begin{pmatrix} U_9(g) & 0 \\ 0 & U_\perp(g) \end{pmatrix},$$

where

$$U_9(g) \in \mathcal{U}(\mathcal{H}_9)$$

and

$$U_\perp(g) \in \mathcal{U}(\mathcal{H}_9^\perp).$$

Proof. The preceding two propositions show that both $\mathcal{H}_9$ and $\mathcal{H}_9^\perp$ are invariant under $U(g)$. Therefore the off-diagonal blocks vanish with respect to their orthogonal direct sum. The diagonal restrictions are unitary because $U(g)$ is unitary on V_9 . □

Corollary 399.5. *The stable projection is conserved by the induced dynamics:*

$$U(g)P_9U(g)^* = P_9 \quad (g \in G).$$

Equivalently,

$$[U(g), P_9] = 0.$$

Proof. The assumed commutation relation gives

$$U(g)P_9 = P_9U(g).$$

Multiplication on the right by $U(g)^*$ yields

$$U(g)P_9U(g)^* = P_9.$$

□

400 Dynamical Invariance of the Canonical Stable State

Let

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}.$$

Theorem 400.1. *The canonical stable state is invariant under the entire induced dynamics:*

$$U_9(g)\rho_9U_9(g)^* = \rho_9 \quad (g \in G).$$

Proof. Since

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9},$$

one has

$$\begin{aligned} U_9(g)\rho_9U_9(g)^* &= \frac{1}{9}U_9(g)I_{\mathcal{H}_9}U_9(g)^* \\ &= \frac{1}{9}U_9(g)U_9(g)^* \\ &= \frac{1}{9}I_{\mathcal{H}_9} \\ &= \rho_9. \end{aligned}$$

□

Corollary 400.2. *For every stable observable*

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

its expectation value in the canonical state is invariant under conjugate evolution:

$$\text{tr}_{\mathcal{H}_9}(\rho_9U_9(g)XU_9(g)^*) = \text{tr}_{\mathcal{H}_9}(\rho_9X).$$

Proof. By cyclicity of the trace,

$$\text{tr}(\rho_9U_9(g)XU_9(g)^*) = \text{tr}(U_9(g)^*\rho_9U_9(g)X).$$

The invariance of ρ_9 implies

$$U_9(g)^*\rho_9U_9(g) = \rho_9.$$

Therefore

$$\text{tr}(\rho_9U_9(g)XU_9(g)^*) = \text{tr}(\rho_9X).$$

□

Proposition 400.3. *If an ambient observable A commutes with P_9 , then its stable compression*

$$A_9 = A|_{\mathcal{H}_9}$$

transforms covariantly under the induced dynamics:

$$U_9(g) A_9 U_9(g)^* = (U(g) A U(g)^*)|_{\mathcal{H}_9}.$$

Proof. For

$$v \in \mathcal{H}_9,$$

one has

$$U_9(g) A_9 U_9(g)^* v = U(g) A U(g)^* v.$$

Indeed,

$$U_9(g)^* v = U(g)^* v$$

belongs to $\mathcal{H}_9$, because the stable sector is invariant under $U(g)^*$. Since A commutes with P_9 , it also preserves $\mathcal{H}_9$. Thus the right-hand side belongs to $\mathcal{H}_9$ and equals the restriction of

$$U(g) A U(g)^*$$

to the stable sector. □

401 Spectral Structure of Stable Observables

Let

$$A = A^* \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be a self-adjoint stable observable.

Theorem 401.1. *There exist distinct real numbers*

$$\lambda_1, \dots, \lambda_q$$

and mutually orthogonal projections

$$E_1, \dots, E_q \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

such that

$$A = \sum_{k=1}^q \lambda_k E_k,$$

with

$$E_k E_\ell = \delta_{k\ell} E_k, \quad \sum_{k=1}^q E_k = I_{\mathcal{H}_9}.$$

The projections E_k are uniquely determined by A .

Proof. Since $\mathcal{H}_9$ is a finite-dimensional complex Hilbert space and A is self-adjoint, the spectral theorem provides an orthonormal basis of eigenvectors of A . Let

$$\lambda_1, \dots, \lambda_q$$

be the distinct eigenvalues of A , and let

$$\mathcal{E}_k = \ker(A - \lambda_k I_{\mathcal{H}_9})$$

be the corresponding eigenspaces.

Self-adjointness implies that eigenspaces corresponding to distinct eigenvalues are orthogonal. Hence

$$\mathcal{H}_g = \mathcal{E}_1 \oplus \cdots \oplus \mathcal{E}_q.$$

Let E_k be the orthogonal projection onto $\mathcal{E}_k$. Then

$$E_k E_\ell = \delta_{k\ell} E_k$$

and

$$\sum_{k=1}^q E_k = I_{\mathcal{H}_g}.$$

Every vector in $\mathcal{E}_k$ is an eigenvector with eigenvalue λ_k , so

$$A = \sum_{k=1}^q \lambda_k E_k.$$

The spectral projections are uniquely determined by the eigenspaces of A . □

Proposition 401.2. *For every self-adjoint stable observable A ,*

$$\tau_g(A) = \frac{1}{9} \sum_{k=1}^q \lambda_k \text{rank}(E_k).$$

Proof. By the spectral decomposition,

$$A = \sum_{k=1}^q \lambda_k E_k.$$

Therefore

$$\begin{aligned} \tau_g(A) &= \frac{1}{9} \text{tr}_{\mathcal{H}_g}(A) \\ &= \frac{1}{9} \sum_{k=1}^q \lambda_k \text{tr}_{\mathcal{H}_g}(E_k). \end{aligned}$$

For an orthogonal projection,

$$\text{tr}(E_k) = \text{rank}(E_k),$$

which gives

$$\tau_g(A) = \frac{1}{9} \sum_{k=1}^q \lambda_k \text{rank}(E_k).$$

□

Theorem 401.3. *The canonical stable state assigns to the spectral event E_k the probability*

$$p_k = \text{tr}_{\mathcal{H}_g}(\rho_g E_k) = \frac{\text{rank}(E_k)}{9}.$$

Consequently,

$$p_k \geq 0$$

and

$$\sum_{k=1}^q p_k = 1.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one obtains

$$\mathrm{tr}(\rho_9 E_k) = \frac{1}{9} \mathrm{tr}(E_k) = \frac{\mathrm{rank}(E_k)}{9}.$$

The nonnegativity follows from the fact that E_k is a positive operator. Moreover,

$$\begin{aligned} \sum_{k=1}^q p_k &= \frac{1}{9} \sum_{k=1}^q \mathrm{rank}(E_k) \\ &= \frac{1}{9} \dim_{\mathbb{C}} \mathcal{H}_9 \\ &= 1. \end{aligned}$$

□

402 Functional Calculus and Stable Expectations

Let

$$A = A^*$$

have spectral decomposition

$$A = \sum_{k=1}^q \lambda_k E_k.$$

For every function

$$f : \{\lambda_1, \dots, \lambda_q\} \longrightarrow \mathbb{C},$$

define

$$f(A) = \sum_{k=1}^q f(\lambda_k) E_k.$$

Proposition 402.1. *The functional calculus satisfies*

$$f(A)g(A) = (fg)(A)$$

and

$$f(A)^* = \overline{f}(A),$$

where

$$\overline{f}(\lambda) = \overline{f(\lambda)}.$$

Proof. Using

$$E_k E_\ell = \delta_{k\ell} E_k,$$

one has

$$\begin{aligned} f(A)g(A) &= \left(\sum_{k=1}^q f(\lambda_k) E_k \right) \left(\sum_{\ell=1}^q g(\lambda_\ell) E_\ell \right) \\ &= \sum_{k=1}^q f(\lambda_k) g(\lambda_k) E_k \\ &= (fg)(A). \end{aligned}$$

Furthermore,

$$\begin{aligned} f(A)^* &= \left(\sum_{k=1}^q f(\lambda_k) E_k \right)^* \\ &= \sum_{k=1}^q \overline{f(\lambda_k)} E_k \\ &= \overline{f}(A), \end{aligned}$$

because each E_k is self-adjoint. □

Theorem 402.2. *For every function*

$$f : \{\lambda_1, \dots, \lambda_q\} \longrightarrow \mathbb{C},$$

the canonical stable expectation is

$$\tau_9(f(A)) = \frac{1}{9} \sum_{k=1}^q f(\lambda_k) \operatorname{rank}(E_k).$$

Equivalently,

$$\tau_9(f(A)) = \sum_{k=1}^q p_k f(\lambda_k), \quad p_k = \frac{\operatorname{rank}(E_k)}{9}.$$

Proof. By definition of the functional calculus,

$$f(A) = \sum_{k=1}^q f(\lambda_k) E_k.$$

Hence

$$\begin{aligned} \tau_9(f(A)) &= \frac{1}{9} \operatorname{tr} \left(\sum_{k=1}^q f(\lambda_k) E_k \right) \\ &= \frac{1}{9} \sum_{k=1}^q f(\lambda_k) \operatorname{tr}(E_k) \\ &= \frac{1}{9} \sum_{k=1}^q f(\lambda_k) \operatorname{rank}(E_k). \end{aligned}$$

Using

$$p_k = \frac{\operatorname{rank}(E_k)}{9}$$

gives the second expression. □

Corollary 402.3. *For every real-valued function*

$$f : \{\lambda_1, \dots, \lambda_q\} \longrightarrow \mathbb{R},$$

the operator $f(A)$ is self-adjoint and its stable expectation is real:

$$\tau_9(f(A)) \in \mathbb{R}.$$

Proof. If f is real-valued, then

$$\overline{f} = f,$$

and therefore

$$f(A)^* = f(A).$$

The expectation is consequently real. □

403 Invariant Observables and the Representation Commutant

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

and define the invariant self-adjoint algebra

$$\mathcal{A}_9^G = \{A \in \mathcal{A}_9 : A = A^*, \quad A\pi_9(g) = \pi_9(g)A \text{ for all } g \in G\}.$$

Proposition 403.1. *If*

$$A \in \mathcal{A}_9^G,$$

then every spectral projection of A commutes with $\pi_9(G)$.

Proof. Let

$$A = \sum_{k=1}^q \lambda_k E_k$$

be the spectral decomposition of A . Since A commutes with $\pi_9(g)$, every eigenspace of A is invariant under $\pi_9(g)$. Indeed, if

$$v \in E_k \mathcal{H}_9,$$

then

$$Av = \lambda_k v,$$

and therefore

$$A\pi_9(g)v = \pi_9(g)Av = \lambda_k \pi_9(g)v.$$

Thus

$$\pi_9(g)v \in E_k \mathcal{H}_9.$$

Because $\pi_9(g)$ is unitary, the orthogonal complement of each eigenspace is invariant as well. Hence the orthogonal projection onto each eigenspace commutes with $\pi_9(g)$, giving

$$E_k \pi_9(g) = \pi_9(g) E_k.$$

□

Theorem 403.2. *If the stable representation is irreducible, then every self-adjoint stable observable invariant under G is scalar:*

$$A \in \mathcal{A}_9^G \implies A = \lambda I_{\mathcal{H}_9}$$

for some

$$\lambda \in \mathbb{R}.$$

Proof. Since A commutes with the representation and the stable representation is irreducible, Schur's lemma gives

$$A = \lambda I_{\mathcal{H}_9}$$

for some

$$\lambda \in \mathbb{C}.$$

Because A is self-adjoint,

$$A^* = A,$$

and hence

$$\bar{\lambda} = \lambda.$$

Therefore

$$\lambda \in \mathbb{R}.$$

□

Corollary 403.3. *If the stable representation is irreducible, the only G -invariant spectral event of a nontrivial self-adjoint observable is the trivial event 0 or the whole stable sector $I_{\mathcal{H}_g}$.*

Proof. By the theorem,

$$A = \lambda I_{\mathcal{H}_g}.$$

A scalar operator has only one spectral value when λ is fixed, and its corresponding spectral projection is

$$I_{\mathcal{H}_g}.$$

The complementary spectral projection is

$$0.$$

Thus no nontrivial spectral decomposition invariant under the full representation can arise in the irreducible case. $\square$

404 The Commutant Criterion for Stable State Uniqueness

Let

$$\mathcal{D}(\mathcal{H}_g) = \{\rho \in \text{End}_{\mathbb{C}}(\mathcal{H}_g) : \rho \geq 0, \text{tr}(\rho) = 1\}$$

denote the state space of the stable sector, and let

$$\mathcal{D}(\mathcal{H}_g)^G = \{\rho \in \mathcal{D}(\mathcal{H}_g) : \pi_g(g)\rho\pi_g(g)^* = \rho \text{ for all } g \in G\}.$$

Proposition 404.1. *A density operator $\rho \in \mathcal{D}(\mathcal{H}_g)$ is G -invariant if and only if*

$$\rho \in \pi_g(G)'.$$

Proof. Suppose first that

$$\pi_g(g)\rho\pi_g(g)^* = \rho$$

for every $g \in G$. Multiplying on the right by $\pi_g(g)$ gives

$$\pi_g(g)\rho = \rho\pi_g(g).$$

Hence

$$\rho \in \pi_g(G)'.$$

Conversely, if

$$\rho \in \pi_g(G)',$$

then

$$\pi_g(g)\rho = \rho\pi_g(g),$$

and therefore

$$\pi_g(g)\rho\pi_g(g)^* = \rho\pi_g(g)\pi_g(g)^* = \rho.$$

Thus ρ is G -invariant. $\square$

Theorem 404.2. *The following statements are equivalent:*

- (i) $\mathcal{D}(\mathcal{H}_g)^G = \left\{ \frac{1}{9} I_{\mathcal{H}_g} \right\};$
- (ii) $\pi_g(G)' = \mathbb{C} I_{\mathcal{H}_g};$
- (iii) every self-adjoint element of $\pi_g(G)'$ is scalar.

Proof. The equivalence of (ii) and (iii) follows from the fact that every operator in a finite-dimensional complex $*$ -algebra is a linear combination of self-adjoint operators.

Assume (ii). Let

$$\rho \in \mathcal{D}(\mathcal{H}_9)^G.$$

By the preceding proposition,

$$\rho \in \pi_9(G)'$$

Hence

$$\rho = \lambda I_{\mathcal{H}_9}$$

for some

$$\lambda \in \mathbb{R}.$$

The normalization condition gives

$$1 = \text{tr}(\rho) = 9\lambda,$$

and therefore

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

Thus (i) holds.

Conversely, assume (i). Suppose that

$$\pi_9(G)' \neq \mathbb{C} I_{\mathcal{H}_9}.$$

Then there exists a non-scalar self-adjoint operator

$$A \in \pi_9(G)'.$$

Subtracting its normalized trace,

$$B = A - \tau_9(A) I_{\mathcal{H}_9},$$

we obtain

$$B = B^*, \quad B \in \pi_9(G)', \quad \tau_9(B) = 0.$$

Moreover,

$$B \neq 0.$$

Let

$$\|B\|_{\text{op}}$$

denote the operator norm. Choose

$$0 < \varepsilon < \frac{1}{9\|B\|_{\text{op}}}.$$

Since B is self-adjoint,

$$-\|B\|_{\text{op}} I_{\mathcal{H}_9} \leq B \leq \|B\|_{\text{op}} I_{\mathcal{H}_9}.$$

Consequently,

$$\frac{1}{9} I_{\mathcal{H}_9} + \varepsilon B \geq \left(\frac{1}{9} - \varepsilon \|B\|_{\text{op}} \right) I_{\mathcal{H}_9} > 0.$$

Define

$$\rho_\varepsilon = \frac{1}{9} I_{\mathcal{H}_9} + \varepsilon B.$$

Then

$$\rho_\varepsilon \geq 0$$

and

$$\begin{aligned}\mathrm{tr}(\rho_\varepsilon) &= 1 + \varepsilon \mathrm{tr}(B) \\ &= 1 + 9\varepsilon \tau_9(B) \\ &= 1.\end{aligned}$$

Since B belongs to the commutant,

$$\pi_9(g)B\pi_9(g)^* = B,$$

and hence

$$\pi_9(g)\rho_\varepsilon\pi_9(g)^* = \rho_\varepsilon.$$

Thus

$$\rho_\varepsilon \in \mathcal{D}(\mathcal{H}_9)^G.$$

But

$$\rho_\varepsilon \neq \frac{1}{9}I_{\mathcal{H}_9}$$

because $B \neq 0$, contradicting (i). Therefore

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

□

405 Isotypic Obstruction to Uniqueness

Retain the isotypic decomposition

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^r (\mathbb{C}^{m_\alpha} \otimes \mathcal{K}_\alpha),$$

where the $\mathcal{K}_\alpha$ are pairwise inequivalent irreducible representations.

Theorem 405.1. *If either*

$$r > 1$$

or

$$m_\alpha > 1$$

for some α , then

$$\pi_9(G)' \neq \mathbb{C}I_{\mathcal{H}_9}.$$

Consequently, the maximally mixed state

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

is not the unique G -invariant density operator.

Proof. If

$$r > 1,$$

then one of the isotypic projections

$$Q_\alpha$$

is nonzero and different from the identity. By construction,

$$Q_\alpha \pi_9(g) = \pi_9(g) Q_\alpha$$

for every $g \in G$. Hence

$$Q_\alpha \in \pi_9(G)'$$

and Q_α is non-scalar.

Now suppose instead that

$$m_\alpha > 1$$

for some α . On the corresponding isotypic component,

$$\mathcal{H}_\alpha \cong \mathbb{C}^{m_\alpha} \otimes \mathcal{K}_\alpha,$$

the commutant contains

$$M_{m_\alpha}(\mathbb{C}) \otimes I_{\mathcal{K}_\alpha}.$$

Since

$$m_\alpha > 1,$$

this algebra contains non-scalar self-adjoint operators. Hence again

$$\pi_9(G)' \neq \mathbb{C}I_{\mathcal{H}_9}.$$

The preceding commutant criterion now implies that

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}$$

cannot be the unique G -invariant density operator. $\square$

Corollary 405.2. *The canonical stable state is uniquely determined by G -invariance if and only if the stable representation is irreducible.*

Proof. If the stable representation is irreducible, Schur's lemma gives

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9},$$

and uniqueness follows from the commutant criterion.

Conversely, if the representation is reducible, complete reducibility gives at least two nonzero orthogonal invariant sectors. Their orthogonal projections are non-scalar elements of the commutant. Hence the commutant is strictly larger than the scalars, and uniqueness fails. $\square$

406 Quantitative Form of the Invariant-State Simplex

For the isotypic decomposition, write

$$d_\alpha = \dim_{\mathbb{C}} \mathcal{K}_\alpha$$

and

$$n_\alpha = m_\alpha d_\alpha.$$

Theorem 406.1. *Every G -invariant density operator on $\mathcal{H}_9$ has the form*

$$\rho = \bigoplus_{\alpha=1}^r \left(\sigma_\alpha \otimes \frac{1}{d_\alpha} I_{\mathcal{K}_\alpha} \right),$$

where

$$\sigma_\alpha \geq 0$$

acts on $\mathbb{C}^{m_\alpha}$ and satisfies

$$\sum_{\alpha=1}^r \text{tr}(\sigma_\alpha) = 1.$$

Proof. By invariance,

$$\rho \in \pi_{\mathfrak{g}}(G)'.$$

The commutant decomposition gives

$$\rho = \bigoplus_{\alpha=1}^r (A_{\alpha} \otimes I_{\mathcal{K}_{\alpha}})$$

for operators

$$A_{\alpha} \in M_{m_{\alpha}}(\mathbb{C}).$$

Since ρ is positive, each A_{α} is positive.

Set

$$\sigma_{\alpha} = d_{\alpha} A_{\alpha}.$$

Then

$$\rho = \bigoplus_{\alpha=1}^r \left(\sigma_{\alpha} \otimes \frac{1}{d_{\alpha}} I_{\mathcal{K}_{\alpha}} \right).$$

Furthermore,

$$\begin{aligned} 1 &= \text{tr}(\rho) \\ &= \sum_{\alpha=1}^r \text{tr}(\sigma_{\alpha}) \text{tr} \left(\frac{1}{d_{\alpha}} I_{\mathcal{K}_{\alpha}} \right) \\ &= \sum_{\alpha=1}^r \text{tr}(\sigma_{\alpha}). \end{aligned}$$

Thus the stated normalization condition holds. □

Corollary 406.2. *The invariant-state space is a singleton precisely when*

$$r = 1 \quad \text{and} \quad m_1 = 1.$$

Equivalently,

$$\mathcal{H}_{\mathfrak{g}}$$

must carry a single irreducible representation with multiplicity one.

Proof. If

$$r = 1 \quad \text{and} \quad m_1 = 1,$$

then

$$\sigma_1$$

is a positive scalar of trace one, so

$$\sigma_1 = 1.$$

The formula for ρ becomes

$$\rho = \frac{1}{d_1} I_{\mathcal{H}_{\mathfrak{g}}}.$$

Since

$$d_1 = \dim_{\mathbb{C}} \mathcal{H}_{\mathfrak{g}} = 9,$$

one obtains

$$\rho = \frac{1}{9} I_{\mathcal{H}_{\mathfrak{g}}}.$$

If either $r > 1$ or some $m_{\alpha} > 1$, the preceding theorem provides nontrivial degrees of freedom in the invariant density operators, so the invariant-state space is not a singleton. □

407 The Ambient–Stable Trace Interface

Let

$$\iota_9 : \mathcal{H}_9 \hookrightarrow V_9$$

denote the canonical isometric inclusion, and let

$$P_9 : V_9 \longrightarrow V_9$$

be the orthogonal projection onto $\mathcal{H}_9$. For

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

define its extension by zero to V_9 by

$$\tilde{X} = \iota_9 X \iota_9^*.$$

Thus

$$\tilde{X} = P_9 \tilde{X} P_9.$$

Proposition 407.1. *The map*

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9) \longrightarrow P_9 \text{End}_{\mathbb{C}}(V_9) P_9, \quad X \longmapsto \tilde{X},$$

is a unital algebra isomorphism, where the identity of the target corner algebra is P_9 .

Proof. For

$$X, Y \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

one has

$$\tilde{X} \tilde{Y} = \iota_9 X \iota_9^* \iota_9 Y \iota_9^* = \iota_9 X Y \iota_9^* = \widetilde{XY}.$$

Similarly,

$$\widetilde{X+Y} = \tilde{X} + \tilde{Y}, \quad \widetilde{\lambda X} = \lambda \tilde{X}.$$

Moreover,

$$\widetilde{I_{\mathcal{H}_9}} = \iota_9 \iota_9^* = P_9.$$

Every element A of the corner algebra

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9$$

vanishes on $\mathcal{H}_9^\perp$ and preserves $\mathcal{H}_9$, hence is uniquely determined by its restriction to $\mathcal{H}_9$. Therefore the map is bijective. $\square$

Theorem 407.2. *For every*

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the intrinsic trace is recovered from the ambient trace by

$$\text{tr}_{\mathcal{H}_9}(X) = \text{tr}_{V_9}(\tilde{X}).$$

Consequently,

$$\tau_9(X) = \frac{1}{9} \text{tr}_{V_9}(\tilde{X}).$$

Proof. Choose an orthonormal basis

$$e_1, \dots, e_9$$

of $\mathcal{H}_9$, and extend it to an orthonormal basis

$$e_1, \dots, e_9, e_{10}, \dots, e_{143}$$

of V_9 . Since $\tilde{X}$ acts as X on $\mathcal{H}_9$ and vanishes on $\mathcal{H}_9^\perp$,

$$\mathrm{tr}_{V_9}(\tilde{X}) = \sum_{j=1}^9 \langle e_j, \tilde{X} e_j \rangle + \sum_{j=10}^{143} \langle e_j, \tilde{X} e_j \rangle = \sum_{j=1}^9 \langle e_j, X e_j \rangle.$$

The last expression is

$$\mathrm{tr}_{\mathcal{H}_9}(X).$$

Dividing by 9 gives the formula for τ_9 . □

Proposition 407.3. *The ambient normalized trace*

$$\tau_{V_9}(A) = \frac{1}{143} \mathrm{tr}_{V_9}(A)$$

does not restrict to the normalized stable trace τ_9 . More precisely, for

$$X \in \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9),$$

one has

$$\tau_{V_9}(\tilde{X}) = \frac{9}{143} \tau_9(X).$$

Proof. Using the preceding theorem,

$$\tau_{V_9}(\tilde{X}) = \frac{1}{143} \mathrm{tr}_{V_9}(\tilde{X}) = \frac{1}{143} \mathrm{tr}_{\mathcal{H}_9}(X).$$

Since

$$\tau_9(X) = \frac{1}{9} \mathrm{tr}_{\mathcal{H}_9}(X),$$

the claimed relation follows:

$$\tau_{V_9}(\tilde{X}) = \frac{9}{143} \tau_9(X).$$

□

Theorem 407.4. *The normalized stable trace is characterized intrinsically on the corner algebra by*

$$\tau_9(A) = \frac{\mathrm{tr}_{V_9}(A)}{\mathrm{tr}_{V_9}(P_9)} \quad (A \in P_9 \mathrm{End}_{\mathbb{C}}(V_9) P_9).$$

Proof. Since

$$\mathrm{rank}(P_9) = 9,$$

one has

$$\mathrm{tr}_{V_9}(P_9) = 9.$$

For

$$A = \tilde{X}$$

with

$$X \in \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the preceding trace identity gives

$$\frac{\operatorname{tr}_{V_9}(A)}{\operatorname{tr}_{V_9}(P_9)} = \frac{\operatorname{tr}_{\mathcal{H}_9}(X)}{9} = \tau_9(X).$$

Under the algebra isomorphism established above, this is precisely the normalized trace on the stable corner. $\square$

Corollary 407.5. *The numerical normalization of the stable trace depends only on the rank of the stable projection and not on the ambient dimension 143. In particular,*

$$\tau_9(P_9) = 1$$

when P_9 is regarded as the identity of the corner algebra

$$P_9 \operatorname{End}_{\mathbb{C}}(V_9) P_9.$$

Proof. The corner identity is P_9 , and

$$\tau_9(P_9) = \frac{\operatorname{tr}_{V_9}(P_9)}{\operatorname{tr}_{V_9}(P_9)} = 1.$$

$\square$

408 Compression and Faithfulness of the Stable Algebra

Define the compression map

$$\mathfrak{C}_9 : \operatorname{End}_{\mathbb{C}}(V_9) \longrightarrow \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9)$$

by

$$\mathfrak{C}_9(A) = P_9 A P_9|_{\mathcal{H}_9}.$$

Proposition 408.1. *The compression map $\mathfrak{C}_9$ is completely positive and unital with respect to the corner identity:*

$$\mathfrak{C}_9(P_9) = I_{\mathcal{H}_9}.$$

Proof. For

$$A \geq 0,$$

one has

$$P_9 A P_9 \geq 0,$$

because for every

$$v \in \mathcal{H}_9,$$

$$\langle v, P_9 A P_9 v \rangle = \langle v, A v \rangle \geq 0.$$

The same argument applied entrywise to matrices over $\operatorname{End}_{\mathbb{C}}(V_9)$ proves complete positivity. Finally,

$$\mathfrak{C}_9(P_9) = P_9|_{\mathcal{H}_9} = I_{\mathcal{H}_9}.$$

$\square$

Theorem 408.2. *For every positive operator*

$$A \in \operatorname{End}_{\mathbb{C}}(V_9),$$

the stable trace of its compression satisfies

$$\tau_9(\mathfrak{C}_9(A)) = \frac{1}{9} \operatorname{tr}_{V_9}(A P_9).$$

Proof. By definition,

$$\tau_9(\mathfrak{C}_9(A)) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(P_9 A P_9|_{\mathcal{H}_9}).$$

The operator

$$P_9 A P_9$$

is supported on $\mathcal{H}_9$, so its trace computed in $\mathcal{H}_9$ agrees with its ambient trace:

$$\operatorname{tr}_{\mathcal{H}_9}(P_9 A P_9|_{\mathcal{H}_9}) = \operatorname{tr}_{V_9}(P_9 A P_9).$$

By cyclicity,

$$\operatorname{tr}_{V_9}(P_9 A P_9) = \operatorname{tr}_{V_9}(A P_9^2).$$

Since $P_9^2 = P_9$,

$$\operatorname{tr}_{V_9}(P_9 A P_9) = \operatorname{tr}_{V_9}(A P_9).$$

Therefore

$$\tau_9(\mathfrak{C}_9(A)) = \frac{1}{9} \operatorname{tr}_{V_9}(A P_9).$$

□

Corollary 408.3. *If*

$$A \geq 0$$

and

$$P_9 A P_9 = 0,$$

then

$$\mathfrak{C}_9(A) = 0$$

and

$$\tau_9(\mathfrak{C}_9(A)) = 0.$$

Proof. The first assertion is the definition of the compression map, and the second follows from the preceding theorem. □

409 The Stable State as the Canonical Trace State

Let

$$\mathcal{A}_9 = P_9 \operatorname{End}_{\mathbb{C}}(V_9) P_9 \cong \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be the stable corner algebra, with identity P_9 . Define

$$\omega_9 : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

by

$$\omega_9(A) = \frac{1}{9} \operatorname{tr}_{V_9}(A).$$

Proposition 409.1. *The functional ω_9 is a faithful tracial state on $\mathcal{A}_9$.*

Proof. Since

$$\operatorname{tr}_{V_9}(P_9) = 9,$$

one has

$$\omega_9(P_9) = 1.$$

If

$$A \in \mathcal{A}_9,$$

then A is supported on $\mathcal{H}_9$, and therefore

$$A^* A \geq 0.$$

Hence

$$\omega_9(A^* A) = \frac{1}{9} \operatorname{tr}_{V_9}(A^* A) \geq 0.$$

If

$$\omega_9(A^* A) = 0,$$

then

$$\operatorname{tr}_{V_9}(A^* A) = 0.$$

The positive operator $A^* A$ consequently has all eigenvalues equal to zero, so

$$A^* A = 0$$

and hence

$$A = 0.$$

Thus ω_9 is faithful.

Finally, for

$$A, B \in \mathcal{A}_9,$$

cyclicity of the finite-dimensional trace gives

$$\omega_9(AB) = \frac{1}{9} \operatorname{tr}_{V_9}(AB) = \frac{1}{9} \operatorname{tr}_{V_9}(BA) = \omega_9(BA).$$

Therefore ω_9 is tracial. □

Theorem 409.2. *Under the canonical identification*

$$\mathcal{A}_9 \cong \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the state ω_9 coincides with the density operator

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

More precisely,

$$\omega_9(A) = \operatorname{tr}_{\mathcal{H}_9}(\rho_9 A)$$

for every

$$A \in \mathcal{A}_9.$$

Proof. For $A \in \mathcal{A}_9$, viewed as an operator on $\mathcal{H}_9$,

$$\omega_9(A) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(A).$$

On the other hand,

$$\operatorname{tr}_{\mathcal{H}_9}(\rho_9 A) = \operatorname{tr}_{\mathcal{H}_9}\left(\frac{1}{9} I_{\mathcal{H}_9} A\right) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(A).$$

The two expressions coincide. □

Corollary 409.3. *For every projection*

$$Q \in \mathcal{A}_9,$$

its stable probability is

$$\omega_9(Q) = \frac{\text{rank}(Q)}{9}.$$

Proof. A finite-dimensional projection has trace equal to its rank. Therefore

$$\omega_9(Q) = \frac{1}{9} \text{tr}_{V_9}(Q) = \frac{\text{rank}(Q)}{9}.$$

□

Theorem 409.4. *Let*

$$Q_1, \dots, Q_r \in \mathcal{A}_9$$

be mutually orthogonal projections satisfying

$$Q_\alpha Q_\beta = 0 \quad (\alpha \neq \beta),$$

and

$$\sum_{\alpha=1}^r Q_\alpha = P_9.$$

Then

$$\sum_{\alpha=1}^r \omega_9(Q_\alpha) = 1$$

and

$$\omega_9(Q_\alpha) = \frac{\text{rank}(Q_\alpha)}{9}.$$

Proof. By linearity,

$$\sum_{\alpha=1}^r \omega_9(Q_\alpha) = \omega_9\left(\sum_{\alpha=1}^r Q_\alpha\right) = \omega_9(P_9) = 1.$$

The individual formula follows from the preceding corollary. □

410 Uniqueness of the Tracial State on the Stable Full Matrix Algebra

Theorem 410.1. *The stable algebra*

$$\mathcal{A}_9 \cong M_9(\mathbb{C})$$

admits a unique normalized tracial state. It is ω_9 .

Proof. Let

$$\tau : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a tracial state. Choose matrix units

$$E_{ij}, \quad 1 \leq i, j \leq 9,$$

for an identification

$$\mathcal{A}_9 \cong M_9(\mathbb{C}).$$

For $i \neq j$,

$$E_{ij} = E_{ii}E_{ij},$$

and traciality gives

$$\tau(E_{ij}) = \tau(E_{ii}E_{ij}) = \tau(E_{ij}E_{ii}) = 0.$$

For the diagonal matrix units, if $i \neq j$, then

$$E_{ij}E_{ji} = E_{ii}, \quad E_{ji}E_{ij} = E_{jj}.$$

Hence

$$\tau(E_{ii}) = \tau(E_{ij}E_{ji}) = \tau(E_{ji}E_{ij}) = \tau(E_{jj}).$$

Thus all diagonal matrix units have the same value, say c . Since

$$P_9 = \sum_{i=1}^9 E_{ii}$$

and

$$\tau(P_9) = 1,$$

one obtains

$$1 = \sum_{i=1}^9 \tau(E_{ii}) = 9c.$$

Therefore

$$c = \frac{1}{9}.$$

Writing

$$A = \sum_{i,j=1}^9 a_{ij} E_{ij},$$

we conclude that

$$\tau(A) = \sum_{i=1}^9 a_{ii} \tau(E_{ii}) = \frac{1}{9} \sum_{i=1}^9 a_{ii} = \frac{1}{9} \operatorname{tr}(A) = \omega_9(A).$$

Hence

$$\tau = \omega_9.$$

□

Corollary 410.2. *The state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is not merely G -invariant. It is the unique normalized tracial state of the full stable operator algebra.

Proof. The preceding theorem identifies the unique normalized trace on

$$\operatorname{End}_{\mathbb{C}}(\mathcal{H}_9)$$

with

$$A \longmapsto \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(A).$$

The density operator representing this functional is

$$\frac{1}{9} I_{\mathcal{H}_9}.$$

□

411 Invariant Observables and the Stable State

Let

$$\mathcal{A}_9^G = \{A \in \mathcal{A}_9 : A\pi_9(g) = \pi_9(g)A \text{ for all } g \in G\}$$

denote the invariant subalgebra.

Proposition 411.1. *The restriction*

$$\omega_9|_{\mathcal{A}_9^G}$$

is a faithful tracial state on the invariant algebra.

Proof. Since $\mathcal{A}_9^G$ is a unital $*$ -subalgebra of $\mathcal{A}_9$, the restriction of a tracial state remains tracial and normalized. If

$$A \in \mathcal{A}_9^G$$

satisfies

$$\omega_9(A^* A) = 0,$$

faithfulness of ω_9 on $\mathcal{A}_9$ implies

$$A = 0.$$

Thus the restriction is faithful. □

Theorem 411.2. *If*

$$\mathcal{A}_9^G = \mathbb{C}P_9,$$

then every invariant self-adjoint observable is a scalar multiple of the stable identity, and its stable expectation is exactly that scalar.

Proof. Let

$$A = A^*$$

belong to $\mathcal{A}_9^G$. By hypothesis there exists

$$a \in \mathbb{R}$$

such that

$$A = aP_9.$$

Therefore

$$\omega_9(A) = a\omega_9(P_9) = a.$$

□

Corollary 411.3. *The condition*

$$\mathcal{A}_9^G = \mathbb{C}P_9$$

is equivalent to the irreducibility of the stable representation π_9 .

Proof. By Schur's lemma, irreducibility implies

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9},$$

which, under the identification of $I_{\mathcal{H}_9}$ with P_9 in the stable corner, gives

$$\mathcal{A}_9^G = \mathbb{C}P_9.$$

Conversely, if

$$\mathcal{A}_9^G = \mathbb{C}P_9,$$

then the commutant of the representation contains no nontrivial projection. A proper nonzero invariant subspace would have an orthogonal projection commuting with every $\pi_9(g)$, producing a nontrivial element of $\mathcal{A}_9^G$. Hence no such invariant subspace exists, and π_9 is irreducible. □

412 The Stable Corner as the Intrinsic Observable Algebra

Let

$$\mathcal{A}_9 = P_9 \operatorname{End}_{\mathbb{C}}(V_9) P_9$$

and identify $\mathcal{A}_9$ with

$$\operatorname{End}_{\mathbb{C}}(\mathcal{H}_9)$$

through the restriction to $\mathcal{H}_9$.

The preceding construction separates the ambient finite state space from the stable observable algebra. The latter is determined by the range of the stable projector and does not depend on a choice of basis of V_9 .

Proposition 412.1. *The stable corner is a finite-dimensional unital $*$ -algebra with identity P_9 , and*

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 81.$$

Proof. Since

$$P_9^2 = P_9, \quad P_9^* = P_9,$$

the subspace

$$P_9 \operatorname{End}_{\mathbb{C}}(V_9) P_9$$

is closed under addition, scalar multiplication, multiplication, and adjunction. Its identity is P_9 .

Under the canonical identification

$$\mathcal{A}_9 \cong \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9),$$

and since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

one obtains

$$\dim_{\mathbb{C}} \mathcal{A}_9 = (\dim_{\mathbb{C}} \mathcal{H}_9)^2 = 81.$$

□

Theorem 412.2. *The stable state defines a positive normalized linear functional*

$$\omega_9 : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

given by

$$\omega_9(A) = \frac{1}{9} \operatorname{tr}_{V_9}(A).$$

It satisfies

$$\omega_9(P_9) = 1$$

and

$$\omega_9(A^* A) > 0$$

for every nonzero

$$A \in \mathcal{A}_9.$$

Proof. Normalization follows from

$$\omega_9(P_9) = \frac{1}{9} \operatorname{tr}_{V_9}(P_9) = \frac{1}{9} \operatorname{rank}(P_9) = 1.$$

For

$$A \in \mathcal{A}_9,$$

the operator $A^* A$ is positive. Hence

$$\operatorname{tr}_{V_9}(A^* A) \geq 0.$$

If

$$\omega_9(A^* A) = 0,$$

then

$$\operatorname{tr}_{V_9}(A^* A) = 0.$$

A positive finite-dimensional operator with vanishing trace is zero. Consequently,

$$A^* A = 0,$$

and therefore

$$A = 0.$$

Thus

$$\omega_9(A^* A) > 0$$

whenever $A \neq 0$. □

Theorem 412.3. *For every projection*

$$Q \in \mathcal{A}_9,$$

the stable state assigns to Q the intrinsic dimension ratio

$$\boxed{\omega_9(Q) = \frac{\operatorname{rank}(Q)}{\operatorname{rank}(P_9)} = \frac{\operatorname{rank}(Q)}{9}.}$$

Proof. Since Q is a projection supported on $\mathcal{H}_9$,

$$\operatorname{tr}_{V_9}(Q) = \operatorname{rank}(Q).$$

Therefore

$$\omega_9(Q) = \frac{1}{9} \operatorname{tr}_{V_9}(Q) = \frac{\operatorname{rank}(Q)}{9}.$$

□

Corollary 412.4. *If*

$$Q_1, \dots, Q_k$$

are pairwise orthogonal projections in $\mathcal{A}_9$, then

$$\omega_9\left(\sum_{j=1}^k Q_j\right) = \sum_{j=1}^k \frac{\operatorname{rank}(Q_j)}{9}.$$

In particular, if

$$\sum_{j=1}^k Q_j = P_9,$$

then

$$\sum_{j=1}^k \text{rank}(Q_j) = 9$$

and

$$\sum_{j=1}^k \omega_9(Q_j) = 1.$$

Proof. Linearity of ω_9 gives

$$\omega_9\left(\sum_{j=1}^k Q_j\right) = \sum_{j=1}^k \omega_9(Q_j).$$

Applying the preceding theorem to each projection yields the first identity.

If

$$\sum_{j=1}^k Q_j = P_9,$$

then taking traces gives

$$\sum_{j=1}^k \text{rank}(Q_j) = \text{rank}(P_9) = 9.$$

Normalization of ω_9 then gives

$$\sum_{j=1}^k \omega_9(Q_j) = \omega_9(P_9) = 1.$$

□

413 The Stable Probability Law on Orthogonal Decompositions

Let

$$Q_1, \dots, Q_k$$

be nonzero pairwise orthogonal projections satisfying

$$\sum_{j=1}^k Q_j = P_9.$$

Set

$$d_j = \text{rank}(Q_j).$$

Then

$$d_j \in \mathbb{N}, \quad \sum_{j=1}^k d_j = 9.$$

Theorem 413.1. *The stable state induces on every finite orthogonal decomposition of the stable sector the probability law*

$$p_j = \frac{d_j}{9}.$$

This law depends only on the ranks of the events and not on their particular realization as subspaces of $\mathcal{H}_9$.

Proof. By the projection formula,

$$p_j = \omega_9(Q_j) = \frac{\text{rank}(Q_j)}{9} = \frac{d_j}{9}.$$

If another orthogonal family

$$R_1, \dots, R_k$$

has

$$\text{rank}(R_j) = d_j$$

for every j , then

$$\omega_9(R_j) = \frac{d_j}{9} = \omega_9(Q_j).$$

Thus the probabilities depend only on the ranks. □

Corollary 413.2. *For a complete rank-one resolution*

$$P_9 = P_1 + \dots + P_9, \quad \text{rank}(P_j) = 1,$$

one obtains

$$p_j = \frac{1}{9} \quad (1 \leq j \leq 9).$$

Proof. The theorem gives

$$p_j = \frac{\text{rank}(P_j)}{9} = \frac{1}{9}.$$

□

Proposition 413.3. *The rank law is compatible with refinement. Suppose*

$$Q = Q_1 + \dots + Q_\ell$$

is an orthogonal decomposition into projections in $\mathcal{A}_9$. Then

$$\omega_9(Q) = \sum_{j=1}^{\ell} \omega_9(Q_j).$$

Equivalently,

$$\frac{\text{rank}(Q)}{9} = \sum_{j=1}^{\ell} \frac{\text{rank}(Q_j)}{9}.$$

Proof. Orthogonality gives

$$\text{rank}(Q) = \sum_{j=1}^{\ell} \text{rank}(Q_j).$$

Dividing by 9 yields the stated identity. Equivalently, it follows directly from linearity of the trace state. □

414 The Stable State and Unitary Equivalence of Contexts

Theorem 414.1. *Let*

$$U \in \mathcal{U}(\mathcal{H}_9)$$

and let

$$Q \in \mathcal{A}_9$$

be a projection. Then

$$\omega_9(UQU^*) = \omega_9(Q).$$

Consequently, unitarily equivalent stable events have identical probabilities.

Proof. Using cyclicity of the trace,

$$\begin{aligned} \omega_9(UQU^*) &= \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(UQU^*) \\ &= \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(QU^*U) \\ &= \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(Q) \\ &= \omega_9(Q). \end{aligned}$$

□

Corollary 414.2. *The probability assigned by ρ_9 to a stable event is invariant under all changes of orthonormal coordinates on $\mathcal{H}_9$.*

Proof. A change of orthonormal coordinates is implemented by a unitary operator on $\mathcal{H}_9$. The preceding theorem therefore applies. □

415 Necessary Separation of the Structural and Computational Claims

The preceding results establish properties of the finite-dimensional stable sector once the identities

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9, \quad P_9 = \Psi_9^4, \quad \rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

have been established.

Accordingly, the derivation of these identities from the underlying arithmetic-relational construction is a logically prior part of the theory.

Theorem 415.1. *Assume that the finite arithmetic-relational construction proves*

$$P_9^2 = P_9, \quad P_9^* = P_9, \quad \operatorname{rank}(P_9) = 9,$$

and that the normalized state induced on $\operatorname{range}(P_9)$ is

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Then all probability statements established in the present section follow without any additional dynamical or probabilistic postulate.

Proof. The projection identities imply that

$$\mathcal{H}_9 = \text{range}(P_9)$$

is a nine-dimensional Hilbert space. The assumed form of ρ_9 defines the normalized trace state

$$\omega_9(A) = \text{tr}_{\mathcal{H}_9}(\rho_9 A) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(A).$$

Every projection therefore has probability equal to its rank divided by 9, and every rank-one projection has probability 1/9.

No further probability assignment is introduced in this argument. $\square$

Remark 415.2. The remaining nontrivial task is therefore not to rederive the finite matrix-algebra consequences, but to establish from the preceding arithmetic-relational construction that the stable projector has the stated rank and that the induced state is precisely the normalized trace state. Those implications must be proved from the defining relations of the construction rather than inserted as assumptions.

416 The Stable Algebra and the Compression of the Ambient Representation

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

be the full finite-dimensional operator algebra of the stable sector. Let

$$\iota: \mathcal{A}_9 \longrightarrow \text{End}_{\mathbb{C}}(V_9)$$

denote the extension-by-zero embedding determined by the orthogonal decomposition

$$V_9 = \mathcal{H}_9 \oplus \mathcal{H}_9^{\perp}.$$

Thus

$$\iota(X) = P_9 X P_9,$$

where X is identified with its action on $\mathcal{H}_9$.

Proposition 416.1. *The map*

$$\iota: \mathcal{A}_9 \longrightarrow \text{End}_{\mathbb{C}}(V_9)$$

is an injective unital $$ -homomorphism onto the corner algebra*

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

Proof. For

$$X, Y \in \mathcal{A}_9,$$

the operators $\iota(X)$ and $\iota(Y)$ vanish on $\mathcal{H}_9^{\perp}$. Hence

$$\iota(X)\iota(Y) = P_9 X P_9 Y P_9 = P_9 X Y P_9 = \iota(XY).$$

Likewise,

$$\iota(X^*) = \iota(X)^*.$$

The identity of the corner algebra is P_9 , since

$$P_9 \iota(X) = \iota(X) = \iota(X) P_9.$$

If

$$\iota(X) = 0,$$

then its restriction to $\mathcal{H}_9 = \text{range}(P_9)$ vanishes, and therefore $X = 0$. Thus ι is injective.

Conversely, every element of

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9$$

vanishes on $\mathcal{H}_9^\perp$ and preserves $\mathcal{H}_9$, hence is the extension by zero of a unique element of $\mathcal{A}_9$. Therefore

$$\iota(\mathcal{A}_9) = P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

□

Theorem 416.2. *The stable operator algebra is canonically identified with the 9-dimensional corner of the ambient operator algebra:*

$$\mathcal{A}_9 \cong P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

In particular,

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 81.$$

Proof. The first assertion is the preceding proposition. Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

one has

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9) \cong M_9(\mathbb{C}).$$

Therefore

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 9^2 = 81.$$

□

Proposition 416.3. *The normalized trace state on the stable algebra is given intrinsically by*

$$\tau_9(X) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(X),$$

and, under the corner embedding,

$$\tau_9(X) = \frac{1}{9} \text{tr}_{V_9}(\iota(X)).$$

Proof. By the definition of the extension-by-zero embedding and the orthogonal decomposition

$$V_9 = \mathcal{H}_9 \oplus \mathcal{H}_9^\perp,$$

the matrix of $\iota(X)$ has the form

$$\begin{pmatrix} X & 0 \\ 0 & 0 \end{pmatrix}.$$

Consequently,

$$\text{tr}_{V_9}(\iota(X)) = \text{tr}_{\mathcal{H}_9}(X).$$

Multiplication by $1/9$ gives the stated identity.

□

Theorem 416.4. *The stable state is the unique tracial state on the full stable operator algebra $\mathcal{A}_9$.*

Proof. Under the canonical identification

$$\mathcal{A}_9 \cong M_9(\mathbb{C}),$$

let

$$\varphi : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a tracial state. Choose matrix units

$$E_{ij}, \quad 1 \leq i, j \leq 9.$$

For $i \neq j$, traciality gives

$$\varphi(E_{ij}) = \varphi(E_{ii}E_{ij}) = \varphi(E_{ij}E_{ii}) = 0.$$

Moreover, for all i, j ,

$$\varphi(E_{ii}) = \varphi(E_{ij}E_{ji}) = \varphi(E_{ji}E_{ij}) = \varphi(E_{jj}).$$

Hence all diagonal matrix units have the same value, say c . Since

$$I_{\mathcal{H}_9} = \sum_{i=1}^9 E_{ii}$$

and $\varphi(I_{\mathcal{H}_9}) = 1$,

$$1 = \sum_{i=1}^9 \varphi(E_{ii}) = 9c.$$

Therefore

$$c = \frac{1}{9}.$$

It follows that

$$\varphi(X) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(X) = \tau_9(X)$$

for every

$$X \in \mathcal{A}_9.$$

Thus τ_9 is the unique tracial state on $\mathcal{A}_9$. □

Corollary 416.5. *The normalized trace on the stable sector is not an additional probabilistic assumption: once the stable operator algebra is identified with $M_9(\mathbb{C})$, it is the unique normalized positive functional satisfying the trace identity*

$$\tau_9(XY) = \tau_9(YX).$$

Proof. This is precisely the uniqueness assertion of the preceding theorem. □

417 Characterization of the Stable Probability Functional

Let

$$\mathcal{A}_9 = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9) \cong M_9(\mathbb{C}),$$

and let

$$N_9 : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a normalized positive linear functional satisfying

$$N_9(I_{\mathcal{H}_9}) = 1.$$

Assume in addition that N_9 is invariant under the full unitary group of the stable algebra:

$$N_9(UXU^*) = N_9(X) \quad (X \in \mathcal{A}_9, U \in \mathcal{U}(\mathcal{H}_9)).$$

Theorem 417.1. *Under the preceding hypotheses,*

$$N_9(X) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(X) \quad (X \in \mathcal{A}_9).$$

Thus

$$N_9 = \tau_9.$$

Proof. Let

$$E_{ij} \quad (1 \leq i, j \leq 9)$$

be a system of matrix units relative to an orthonormal basis of $\mathcal{H}_9$.

For every pair i, j , there exists a unitary operator exchanging the corresponding basis vectors. By unitary invariance,

$$N_9(E_{ii}) = N_9(E_{jj})$$

for all i, j . Denote their common value by c . Since

$$I_{\mathcal{H}_9} = \sum_{j=1}^9 E_{jj},$$

normalization gives

$$1 = N_9(I_{\mathcal{H}_9}) = \sum_{j=1}^9 N_9(E_{jj}) = 9c.$$

Hence

$$c = \frac{1}{9}.$$

It remains to determine the off-diagonal matrix units. For $i \neq j$, consider the diagonal unitary

$$U_\theta = I_{\mathcal{H}_9} + (e^{i\theta} - 1)E_{ii}.$$

A direct computation gives

$$U_\theta E_{ij} U_\theta^* = e^{i\theta} E_{ij}.$$

Unitary invariance therefore implies

$$N_9(E_{ij}) = e^{i\theta} N_9(E_{ij})$$

for every

$$\theta \in \mathbb{R}.$$

Choosing θ such that

$$e^{i\theta} \neq 1$$

forces

$$N_9(E_{ij}) = 0.$$

Consequently, if

$$X = \sum_{i,j=1}^9 x_{ij} E_{ij},$$

then

$$N_9(X) = \sum_{i,j=1}^9 x_{ij} N_9(E_{ij}) = \frac{1}{9} \sum_{i=1}^9 x_{ii}.$$

Therefore

$$N_9(X) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(X) = \tau_9(X).$$

□

Corollary 417.2. *For every orthogonal projection*

$$P \in \mathcal{A}_9$$

of rank

$$k,$$

one has

$$N_9(P) = \frac{k}{9}.$$

Proof. Since the trace of a finite-dimensional orthogonal projection equals its rank,

$$\mathrm{tr}_{\mathcal{H}_9}(P) = k.$$

The theorem therefore gives

$$N_9(P) = \frac{1}{9} \mathrm{tr}_{\mathcal{H}_9}(P) = \frac{k}{9}.$$

□

Corollary 417.3. *Every rank-one projection*

$$P_\psi = |\psi\rangle\langle\psi|, \quad \|\psi\| = 1,$$

has probability

$$N_9(P_\psi) = \frac{1}{9}.$$

Proof. A rank-one orthogonal projection has trace equal to 1. Hence

$$N_9(P_\psi) = \frac{1}{9} \mathrm{tr}_{\mathcal{H}_9}(P_\psi) = \frac{1}{9}.$$

□

Theorem 417.4. *The value of the stable probability functional on projections is determined solely by their rank. In particular, if*

$$P, Q \in \mathcal{A}_9$$

are orthogonal projections satisfying

$$\mathrm{rank}(P) = \mathrm{rank}(Q),$$

then

$$N_9(P) = N_9(Q).$$

Proof. Two orthogonal projections on a finite-dimensional Hilbert space have the same rank if and only if their ranges have the same dimension. Hence there exists a unitary operator

$$U \in \mathcal{U}(\mathcal{H}_9)$$

such that

$$Q = UPU^*.$$

By unitary invariance of N_9 ,

$$N_9(Q) = N_9(UPU^*) = N_9(P).$$

□

Proposition 417.5. *Let*

$$P_1, \dots, P_r$$

be pairwise orthogonal projections satisfying

$$\sum_{j=1}^r P_j = I_{\mathcal{H}_9}.$$

Then

$$\sum_{j=1}^r N_9(P_j) = 1,$$

and more precisely

$$\sum_{j=1}^r \frac{\text{rank}(P_j)}{9} = 1.$$

Proof. Linearity of N_9 and normalization give

$$\sum_{j=1}^r N_9(P_j) = N_9\left(\sum_{j=1}^r P_j\right) = N_9(I_{\mathcal{H}_9}) = 1.$$

By the preceding corollary,

$$N_9(P_j) = \frac{\text{rank}(P_j)}{9}.$$

Therefore

$$\sum_{j=1}^r \frac{\text{rank}(P_j)}{9} = 1.$$

Equivalently,

$$\sum_{j=1}^r \text{rank}(P_j) = 9,$$

as follows also directly from the orthogonal decomposition of $\mathcal{H}_9$. □

418 Unitary Invariance and the Stable Born Rule

For a unit vector

$$\psi \in \mathcal{H}_9,$$

define the associated rank-one event by

$$P_\psi = |\psi\rangle\langle\psi|.$$

Theorem 418.1. *Let*

$$U \in \mathcal{U}(\mathcal{H}_9).$$

Then

$$P_{U\psi} = UP_\psi U^*,$$

and consequently

$$N_9(P_{U\psi}) = N_9(P_\psi).$$

Proof. For every

$$v \in \mathcal{H}_9,$$

one has

$$UP_\psi U^* v = U\psi \langle \psi, U^* v \rangle = U\psi \langle U\psi, v \rangle.$$

Hence

$$UP_\psi U^* = |U\psi\rangle \langle U\psi| = P_{U\psi}.$$

The invariance of N_9 then yields

$$N_9(P_{U\psi}) = N_9(UP_\psi U^*) = N_9(P_\psi).$$

□

Corollary 418.2. *The probability of a rank-one stable event is independent of the orthonormal coordinates used to represent its defining vector.*

Proof. Any two unit vectors related by a unitary transformation determine unitarily conjugate rank-one projections. The preceding theorem shows that N_9 assigns the same value to such projections. □

Theorem 418.3. *The stable probability functional satisfies the finite-dimensional Born rule*

$$N_9(P_\psi) = \langle \psi, \rho_9 \psi \rangle$$

with

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Hence

$$N_9(P_\psi) = \frac{1}{9}$$

for every unit vector

$$\psi \in \mathcal{H}_9.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\langle \psi, \rho_9 \psi \rangle = \frac{1}{9} \langle \psi, \psi \rangle = \frac{1}{9}$$

for every normalized vector ψ .

On the other hand, the preceding characterization gives

$$N_9(P_\psi) = \frac{1}{9} \text{tr}(P_\psi) = \frac{1}{9}.$$

Therefore

$$N_9(P_\psi) = \langle \psi, \rho_9 \psi \rangle.$$

□

419 Density-Operator Representation of Stable States

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9) \cong M_9(\mathbb{C}),$$

and let

$$\omega : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be a positive normalized linear functional, that is,

$$\omega(X^* X) \geq 0$$

for every

$$X \in \mathcal{A}_9,$$

and

$$\omega(I_{\mathcal{H}_9}) = 1.$$

Theorem 419.1. *There exists a unique density operator*

$$\rho_\omega \in \mathcal{A}_9$$

such that

$$\rho_\omega \geq 0, \quad \text{tr}_{\mathcal{H}_9}(\rho_\omega) = 1,$$

and

$$\omega(X) = \text{tr}_{\mathcal{H}_9}(\rho_\omega X)$$

for every

$$X \in \mathcal{A}_9.$$

Proof. Fix an orthonormal basis

$$e_1, \dots, e_9$$

of $\mathcal{H}_9$, and let

$$E_{ij} = |e_i\rangle\langle e_j|$$

be the associated matrix units.

Define

$$(\rho_\omega)_{ji} = \omega(E_{ij}).$$

Thus

$$\rho_\omega = \sum_{i,j=1}^9 \omega(E_{ij}) E_{ji}.$$

For arbitrary

$$v = \sum_{i=1}^9 v_i e_i,$$

one obtains

$$\langle v, \rho_\omega v \rangle = \sum_{i,j=1}^9 \overline{v_i} v_j \omega(E_{ji}).$$

Let

$$X_v = \sum_{i=1}^9 v_i E_{i1}.$$

Then

$$X_\nu^* X_\nu = \sum_{i,j=1}^9 \overline{v_i} v_j E_{1i} E_{j1} = \sum_{i,j=1}^9 \overline{v_i} v_j \delta_{ij} E_{11}.$$

More directly, positivity of ω applied to rank-one operators determines the positive semidefinite matrix

$$[\omega(E_{ij})]_{i,j=1}^9.$$

Equivalently, for every

$$z = (z_1, \dots, z_9) \in \mathbb{C}^9,$$

positivity gives

$$\sum_{i,j=1}^9 \overline{z_i} z_j \omega(E_{ji}) \geq 0.$$

Hence

$$\rho_\omega \geq 0.$$

Furthermore,

$$\mathrm{tr}(\rho_\omega) = \sum_{i=1}^9 \omega(E_{ii}) = \omega\left(\sum_{i=1}^9 E_{ii}\right) = \omega(I_{\mathcal{H}_9}) = 1.$$

Finally, for

$$X = \sum_{i,j=1}^9 x_{ij} E_{ij},$$

linearity gives

$$\omega(X) = \sum_{i,j=1}^9 x_{ij} \omega(E_{ij}).$$

On the other hand,

$$\mathrm{tr}(\rho_\omega X) = \sum_{i,j=1}^9 x_{ij} \omega(E_{ij}).$$

Therefore

$$\omega(X) = \mathrm{tr}(\rho_\omega X).$$

For uniqueness, suppose that ρ and σ both represent ω . Then

$$\mathrm{tr}((\rho - \sigma)X) = 0$$

for every

$$X \in \mathcal{A}_9.$$

Taking $X = E_{ji}$ for all i, j shows that every matrix coefficient of $\rho - \sigma$ vanishes. Hence

$$\rho = \sigma.$$

□

Corollary 419.2. *Every normalized positive functional on the stable algebra is uniquely determined by its density operator.*

Proof. This is the uniqueness assertion of the preceding theorem. □

420 Pure and Mixed Stable States

Let

$$\omega_\rho(X) = \text{tr}_{\mathcal{H}_g}(\rho X)$$

be the state associated with a density operator

$$\rho \geq 0, \quad \text{tr}(\rho) = 1.$$

Definition 420.1. The state ω_ρ is called pure if

$$\rho = P_\psi$$

for some unit vector

$$\psi \in \mathcal{H}_g.$$

It is called mixed otherwise.

Proposition 420.2. *If*

$$\rho = P_\psi, \quad \|\psi\| = 1,$$

then

$$\omega_\rho(X) = \langle \psi, X\psi \rangle$$

for every

$$X \in \mathcal{A}_g.$$

Proof. Using the rank-one form of P_ψ ,

$$\begin{aligned} \omega_\rho(X) &= \text{tr}(P_\psi X) \\ &= \text{tr}(|\psi\rangle\langle\psi|X) \\ &= \langle \psi, X\psi \rangle. \end{aligned}$$

□

Theorem 420.3. *A density operator ρ defines a pure state if and only if*

$$\rho^2 = \rho.$$

In that case,

$$\text{rank}(\rho) = 1.$$

Proof. Suppose first that

$$\rho = P_\psi.$$

Since P_ψ is an orthogonal projection,

$$\rho^2 = P_\psi^2 = P_\psi = \rho,$$

and

$$\text{rank}(\rho) = 1.$$

Conversely, suppose

$$\rho^2 = \rho.$$

Because ρ is positive, it is self-adjoint. Its spectrum therefore consists of real nonnegative eigenvalues. The equation

$$\rho^2 = \rho$$

forces every eigenvalue λ to satisfy

$$\lambda^2 = \lambda,$$

hence

$$\lambda \in \{0, 1\}.$$

Since

$$\text{tr}(\rho) = 1,$$

the sum of the eigenvalues equals 1. Consequently exactly one eigenvalue is equal to 1, while all remaining eigenvalues vanish. Thus

$$\text{rank}(\rho) = 1,$$

and therefore

$$\rho = P_\psi$$

for a unit vector ψ . □

421 The Maximally Mixed Stable State

Theorem 421.1. *The state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is mixed and has spectrum

$$\text{spec}(\rho_9) = \left\{ \frac{1}{9}, \dots, \frac{1}{9} \right\},$$

with multiplicity 9.

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

every nonzero vector of $\mathcal{H}_9$ is an eigenvector with eigenvalue $1/9$. Hence

$$\text{spec}(\rho_9) = \left\{ \frac{1}{9} \right\}$$

with multiplicity 9.

Moreover,

$$\rho_9^2 = \frac{1}{81} I_{\mathcal{H}_9} \neq \frac{1}{9} I_{\mathcal{H}_9} = \rho_9.$$

Therefore ρ_9 is not a projection and, by the preceding theorem, the corresponding state is mixed. □

Proposition 421.2. *The purity of the canonical stable state is*

$$\text{tr}(\rho_9^2) = \frac{1}{9}.$$

Proof. One has

$$\rho_9^2 = \frac{1}{81} I_{\mathcal{H}_9}.$$

Therefore

$$\text{tr}(\rho_9^2) = \frac{1}{81} \text{tr}(I_{\mathcal{H}_9}) = \frac{9}{81} = \frac{1}{9}.$$

□

Corollary 421.3. *For every density operator ρ on the nine-dimensional stable sector,*

$$\mathrm{tr}(\rho^2) \geq \frac{1}{9},$$

with equality if and only if

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. Let

$$\lambda_1, \dots, \lambda_9$$

be the eigenvalues of ρ . Positivity and normalization imply

$$\lambda_j \geq 0, \quad \sum_{j=1}^9 \lambda_j = 1.$$

By Cauchy–Schwarz,

$$\left(\sum_{j=1}^9 \lambda_j \right)^2 \leq 9 \sum_{j=1}^9 \lambda_j^2.$$

Hence

$$1 \leq 9 \mathrm{tr}(\rho^2),$$

so

$$\mathrm{tr}(\rho^2) \geq \frac{1}{9}.$$

Equality holds precisely when

$$\lambda_1 = \dots = \lambda_9 = \frac{1}{9}.$$

The spectral theorem then gives

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

422 Rigidity of the Canonical Stable State

Let

$$\mathcal{A}_9 = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_9)$$

and let

$$\mathcal{A}_9^G = \{X \in \mathcal{A}_9 : \pi_9(g) X \pi_9(g)^* = X \text{ for all } g \in G\}$$

denote the algebra of G -invariant operators.

Proposition 422.1. *One has*

$$\mathcal{A}_9^G = \pi_9(G)'.$$

Proof. For

$$X \in \mathcal{A}_9,$$

the condition

$$\pi_9(g) X \pi_9(g)^* = X$$

is equivalent, after multiplication on the right by $\pi_9(g)$, to

$$\pi_9(g) X = X \pi_9(g).$$

Thus X is invariant under conjugation by every $\pi_9(g)$ if and only if it commutes with every element of $\pi_9(G)$. Hence

$$\mathcal{A}_9^G = \pi_9(G)'. \quad \square$$

Theorem 422.2. *The canonical state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is the unique density operator whose expectation functional is invariant under every unitary operator on $\mathcal{H}_9$.

Proof. Let ρ be a density operator satisfying

$$U\rho U^* = \rho$$

for every

$$U \in \mathcal{U}(\mathcal{H}_9).$$

In particular, ρ commutes with every unitary operator.

Let

$$v, w \in \mathcal{H}_9$$

be unit vectors. There exists a unitary operator U such that

$$Uv = w.$$

Since

$$U\rho = \rho U,$$

one has

$$\langle w, \rho w \rangle = \langle Uv, \rho Uv \rangle = \langle v, U^* \rho Uv \rangle = \langle v, \rho v \rangle.$$

Thus the quadratic form

$$v \longmapsto \langle v, \rho v \rangle$$

has the same value on every unit vector. Hence there exists

$$c \in \mathbb{R}$$

such that

$$\langle v, \rho v \rangle = c$$

for every unit vector v .

By the polarization identity,

$$\langle x, \rho y \rangle = c \langle x, y \rangle$$

for all

$$x, y \in \mathcal{H}_9.$$

Therefore

$$\rho = c I_{\mathcal{H}_9}.$$

Taking traces gives

$$1 = \text{tr}(\rho) = 9c,$$

and consequently

$$c = \frac{1}{9}.$$

Thus

$$\rho = \frac{1}{9} I_{\mathcal{H}_9} = \rho_9. \quad \square$$

423 Invariant Averaging on the Stable Operator Algebra

Assume from now on that G is compact and let

$$dg$$

denote its normalized Haar measure, so that

$$\int_G dg = 1.$$

For

$$X \in \mathcal{A}_9,$$

define the averaging operator

$$\mathcal{E}_G(X) = \int_G \pi_9(g) X \pi_9(g)^* dg.$$

Theorem 423.1. *The map*

$$\mathcal{E}_G : \mathcal{A}_9 \longrightarrow \mathcal{A}_9$$

is a linear projection onto the invariant algebra

$$\mathcal{A}_9^G = \pi_9(G)'. \quad \square$$

In particular,

$$\mathcal{E}_G^2 = \mathcal{E}_G,$$

and

$$\text{range}(\mathcal{E}_G) = \mathcal{A}_9^G.$$

Proof. Linearity follows from linearity of the integral.

Let

$$h \in G.$$

Then

$$\pi_9(h) \mathcal{E}_G(X) \pi_9(h)^* = \int_G \pi_9(hg) X \pi_9(hg)^* dg.$$

By left invariance of Haar measure, the substitution

$$k = hg$$

leaves the measure unchanged. Hence

$$\pi_9(h) \mathcal{E}_G(X) \pi_9(h)^* = \int_G \pi_9(k) X \pi_9(k)^* dk = \mathcal{E}_G(X).$$

Therefore

$$\mathcal{E}_G(X) \in \mathcal{A}_9^G.$$

Conversely, if

$$X \in \mathcal{A}_9^G,$$

then

$$\pi_9(g) X \pi_9(g)^* = X$$

for every $g \in G$, and therefore

$$\mathcal{E}_G(X) = \int_G X dg = X. \quad \square$$

Thus every invariant operator belongs to the range of $\mathcal{E}_G$.

It follows that

$$\text{range}(\mathcal{E}_G) = \mathcal{A}_9^G.$$

Applying the preceding identity to $\mathcal{E}_G(X)$, which is already invariant, gives

$$\mathcal{E}_G^2(X) = \mathcal{E}_G(X).$$

Hence

$$\mathcal{E}_G^2 = \mathcal{E}_G.$$

□

Proposition 423.2. *The averaging map preserves the normalized trace:*

$$\tau_9(\mathcal{E}_G(X)) = \tau_9(X)$$

for every

$$X \in \mathcal{A}_9.$$

Proof. Using linearity of the trace and its invariance under unitary conjugation,

$$\begin{aligned} \tau_9(\mathcal{E}_G(X)) &= \int_G \tau_9(\pi_9(g)X\pi_9(g)^*) dg \\ &= \int_G \tau_9(X) dg \\ &= \tau_9(X). \end{aligned}$$

□

Proposition 423.3. *If*

$$X \geq 0,$$

then

$$\mathcal{E}_G(X) \geq 0.$$

Moreover,

$$X = X^* \implies \mathcal{E}_G(X) = \mathcal{E}_G(X)^*.$$

Proof. For every $g \in G$,

$$\pi_9(g)X\pi_9(g)^* \geq 0$$

whenever $X \geq 0$. An integral of positive operators is positive, so

$$\mathcal{E}_G(X) \geq 0.$$

If $X = X^*$, then

$$(\pi_9(g)X\pi_9(g)^*)^* = \pi_9(g)X\pi_9(g)^*.$$

Integration therefore yields

$$\mathcal{E}_G(X)^* = \mathcal{E}_G(X).$$

□

424 Invariantization of Stable States

Let

$$\rho$$

be an arbitrary density operator on $\mathcal{H}_9$, and define

$$\bar{\rho} = \mathcal{E}_G(\rho).$$

Theorem 424.1. *The operator*

$$\bar{\rho}$$

is a G -invariant density operator.

Proof. Since

$$\rho \geq 0,$$

the preceding proposition gives

$$\bar{\rho} \geq 0.$$

Furthermore,

$$\mathrm{tr}(\bar{\rho}) = 9\tau_9(\bar{\rho}) = 9\tau_9(\rho) = \mathrm{tr}(\rho) = 1.$$

By construction,

$$\pi_9(h)\bar{\rho}\pi_9(h)^* = \bar{\rho} \quad (h \in G).$$

Hence $\bar{\rho}$ is a G -invariant density operator. □

Corollary 424.2. *If*

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9},$$

then every density operator ρ satisfies

$$\mathcal{E}_G(\rho) = \frac{1}{9}I_{\mathcal{H}_9}.$$

Proof. The operator $\mathcal{E}_G(\rho)$ is invariant and hence belongs to

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

Thus

$$\mathcal{E}_G(\rho) = cI_{\mathcal{H}_9}$$

for some $c \geq 0$. Since its trace is one,

$$9c = 1.$$

Therefore

$$c = \frac{1}{9},$$

which proves

$$\mathcal{E}_G(\rho) = \frac{1}{9}I_{\mathcal{H}_9}.$$

□

425 Invariant Observables and Stable Probabilities

Let

$$Q \in \mathcal{A}_g$$

be an orthogonal projection satisfying

$$\pi_g(g)Q\pi_g(g)^* = Q \quad (g \in G).$$

Theorem 425.1. *If*

$$\pi_g(G)' = \mathbb{C}I_{\mathcal{H}_g},$$

then every G -invariant orthogonal projection on $\mathcal{H}_g$ is either

$$0$$

or

$$I_{\mathcal{H}_g}.$$

Proof. By invariance,

$$Q \in \pi_g(G)'.$$

The assumed commutant condition therefore implies

$$Q = \lambda I_{\mathcal{H}_g}$$

for some

$$\lambda \in \mathbb{C}.$$

Since Q is an orthogonal projection,

$$Q^2 = Q.$$

Hence

$$\lambda^2 = \lambda,$$

so

$$\lambda \in \{0, 1\}.$$

Consequently

$$Q = 0$$

or

$$Q = I_{\mathcal{H}_g}.$$

□

Corollary 425.2. *Under the same commutant condition, the stable representation admits no nontrivial G -invariant sharp event.*

Proof. A sharp event is represented by an orthogonal projection. The preceding theorem shows that a G -invariant such projection can only be the zero event or the certain event. □

426 Faithfulness of the Stable Probability Functional

Define

$$\omega_9(X) = \text{tr}(\rho_9 X) = \frac{1}{9} \text{tr}(X).$$

Theorem 426.1. *The state ω_9 is faithful on*

$$\mathcal{A}_9.$$

That is,

$$X \geq 0, \quad \omega_9(X) = 0$$

imply

$$X = 0.$$

Proof. Let

$$X \geq 0$$

and suppose

$$\omega_9(X) = 0.$$

Then

$$0 = \frac{1}{9} \text{tr}(X),$$

so

$$\text{tr}(X) = 0.$$

All eigenvalues of the positive operator X are nonnegative, while their sum is zero. Hence every eigenvalue vanishes. Therefore

$$X = 0.$$

□

Corollary 426.2. *For every nonzero orthogonal projection*

$$P \in \mathcal{A}_9,$$

one has

$$\omega_9(P) > 0.$$

Proof. A nonzero projection is positive and nonzero. Faithfulness therefore gives

$$\omega_9(P) \neq 0.$$

Since $\omega_9(P) \geq 0$,

$$\omega_9(P) > 0.$$

□

427 The Stable Probability Measure on the Projection Lattice

Let

$$\mathcal{P}(\mathcal{H}_9)$$

denote the set of orthogonal projections on $\mathcal{H}_9$, partially ordered by

$$P \leq Q \iff PQ = P.$$

Define

$$\mu_9 : \mathcal{P}(\mathcal{H}_9) \longrightarrow [0, 1]$$

by

$$\mu_9(P) = \text{tr}(\rho_9 P) = \frac{1}{9} \text{tr}(P).$$

Proposition 427.1. *The function μ_9 is normalized and finitely additive on mutually orthogonal projections. More precisely,*

$$\mu_9(0) = 0, \quad \mu_9(I_{\mathcal{H}_9}) = 1,$$

and whenever

$$P_i P_j = 0 \quad (i \neq j),$$

one has

$$\mu_9 \left(\sum_{j=1}^n P_j \right) = \sum_{j=1}^n \mu_9(P_j).$$

Proof. Since

$$\text{tr}(0) = 0,$$

one has

$$\mu_9(0) = 0.$$

Moreover,

$$\mu_9(I_{\mathcal{H}_9}) = \frac{1}{9} \text{tr}(I_{\mathcal{H}_9}) = 1.$$

For mutually orthogonal projections,

$$\sum_{j=1}^n P_j$$

is again a projection and linearity of the trace gives

$$\begin{aligned} \mu_9 \left(\sum_{j=1}^n P_j \right) &= \frac{1}{9} \text{tr} \left(\sum_{j=1}^n P_j \right) \\ &= \frac{1}{9} \sum_{j=1}^n \text{tr}(P_j) \\ &= \sum_{j=1}^n \mu_9(P_j). \end{aligned}$$

□

Theorem 427.2. *For every orthogonal projection*

$$P \in \mathcal{P}(\mathcal{H}_9),$$

the stable probability is determined solely by its rank:

$$\mu_9(P) = \frac{\text{rank}(P)}{9}.$$

Proof. The trace of a finite-dimensional orthogonal projection equals its rank. Therefore

$$\mu_9(P) = \frac{1}{9} \text{tr}(P) = \frac{\text{rank}(P)}{9}.$$

□

Corollary 427.3. *If*

$$P, Q \in \mathcal{P}(\mathcal{H}_9)$$

have equal rank, then

$$\mu_9(P) = \mu_9(Q).$$

Proof. By the preceding theorem,

$$\mu_9(P) = \frac{\text{rank}(P)}{9} = \frac{\text{rank}(Q)}{9} = \mu_9(Q).$$

□

Theorem 427.4. *The values of μ_9 on the projection lattice are precisely*

$$\left\{0, \frac{1}{9}, \frac{2}{9}, \dots, \frac{8}{9}, 1\right\}.$$

Proof. Every projection on the nine-dimensional space has rank

$$k \in \{0, 1, \dots, 9\}.$$

The preceding rank formula gives

$$\mu_9(P) = \frac{k}{9}.$$

Conversely, for every

$$k \in \{0, \dots, 9\},$$

an orthogonal projection of rank k exists on $\mathcal{H}_9$. Hence every value in the displayed set is attained.

□

428 Unitary Covariance of the Stable Probability Measure

Theorem 428.1. *For every*

$$U \in \mathcal{U}(\mathcal{H}_9)$$

and every

$$P \in \mathcal{P}(\mathcal{H}_9),$$

one has

$$\mu_9(UPU^*) = \mu_9(P).$$

Proof. The operator

$$UPU^*$$

is an orthogonal projection of the same rank as P . Hence

$$\text{rank}(UPU^*) = \text{rank}(P).$$

The rank formula therefore yields

$$\mu_9(UPU^*) = \frac{\text{rank}(UPU^*)}{9} = \frac{\text{rank}(P)}{9} = \mu_9(P).$$

□

Corollary 428.2. *The stable probability measure is invariant under the full unitary group of the stable sector, and therefore in particular under the restricted representation $\pi_9(G)$.*

Proof. The restricted operators

$$\pi_9(g)$$

are unitary on $\mathcal{H}_9$. Apply the preceding theorem with

$$U = \pi_9(g).$$

□

429 Orthogonal Resolutions and Rank Conservation

Let

$$P_1, \dots, P_n$$

be pairwise orthogonal projections satisfying

$$\sum_{j=1}^n P_j = I_{\mathcal{H}_9}.$$

Theorem 429.1. *The ranks of the projections in every finite orthogonal resolution satisfy*

$$\sum_{j=1}^n \text{rank}(P_j) = 9,$$

and their stable probabilities satisfy

$$\sum_{j=1}^n \mu_9(P_j) = 1.$$

Proof. Since the projections are pairwise orthogonal, their ranges form an orthogonal direct sum:

$$\mathcal{H}_9 = \text{range}(P_1) \oplus \dots \oplus \text{range}(P_n).$$

Taking dimensions gives

$$9 = \sum_{j=1}^n \dim \text{range}(P_j) = \sum_{j=1}^n \text{rank}(P_j).$$

The probability identity follows from

$$\mu_9(P_j) = \frac{\text{rank}(P_j)}{9}$$

and therefore

$$\sum_{j=1}^n \mu_9(P_j) = \frac{1}{9} \sum_{j=1}^n \text{rank}(P_j) = 1.$$

□

Corollary 429.2. *If*

$$P_1, \dots, P_9$$

are rank-one projections forming a resolution of the identity, then

$$\mu_9(P_j) = \frac{1}{9} \quad (1 \leq j \leq 9).$$

Proof. Each P_j has rank one, so the rank formula gives

$$\mu_9(P_j) = \frac{1}{9}.$$

□

430 The Lattice-Theoretic Characterization of the Stable Measure

Theorem 430.1. *Let*

$$\mu: \mathcal{P}(\mathcal{H}_9) \longrightarrow [0, 1]$$

be a finitely additive probability measure satisfying

$$\mu(UPU^*) = \mu(P)$$

for every

$$U \in \mathcal{U}(\mathcal{H}_9)$$

and every

$$P \in \mathcal{P}(\mathcal{H}_9).$$

Then

$$\mu(P) = \frac{\text{rank}(P)}{9}$$

for every projection P .

Proof. Any two rank-one projections are unitarily conjugate. Hence unitary invariance implies that there exists

$$c \geq 0$$

such that

$$\mu(P) = c$$

for every rank-one projection P .

Let

$$P_1, \dots, P_9$$

be the rank-one projections associated with an arbitrary orthonormal basis. Then

$$\sum_{j=1}^9 P_j = I_{\mathcal{H}_9}.$$

Finite additivity gives

$$1 = \mu(I_{\mathcal{H}_9}) = \sum_{j=1}^9 \mu(P_j) = 9c.$$

Therefore

$$c = \frac{1}{9}.$$

Now let P have rank k . Choose an orthonormal basis

$$e_1, \dots, e_9$$

such that

$$P = \sum_{j=1}^k |e_j\rangle\langle e_j|.$$

Finite additivity then gives

$$\mu(P) = \sum_{j=1}^k \mu(|e_j\rangle\langle e_j|) = \frac{k}{9}.$$

Thus

$$\mu(P) = \frac{\text{rank}(P)}{9}.$$

□

431 Compatibility with the Arithmetic State Functional

Let

$$N_F$$

denote the normalized arithmetic-relational state functional on the stable projection algebra, and suppose that its restriction to

$$\mathcal{P}(\mathcal{H}_9)$$

is represented by the density operator

$$\rho_F.$$

Theorem 431.1. *If*

$$\rho_F = \frac{1}{9} I_{\mathcal{H}_9},$$

then

$$N_F(P) = \mu_9(P) = \frac{\text{rank}(P)}{9}$$

for every

$$P \in \mathcal{P}(\mathcal{H}_9).$$

Proof. By the density-operator representation,

$$N_F(P) = \text{tr}(\rho_F P).$$

Substituting

$$\rho_F = \frac{1}{9} I_{\mathcal{H}_9}$$

gives

$$N_F(P) = \frac{1}{9} \text{tr}(P).$$

Since P is an orthogonal projection,

$$\text{tr}(P) = \text{rank}(P).$$

Consequently,

$$N_F(P) = \frac{\text{rank}(P)}{9} = \mu_9(P).$$

□

Corollary 431.2. *On the stable projection lattice, the arithmetic-relational probability functional and the intrinsic normalized trace state coincide:*

$$N_F|_{\mathcal{P}(\mathcal{H}_9)} = \tau_9|_{\mathcal{P}(\mathcal{H}_9)}.$$

Proof. For every

$$P \in \mathcal{P}(\mathcal{H}_9),$$

one has

$$N_F(P) = \frac{1}{9} \operatorname{tr}(P) = \tau_9(P).$$

□

432 Separation of the Stable Probability Law from the Ambient Dimension

Let

$$P_9 : V_9 \longrightarrow V_9$$

be the orthogonal projection onto

$$\mathcal{H}_9,$$

with

$$\dim_{\mathbb{C}} V_9 = 143, \quad \operatorname{rank}(P_9) = 9.$$

Proposition 432.1. *The intrinsic stable normalization and the ambient normalization assign different weights to the stable sector:*

$$\frac{\operatorname{tr}_{V_9}(P_9)}{\dim_{\mathbb{C}} V_9} = \frac{9}{143},$$

whereas

$$\tau_9(I_{\mathcal{H}_9}) = 1.$$

Proof. Since P_9 has rank nine,

$$\operatorname{tr}_{V_9}(P_9) = 9.$$

Therefore the ambient normalized trace gives

$$\frac{1}{143} \operatorname{tr}_{V_9}(P_9) = \frac{9}{143}.$$

By contrast, the identity of the stable algebra is

$$I_{\mathcal{H}_9},$$

and hence

$$\tau_9(I_{\mathcal{H}_9}) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(I_{\mathcal{H}_9}) = 1.$$

□

Theorem 432.2. *The probability normalization on the stable sector is intrinsic to $\mathcal{H}_9$ and cannot be replaced by the ambient normalized trace without changing the probability space.*

Proof. The stable event represented by the entire stable sector is the identity of the stable operator algebra. Its intrinsic probability is therefore

$$\tau_9(I_{\mathcal{H}_9}) = 1.$$

If instead one uses the ambient normalized trace, the corresponding projection is P_9 , whose value is

$$\frac{9}{143} \neq 1.$$

Thus the ambient normalization does not define the normalized probability measure on the stable sector. The two normalizations describe different trace states and coincide only after the stable algebra has been endowed with its intrinsic identity. $\square$

433 The Stable Sector as a Finite Quantum Probability Space

Define the pair

$$(\mathcal{A}_9, \omega_9)$$

by

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9), \quad \omega_9(X) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(X).$$

Theorem 433.1. *The pair*

$$(\mathcal{A}_9, \omega_9)$$

is a finite-dimensional noncommutative probability space. Its state ω_9 is faithful, tracial, and invariant under every inner automorphism of $\mathcal{A}_9$.

Proof. The algebra $\mathcal{A}_9$ is a unital finite-dimensional $\mathbb{C}$ -algebra with involution $X \mapsto X^*$. The functional ω_9 is normalized because

$$\omega_9(I_{\mathcal{H}_9}) = \frac{1}{9} \text{tr}(I_{\mathcal{H}_9}) = 1.$$

For every $X \in \mathcal{A}_9$,

$$\omega_9(X^* X) = \frac{1}{9} \text{tr}(X^* X) \geq 0.$$

Faithfulness was established previously.

For arbitrary $X, Y \in \mathcal{A}_9$,

$$\omega_9(XY) = \frac{1}{9} \text{tr}(XY) = \frac{1}{9} \text{tr}(YX) = \omega_9(YX).$$

Finally, if U is unitary, then

$$\omega_9(UXU^*) = \frac{1}{9} \text{tr}(UXU^*) = \frac{1}{9} \text{tr}(XU^*U) = \omega_9(X).$$

Thus all stated properties hold. $\square$

Proposition 433.2. *The G -action on the stable algebra is given by the $*$ -automorphisms*

$$\alpha_g(X) = \pi_9(g)X\pi_9(g)^*, \quad g \in G.$$

These automorphisms satisfy

$$\alpha_{g_1 g_2} = \alpha_{g_1} \circ \alpha_{g_2}$$

and

$$\omega_9 \circ \alpha_g = \omega_9.$$

Proof. For

$$X, Y \in \mathcal{A}_9,$$

one has

$$\alpha_g(XY) = \pi_9(g)XY\pi_9(g)^* = \alpha_g(X)\alpha_g(Y),$$

and

$$\alpha_g(X^*) = \alpha_g(X)^*.$$

Since π_9 is a representation,

$$\begin{aligned} \alpha_{g_1 g_2}(X) &= \pi_9(g_1 g_2)X\pi_9(g_1 g_2)^* \\ &= \pi_9(g_1)\pi_9(g_2)X\pi_9(g_2)^*\pi_9(g_1)^* \\ &= \alpha_{g_1}(\alpha_{g_2}(X)). \end{aligned}$$

The invariance of ω_9 follows from the preceding theorem. □

434 Invariant Observables and the Fixed-Point Algebra

Define the fixed-point algebra

$$\mathcal{A}_9^G = \{X \in \mathcal{A}_9 : \alpha_g(X) = X \text{ for all } g \in G\}.$$

Theorem 434.1. *The fixed-point algebra is a unital $*$ -subalgebra of $\mathcal{A}_9$, and*

$$\mathcal{A}_9^G = \pi_9(G)'. \quad \square$$

Proof. If $X, Y \in \mathcal{A}_9^G$, then for every $g \in G$,

$$\alpha_g(XY) = \alpha_g(X)\alpha_g(Y) = XY,$$

so $XY \in \mathcal{A}_9^G$. Similarly,

$$\alpha_g(X^*) = \alpha_g(X)^* = X^*,$$

hence $X^* \in \mathcal{A}_9^G$. The identity is fixed by every α_g , so

$$I_{\mathcal{H}_9} \in \mathcal{A}_9^G.$$

The equality with the commutant follows from

$$\alpha_g(X) = X$$

if and only if

$$\pi_9(g)X\pi_9(g)^* = X,$$

which, by multiplication on the right by $\pi_9(g)$, is equivalent to

$$\pi_9(g)X = X\pi_9(g).$$

Thus

$$\mathcal{A}_9^G = \pi_9(G)'. \quad \square$$

Theorem 434.2. *Suppose that the stable representation is irreducible. Then*

$$\mathcal{A}_9^G = \mathbb{C}I_{\mathcal{H}_9}.$$

Consequently, every self-adjoint G -invariant observable is constant:

$$X = X^*, \quad \alpha_g(X) = X \quad (g \in G)$$

implies

$$X = \lambda I_{\mathcal{H}_9}$$

for some

$$\lambda \in \mathbb{R}.$$

Proof. Irreducibility and Schur's lemma give

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

The fixed-point characterization therefore yields

$$\mathcal{A}_9^G = \mathbb{C}I_{\mathcal{H}_9}.$$

If $X = X^*$, then

$$X = \lambda I_{\mathcal{H}_9}$$

with

$$\lambda = \overline{\lambda},$$

so

$$\lambda \in \mathbb{R}.$$

□

435 Conditional Expectations onto the Invariant Algebra

Assume again that G is compact and let

$$dg$$

be normalized Haar measure. Define

$$\mathcal{E}_G(X) = \int_G \alpha_g(X) dg.$$

Theorem 435.1. *The averaging map*

$$\mathcal{E}_G : \mathcal{A}_9 \longrightarrow \mathcal{A}_9^G$$

is trace-preserving, unital, positive, and idempotent:

$$\mathcal{E}_G(I) = I,$$

$$X \geq 0 \implies \mathcal{E}_G(X) \geq 0,$$

$$\omega_9(\mathcal{E}_G(X)) = \omega_9(X),$$

and

$$\mathcal{E}_G^2 = \mathcal{E}_G.$$

Proof. Unitality follows from

$$\alpha_g(I) = I$$

for every $g \in G$. Positivity follows because every

$$\alpha_g$$

is implemented by a unitary operator and hence preserves the positive cone.

Trace preservation follows from invariance of ω_g :

$$\begin{aligned} \omega_g(\mathcal{E}_G(X)) &= \int_G \omega_g(\alpha_g(X)) dg \\ &= \int_G \omega_g(X) dg \\ &= \omega_g(X). \end{aligned}$$

The range of $\mathcal{E}_G$ lies in $\mathcal{A}_g^G$ by Haar invariance. If

$$X \in \mathcal{A}_g^G,$$

then

$$\mathcal{E}_G(X) = \int_G X dg = X.$$

Therefore

$$\mathcal{E}_G^2 = \mathcal{E}_G.$$

□

Corollary 435.2. *If*

$$\mathcal{A}_g^G = \mathbb{C}I_{\mathcal{H}_g},$$

then

$$\mathcal{E}_G(X) = \omega_g(X)I_{\mathcal{H}_g}$$

for every

$$X \in \mathcal{A}_g.$$

Proof. The averaged operator belongs to

$$\mathcal{A}_g^G = \mathbb{C}I_{\mathcal{H}_g}.$$

Hence

$$\mathcal{E}_G(X) = cI_{\mathcal{H}_g}$$

for some $c \in \mathbb{C}$. Applying ω_g and using trace preservation gives

$$c = \omega_g(\mathcal{E}_G(X)) = \omega_g(X).$$

Thus

$$\mathcal{E}_G(X) = \omega_g(X)I_{\mathcal{H}_g}.$$

□

436 The Ergodic Consequence of Irreducibility

Theorem 436.1. *Assume that π_g is irreducible. If*

$$X = X^*$$

and

$$\mathcal{E}_G(X) = X,$$

then

$$X = \omega_g(X) I_{\mathcal{H}_g}.$$

Proof. The identity

$$\mathcal{E}_G(X) = X$$

means precisely that X belongs to the fixed-point algebra:

$$X \in \mathcal{A}_g^G.$$

Irreducibility gives

$$\mathcal{A}_g^G = \mathbb{C} I_{\mathcal{H}_g}.$$

Therefore

$$X = \lambda I_{\mathcal{H}_g}.$$

Applying ω_g yields

$$\lambda = \omega_g(X),$$

because

$$\omega_g(I_{\mathcal{H}_g}) = 1.$$

Hence

$$X = \omega_g(X) I_{\mathcal{H}_g}.$$

□

Corollary 436.2. *For an irreducible stable representation, the only invariant sharp events are*

$$0 \quad \text{and} \quad I_{\mathcal{H}_g},$$

and every invariant self-adjoint observable is completely determined by its stable expectation value.

Proof. The assertion about projections follows from the scalar form of the fixed-point algebra together with

$$P^2 = P.$$

The assertion about self-adjoint observables follows from the preceding theorem.

□

437 Entropy of the Canonical Stable State

Let

$$S(\rho) = -\operatorname{tr}(\rho \log \rho)$$

denote the von Neumann entropy of a density operator on $\mathcal{H}_9$.

Theorem 437.1. *The canonical stable state has entropy*

$$S(\rho_9) = \log 9.$$

Moreover, for every density operator ρ on $\mathcal{H}_9$,

$$S(\rho) \leq \log 9,$$

with equality if and only if

$$\rho = \rho_9.$$

Proof. The eigenvalues of ρ_9 are all equal to $1/9$. Therefore

$$\begin{aligned} S(\rho_9) &= -\sum_{j=1}^9 \frac{1}{9} \log \left(\frac{1}{9} \right) \\ &= -\log \left(\frac{1}{9} \right) \\ &= \log 9. \end{aligned}$$

For an arbitrary density operator ρ , let

$$\lambda_1, \dots, \lambda_9$$

be its eigenvalues. Then

$$\lambda_j \geq 0, \quad \sum_{j=1}^9 \lambda_j = 1,$$

and

$$S(\rho) = -\sum_{j=1}^9 \lambda_j \log \lambda_j.$$

The function

$$x \mapsto -x \log x$$

is strictly concave on $(0, \infty)$. Jensen's inequality therefore gives

$$\frac{1}{9} \sum_{j=1}^9 (-\lambda_j \log \lambda_j) \leq -\frac{1}{9} \log \frac{1}{9}.$$

Hence

$$S(\rho) \leq \log 9.$$

Equality holds precisely when

$$\lambda_1 = \dots = \lambda_9 = \frac{1}{9}.$$

By the spectral theorem this is equivalent to

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

□

438 The Maximally Mixed State as the Unique Entropy Maximizer

Let

$$\mathfrak{D}(\mathcal{H}_9) = \{\rho \in \text{End}_{\mathbb{C}}(\mathcal{H}_9) : \rho \geq 0, \text{tr}(\rho) = 1\}$$

denote the state space of the stable sector.

Theorem 438.1. *The state*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is the unique maximizer of the von Neumann entropy on $\mathfrak{D}(\mathcal{H}_9)$. Equivalently,

$$S(\rho) \leq S(\rho_9) = \log 9$$

for every

$$\rho \in \mathfrak{D}(\mathcal{H}_9),$$

with equality if and only if

$$\rho = \rho_9.$$

Proof. The entropy formula established above gives

$$S(\rho) = - \sum_{j=1}^9 \lambda_j \log \lambda_j,$$

where

$$\lambda_j \geq 0, \quad \sum_{j=1}^9 \lambda_j = 1.$$

The strict concavity of the Shannon entropy on the probability simplex implies that its unique maximizer is the uniform distribution

$$\lambda_1 = \cdots = \lambda_9 = \frac{1}{9}.$$

The spectral theorem then gives

$$\rho = \sum_{j=1}^9 \lambda_j P_j = \frac{1}{9} \sum_{j=1}^9 P_j = \frac{1}{9} I_{\mathcal{H}_9}.$$

Thus the unique entropy maximizer is ρ_9 . □

Proposition 438.2. *For every density operator*

$$\rho \in \mathfrak{D}(\mathcal{H}_9),$$

the relative entropy with respect to the canonical stable state is

$$D(\rho \| \rho_9) = \log 9 - S(\rho).$$

In particular,

$$D(\rho \| \rho_9) \geq 0,$$

and

$$D(\rho \| \rho_9) = 0 \iff \rho = \rho_9.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\log \rho_9 = -\log 9 I_{\mathcal{H}_9}.$$

Therefore

$$\begin{aligned} D(\rho \| \rho_9) &= \text{tr}(\rho(\log \rho - \log \rho_9)) \\ &= \text{tr}(\rho \log \rho) + \log 9 \text{tr}(\rho) \\ &= -S(\rho) + \log 9. \end{aligned}$$

Non-negativity of quantum relative entropy yields

$$D(\rho \| \rho_9) \geq 0,$$

and equality holds precisely when

$$\rho = \rho_9.$$

□

439 Basis Independence of the Stable Probability Law

Let

$$\mathcal{B} = \{\psi_1, \dots, \psi_9\}$$

and

$$\mathcal{B}' = \{\phi_1, \dots, \phi_9\}$$

be arbitrary orthonormal bases of $\mathcal{H}_9$. Define

$$P_j = |\psi_j\rangle\langle\psi_j|, \quad R_j = |\phi_j\rangle\langle\phi_j|.$$

Theorem 439.1. *The canonical stable state induces the same probability vector*

$$\left(\frac{1}{9}, \dots, \frac{1}{9}\right)$$

for every orthonormal basis of $\mathcal{H}_9$.

Proof. For every j ,

$$\text{tr}(\rho_9 P_j) = \frac{1}{9} \text{tr}(P_j) = \frac{1}{9}.$$

The same argument gives

$$\text{tr}(\rho_9 R_j) = \frac{1}{9}.$$

Since the argument depends only on the rank of the projection and not on the vectors defining the basis, the resulting probability vector is basis-independent. □

Corollary 439.2. *If*

$$U \in \mathcal{U}(\mathcal{H}_9),$$

then

$$\rho_9 = U \rho_9 U^*,$$

and consequently

$$\text{tr}(\rho_9 U P_j U^*) = \text{tr}(\rho_9 P_j) = \frac{1}{9}.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

unitary conjugation leaves ρ_9 unchanged. Hence

$$\begin{aligned} \operatorname{tr}(\rho_9 U P_j U^*) &= \operatorname{tr}(U^* \rho_9 U P_j) \\ &= \operatorname{tr}(\rho_9 P_j) \\ &= \frac{1}{9}. \end{aligned}$$

□

440 Uniformity on Arbitrary Finite Orthogonal Resolutions

Let

$$\mathcal{C} = \{E_1, \dots, E_r\}$$

be an orthogonal resolution of the identity on $\mathcal{H}_9$:

$$E_i E_j = 0 \quad (i \neq j),$$

$$E_j^* = E_j,$$

and

$$\sum_{j=1}^r E_j = I_{\mathcal{H}_9}.$$

Theorem 440.1. *For the canonical stable state,*

$$\operatorname{tr}(\rho_9 E_j) = \frac{\operatorname{rank}(E_j)}{9}$$

for every

$$j = 1, \dots, r.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

one has

$$\operatorname{tr}(\rho_9 E_j) = \frac{1}{9} \operatorname{tr}(E_j).$$

Because E_j is an orthogonal projection,

$$\operatorname{tr}(E_j) = \operatorname{rank}(E_j).$$

Therefore

$$\operatorname{tr}(\rho_9 E_j) = \frac{\operatorname{rank}(E_j)}{9}.$$

□

Corollary 440.2. *If every projection in the resolution has the same rank d , then*

$$r d = 9$$

and

$$\operatorname{tr}(\rho_9 E_j) = \frac{1}{r}$$

for every j .

Proof. The orthogonal resolution implies

$$9 = \text{tr}(I_{\mathcal{H}_9}) = \sum_{j=1}^r \text{tr}(E_j) = rd.$$

The preceding theorem then gives

$$\text{tr}(\rho_9 E_j) = \frac{d}{9} = \frac{1}{r}.$$

□

441 Spectral Decomposition of Stable Observables

Let

$$A = A^* \in \text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

By the finite-dimensional spectral theorem, there exist distinct real numbers

$$a_1, \dots, a_s$$

and pairwise orthogonal projections

$$E_1, \dots, E_s$$

such that

$$A = \sum_{\alpha=1}^s a_{\alpha} E_{\alpha},$$

$$E_{\alpha} E_{\beta} = \delta_{\alpha\beta} E_{\alpha},$$

and

$$\sum_{\alpha=1}^s E_{\alpha} = I_{\mathcal{H}_9}.$$

Theorem 441.1. *The expectation of every self-adjoint stable observable in the canonical state is*

$$\omega_9(A) = \frac{1}{9} \sum_{\alpha=1}^s a_{\alpha} \text{rank}(E_{\alpha}).$$

Proof. Using the spectral decomposition,

$$\begin{aligned} \omega_9(A) &= \frac{1}{9} \text{tr}(A) \\ &= \frac{1}{9} \text{tr} \left(\sum_{\alpha=1}^s a_{\alpha} E_{\alpha} \right) \\ &= \frac{1}{9} \sum_{\alpha=1}^s a_{\alpha} \text{tr}(E_{\alpha}). \end{aligned}$$

Since every E_{α} is an orthogonal projection,

$$\text{tr}(E_{\alpha}) = \text{rank}(E_{\alpha}).$$

Substitution gives the claimed formula. □

Corollary 441.2. *The variance of A in the canonical stable state is*

$$\text{Var}_{\rho_9}(A) = \frac{1}{9} \sum_{\alpha=1}^s a_{\alpha}^2 \text{rank}(E_{\alpha}) - \left(\frac{1}{9} \sum_{\alpha=1}^s a_{\alpha} \text{rank}(E_{\alpha}) \right)^2.$$

Proof. By definition,

$$\text{Var}_{\rho_9}(A) = \omega_9(A^2) - \omega_9(A)^2.$$

The spectral decomposition gives

$$A^2 = \sum_{\alpha=1}^s a_\alpha^2 E_\alpha.$$

Applying the preceding theorem to A^2 gives

$$\omega_9(A^2) = \frac{1}{9} \sum_{\alpha=1}^s a_\alpha^2 \text{rank}(E_\alpha).$$

Subtracting $\omega_9(A)^2$ yields the result. $\square$

442 Characterization by Unitary Invariance

Theorem 442.1. *Let*

$$\rho \in \mathcal{D}(\mathcal{H}_9).$$

Suppose that

$$U\rho U^* = \rho \quad \text{for every } U \in \mathcal{U}(\mathcal{H}_9).$$

Then

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. The hypothesis implies that ρ commutes with every unitary operator on $\mathcal{H}_9$. Every operator on a finite-dimensional complex Hilbert space is a linear combination of unitary operators. Hence ρ commutes with every element of

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

The commutant of the full matrix algebra consists only of scalar operators. Thus

$$\rho = \lambda I_{\mathcal{H}_9}$$

for some

$$\lambda \geq 0.$$

The normalization condition gives

$$1 = \text{tr}(\rho) = 9\lambda,$$

and therefore

$$\lambda = \frac{1}{9}.$$

Hence

$$\rho = \frac{1}{9} I_{\mathcal{H}_9}.$$

$\square$

Corollary 442.2. *The canonical stable state is the unique density operator invariant under the full unitary group*

$$\mathcal{U}(\mathcal{H}_9).$$

Proof. The state

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is invariant under every unitary conjugation, while the preceding theorem proves uniqueness. $\square$

443 The Stable State as a Unitary Invariant of the Finite Construction

Let

$$\mathfrak{D}(\mathcal{H}_9) = \{\rho \in \text{End}_{\mathbb{C}}(\mathcal{H}_9) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

Theorem 443.1. *The assignment*

$$\mathcal{H}_9 \longmapsto \rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is invariant under unitary equivalence of the stable sector.

More precisely, if

$$U : \mathcal{H}_9 \longrightarrow \mathcal{H}'_9$$

is unitary, then

$$\rho'_9 = U \rho_9 U^* = \frac{1}{9} I_{\mathcal{H}'_9}.$$

Proof. Since

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9},$$

unitary conjugation gives

$$U \rho_9 U^* = \frac{1}{9} U I_{\mathcal{H}_9} U^* = \frac{1}{9} I_{\mathcal{H}'_9}.$$

Thus the normalized state is independent of the particular orthonormal realization of the nine-dimensional Hilbert space. $\square$

Proposition 443.2. *Let*

$$\Phi : \text{End}_{\mathbb{C}}(\mathcal{H}_9) \longrightarrow \text{End}_{\mathbb{C}}(\mathcal{H}'_9)$$

be the $$ -isomorphism induced by a unitary map U ,*

$$\Phi(X) = U X U^*.$$

Then

$$\omega'_9(\Phi(X)) = \omega_9(X)$$

for every

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

where

$$\omega_9(X) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(X), \quad \omega'_9(Y) = \frac{1}{9} \text{tr}_{\mathcal{H}'_9}(Y).$$

Proof. Unitary equivalence preserves the ordinary finite-dimensional trace:

$$\text{tr}_{\mathcal{H}'_9}(U X U^*) = \text{tr}_{\mathcal{H}_9}(X).$$

Therefore

$$\omega'_9(\Phi(X)) = \frac{1}{9} \text{tr}_{\mathcal{H}'_9}(U X U^*) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(X) = \omega_9(X).$$

$\square$

444 Functoriality of the Stable Probability Assignment

For every orthogonal projection

$$P \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

define

$$\mu_9(P) = \omega_9(P) = \frac{1}{9} \text{tr}(P).$$

Theorem 444.1. *The probability assignment*

$$\mu_9 : \text{Proj}(\mathcal{H}_9) \longrightarrow [0, 1]$$

is characterized by

$$\mu_9(P) = \frac{\text{rank}(P)}{9}.$$

It is additive on mutually orthogonal projections:

$$PQ = 0 \implies \mu_9(P + Q) = \mu_9(P) + \mu_9(Q).$$

Proof. For an orthogonal projection P ,

$$\text{tr}(P) = \text{rank}(P).$$

Hence

$$\mu_9(P) = \frac{1}{9} \text{rank}(P).$$

If

$$PQ = 0,$$

then $P + Q$ is again an orthogonal projection and

$$\text{tr}(P + Q) = \text{tr}(P) + \text{tr}(Q).$$

Consequently,

$$\mu_9(P + Q) = \frac{1}{9} (\text{tr}(P) + \text{tr}(Q)) = \mu_9(P) + \mu_9(Q).$$

□

Corollary 444.2. *For every finite orthogonal family*

$$P_1, \dots, P_r$$

satisfying

$$P_i P_j = 0 \quad (i \neq j),$$

one has

$$\mu_9 \left(\sum_{j=1}^r P_j \right) = \sum_{j=1}^r \mu_9(P_j).$$

In particular, if

$$\sum_{j=1}^r P_j = I_{\mathcal{H}_9},$$

then

$$\sum_{j=1}^r \mu_9(P_j) = 1.$$

Proof. Repeated application of finite additivity gives

$$\mu_9\left(\sum_{j=1}^r P_j\right) = \sum_{j=1}^r \mu_9(P_j).$$

For a resolution of the identity,

$$\mu_9(I_{\mathcal{H}_9}) = \frac{1}{9} \operatorname{tr}(I_{\mathcal{H}_9}) = 1.$$

□

445 The Rank-One Probability Law

Let

$$\mathbb{P}(\mathcal{H}_9) = \{P \in \operatorname{End}_{\mathbb{C}}(\mathcal{H}_9) : P^2 = P = P^*\}$$

denote the projection lattice of the stable sector.

Theorem 445.1. *The restriction of μ_9 to rank-one projections is constant:*

$$\operatorname{rank}(P) = 1 \implies \mu_9(P) = \frac{1}{9}.$$

Conversely, if a density operator

$$\rho \in \mathfrak{D}(\mathcal{H}_9)$$

satisfies

$$\operatorname{tr}(\rho P) = \frac{1}{9}$$

for every rank-one projection P , then

$$\rho = \rho_9.$$

Proof. The first assertion follows from

$$\mu_9(P) = \frac{\operatorname{rank}(P)}{9}.$$

For the converse, let

$$P_\psi = |\psi\rangle\langle\psi|$$

for every unit vector

$$\psi \in \mathcal{H}_9.$$

The hypothesis gives

$$\langle\psi, \rho\psi\rangle = \frac{1}{9} = \left\langle\psi, \frac{1}{9} I_{\mathcal{H}_9} \psi\right\rangle.$$

Therefore

$$\left\langle\psi, \left(\rho - \frac{1}{9} I_{\mathcal{H}_9}\right) \psi\right\rangle = 0$$

for every unit vector ψ . Since the operator

$$\rho - \frac{1}{9} I_{\mathcal{H}_9}$$

is self-adjoint, the polarization identity implies that all its matrix coefficients vanish. Hence

$$\rho - \frac{1}{9} I_{\mathcal{H}_9} = 0,$$

which proves

$$\rho = \rho_9.$$

□

446 Compatibility with the Stable Projection

Let

$$P_9 = \Psi_9^4$$

be the stable projection in the ambient finite-dimensional space V_9 , and assume

$$\text{rank}(P_9) = 9.$$

Define

$$\iota_9 : \mathcal{H}_9 \hookrightarrow V_9$$

as the canonical inclusion.

Proposition 446.1. *For every*

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the zero extension

$$\tilde{X} = \iota_9 X \iota_9^*$$

satisfies

$$P_9 \tilde{X} = \tilde{X} = \tilde{X} P_9.$$

Proof. The range of ι_9 is

$$\mathcal{H}_9 = \text{range}(P_9).$$

Hence

$$P_9 \iota_9 = \iota_9$$

and

$$\iota_9^* P_9 = \iota_9^*.$$

Therefore

$$P_9 \tilde{X} = P_9 \iota_9 X \iota_9^* = \iota_9 X \iota_9^* = \tilde{X},$$

and similarly

$$\tilde{X} P_9 = \iota_9 X \iota_9^* P_9 = \iota_9 X \iota_9^* = \tilde{X}.$$

□

Theorem 446.2. *The stable algebra is canonically identified with the corner algebra*

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

More precisely,

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9) \cong P_9 \text{End}_{\mathbb{C}}(V_9) P_9.$$

Proof. Every element of

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9$$

has range contained in

$$\text{range}(P_9) = \mathcal{H}_9$$

and vanishes on

$$\ker(P_9) = \mathcal{H}_9^{\perp}.$$

Thus its restriction to $\mathcal{H}_9$ determines an element of

$$\text{End}_{\mathbb{C}}(\mathcal{H}_9).$$

Conversely, every

$$X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9)$$

has the zero extension

$$\tilde{X} = \iota_9 X \iota_9^*$$

constructed above, and

$$P_9 \tilde{X} P_9 = \tilde{X}.$$

The two constructions are inverse to one another, establishing the claimed algebraic $*$ -isomorphism. $\square$

447 Intrinsic Characterization of the Stable Algebra

Theorem 447.1. *The corner algebra*

$$P_9 \text{End}_{\mathbb{C}}(V_9) P_9$$

is a full matrix algebra of dimension

$$9^2 = 81$$

over $\mathbb{C}$. In particular,

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 81.$$

Proof. Since

$$\dim_{\mathbb{C}} \mathcal{H}_9 = 9,$$

one has

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9) \cong M_9(\mathbb{C}).$$

The complex dimension of $M_9(\mathbb{C})$ is

$$9^2 = 81.$$

The preceding corner-algebra identification therefore gives

$$\dim_{\mathbb{C}} (P_9 \text{End}_{\mathbb{C}}(V_9) P_9) = 81.$$

$\square$

Corollary 447.2. *The stable sector carries a canonical finite-dimensional operator algebra whose algebraic dimension is determined solely by*

$$\text{rank}(P_9) = 9.$$

No choice of basis of $\mathcal{H}_9$ enters this identification.

Proof. The identification

$$\mathcal{A}_9 \cong M_9(\mathbb{C})$$

is intrinsic to the nine-dimensional Hilbert space. A change of orthonormal basis acts by an inner $*$ -automorphism and therefore does not change the algebra or its dimension. $\square$

448 Separation of the Stable Sector from the Ambient Multiplicity

Theorem 448.1. *The numerical value*

$$\frac{1}{9}$$

associated with a rank-one stable event is determined by the intrinsic dimension of the stable sector and not by the ambient dimension

$$\dim_{\mathbb{C}} V_9 = 143.$$

Proof. For every rank-one projection

$$P \in \text{End}_{\mathbb{C}}(\mathcal{H}_9),$$

the intrinsic stable state gives

$$\omega_9(P) = \frac{1}{9} \text{tr}_{\mathcal{H}_9}(P) = \frac{1}{9}.$$

The ambient dimension does not occur in this expression.

If P is instead regarded as its zero extension to V_9 , its ambient trace remains

$$\text{tr}_{V_9}(P) = 1,$$

while the normalized ambient trace is

$$\frac{1}{143} \text{tr}_{V_9}(P) = \frac{1}{143}.$$

Thus the two normalized functionals describe different normalizations: the first is intrinsic to the stable sector, whereas the second is intrinsic to the full ambient space.

Consequently,

$$\frac{1}{9} \neq \frac{1}{143}$$

is not a contradiction but a consequence of restricting the state space from V_9 to the rank-nine stable sector. $\square$

449 The Stable Sector as a Full Matrix Factor

Let

$$\mathcal{A}_9 = \text{End}_{\mathbb{C}}(\mathcal{H}_9) \cong M_9(\mathbb{C}).$$

Theorem 449.1. *The algebra $\mathcal{A}_9$ is a simple finite-dimensional $*$ -algebra. Its center is*

$$Z(\mathcal{A}_9) = \mathbb{C}I_{\mathcal{H}_9}.$$

Proof. Choose an orthonormal basis of $\mathcal{H}_9$. Then

$$\mathcal{A}_9 \cong M_9(\mathbb{C}).$$

Let

$$A = (a_{ij}) \in M_9(\mathbb{C})$$

commute with every matrix unit

$$E_{kl}.$$

The relation

$$AE_{kl} = E_{kl}A$$

implies

$$a_{ik}\delta_{lj} = \delta_{ik}a_{lj}$$

for all indices. Taking suitable values of the indices first forces all off-diagonal entries of A to vanish and then forces all diagonal entries to coincide. Hence

$$A = \lambda I$$

for some

$$\lambda \in \mathbb{C}.$$

Thus

$$Z(\mathcal{A}_9) = \mathbb{C}I_{\mathcal{H}_9}.$$

The algebra $M_9(\mathbb{C})$ is simple: if J is a nonzero two-sided ideal and

$$A = (a_{ij}) \in J$$

has some nonzero entry a_{kl} , then

$$E_{ik}AE_{lj} = a_{kl}E_{ij} \in J$$

for all i, j . Hence every matrix unit belongs to J , so

$$J = M_9(\mathbb{C}).$$

Therefore $\mathcal{A}_9$ is simple. □

Corollary 449.2. *The stable operator algebra has no nontrivial two-sided ideals and no nontrivial central projections.*

Proof. Simplicity excludes nonzero proper two-sided ideals. A central projection generates a two-sided ideal, so simplicity forces every central projection to be either

$$0$$

or

$$I_{\mathcal{H}_9}.$$

□

450 Minimal Projections and Rank-One Stable Events

Definition 450.1. A nonzero projection

$$P \in \mathcal{A}_9$$

is called minimal if there is no nonzero projection

$$Q \in \mathcal{A}_9$$

such that

$$Q \leq P, \quad Q \neq P.$$

Theorem 450.2. *A nonzero projection*

$$P \in \mathcal{A}_9$$

is minimal if and only if

$$\text{rank}(P) = 1.$$

Proof. Suppose first that

$$\text{rank}(P) = 1.$$

Then

$$\text{range}(P)$$

is one-dimensional. If Q is a nonzero projection satisfying

$$Q \leq P,$$

then

$$\text{range}(Q) \subseteq \text{range}(P).$$

Since the latter space is one-dimensional and $Q \neq 0$,

$$\text{range}(Q) = \text{range}(P),$$

hence

$$Q = P.$$

Thus P is minimal.

Conversely, suppose

$$\text{rank}(P) \geq 2.$$

Choose a unit vector

$$\psi \in \text{range}(P)$$

and define

$$Q_\psi = |\psi\rangle\langle\psi|.$$

Then

$$0 \neq Q_\psi \leq P$$

and

$$Q_\psi \neq P$$

because P has rank at least two. Hence P is not minimal. □

Corollary 450.3. *The minimal events of the stable operator algebra are precisely the rank-one projections*

$$P_\psi = |\psi\rangle\langle\psi|, \quad \|\psi\| = 1.$$

451 The Stable State on Minimal Events

Theorem 451.1. *The canonical stable state assigns the same probability to every minimal projection:*

$$\omega_9(P) = \frac{1}{9}$$

for every minimal projection

$$P \in \mathcal{A}_9.$$

Proof. By the preceding characterization, every minimal projection has rank one. Therefore

$$\omega_9(P) = \frac{1}{9} \operatorname{tr}(P) = \frac{1}{9}.$$

□

Proposition 451.2. *Let*

$$P_1, \dots, P_9$$

be pairwise orthogonal minimal projections satisfying

$$\sum_{j=1}^9 P_j = I_{\mathcal{H}_9}.$$

Then

$$\omega_9(P_j) = \frac{1}{9}$$

for every j , and

$$\sum_{j=1}^9 \omega_9(P_j) = 1.$$

Proof. Each P_j is minimal and hence rank one. The preceding theorem gives

$$\omega_9(P_j) = \frac{1}{9}.$$

Therefore

$$\sum_{j=1}^9 \omega_9(P_j) = 9 \cdot \frac{1}{9} = 1.$$

□

452 The State as a Faithful Trace

Theorem 452.1. *For*

$$X \in \mathcal{A}_9,$$

the following are equivalent:

$$X = 0,$$

and

$$\omega_9(X^* X) = 0.$$

Thus ω_9 is faithful.

Proof. If $X = 0$, then clearly

$$\omega_9(X^* X) = 0.$$

Conversely,

$$\omega_9(X^* X) = \frac{1}{9} \operatorname{tr}(X^* X).$$

Let

$$s_1, \dots, s_9$$

be the singular values of X . Then

$$\operatorname{tr}(X^* X) = \sum_{j=1}^9 s_j^2.$$

If this quantity vanishes, every

$$s_j = 0.$$

Hence

$$X = 0.$$

□

Corollary 452.2. *The sesquilinear form*

$$\langle X, Y \rangle_{\omega_9} = \omega_9(X^* Y)$$

defines an inner product on $\mathcal{A}_9$.

Proof. Sesquilinearity and conjugate symmetry follow from the corresponding properties of the trace. Positive definiteness follows from the faithfulness established above:

$$\langle X, X \rangle_{\omega_9} = \omega_9(X^* X) = 0 \implies X = 0.$$

□

453 The Hilbert–Schmidt Geometry of the Stable Algebra

Theorem 453.1. *The inner product*

$$\langle X, Y \rangle_{\omega_9} = \frac{1}{9} \operatorname{tr}(X^* Y)$$

induces the normalized Hilbert–Schmidt norm

$$\|X\|_{\omega_9} = \left(\frac{1}{9} \operatorname{tr}(X^* X) \right)^{1/2}.$$

The space

$$\mathcal{A}_9$$

is complete with respect to this norm.

Proof. Since

$$\mathcal{A}_9$$

is finite-dimensional, every inner-product norm on it is complete. The displayed formula follows directly from the definition of the inner product. □

Proposition 453.2. *For every*

$$g \in G$$

and

$$X \in \mathcal{A}_9,$$

one has

$$\|\alpha_g(X)\|_{\omega_9} = \|X\|_{\omega_9}.$$

Proof. Using

$$\alpha_g(X) = \pi_g(g)X\pi_g(g)^*,$$

one obtains

$$\begin{aligned}\|\alpha_g(X)\|_{\omega_g}^2 &= \frac{1}{9} \operatorname{tr}(\alpha_g(X)^* \alpha_g(X)) \\ &= \frac{1}{9} \operatorname{tr}(\pi_g(g)X^*X\pi_g(g)^*) \\ &= \frac{1}{9} \operatorname{tr}(X^*X) \\ &= \|X\|_{\omega_g}^2.\end{aligned}$$

Taking square roots gives the result. $\square$

454 Orthogonal Decomposition into Scalar and Traceless Parts

Define

$$\mathcal{A}_g^0 = \{X \in \mathcal{A}_g : \omega_g(X) = 0\}.$$

Theorem 454.1. *The stable algebra decomposes orthogonally as*

$$\mathcal{A}_g = \mathbb{C}I_{\mathcal{H}_g} \oplus \mathcal{A}_g^0.$$

More precisely, every

$$X \in \mathcal{A}_g$$

has the unique decomposition

$$X = \omega_g(X)I_{\mathcal{H}_g} + X_0,$$

where

$$X_0 \in \mathcal{A}_g^0.$$

Proof. Define

$$X_0 = X - \omega_g(X)I_{\mathcal{H}_g}.$$

Then

$$\omega_g(X_0) = \omega_g(X) - \omega_g(X)\omega_g(I_{\mathcal{H}_g}) = 0.$$

Hence

$$X_0 \in \mathcal{A}_g^0.$$

Thus the decomposition exists.

If

$$X = \lambda I + Y, \quad Y \in \mathcal{A}_g^0,$$

then applying ω_g gives

$$\omega_g(X) = \lambda.$$

Therefore the decomposition is unique.

Finally,

$$\langle I_{\mathcal{H}_g}, X_0 \rangle_{\omega_g} = \omega_g(X_0) = 0,$$

so the two summands are orthogonal. $\square$

Corollary 454.2. *The traceless subspace has complex dimension*

$$\dim_{\mathbb{C}} \mathcal{A}_g^0 = 80.$$

Proof. Since

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 81$$

and

$$\dim_{\mathbb{C}} \mathbb{C} I_{\mathcal{H}_9} = 1,$$

the orthogonal decomposition gives

$$\dim_{\mathbb{C}} \mathcal{A}_9^0 = 81 - 1 = 80.$$

□

455 Vanishing of the Invariant Traceless Part

Theorem 455.1. *If the stable representation is irreducible, then*

$$\mathcal{A}_9^0 \cap \mathcal{A}_9^G = \{0\}.$$

Proof. Irreducibility gives

$$\mathcal{A}_9^G = \mathbb{C} I_{\mathcal{H}_9}.$$

Let

$$X \in \mathcal{A}_9^0 \cap \mathcal{A}_9^G.$$

Then

$$X = \lambda I_{\mathcal{H}_9}$$

for some $\lambda \in \mathbb{C}$. Since X is traceless,

$$0 = \omega_9(X) = \lambda \omega_9(I_{\mathcal{H}_9}) = \lambda.$$

Therefore

$$X = 0.$$

□

Corollary 455.2. *Under irreducibility, every non-scalar stable observable has a nontrivial component orthogonal to the invariant algebra.*

Proof. The preceding decomposition gives

$$X = \omega_9(X) I_{\mathcal{H}_9} + X_0, \quad X_0 \in \mathcal{A}_9^0.$$

If X is not scalar, then

$$X_0 \neq 0.$$

The preceding theorem shows that this nonzero component cannot be G -invariant.

□

456 The Fixed-Point Algebra and the Ergodic Alternative

Define the fixed-point algebra of the stable adjoint action by

$$\mathcal{A}_9^G = \{X \in \mathcal{A}_9 : \pi_9(g)X\pi_9(g)^* = X \quad \forall g \in G\}.$$

Proposition 456.1. *The fixed-point algebra coincides with the commutant:*

$$\mathcal{A}_9^G = \pi_9(G)'. \quad \square$$

Proof. Let $X \in \mathcal{A}_9^G$. Then

$$\pi_9(g)X\pi_9(g)^* = X.$$

Multiplying on the right by $\pi_9(g)$ yields

$$\pi_9(g)X = X\pi_9(g),$$

and hence

$$X \in \pi_9(G)'.$$

Conversely, if $X \in \pi_9(G)'$, then

$$\pi_9(g)X = X\pi_9(g),$$

so

$$\pi_9(g)X\pi_9(g)^* = X\pi_9(g)\pi_9(g)^* = X.$$

Thus

$$X \in \mathcal{A}_9^G.$$

Theorem 456.2. *The following conditions are equivalent:*

$$(i) \quad \mathfrak{S}_G = \left\{ \frac{1}{9} I_{\mathcal{H}_9} \right\},$$

$$(ii) \quad \mathcal{A}_9^G = \mathbb{C} I_{\mathcal{H}_9},$$

$$(iii) \quad \pi_9(G)' = \mathbb{C} I_{\mathcal{H}_9},$$

$$(iv) \quad \pi_9 \text{ is irreducible.}$$

Proof. By the preceding proposition,

$$\mathcal{A}_9^G = \pi_9(G)'.$$

Hence (ii) and (iii) are equivalent.

Assume (iii). If π_9 were reducible, there would exist a nonzero proper G -invariant subspace

$$W \subsetneq \mathcal{H}_9.$$

Since π_9 is unitary, $W^\perp$ is also G -invariant. Let Q_W be the orthogonal projection onto W . Then

$$Q_W \pi_9(g) = \pi_9(g) Q_W \quad (g \in G),$$

so

$$Q_W \in \pi_9(G)'.$$

But

$$Q_W \neq 0, \quad Q_W \neq I_{\mathcal{H}_9},$$

which contradicts

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

Therefore π_9 is irreducible.

Conversely, assume (iv). Schur's lemma gives

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9},$$

so (iii) follows.

It remains to relate (i) and (iii). If

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9},$$

then every invariant density operator belongs to the commutant and therefore has the form

$$\rho = \lambda I_{\mathcal{H}_9}.$$

The condition

$$\text{tr}(\rho) = 1$$

gives

$$9\lambda = 1,$$

hence

$$\rho = \frac{1}{9}I_{\mathcal{H}_9}.$$

Conversely, suppose that ρ_9 is the unique invariant density operator. If the commutant were larger than

$$\mathbb{C}I_{\mathcal{H}_9},$$

it would contain a non-scalar self-adjoint element A . Replacing A by

$$A - \frac{1}{9}\text{tr}(A)I_{\mathcal{H}_9},$$

we may assume

$$A = A^*, \quad \text{tr}(A) = 0, \quad A \neq 0.$$

Choose

$$0 < \varepsilon < \frac{1}{9\|A\|}.$$

Then

$$\rho_\varepsilon = \frac{1}{9}I_{\mathcal{H}_9} + \varepsilon A$$

is positive definite, because

$$\lambda_{\min}(\rho_\varepsilon) \geq \frac{1}{9} - \varepsilon\|A\| > 0.$$

Moreover,

$$\text{tr}(\rho_\varepsilon) = 1 + \varepsilon\text{tr}(A) = 1,$$

and A commutes with $\pi_9(G)$, so ρ_ε is G -invariant. This contradicts uniqueness. Therefore

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

□

457 The Commutant Criterion for Stable Uniqueness

Theorem 457.1. *Let*

$$A = A^* \in \mathcal{A}_g$$

satisfy

$$\pi_g(g)A = A\pi_g(g) \quad (g \in G).$$

If

$$\omega_g(A) = 0$$

and π_g is irreducible, then

$$A = 0.$$

Proof. The commutation relation gives

$$A \in \pi_g(G)'$$

Irreducibility and Schur's lemma imply

$$A = \lambda I_{\mathcal{H}_g}$$

for some

$$\lambda \in \mathbb{C}.$$

Since $A = A^*$,

$$\lambda \in \mathbb{R}.$$

Furthermore,

$$0 = \omega_g(A) = \lambda \omega_g(I_{\mathcal{H}_g}) = \lambda.$$

Thus

$$A = 0.$$

□

Corollary 457.2. *If π_g is irreducible, there is no nonzero self-adjoint G -invariant tangent direction to the affine hyperplane of normalized states at ρ_g .*

Proof. A self-adjoint tangent direction preserving the trace has

$$\text{tr}(A) = 0,$$

equivalently

$$\omega_g(A) = 0.$$

If it is G -invariant, then

$$A \in \pi_g(G)'.$$

The theorem gives

$$A = 0.$$

□

458 Reducibility Produces Additional Invariant States

Theorem 458.1. *If π_9 is reducible, then*

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}$$

is not the unique G -invariant density operator.

Proof. Choose a nonzero proper invariant subspace

$$W \subsetneq \mathcal{H}_9.$$

Let Q_W be the orthogonal projection onto W . Unitarity implies

$$Q_W \pi_9(g) = \pi_9(g) Q_W.$$

Set

$$d_W = \text{tr}(Q_W) = \dim_{\mathbb{C}} W.$$

Then

$$1 \leq d_W \leq 8.$$

The operator

$$\rho_W = \frac{1}{d_W} Q_W$$

satisfies

$$\rho_W \geq 0, \quad \text{tr}(\rho_W) = 1,$$

and

$$\pi_9(g) \rho_W \pi_9(g)^* = \rho_W.$$

Therefore

$$\rho_W \in \mathfrak{S}_G.$$

Since

$$\text{rank}(\rho_W) = d_W < 9,$$

whereas ρ_9 has rank 9, one has

$$\rho_W \neq \rho_9.$$

Thus uniqueness fails. □

Corollary 458.2. *The canonical stable state is unique among G -invariant density operators if and only if the stable representation is irreducible.*

Proof. The forward implication follows from the equivalence theorem above. The reverse implication follows from Schur's lemma and the same theorem. □

459 Invariant States and the Isotypic Commutant

Let

$$\mathcal{H}_9 = \bigoplus_{\alpha=1}^r \mathcal{H}_\alpha$$

be the isotypic decomposition, with

$$\mathcal{H}_\alpha \cong \mathbb{C}^{m_\alpha} \otimes \mathcal{K}_\alpha,$$

where the $\mathcal{K}_\alpha$ are pairwise inequivalent irreducible representations. Then

$$\pi_9(g) \cong \bigoplus_{\alpha=1}^r (I_{m_\alpha} \otimes \pi_\alpha(g)).$$

Theorem 459.1. *The positive elements of the commutant are precisely the operators of the form*

$$\rho = \bigoplus_{\alpha=1}^r (B_\alpha \otimes I_{\mathcal{K}_\alpha}),$$

where

$$B_\alpha \in M_{m_\alpha}(\mathbb{C}), \quad B_\alpha \geq 0.$$

Such an operator is a density operator if and only if

$$\sum_{\alpha=1}^r \operatorname{tr}(B_\alpha) \dim_{\mathbb{C}} \mathcal{K}_\alpha = 1.$$

Proof. The commutant decomposition gives

$$\pi_9(G)' \cong \bigoplus_{\alpha=1}^r (M_{m_\alpha}(\mathbb{C}) \otimes I_{\mathcal{K}_\alpha}).$$

Hence every positive element of the commutant has the stated form.

For each α ,

$$B_\alpha \geq 0$$

implies

$$B_\alpha \otimes I_{\mathcal{K}_\alpha} \geq 0.$$

Furthermore,

$$\operatorname{tr}(B_\alpha \otimes I_{\mathcal{K}_\alpha}) = \operatorname{tr}(B_\alpha) \dim_{\mathbb{C}} \mathcal{K}_\alpha.$$

Therefore

$$\operatorname{tr}(\rho) = \sum_{\alpha=1}^r \operatorname{tr}(B_\alpha) \dim_{\mathbb{C}} \mathcal{K}_\alpha.$$

The normalization condition is consequently exactly the displayed condition. □

Corollary 459.2. *If*

$$\pi_9$$

is irreducible, then

$$r = 1, \quad m_1 = 1, \quad \dim_{\mathbb{C}} \mathcal{K}_1 = 9,$$

and the invariant density operator is uniquely

$$\rho_9 = \frac{1}{9} I_{\mathcal{H}_9}.$$

Proof. Irreducibility means that there is exactly one irreducible summand and that it occurs with multiplicity one. Thus

$$\pi_9(G)' = \mathbb{C}I_{\mathcal{H}_9}.$$

The trace normalization then fixes the scalar uniquely:

$$\rho_9 = \frac{1}{9}I_{\mathcal{H}_9}.$$

□

460 The Stable State as a Central Normalized Trace

Theorem 460.1. *The functional*

$$\omega_9(X) = \frac{1}{9} \operatorname{tr}_{\mathcal{H}_9}(X)$$

is the unique normalized tracial state on

$$\mathcal{A}_9 = M_9(\mathbb{C}).$$

Proof. The normalized trace is a positive unital tracial functional. Let

$$\varphi : \mathcal{A}_9 \longrightarrow \mathbb{C}$$

be another normalized tracial state. Let

$$E_{ij}$$

be matrix units. By traciality,

$$\varphi(E_{ij}) = \varphi(E_{ji}E_{ij}) = \varphi(E_{ij}E_{ji}) = \varphi(E_{ji}).$$

More directly, for $i \neq j$,

$$E_{ij} = E_{ii}E_{ij}E_{jj},$$

and cyclicity gives

$$\varphi(E_{ij}) = \varphi(E_{jj}E_{ii}E_{ij}) = 0,$$

because

$$E_{jj}E_{ii} = 0.$$

Thus

$$\varphi(E_{ij}) = 0 \quad (i \neq j).$$

For the diagonal matrix units,

$$\varphi(E_{ii}) = \varphi(E_{ij}E_{ji}) = \varphi(E_{ji}E_{ij}) = \varphi(E_{jj}),$$

so all diagonal values are equal. Since

$$\sum_{i=1}^9 E_{ii} = I$$

and

$$\varphi(I) = 1,$$

one obtains

$$\varphi(E_{ii}) = \frac{1}{9}.$$

Therefore, for

$$X = \sum_{i,j=1}^9 x_{ij} E_{ij},$$

one has

$$\varphi(X) = \frac{1}{9} \sum_{i=1}^9 x_{ii} = \frac{1}{9} \operatorname{tr}(X) = \omega_9(X).$$

Hence

$$\varphi = \omega_9.$$

□

461 The Stable Adjoint Action Preserves the Full Operator Geometry

Define

$$\alpha_g(X) = \pi_9(g) X \pi_9(g)^*.$$

Proposition 461.1. *For every $g \in G$, the map*

$$\alpha_g : \mathcal{A}_9 \longrightarrow \mathcal{A}_9$$

is a trace-preserving $$ -automorphism.*

Proof. For

$$X, Y \in \mathcal{A}_9,$$

one has

$$\alpha_g(XY) = \pi_9(g) XY \pi_9(g)^* = \alpha_g(X) \alpha_g(Y).$$

Moreover,

$$\alpha_g(X^*) = \pi_9(g) X^* \pi_9(g)^* = \alpha_g(X)^*.$$

The inverse map is

$$\alpha_{g^{-1}},$$

so α_g is an automorphism.

Finally,

$$\omega_9(\alpha_g(X)) = \frac{1}{9} \operatorname{tr}(\pi_9(g) X \pi_9(g)^*) = \frac{1}{9} \operatorname{tr}(X) = \omega_9(X).$$

Thus the normalized trace is preserved. □

Theorem 461.2. *The adjoint representation*

$$\alpha : G \longrightarrow \operatorname{Aut}(\mathcal{A}_9)$$

preserves the decomposition

$$\mathcal{A}_9 = \mathbb{C} I_{\mathcal{H}_9} \oplus \mathcal{A}_9^0.$$

Proof. For every $g \in G$,

$$\alpha_g(I_{\mathcal{H}_9}) = \pi_9(g) I_{\mathcal{H}_9} \pi_9(g)^* = I_{\mathcal{H}_9}.$$

Thus the scalar summand is invariant.

If

$$X \in \mathcal{A}_9^0,$$

then

$$\omega_9(X) = 0.$$

Trace preservation gives

$$\omega_9(\alpha_g(X)) = \omega_9(X) = 0.$$

Hence

$$\alpha_g(X) \in \mathcal{A}_9^0.$$

Therefore both summands are invariant. $\square$

Corollary 461.3. *The non-scalar operator content of the stable sector is contained in the 80-dimensional space*

$$\mathcal{A}_9^0 = \{X \in \text{End}_{\mathbb{C}}(\mathcal{H}_9) : \text{tr}(X) = 0\},$$

and this space is invariant under the adjoint action of G .

Proof. The dimension statement follows from

$$\dim_{\mathbb{C}} \mathcal{A}_9 = 9^2 = 81$$

and the one-dimensional scalar subspace. Invariance was established above. $\square$

Epigraph

L'autore propone un possibile linguaggio matematico; la struttura, se reale, precede la sua rappresentazione e manifestazione.

Author's Claim

Prima parlano i dati. Poi, attraverso i dati, parla Alain Faure.

Author Identity

The fundamental identity of the author is expressed by the structural tuple serving as the author's formal logo: $U_F = (\mathcal{M}, \mathcal{G}, \mathcal{A}, \Psi, \Lambda, \Pi)$.

Author Information

Alain Faure

Master Architect — Singlepoint Mind — Q.E.D.

Formal Identity (Logo): $U_F = (\mathcal{M}, \mathcal{G}, \mathcal{A}, \Psi, \Lambda, \Pi)$

Official Repository: AUF2026 (GitHub)

Official Publications: @auf2026 (Medium)

ORCID: 0009-0009-5333-1181

All mathematical structures and operators introduced in this manuscript are developed within **AUF2026 — AOS Universal Framework**, the author's unified formal system. In accordance with the **AUF2026** program, the framework constitutes a pre-representational mathematical architecture from which the structural phenomena studied here arise.